PHYSICS WALLAH

ICSE BOARD ENDGAME

Complete Revision Notes

Chapter-wise • Exam Oriented

SCIENCE

MATHS

REVISION SERIES BOOK

VIDYAPEETH
IIT-JEE | NEET | FOUNDATION

CLASS 11 - JEE/NEET OFFLINE COACHING

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• 2 YEARS MOST STRUCTURED •
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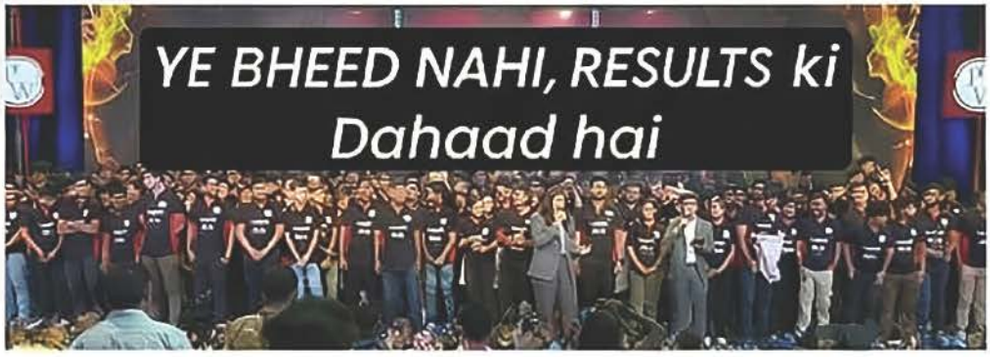

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PHYSICS

1 CHAPTER

Force

Force

A force is that physical cause which change (or tends to change) either the size or shape or the state of rest or motion of the body.

S.I. unit = Newton (N)

CGS unit = dyne

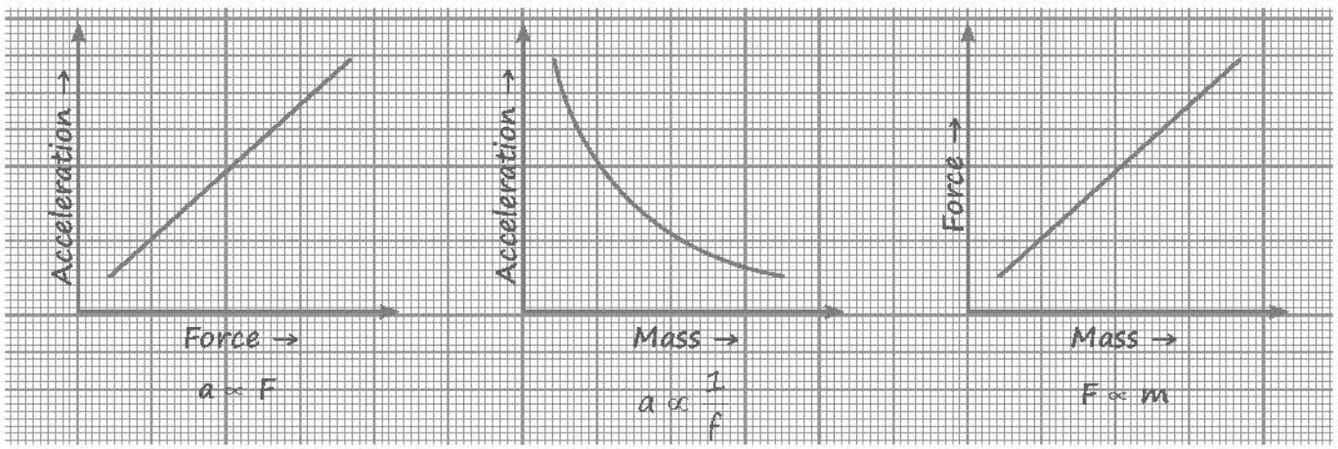
$1N = 10^5$ dynet

Kinds of force

- Contact
- Non-Contact

Newtons Laws of Motion

FIRST LAW	SECOND LAW	THIRD LAW
A body continues to be in its state of rest or of uniform motion is a straight line unless an external force is applied on it.	The rate of change of momentum of a body is directly proportional to the force applied on it and this change in momentum takes place in the direction of applied force.	To every action there is alway an equal and opposite reaction.
This law is also called "LAW OF INERTIA"	$F \propto \frac{\Delta p}{\Delta t} = \frac{m\Delta v}{\Delta t}$ $F = ma$	$\overrightarrow{F_{12}} = -\overrightarrow{F_{21}}$



Momentum

The momentum of a body in linear motion is the product of the mass of the body and its velocity.

$$p = mv$$

It is vector quantify.

S.I. unit = kg m s^{-1}

C.G.S. unit = g cm s^{-1}

Equations of Motion

(i) $v = u + at$

(ii) $s = ut + \frac{1}{2}at^2$

(iii) $v^2 = u^2 + 2as$

Moment of Force

The turning effect produced by a force on a rigid body about a point, pivot or fulcrum is called moment of force.

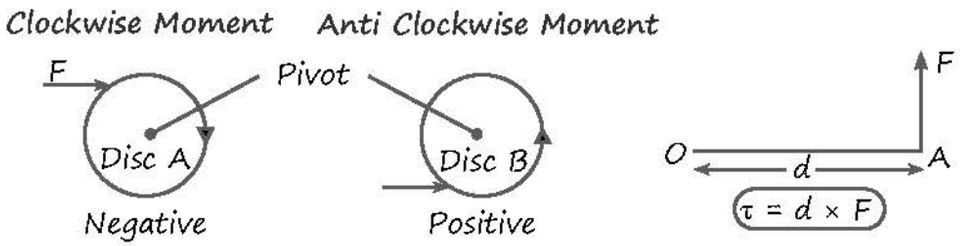
S.I unit ⇒ Nm

C.G.S unit ⇒ Dyne cm

1Nm = 10^7 Dyne cm

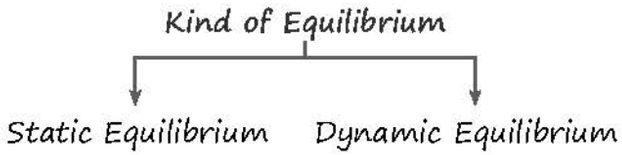
1kgf m = 9.8 N m

1gf cm = 980 dyne cm



Equilibrium of Bodies

When a number of force acting an a body produce no change in its state of rest or of motion the body is said to be in equilibrium.



Conditions for Equilibrium

(i) The resultant of all the forces acting an a body should be 0.

(ii) Algebraic sum of moments acting on a body should be 0.

Principle of Moments

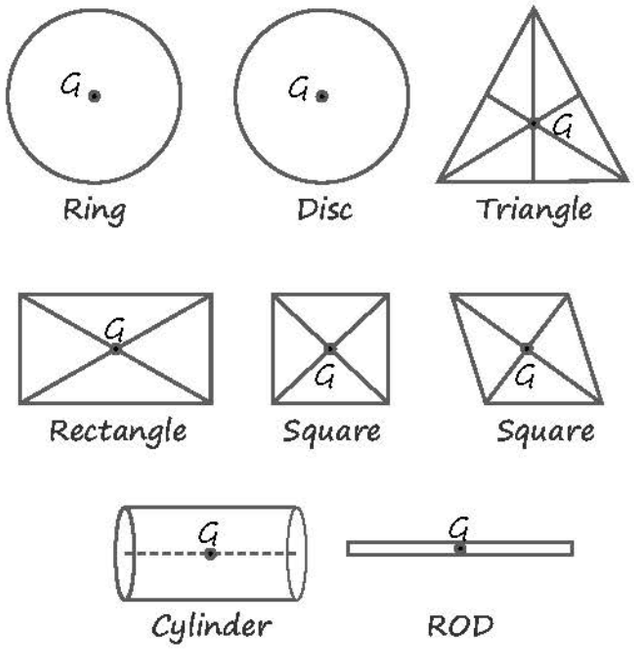
According to principle of moments in equilibrium.

Sum of anticlockwise moment = Sum of clockwise moment

Centre of Gravity

The centre of gravity (C.G.) of a body is the point about which the algebraic sum of moments of weights of all the particles constituting the body is zero.

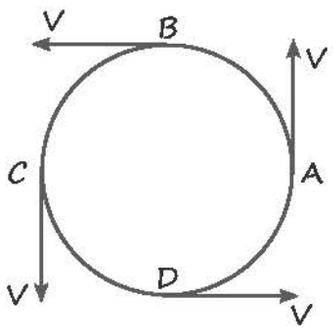
The entire weight of the body can be considered to act at this point.



Uniform Circular Motion

When a particle moves with constant speed in a circular path, its motion is said to be the uniform circular motion.

It is accelerated motion.



Centripetal Force	Centrifugal Force
A force that acts towards the center of circle, keeping object moving in circular path.	A force that acts away from center of a circle.
It is a real force.	It doesn't really exist.

Formula sheet

1. Momentum (p)

   $p = mv$

   p-momentum, m-mass, v-velocity

2. Force (F)

   $F = ma$

   F-force, m-mass, a-acceleration

3. Equations of motion

   u-initial velocity

   v-final velocity

   t-time

   a-acceleration

s-distance of travelled

$$\boxed{\begin{aligned} v &= u + at \\ s &= ut + \frac{1}{2}at^2 \\ v^2 &= u^2 + 2as \end{aligned}}$$

4. Moment of force (τ)

$\boxed{\tau = d \times F}$

d-perpendicular distance, F-force from pivot

2 CHAPTER

Work, Energy and Power

Work

Work is said to be done only when the force applied on a body makes the body move.

It is scalar quantity

F – applied force

s – displacement

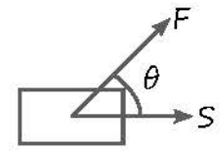
θ – between force and displacement

S.I. unit = joule(J)

C.G.S. unit = erg

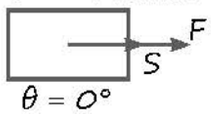
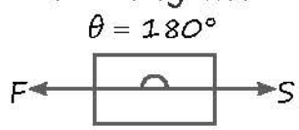
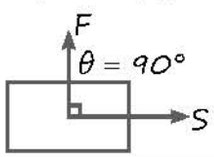
$1J = 10^7$ erg

$1kJ = 10^3$ J, $1MJ = 10^6$ J, $1GJ = 10^9$ J



$W = FS\cos\theta$

Different Cases of Work Done

W → Positive	W → Negative	W → Zero
		

Power

The rate of doing work is called power. It is a scalar quantity.

$$\text{Power } P = \frac{\text{Work done } w}{\text{Time taken } t}$$

$$P = \frac{w}{t}$$

$$P = \frac{F \times s}{t} = F \times v$$

F → force v → Average speed

S.I. unit = watt (w)

C.G.S. unit = erg s^{-1}

$1W = 10^7$ erg s^{-1}

1 H.P. = 746W = 0.746 kW

Energy

The energy of a body is its capacity to do work.

S.I. unit = joule (J)

C.G.S. unit = erg

Bigger unit of energy kilowatt hour (kWh)

1kWh = 3.6×10^6J = 3.6 MJ

Heat energy is usually measured in calorie.

1J = 0.24 calories, 1 calorie = 4.18 J

The energy of atomic particles is measured in electron volt (eV).

1eV = 1.6×10^{-19}J

WORK	ENERGY	POWER
It is equal to product of force & displacement in direction of force.	Capacity to do work.	The rate of doing work.
Does not depend on time.	Does not depend on time.	Depends on time.
S.I. unit → joule	S.I. unit → joule	S.I. unit → watt

Different forms of Energy

Mechanical Energy- The energy possessed by a body due to its state of rest or of motion is caused mechanical energy. It is in two forms:

(i) Kinetic energy

(ii) Potential energy

Kinetic energy- Energy possessed by a body by virtue of its state of motion.

$$K = \frac{1}{2}mv^2$$

k → kinetic energy

m → mass

v → velocity

There are 3 forms of K.E.

1. Translational K.E.
2. Rotational K.E.
3. Vibrational K.E.

Relation between K.E. & momentum

$$p = \sqrt{2mk} \quad \text{or} \quad K = \frac{p^2}{2m}$$

Work energy theorem- The work done by a force on moving body is equal to the increase in its kinetic energy.

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Potential energy- Energy possessed by a body by virtue of its specific position (or changed configuration) is called potential energy.

It is usually denoted by symbol U.

It is mainly of two types-

(i) Gravitational P.E.

(ii) Elastic P.E.

+ **Gravitational P.E.** is the amount of work done in raising a body from ground to a point at height h, against gravity.

  $u = mgh$

  m - mass of object, g - acceleration clue to gravity h - height through which object is raised

  h - height through which object is raised

1. **Solar energy**- energy radiated by Sun.
2. **Heat energy**- energy released on burning coal, oil, wood, gas.
3. **Light energy**- energy in presence of which other objects are seen.
4. **Chemical energy**- Energy possessed by fossil fuels.
5. **Hydro energy**- Energy of fast moving water.
6. **Electrical energy**- Energy possessed by charged body due to movement of free electrons.
7. **Nuclear energy**- Energy released due to process of nuclear fusion or fusion.

Energy Conversion

(i) Mechanical → electrical = Electric generator (dynamo)

(ii) Electrical → mechanical = fan, washing machine, mixer

(iii) Electrical → heat = heater, oven, geyser, toaster

(iv) heat → electrical = thermocouple

(v) electrical → sound = loudspeaker

(vi) sound → electrical = microphone

(vii) light → chemical = photosynthesis

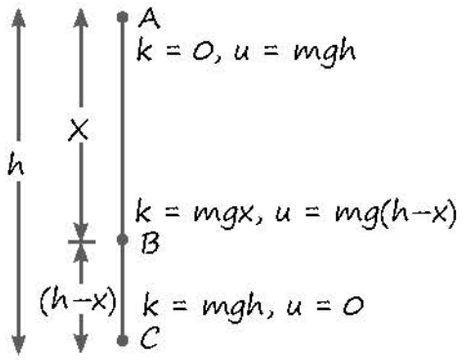
(viii) light → electrical = photoelectric cell

Energy Sources

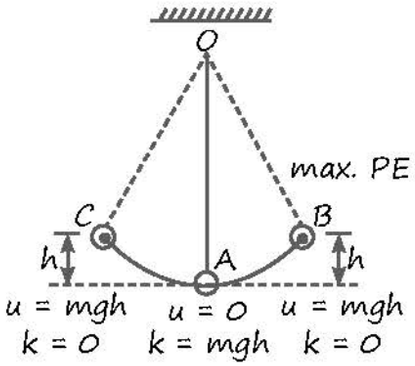
Renewable or non conventional source	Non-Renewable or conventional source
A natural source providing us energy continuously is called renewable source of energy.	These are the sources from which energy cannot be continuously obtained over a long period of time.
These sources can be regenerated.	These sources cannot be regenerated.
E.g. Sun, wind, flowing water, tides, Oceans, geothermal spots, nuclear fuel.	E.g. - Coal, petroleum, natural gas

Law of Conservation of Energy

Energy can neither be created nor can it be destroyed. It only changes from one form to another.



Free fall of body



motion of simple pendulum

Formula Sheet

(i) Work(w)

$$W = FS\cos\theta$$

(ii) Power(p)

$$P = \frac{w}{t} \qquad P = F \times V$$

p → power, w → work, t → time f → force, v → avg. speed

(iii) Kinetic energy (K)

$$K = \frac{1}{2}mv^2 \qquad K = \frac{p^2}{2m}$$

K-K.E., m → mass, v → velocity, p → momentum

(iv) Gravitation P.E. (U)

$$U = mgh$$

m → mass, g → acceleration due to gravity, h → height

(v) Work energy theorem.

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

3 CHAPTER

Machine

Machine

It is a device by which we can either overcome a large resistive force (or load) at some point by applying a small force (or effort) at a convenient point and in a desired direction or by which we can obtain a gain in speed.

Uses of Machine

Multiply force

To change direction of the force

To increase or decrease speed

Terms Related to Machine

(i) **Load**: The resistive or opposing force to be overcome, by machine.

(ii) **Effort**: The force applied on machine to overcome load.

(iii) **Mechanical advantage (M.A.)**: Ratio of the load (L) to the effort (E)

$$\text{M.A.} = \frac{\text{Load}(L)}{\text{Effort}(E)}$$

It has no unit

M.A. > 1 $\Rightarrow$ force multiplier

M.A. < 1 $\Rightarrow$ gains speed

M.A. = 1 $\Rightarrow$ change in direction of effort.

(iv) **Velocity Ratio (V.R.)**: Ratio of distance moved by effort (d_E) to the distance moved by load (d_L).

$$\text{V.R.} = \frac{\text{Velocity of effort } (V_E)}{\text{Velocity of load } (V_L)}$$

$$\text{V.R.} = \frac{V_E}{V_L} = \frac{d_E}{d_L}$$

It has no unit

V.R. < 1 $\Rightarrow$ gains speed

V.R. > 1 $\Rightarrow$ force multiplier

V.R. = 1 $\Rightarrow$ changes direction of effort

(v) **Work Input Winput:** The work done on the machine by the effort.

(vi) **Work Output Woutput:** The work done by machine on the load.

(vii) **Efficiency (η):** It is the ratio of work done on load by machine to the work done on machine by effort.

$$\text{Efficiency}(\eta) = \frac{\text{Work output}}{\text{Work input}}$$

It can be expressed in percentage, it has no unit.

$$\eta = \frac{W_{output}}{W_{input}} \times 100$$

Ideal Machine

An ideal machine is that in which there is no loss of energy in any manner. The efficiency of ideal machine is 100%.

Relation Between η, M.A. and V.R.

$$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$

Machine

Lever Pulley

Levers

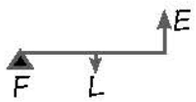
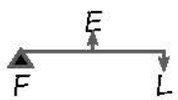
A lever is a rigid straight (or bent) bar which is capable of turning about a fixed axis.

According to principle of lever,

Movement of load about fulcrum = Movement of effort about fulcrum

$$L \times d_L = E \times d_E$$

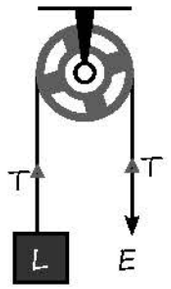
Kinds of Levers

Class I	Class II	Class III
		
M.A. = 1, M.A. < 1, M.A. > 1	M.A. > 1	M.A. < 1
V.R. = 1, V.R. < 1, V.R. > 1	V.R. > 1	V.R. < 1

E.g., A pair of scissors, handle of water pump, a catapult, nodding of human head	E.g., Bottle opener, wheel barrow, paper cutter, raising weight of human body on toes.	E.g., Sugar tongs, foot treadle, spade lifting load, forearm used for lifting a load.

Pulley: It is a wheel on an axle that is designed to support movement and change the direction of a cable or belt along its circumference. A set of pulleys assembled so that they rotate independently on the same axle to form a block is called pulley system.

Single Fixed Pulley: A pulley which has its axis of rotation fixed in position.



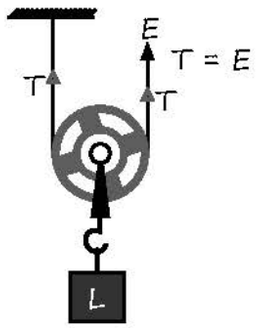
$$M.A. = \frac{Load}{Effort} = \frac{T}{T} = 1$$

$$V.R. = \frac{d_E}{d_L} = \frac{d}{d} = 1$$

$$\eta = \frac{M.A.}{V.R.} = 1 \text{ or } 100\%$$

with the use of single fixed pulley effort can be applied in more convenient direction.

Single Movable Pulley: A pulley whose axis of rotation is not fixed.



It is used as force multiplier

$$M.A. = \frac{Load}{Effort} = \frac{2T}{T} = 2$$

$$V.R. = \frac{d_E}{d_L} = \frac{2d}{d} = 2$$

$$\eta = \frac{M.A.}{V.R.} = \frac{2}{2} = 1 \text{ or } 100\%$$

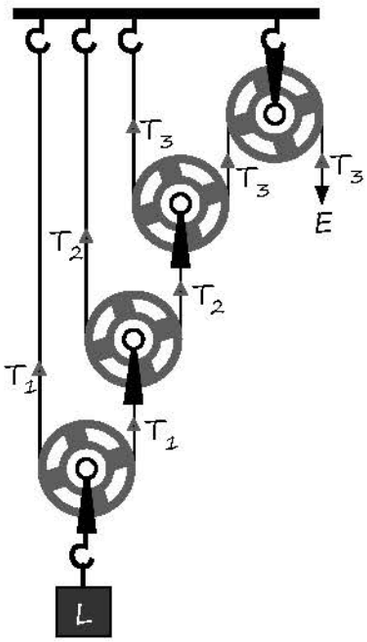
Combination of pulleys

(1) using one fixed pulley and other movable pulleys:

$M.A. = 2^n$

$V.R. = 2^n$

$n \Rightarrow$ number of movable pulleys



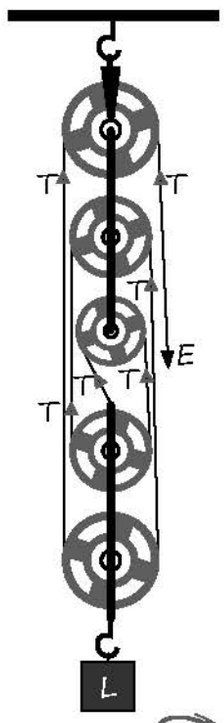
(2) Using several pulleys in two blocks (block and tackle system)

$$M.A. = \frac{Load}{Effort} = \frac{nT}{T} = n$$

$$V.R = \frac{d_E}{d_L} = \frac{nd}{d} = n$$

$$\text{Efficiency } \eta = \frac{M.A.}{V.R.} = \frac{n}{n} = 1 \text{ or } 100\%$$

n = total number of pulleys



Effect of weight of pulleys on M.A., V.R. and η

$$M.A. = \frac{L}{E} = \frac{nE - W}{E} = n - \frac{W}{E}$$

$n \Rightarrow$ no. of pulleys, $w \Rightarrow$ weight of lower block (total)

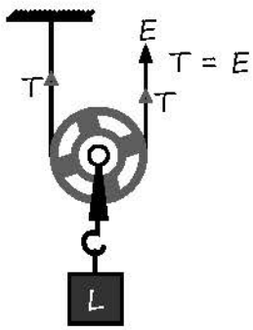
Formula Sheet

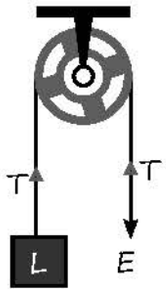
(1) Mechanical advantage:

$$M.A. = \frac{Load(L)}{Effort(E)}$$

(2) Velocity Ratio:

$$V.R. = \frac{\text{velocity of effort } (V_E)}{\text{velocity of load } (V_L)} = \frac{d_E}{d_L}$$





(3) Efficiency

$$\eta = \frac{W_{output}}{W_{input}}$$

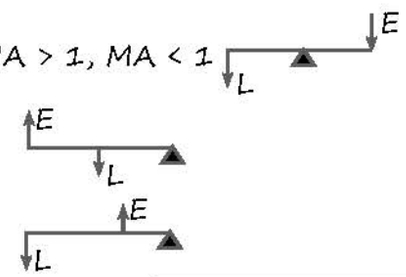
$$In\%\ \eta = \frac{W_{output}}{W_{input}} \times 100$$

(4) $\eta = \frac{M.A.}{V.R.}$

(5) Principle of Lever $\Rightarrow L \times d_L = E \times d_E$

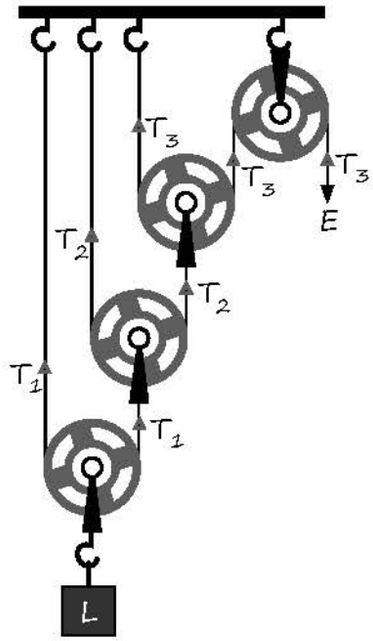
(6)

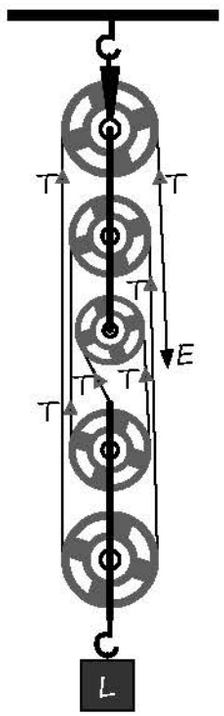
Kinds of Lever
- Class I MA = 1, MA > 1, MA < 1
- Class II MA > 1
- Class III MA < 1



$$M.A. = \frac{\text{Load Arm}}{\text{Effort Arm}}$$

$$V.R. = \frac{d_E}{d_L}$$





(7) Single Fixed Pulley

M.A. = 1

V.R. = 1

η = 1 or 100%

(8) Single Movable Pulley

M.A. = 2

V.R. = 2

η = 1 or 100%

(9) Combination of pulleys

1 fixed & n movable pulleys

M.A. = 2^n

V.R. = 2^n

(10) Block and tackle system

n number of pulleys

M.A. = n

V.R. = n

If total weight of pulleys in lower back is W

$$M.A. = n - \frac{W}{E}, \quad V.R. = n$$

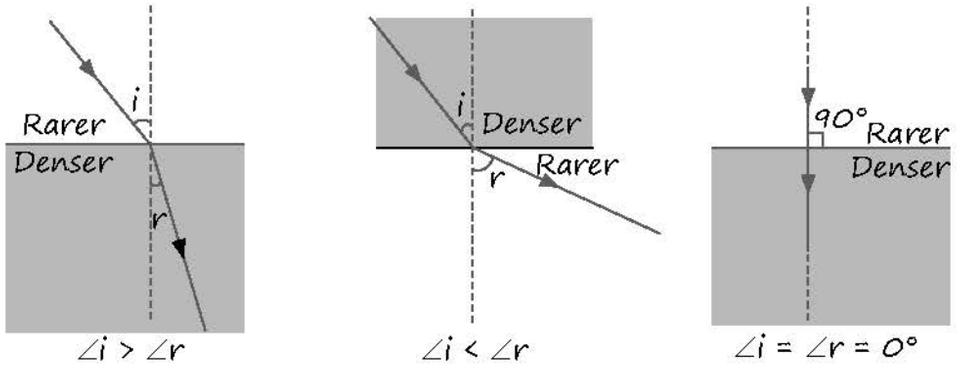
$$\eta = 1 - \frac{W}{nE}$$

4 CHAPTER

Refraction of Light at Plane Surface

Refraction of Light

The change in the direction of the path o light, when it passes from one transparent medium to another transparent medium is called refraction.



Refracted beam has change in speed (v), wavelength (λ), direction but frequency (υ)remains constant.

Laws of Refraction

(1) The incident ray, the refracted ray and the normal at the point of incidence all lie in same plane.

(2) The ratio of sine of angle of incidence (i) to the sine of the angle of refraction (r) is constant for given pair of media.

$$\frac{\sin i}{\sin r} = \text{constant} = {}_1\mu_2$$ (R.I. of second medium w.r.t. first medium)

This is also known as Snell's law.

Refractive Index

The refractive index of second medium with respect to the first medium is defined as the ratio of sine of angle of incidence in first medium to the sine of angle of refraction in the second medium.

R.I. or absolute R.I. has no unit.

$$\mu = \frac{\text{Speed of light in vacuum or air (c)}}{\text{Speed of light in that medium (v)}}$$

$$\mu = \frac{c}{v} = \frac{f\lambda_{air}}{f\lambda_{medium}} = \frac{\lambda_{air}}{\lambda_{medium}}$$

Relative refractive index

It is the ratio of speed of light in one medium to the speed of light in second medium.

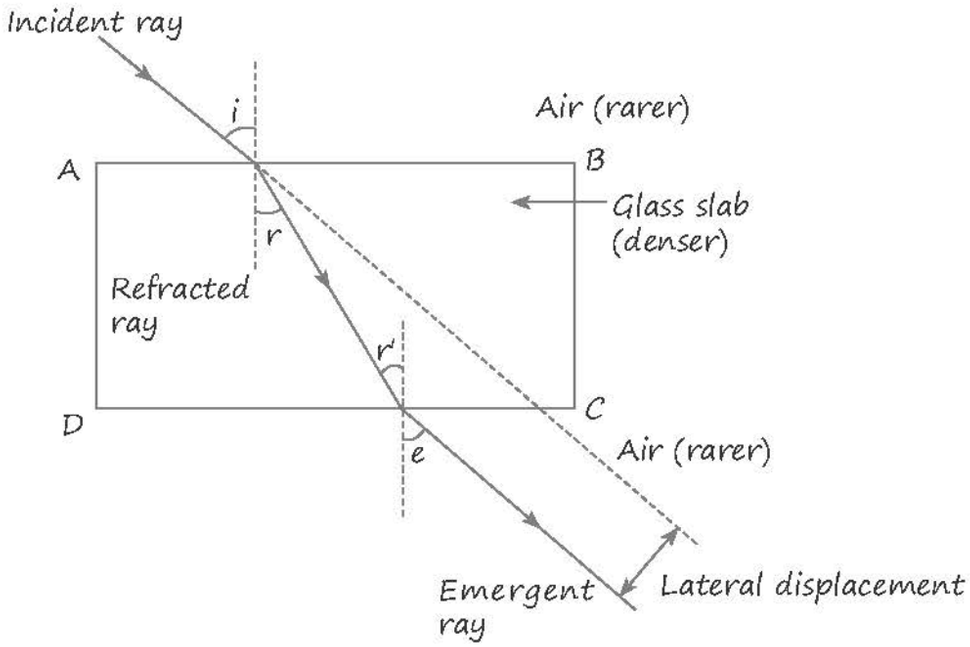
$${}_1\mu_2 \text{ or } n_{21} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$$

Note: $${}_1\mu_2 = \frac{1}{{}_2\mu_1}$$

Factors affecting refractive index of a medium

(1) Nature of medium (optical density)
(2) Temperature of medium (temperature ↑ v↑ μ↓)
(3) Colour or wavelength of light ($\mu_v > \mu_R$)

Refraction of Light Through a Glass Slab

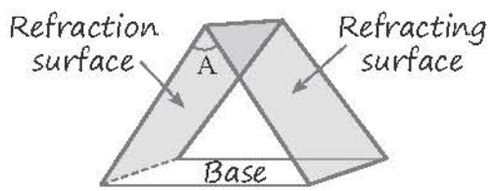


Lateral displacement depends upon

(1) Thickness of glass block
(2) Angle of incidence
(3) Refractive index of glass

$$\text{Lateral displacement} = \frac{t\sin(i-r)}{\cos r}$$

Refraction of light through prism



Prism: It is a transparent medium bounded by five plane surfaces with a triangular cross section.

The different relation $\angle$ in prism

(i) $\delta = (\angle i + \angle e) - (\angle r_1 + \angle r_2)$

(ii) $\angle i + \angle e = \angle A + \delta$

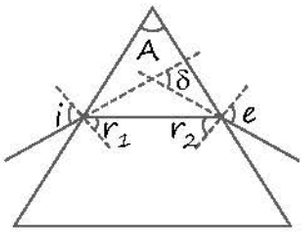
(iii) $\angle r_1 + \angle r_2 = \angle A$

i = angle of incidence

e = angle of emergence

δ = angle of deviation

r_1, r_2 = angle of refraction



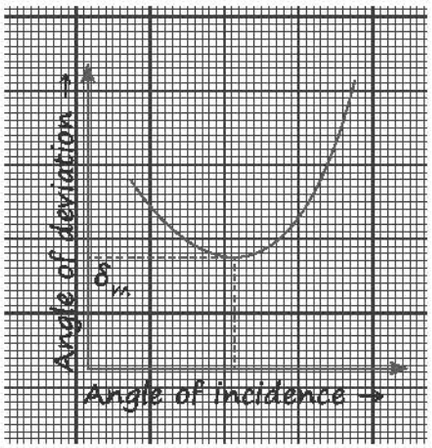
Factors affecting δ:

(1) angle of incidence ($\angle i$)

(2) material of prism (μ)

(3) angle of prism ($\angle A$)

(4) colour of wavelength of light (λ)



Relation between ∠i and ∠s:

In the position of minimum deviation the refracted ray inside the prism is parallel to its base, if prism is equilateral.

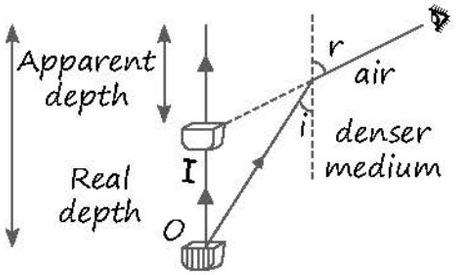
$\delta = \delta_m$, $\angle i = \angle e$, $\delta_m = 2i - A$, $\delta = (\mu - 1)A$

Application of Refraction of Light

Real and apparent depth: An object placed in a denser medium when view from a rarer medium appears to be at a depth lesser than its real depth.

Shift OI = Real depth − Apparent depth

$$\text{Shift} = \text{real depth}\left(1 - \frac{1}{{}_a\mu_m}\right)$$



Shift by which the object appears to be raised depends:

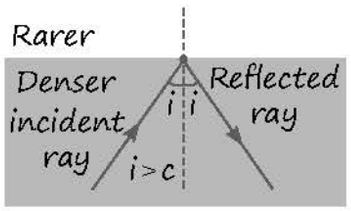
(1) Refractive index of medium $\mu\uparrow$ Shift$\uparrow$

(2) Thickness of medium$\uparrow$ Shift$\uparrow$

(3) Wavelength (color) of light $\lambda\uparrow$ shift$\downarrow$ $\mu_V > \mu_R$ shift V > shift R

Total Internal Reflection

When a ray of light travelling in a denser medium is incident at the surface of rarer medium at the angle of incidence greater than critical angle for the given pair of media, the ray is totally reflected back into the denser medium.



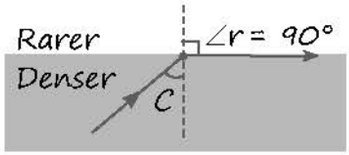
Condition for TIR

(1) Light must travel from denser to rarer medium

(2) $\angle i > \angle c$

Critical angle

The angle of incidence in denser medium for which the angle of refraction in rarer medium is 90°. It is denoted by c.



Factors affecting c

(1) Colour of light $\lambda\uparrow$ $c\uparrow$

(2) Temperature$\uparrow$ $c\uparrow$

Relation between critical angel and refractive index

R.I. of rarer medium w.r.t. denser medium $= {}_d\mu_r = \frac{\sin i}{\sin r}$

If $\angle i = \angle c, \angle r = 90°$ ${}_d\mu_r = \frac{\sin c}{\sin 90°} \Rightarrow {}_d\mu_r = \sin c$

R.I. of denser medium w.r.t. rarer medium

For glass ${}_a\mu_g = \frac{3}{2}$

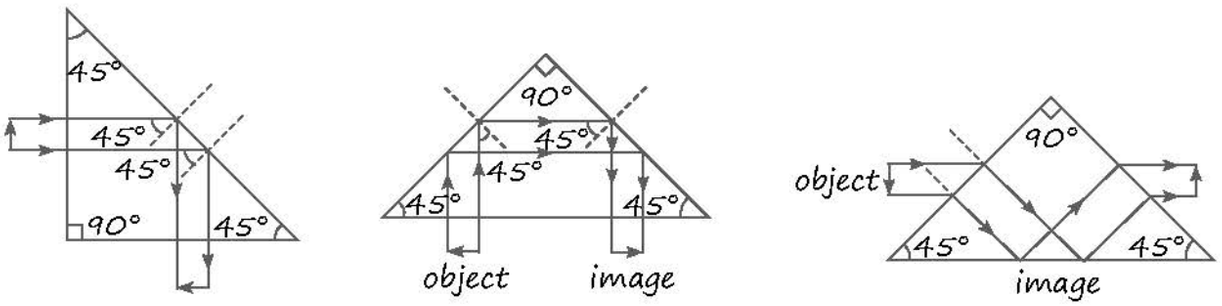
${}_r\mu_d = \frac{1}{\sin c}$, $c = \sin^{-1}\left(\frac{2}{3}\right) = 42°$

Total internal reflection in a prism

(a) To deviate light through 90°: This action of prism is used in periscope, where a total reflecting prism is preferred over a plane mirror.

(b) To deviate light through 180°: This action is used in binocular and camera to invert the image without the loss intensity.

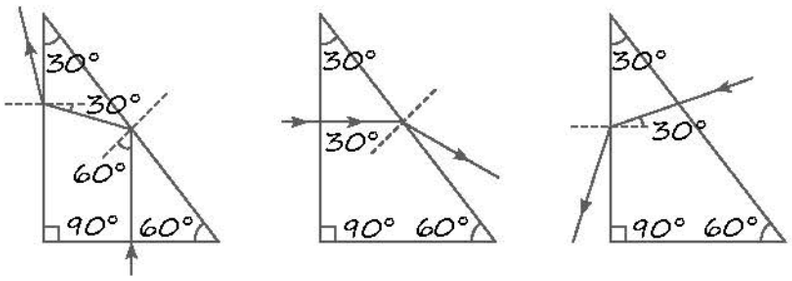
(c) To erect the inverted image with deviation: A prism used in this manner is called erecting prism. This action of prism is used in a slide projector.



(a) (b) (c)

Prim of angle (30°, 60° and 90°)

It can be used to deviate a light ray through an angle less than 60° by TIR.



Some Consequences of Total Internal Reflection

(1) On hot sunny day, a driver may see a pool of water (or wet road) in front of him at some distance. (Mirage)

(2) A crack in glass vessel often shines like mirror.

(3) A piece of diamond sparkles when viewed from certain directions.

(4) An optical fibre is used to transmit a light signal over a long distance without much loss of energy.

Formula Sheet

(1) $_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2}$, $\mu = \frac{c}{v}$, $\mu = \frac{\text{Real depth}}{\text{Apparent depth}}$

(2) $\text{Lateral displacement} = \frac{t\sin(i-r)}{\cos r}$

(3) R.I. of prism $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$, $\delta_{min} = 2i - A$, $\delta = (\mu - 1)A$

(4) $\text{Shift} = \text{real depth}\left[1 - \frac{1}{{}_a\mu_m}\right]$

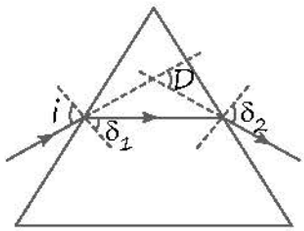
(5) $\mu = \frac{1}{\sin c}$

5 CHAPTER

Spectrum

Deviation Produced by Triangular Prism

Light ray while passing through a prism gets deviated to time while passing through the two refracting surface. The total deviation (D) with respect to incident ray is given as:



$$D = \delta_1 + \delta_2$$

Depends on

1. $\angle i$ at first surface
2. Angle of prism(A)
3. R.I. of prims (color of light)

Dependence of angle deviation on colour of light (λ)

For different colours of light the R.I. of a given transparent medium is different.

$$\lambda\uparrow \ \mu\downarrow$$

$$\lambda_{red} > \lambda_{violet} \quad \therefore \mu_{violet} > \mu_{red}$$

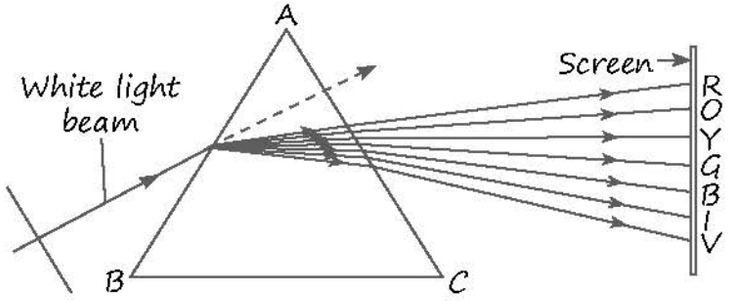
As a result prism deviate violet light most and red light least.

$$\delta_{violet} > \delta_{red}$$

Dispersion of White Light by Glass Prism

The phenomenon of splitting of white light into its constituent colours when it passes through a prism is called dispersion.

This band of seven colours obtained VIBGYOR, (V = violet, I = indigo, B = blue, G = green, Y = yellow, O = orange, R = red) is called spectrum.



Dispersion of white light occurs at the first surface of prism.

Deviation of light occurs at both the surfaces of prism.

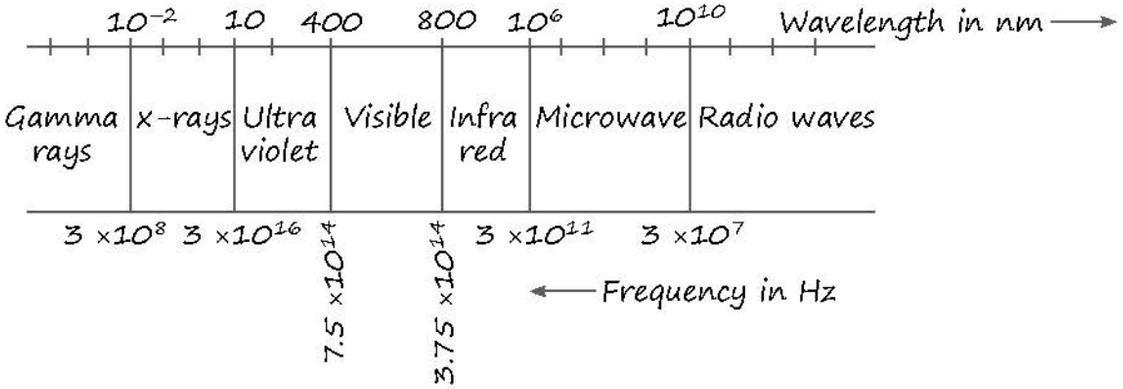
The prism does not produce colours but it only splits the various colours present in the light incident on it.

Wavelength and frequency of different colours of white light

Colour	Wavelength range (Å)	Frequency range (1014)
Violet	4000 – 4460	7.5 – 6.73
Indigo	4460 – 4640	6.73 – 6.47
Blue	4640 – 5000	6.47 – 6.01
Green	5000 – 5780	6.01 – 5.19
Yellow	5780 – 5920	5.19 – 5.07
Orange	5920 – 6200	5.07 – 4.84
Red	6200 – 8000	4.84 – 3.75

Electromagnetic Spectrum and Its Broad Classification

The orderly distribution of electromagnetic radiations according to their wavelength or frequency is called electromagnetic spectrum.



(1) Electromagnetic wave do not require material medium for their propagation.
(2) They travel with same speed in vacuum ($c = 3 \times 10^8$ m/s)
(3) Exhibit property of reflection and refraction.
(4) They do not deflect by electric and magnetic fields.
(5) These are transverse waves.

Properties and uses of different rays

Infrared rays:

- Heat waves, i.e., produced from heat radiation bodies and molecules.
- High penetration power – frequency $\Rightarrow 3 \times 10^{11}$ Hz to 4×10^{14}Hz

Uses:

- Physical therapy
- Weather forecasting
- Solar water heat, solar cells and cooker
- Satellite for army purpose
- Producing dehydrated fruits

Ultraviolet rays:

- Produced by special lamps and very hot bodies.
- Sun is important source of UV rays.
- Frequency 10^{14} to 10^{16} Hz

Uses:

- Burglar alarm
- To study molecular structure
- Surgical instruments (to sterilise)
- Checking mineral sample
- To kill germs in water

Relation between frequency, speed and wavelength of electromagnetic spectrum

$$\text{frequency}(\nu) = \frac{\text{speed of EM wave}(v)}{\text{wavelength}(\lambda)}$$

Scattering of Light

Scattering is the process of absorption and then re-emission of light energy.

The air molecules of size smaller than the wavelength of incident light absorb the energy of incident light and then re-emit it without change in its wavelength. The air molecules of size bigger than wavelength of incident light scatter the light of all wavelengths of white light to he same extent.

$$\boxed{I \propto \frac{1}{\lambda^4}}$$

$I \rightarrow$ Intensity of scattered light, $\lambda \rightarrow$ Wavelength of light

Blue Colour of Sky

The sky appears blue during the day because the particles in the atmosphere scatter shorter wavelengths of light particularly blue more than the longer wavelengths like red. This scattered blue light reaches our eyes.

Formula Sheet

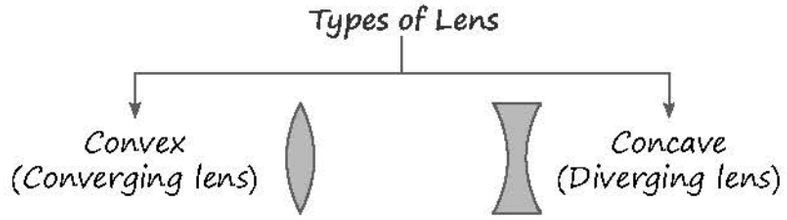
- $\text{frequency} = \frac{\text{speed}}{\text{wavelength}}$
- $I \propto \frac{1}{\lambda^4}$
- $\lambda\uparrow \Rightarrow \nu\uparrow \Rightarrow \mu\downarrow \Rightarrow \delta\uparrow$

6 CHAPTER

Refraction Through a Lens

Lens

A lens is a transparent refracting medium bounded by either the two spherical surfaces or one surface spherical and other surface plane.



Terms related to a Lens

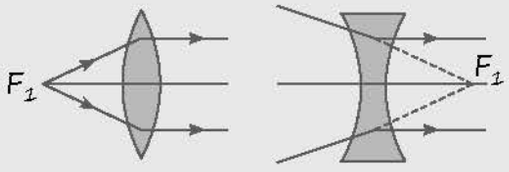
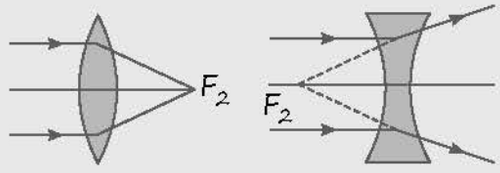
(1) **Centre of curvature:** It is the centre of sphere whose part is a lens surface. It is denoted by C.

(2) **Radius of curvature:** The radius of the sphere whose part is the lens surface.

(3) **Principal axis:** Line joining the centre of curvature of the two surfaces of lens.

(4) **Optical centre:** Point on the principal axis of lens such that a ray of light passing through this point emerges parallel to its direction of incidence.

(5) **Principal focus:**

Principal focus	Second principal focus
Point on principal axis of lens at which the rays starting from or directed to become parallel to principal axis after refraction.	Point on principal axis as which the rays coming parallel to principal axis converge or appears to diverge appears to diverge from the lens.
	

(6) **Focal length:** The distance between focus and optical centre of lens.

(7) **Focal plane:** The plane passing through the focus and perpendicular to the principal axis.

Principal Rays for Ray Diagram

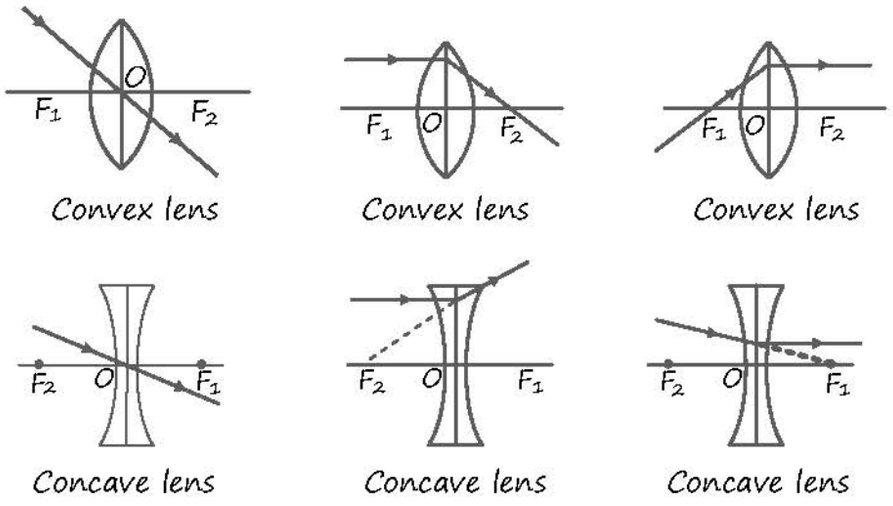
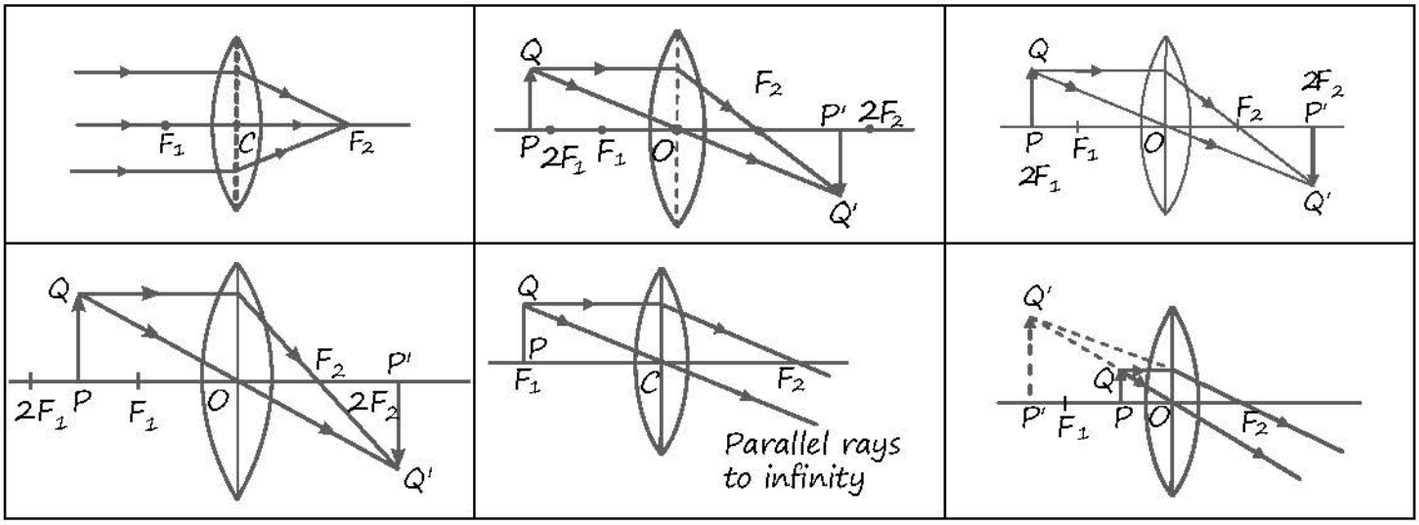


Image Formation by Convex Lens



S.No.	Position of object	Position of image	Size of image	Nature of image
1.	At infinity	At F	Point sized	Real & inverted
2.	Beyond 2F	Between F & 2F	Diminished	Real & inverted
3.	At 2F	At 2F	Same size	Real & inverted
4.	Between 2F & F	Beyond 2F	Enlarged	Real & inverted
5.	At F	At infinity	Highly enlarge	Real & inverted
6.	Between F & O	On the same side of lens as the object	Enlarged	Virtual & erect

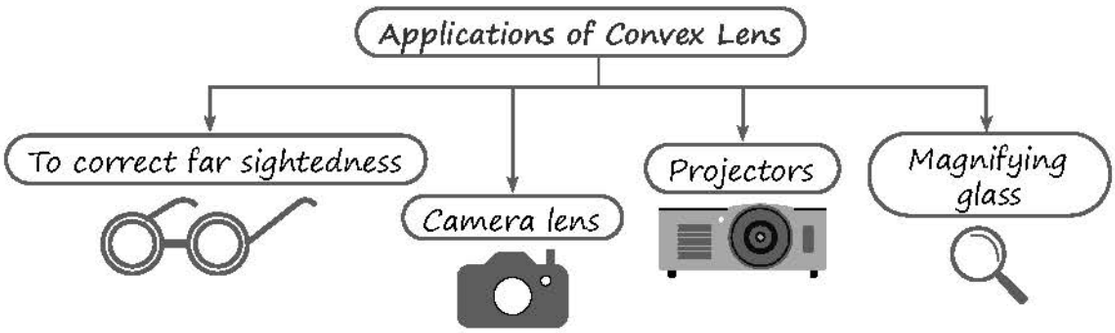
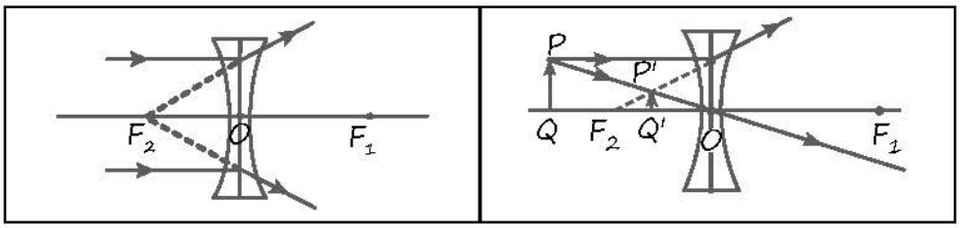
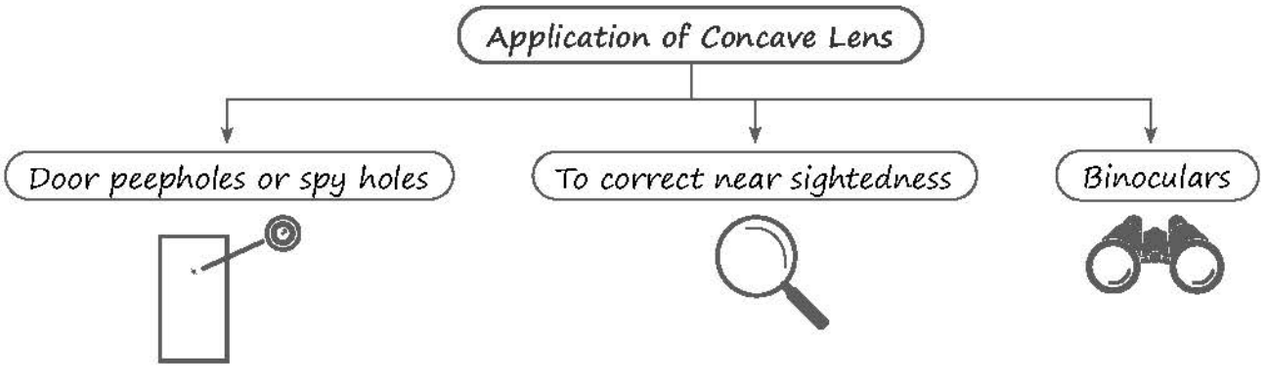


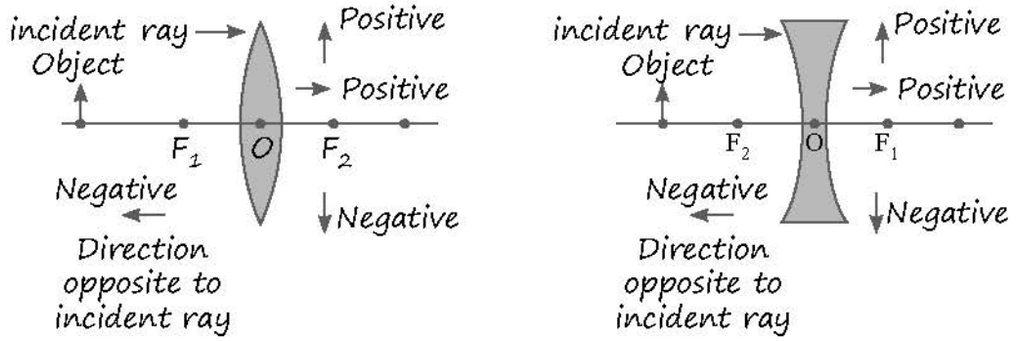
Image Formation by Concave Lens



S.No.	Position of object	Position of image	Size of image	Nature of image
1.	At infinity	At F, same side as the object	Diminished	Virtual & erect
2.	Between lens & infinity	Between F & O, on same side as the object	Diminished	Virtual & erect



Sign Convention and Lens Formula



Lens Formula

$$\boxed{\frac{1}{f}=\frac{1}{v}-\frac{1}{u}}$$

$f \rightarrow$ focal length
$u \rightarrow$ object distance
$v \rightarrow$ image distance

Linear Magnification

The ratio of length of image to the length of object (both perpendicular to principal axis)

$$\boxed{m=\frac{h_I}{h_O}=\frac{v}{u}}$$

Power of a Lens

The deviation of the incident light rays produced by a lens on refraction through it, is a measure of its power.

$$\boxed{\text{Power of lens}=\frac{1}{\text{focal length}}}$$

It is measured in diopter (D).

Formula Sheet

(1) $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$

(2) $m=\frac{h_I}{h_O}=\frac{v}{u}$

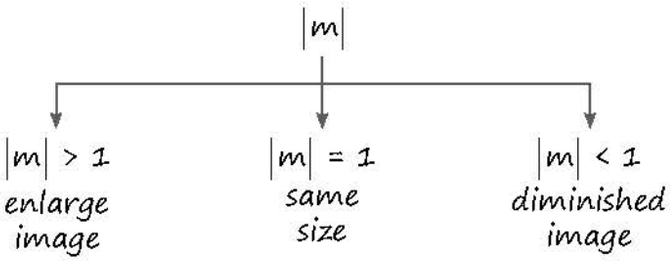
$m \Rightarrow + \Rightarrow$ virtual, erect image

$m \Rightarrow - \Rightarrow$ real, inverted image

(3) $P=\frac{1}{f}$

$P \Rightarrow + \Rightarrow$ Convex lens

$P \Rightarrow - \Rightarrow$ Concave lens



7 CHAPTER

Sound

Sound is a form of energy which produces a sensation of hearing in our ears.

It is produced by vibrating objects. It requires a material medium (solid, liquid or gas) for its propagation.

Sound waves are also called mechanical waves.

Type of Waves

(1) **Longitudinal wave:** Individual particles of the medium vibrate in the direction parallel to the direction of propagation of wave.

(2) **Transverse wave:** Individual particles of the medium vibrate perpendicular to the direction of wave propagation.

Light is a transverse wave (but not a mechanical wave)

Propagation of Sound in Air

When a vibrating object moves forwards, it compresses the air, creating compression (high pressure) and when it moves backward it created rarefaction (low pressure). Compression (C) is where particles are close together and Rarefaction (R) is where particles are father apart.

Relationship between speed, frequency and wavelength of sound wave:

$$\text{Speed} = \text{frequency} \times \text{wavelength}$$

Range of Hearing

The average frequency range over which the human ear is sensitive is called audible range (20 Hz to 20,000 Hz)

Infrasonic sound: Sound of frequency less than 20 Hz, cannot be heard by humans.

Ultrasonic sound: Sound of frequency higher than 20,000 Hz. Bats, dolphins, tortoise and rats can produce as well as hear ultrasound.

Reflection of Sound Waves

The returning back of sound waves on striking a surface such as wall, metal sheet etc.

Sound does not require a smooth and shining surface like mirror for reflection.

Condition for reflection of sound wave is that the size of reflecting surface must be bigger than the wavelength of sound wave.

Echo

The sound heard after reflection from a distant obstacle (such as a cliff, hillside, wall of a building, edge of a forest etc.) after the original sound has ceased is called echo.

Condition for hearing an echo

(1) The size of obstacle/reflector must be large compared to the wavelength of the incident sound.

(2) The distance between the source of sound and the reflector should be at least 17 m.

(3) Intensity of sound must be high enough for echo to be heard clearly.

Note: If there are repeated reflections at the reflecting surface the sound gets prolonged. This effect is known as reverberation.

Determination of speed of sound by method of echo

$$V = \frac{\text{Total distance travelled}}{\text{time interval}} = \frac{2d}{t}\left(ms^{-1}\right)$$

V = speed of sound

Uses of echo

Ehoes find their application in sound ranging and echo depth sounding by using ultrasonic waves.
Reason for using ultrasonic waves:

(1) They can travel undeviated through a long distance.

(2) Can be confined to a narrow beam.

(3) They are not easily absorbed in a medium.

Uses of echo by:

- Bats, dolphins, fisherman
- SONAR (Sound navigation and raging)
- Medical field – i.e., ultra sonography

Natural, Damped and Force Vibrations

Natural vibrations

The periodic vibrations of a body in the absence of any external force on it are called the natural (or free) vibrations.

E.g.,

(i) if bob of a simple pendulum is displaced slightly.

$$f = 2\pi\sqrt{\frac{g}{l}}$$

$f \rightarrow$ frequency
$g \rightarrow$ acceleration due to gravity
$l \rightarrow$ length of pendulum

(ii) A load suspended from spring when stretched.

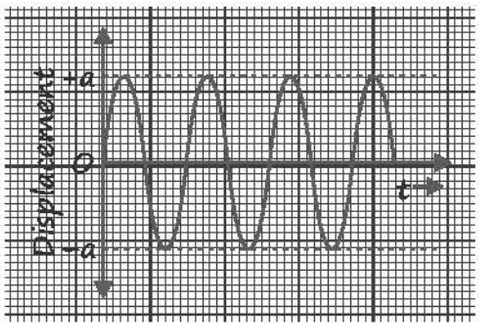
$$f = 2\pi\sqrt{\frac{k}{m}}$$

$f \rightarrow$ frequency
$k \rightarrow$ force constant
$m \rightarrow$ mass of load

(iii) Tuning fork struck against rubber pad.
(iv) Striking keys of piano.
(v) When air column in a flute vibrates.
(vi) When stringed instrument (guitar, sitar, violin) is plucked

$$f = 2\pi\sqrt{\frac{T}{\pi r^2 d}}$$

$f \rightarrow$ frequency
$T \rightarrow$ Tension
$r \rightarrow$ radius of string
$d \rightarrow$ density of string

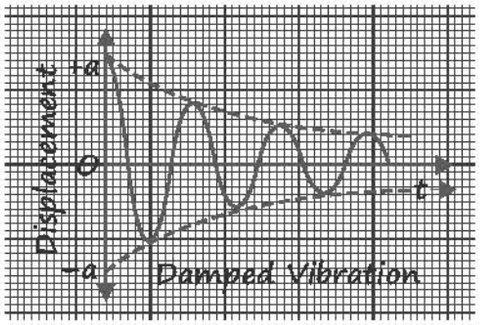


Note: In ideal condition once a body start vibrating it continues to vibrate with same amplitude & frequency forever.

Damped vibrations

The periodic vibration of a body of decreasing amplitude in presence resistive force.
In damped vibrations two forces act on the body (1) restoring force (2) frictional (resistive) force.
E.g.,

(i) Branch of tree when pulled and released
(ii) Tuning fork vibrating in air
(iii) Oscillating pendulum in air
(iv) Spring (load) vibration in air



Forced vibrations

The vibrations of a body which takes place under the influence of an external periodic force acting on it. The vibrating body is acted upon by 3 forces:

(1) restoring force
(2) resistive force
(3) external periodic force

Amplitude of the forced vibrations depends on the frequency of external force.

E.g.,

(i) Swing pushed at regular interval
(ii) While playing a guitar, artist forces the strings of the guitar to execute forced vibrations.

Resonance: It is special case of forced vibration when the frequency of an externally applied periodic force on a body is equal to its natural frequency, the body readily begins to vibrated with

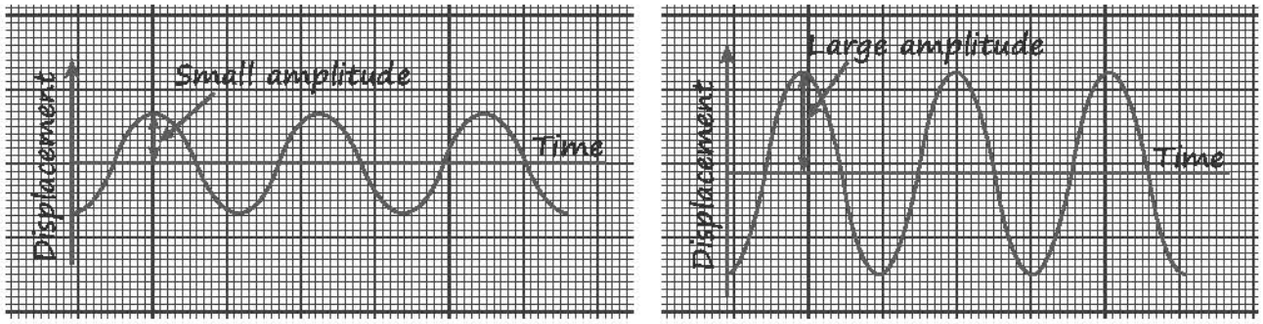
increased aplitude.

E.g.,

(i) Sympathetic vibrations of pendulums
(ii) Resonance in radio and TV receiver
(iii) Resonance in bridge
(iv) Resonance in air column and tuning fork.

Characteristics of Sound

(1) **Loudness:** It is a characteristic by virtue of which a loud sound can be distinguished from a faint one, both having same pitch and quality. It depends on the amplitude of the wave.



Factors affecting loudness:

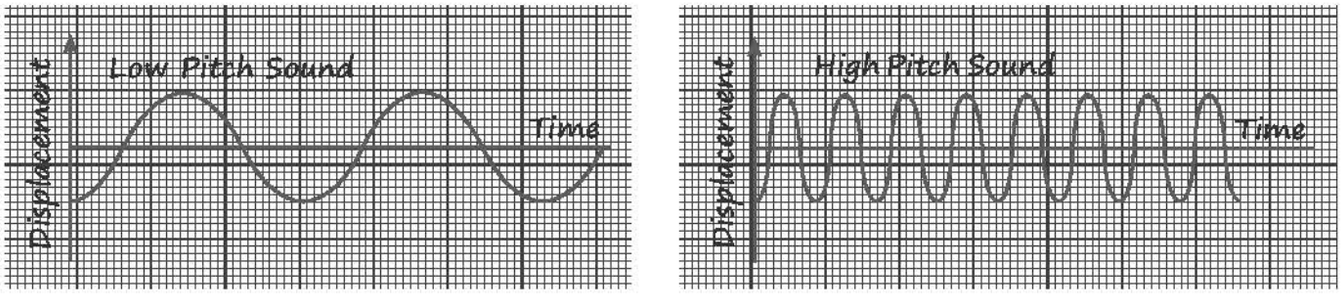
(i) Proportional to square of amplitude
(ii) Inversely proportional to the square of distance from sound
(iii) Surface area of vibrating body
(iv) Density of medium
(v) Presence of resonant bodies

(2) **Intensity:** The amount of sound energy passing each second through unit area.

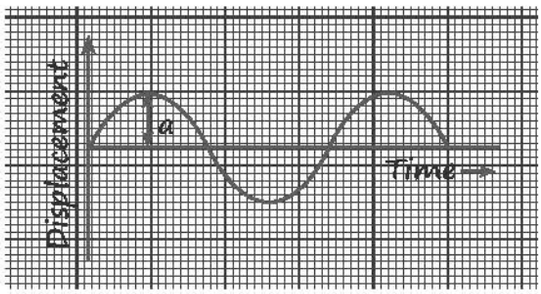
S.I. unit $\Rightarrow$ Watt per square metre (W/m^2)

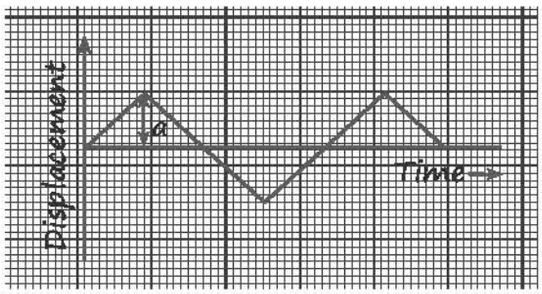
$$\boxed{L = k\log_{10} I}$$

(3) **Pitch (or shrillness):** Pitch is the characteristic of sound by which an acute (shrill) note can be distinguished from a grave or flat note pitch depends on its frequency.



(4) **Quality (or timbre) and waveform:** Quality of a sound is that characteristic which distinguishes the two sounds of the same loudness and same pitch but emitted by two different instruments. The quality of a musical sound depends on the waveform.





Music and Noise

Music		Noise	
(1)	It is regular, smooth and pleasant to ears.	(1)	It is harsh and unpleasant to ears.
(2)	Produced by periodic vibrations.	(2)	Produced by irregular succession of disturbance.
(3)	All the component waves are similar without any sudden change in wavelength & amplitude	(3)	Wave changes their character suddenly and they are short duration.
(4)	Sound level is low (10 dB to 30 dB)	(4)	Sound level is high (about 120 dB)
(5)	Wave form is regular	(5)	Wave form is irregular

Formula Sheet

(1) $v = f\lambda$

$v \rightarrow$ velocity

$f \rightarrow$ frequency

$\lambda \rightarrow$ wavelength

$T \rightarrow$ time period

$$f = \frac{1}{T}$$

(2) Echo

$$D = \frac{vt}{2}$$

$D \rightarrow$ distance between source and obstacle

$v \rightarrow$ speed

$t \rightarrow$ time

(3) Frequency of oscillating pendulum

$$f = 2\pi\sqrt{\frac{g}{l}}$$

Frequency of loaded spring (oscillatin)

$$f = 2\pi\sqrt{\frac{k}{m}}$$

Frequency of oscillating string

$$f = 2\pi\sqrt{\frac{T}{\pi r^2 d}}$$

(4) Intensity & loudness

$$L = K\log_{10} I$$

$$L = 10\log_{10}\left[\frac{I_1}{I_0}\right] dB$$

$$I_0 = 10^{-12}\ W/m^2$$

I_1 = intensity of sound

8 CHAPTER

Current Electricity

Electric Charge

It is a physical entity which is defined by excess or deficiency of electrons on a body.

S.I. unit $\Rightarrow$ coulomb (C)

The total charge acquired by a body is an integral multiple of magnitude of change on a single electron. This principle is called quantisation of charge.

$\boxed{q = ne}$ q = total charge, n = number of electrons, e = charge on 1 electron = $1.6 \times 10^{-19}C$

$1mC = 10^{-3}C$, $1\mu C = 10^{-6}C$, $1nC = 10^{-9}C$

Current

Rate of flow of charge

$$\boxed{I = \frac{Q}{t}} \quad I = \frac{ne}{t}$$ I = current, Q = charge, t = time

S.I. unit $\Rightarrow$ ampere (A), $1mA = 10^{-3}A$, $1\mu A = 10^{-6}$ A

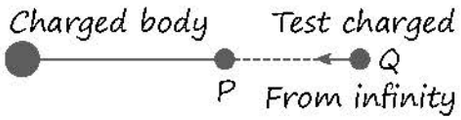
Note: Direction of current is taken opposite to the flow of electrons.

Electric Potential

Potential at a point is defined as the amount of work done in bringing a unit positive charge from infinitely to that point. It is a scalar quantity.

$$\boxed{V = \frac{W}{Q}}$$ V = potential W = work done Q = charge

S.I. unit $\Rightarrow JC^{-1}$ or volt (V)



Potential Difference (p.d.)

The potential difference between two points is equal to the work done in moving a unit positive charge from one point to the other.

$$\boxed{V_A - V_B = \frac{W}{Q}}$$

S.I. unit $\Rightarrow$ Volt (V)

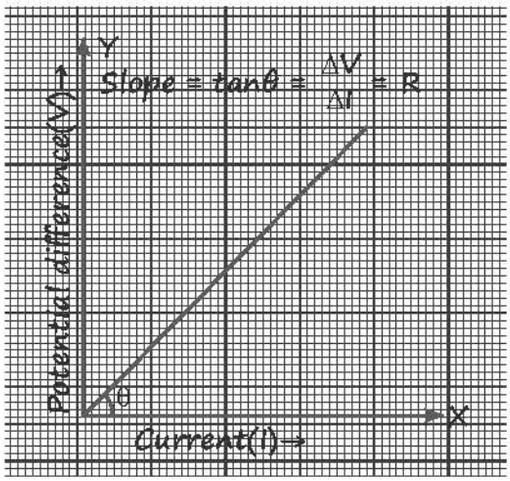
Resistance

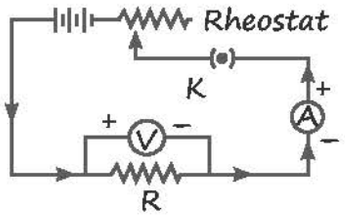
The obstruction offered to the flow of current by the conductor (or wire) is called its resistance.

Ohm's Law

According to Ohm's law the current flowing in a conductor is directly proportional to the potential difference applied across its ends provided the physical conditions and the temperature of conductor remains constant.

$$I \propto V, \quad \frac{V}{I} = \text{constant}, \quad \boxed{R = \frac{V}{I}}, \quad \boxed{V = IR}$$





Resistance of a conductor is numerically equal to the potential difference across its ends when unit current flows through it.

S.I. unit is ⇒ ohm (Ω)

Ohmic and Non-ohmic resistors:

Ohmic resistor		Non ohmic resistor	
(1)	Obey's ohm's law	(1)	Do not obey ohm's law
(2)	V-I graph is a straight line	(2)	V-I graph is not straight line
(3)	Slope of V-I graph is same at all values	(3)	Slope of V-I graph is different for different values
	e.g., silver, iron, copper, nichrome		e.g., junction diode, L.E.D., transistor, filament bulb.

Factors affecting the resistance of a conductor

(1) Nature of conductor

(2) Length of conductor, $R \propto l$

(3) Area of cross section of conductor, $R \propto \frac{1}{A}$

(4) Temperature,

$$R \propto l, \; R \propto \frac{1}{A} \text{ or } R \propto \frac{1}{\pi r^2}, \quad \boxed{R \propto \frac{l}{A} \text{ or } R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}}$$

Specific resistance or Resistivity (ρ): It is the resistance of a wire of that material of unit length and unit area of cross section. S.I. unit ⇒ Ωm

Conductivity

The reciprocal of specific resistance is known as conductivity (σ), $\boxed{\sigma = \frac{1}{\rho} = \frac{l}{RA}}$ *S.I. unit ⇒* $\Omega^{-1}m^{-1}$

Conductance

Reciprocal or resistance, $\boxed{\text{Conductance} = \frac{1}{R}}$ *S.I. unit ⇒* Ω^{-1}

Superconductor

It is a substance of zero resistance (or infinite conductance) at a very low temperature.

Electromotive force (E.M.F.) of a cell

The E.M.F. of a cell is defined as the energy spent (or the work done) per unit charge in taking a positive test charge around the complete circuit of cell.

$$\boxed{E = \frac{W}{q}}$$

Factors affecting the E.M.F. of a cell:

(i) *shape of electrodes*

(ii) *distance between electrodes*

(iii) *amount of electrolyte*

Terminal voltage of a cell

The terminal voltage of a cell is defined as the work done per unit charge in carrying a positive test charge around the circuit, connected across the terminals of cell.

$$\boxed{V = \frac{W'}{q}}$$

Voltage drop in a cell

Work done in carrying a unit charge through the electrolyte is called voltage drop in the cell.

$$\boxed{V = \frac{W}{q}}$$

Relationship between E.M.F. and terminal voltage of a cell:

$$\boxed{E = V + v}$$

Internal resistance of a cell

Resistance offered by the electrolyte inside the cell, to the flow of current is called internal resistance of cell.

$$\boxed{V = Ir}$$

Factors affecting internal resistance of a cell:

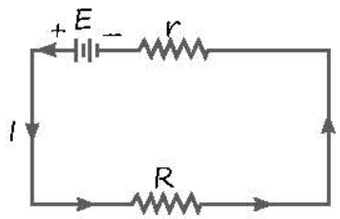
(1) *Surface area of electrodes*

(2) Distance between the electrodes
(3) Nature & concentration of electrolyte
(4) Temperature of electrolyte

Relationship between E.M.F., terminal voltage and internal resistance:

Total resistance = R + r

Current drawn from cell, $I = \frac{E}{R+r}$



E.M.F. (E) = I(R + r)

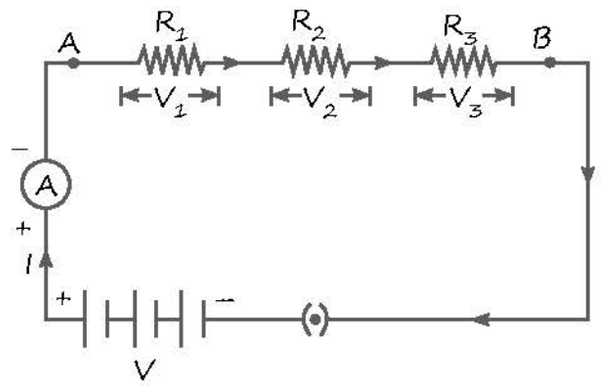
Terminal voltage of cell, V = IR = E – Ir

Voltage drop due to internal resistance = Ir

Internal resistance, $r = \frac{E-V}{I} = \frac{E-V}{\frac{V}{R}} = \left[\frac{E}{V} - 1\right]R$

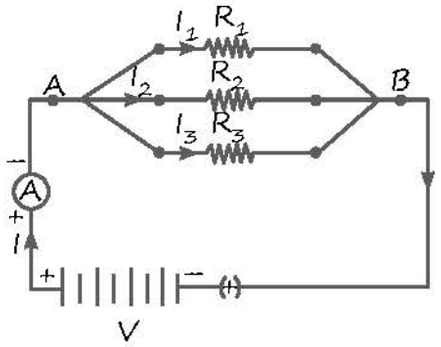
Combination of Resistors

Series



$$R = R_1 + R_1 + R_3$$

Parallel



$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Measurement of Electrical Energy (E)

E = work done

$$W = QV = VIt = I^2Rt = \frac{V^2t}{R}$$

S.I. unit ⇒ joule (J)

Electrical power: The rate at which electrical energy is supplied by the source.

$$P = \frac{W}{t} = VI = \frac{V^2}{R} = I^2R$$

S.I. unit ⇒ watt (W) or Js^{-1}

Bigger units ⇒ 1 kW = 1000 W

$1\ MW = 10^6\ W$

$1\ HP = 746\ W$

Commercial unit of electric energy: $1\ kWh = 3.6 \times 10^6\ J$

Power rating of common electrical appliances:

(i) Resistance of filament of bulb, $R = \frac{V^2}{P}$

(ii) Safe current limit through the filament of the bulb, $I = \frac{P}{V}$

Household consumption of electrical energy: Cost of electricity = electrical energy (in kWh) × cost per kWh

Heating effect of electric current

The heat produced in a wire on passing current through it is called heating effect of current. This is known as Joule's law of heating.

$$\boxed{H = I^2Rt(\text{in joule}) = 0.24\, I^2Rt(\text{in cal})}$$

Formula Sheet

(1) $I = \frac{Q}{t} = \frac{ne}{t}$

(2) $V = \frac{W}{Q}, V_A - V_B = \frac{W}{Q}$

(3) $V = IR$

(4) $\text{Conductance} = \frac{1}{R}$

(5) $R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}$

(6) $\text{Conductivity}\ (\sigma) = \frac{1}{\rho} = \frac{l}{RA}$

(7) $\text{EMF}(E) = \frac{W}{q}$

(8) $\text{Terminal voltage}\ (V) = \frac{W'}{q}$

(9) $E = I(R + r)$

(10) $r = \left(\frac{E}{V} - 1\right)R$

(11) Series combination

$$R = R_1 + R_2 + \ldots\ldots + R_n$$

(12) Parallel combination

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots\ldots + \frac{1}{R_n}$$

(13) Electrical energy

$$W = QV = VIt = I^2Rt = \frac{V^2t}{R}$$

(14) Electrical power

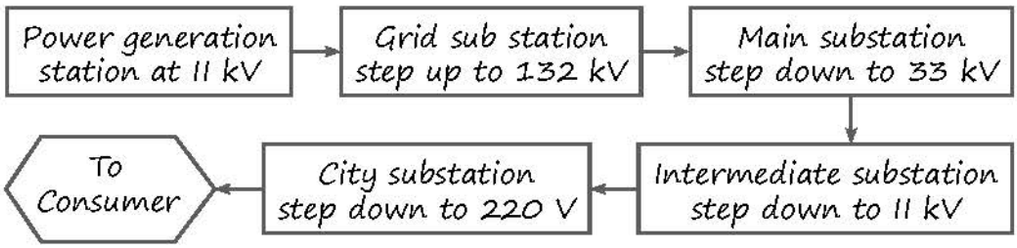
$$P = \frac{W}{t} = VI = \frac{V^2}{R} = I^2R$$

(15) $H = I^2Rt$ (in joule)

$H = 0.24 I^2Rt$ (in calorie)

9 CHAPTER Household Circuits

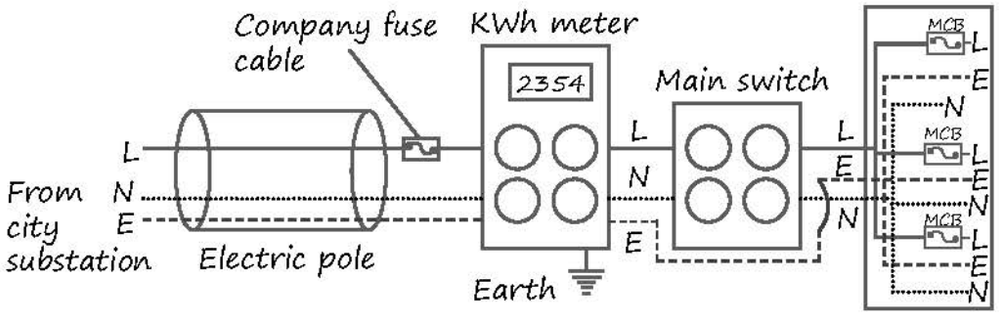
Power Distribution



The power is transmitted over a long distance at a high voltage to minimize the loss of energy in the form of heat in the line wire. At high voltage, the value of current is very less, therefore heat produced, $H = I^2Rt$, will also be less.

Power Distribution to House (Components of Household Electric Circuit)

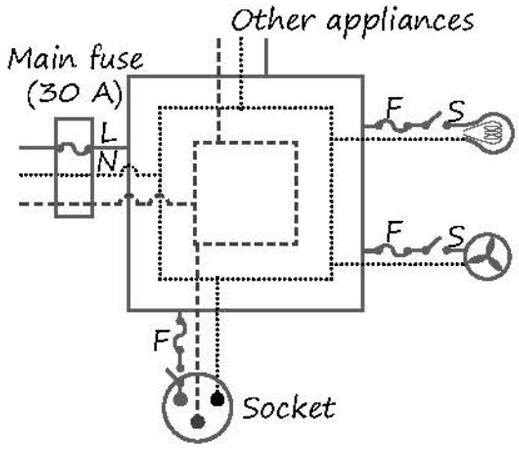
(1) **Main circuit:** Electricity is transmitted to home via two wires, the live wire (220 V) and the neutral wire (0 V). The live wire carries current, while the neutral wire returns it. An earth wire ensures safety. The main fuse is in live wire and the main switch controls both live and neutral wires. Homes have two separate circuits; a 5A fuse for lighting and a 15A fuse for power devices.



Main Circuit

Main switch: It controls the flow of electricity to the home. It is the primary connection between the external power supply and home's wiring. The main switch is often a double pole switch, which means it can break the connection to both the neutral wire and live wire at the same time.

House wiring (Ring system): In this system wires starting from main fuse box run around all the rooms of house and then come back to fuse box again forming the ring.



Advantage:

(i) Each appliance can operate independently without affecting the other appliances connected in the system.

(ii) New appliance can be connected directly without new wiring.

Fuse

An electric fuse is a safety device which is used to limit the current in an electric circuit, fuse safeguards the circuit and the appliances connected in that circuit from being damaged.

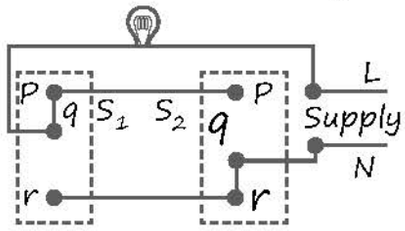
- Thin wire of tin (25%) & lead (75%) alloy
- Low melting point ($\approx 200°C$)
- Works on the principle of heating effect of current
- Always connected to live wire before appliance.

$$\text{Current rating of fuse} = \frac{\text{Total power of appliances in circuit}}{\text{Voltage of supply}}$$

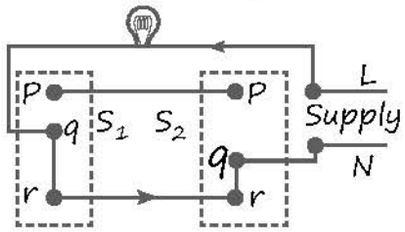
Switch

A switch is an on-off device for current in a circuit. It is connected to live wire in series.

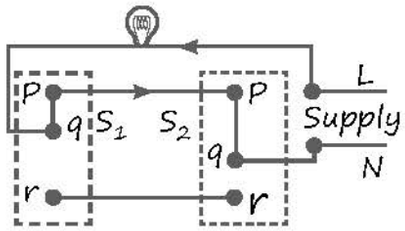
Dual control switch (staircase wiring): Switch that controls a load from two different locations. It is also known as multiway switch or staircase wiring.



Bulb off



Bulb on through S_1



Bulb on through S_2

MCB (Miniature Circuit Breaker)

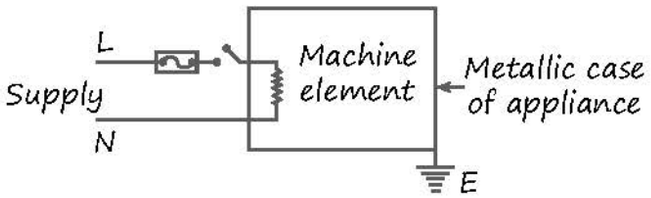
- Safety device automatically shuts off an electrical circuit when there is power surge or short circuit.

Advantage: More reliable than fuses because they can be reset instead of replaced.

Earthing

Process of connecting an electrical appliance's metal case to the ground using earth wire.

It is a safety measure that protects people from electric shocks.



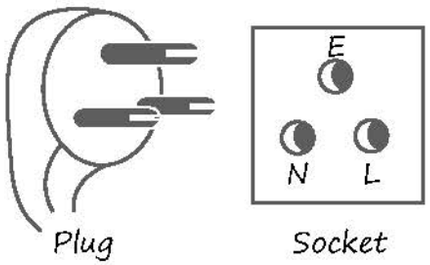
Three pin plug and socket

Top pin: Earthing

Left pin: Live

Right Pin: Neutral

The earth pin is thicker and loner than other two.



Safety precautions

- Do not touch switch with wet hands.
- Switch should always be connected to live wire.
- Appliances should be earthed.
- Use appropriate fuse rating in live wire of circuit.

Colour Coding of Wires		
Wire	Colour	
	Old	New convention
Live	Red	Brown
Neutral	Black	Blue
Earth	Green	Green or Yellow

10 CHAPTER Electromagnetism

Oersted's Experiment

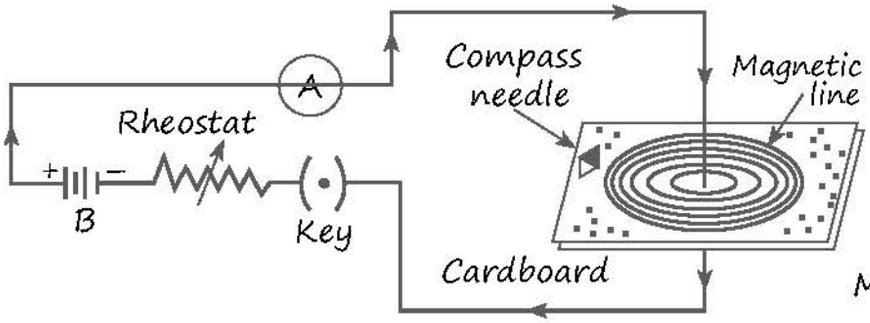
When an electric current is passed through a conducting wire, a magnetic field is produced around it, which is detected using compass needle.

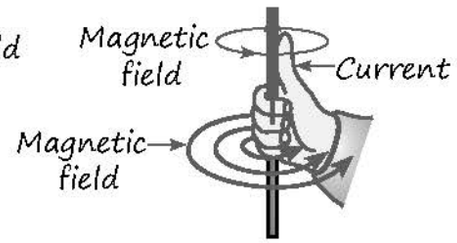
Magnetic field

The region around a magnet in which its effect can be experienced. It is denoted by B.

- Vector quantity
- S.I. unit ⇒ tesla (T)

Magnetic field due to straight current carrying conductor

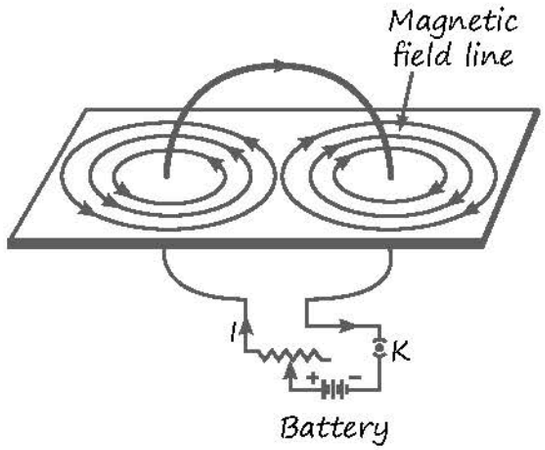


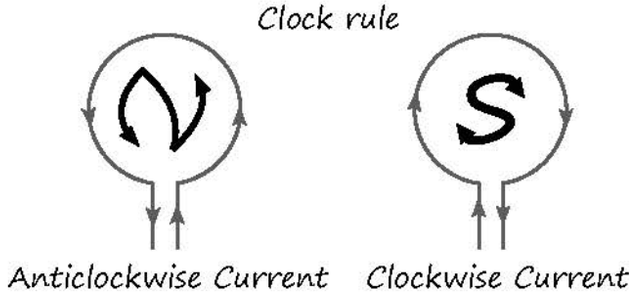


Maxwell's right hand rule-current

If we hold the current carrying conductor in our right hand such that the thumb points in the direction of flow of current, then the fingers encircle the wire in the direction of magnetic field lines.

Magnetic field due to current in a loop





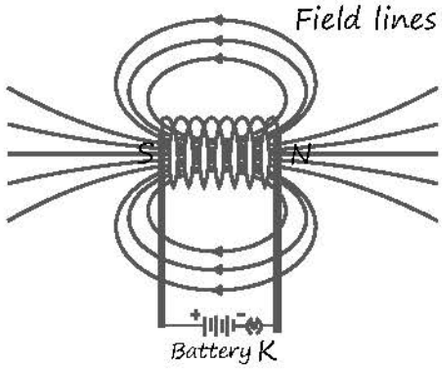
Magnetic field lines become dense if:

(1) the strength of current in the loop is increased.

(2) the number of turns in loop are increased.

Magnetic field due to solenoid

If a conducting wire is wound in form of a cylindrical coil whose diameter is less than comparison to its length, the coil is called solenoid.



Magnetic field depends on:

(1) No. of turns of solenoid $B \propto N$

(2) Strength of current, $B \propto I$

(3) Nature of core material used to make solenoid

Comparison of electromagnet and permanent magnet

Electromagnet		Permanent magnet	
(1)	It is made of soft iron.	(1)	It is made of steel
(2)	Produces temporary magnetic field	(2)	Produces permanent magnetic field
(3)	Magnetic field strength can be changed	(3)	Magnetic field strength cannot be changed
(4)	Polarity can be reversed	(4)	Polarity cannot be reversed
(5)	Can be easily demagnetized	(5)	Cannot be demagnetized

Uses of electromagnet

(1) Lifting and transporting large masses of iron scrap.

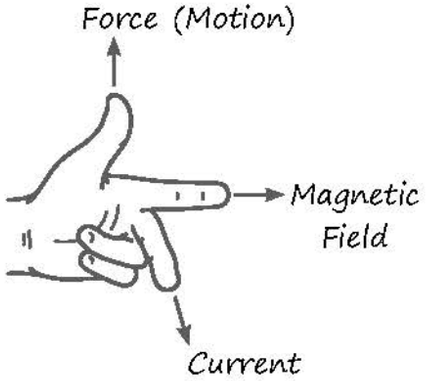
(2) Loading furnace with iron.

(3) Magnetic separation.

(4) Removing iron pieces from wounds.

Force on a current carrying conductor

Fleming's Left hand Rule: Stretch the forefinger central finger and thumb of your left hand mutually perpendicular to each other. If forefinger indicates the direction of magnetic field and the central finger indicates the direction of current, then the thumb will indicate the direction of motion of conductor. (i.e. force on conductor)



Simple D.C. motor

An electric motor is a device which converts the electrical energy into mechanical energy.

Principle: When current is passed through a conductor placed normally in a magnetic field, force acts on the conductor.

Parts of motor:

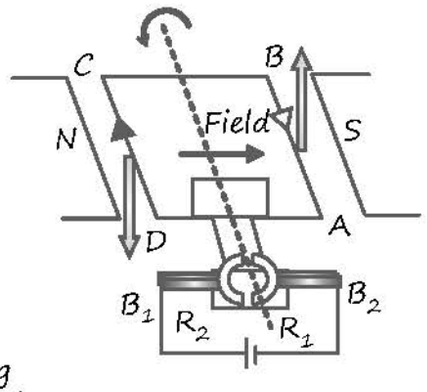
(1) Armature (2) Split rings (3) Brushes

(4) Magnet (5) DC source

Speed rotation of motor can be increased by:

(i) Current (ii) No. of turns

(iii) Area of coil (iv) Magnetic field strength



Electromagnetic induction

The phenomenon of production of emf in a conductor on changing the magnetic field around it. The emf produced is called induced emf.

The factors affecting the magnitude of induced emf are:

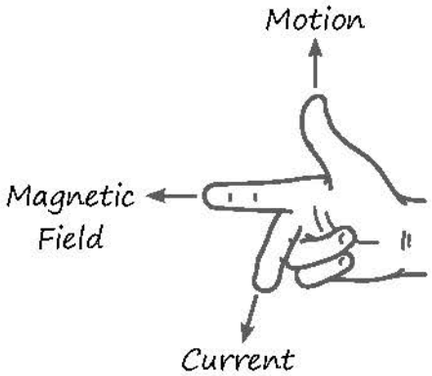
(i) change in the magnetic flux

(ii) Time in which the magnetic flux changes.

Fleming's right hand rule

Stretch the thumb, central finger and forefinger of your right hand mutually perpendicular to each other. If the forefinger indicates the direction of magnetic field and thumb indicates the direction of motion of conductor then the central finger will indicate the direction of induced current.

Fleming's Right Hand Rule



A.C. generator

Device which converts mechanical energy into electrical energy is called generator.

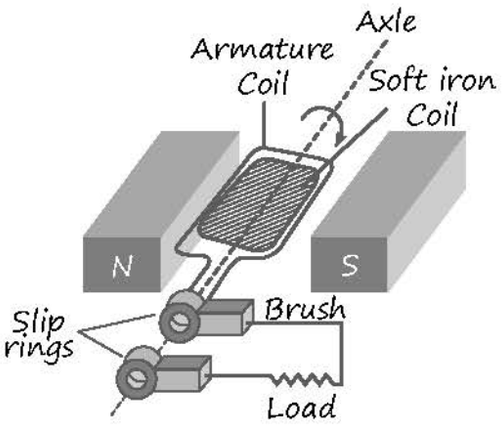
Principle: Electromagnetic induction.

Parts of generator:

(1) Field magnet (2) Armature coil

(3) Slip rings (4) Brushes

Frequency of alternating current: If the coil makes n rotations per second, then the magnitude of induced emf is given as: $e = e_o \sin 2\pi nt$ and current is: $i = i_o \sin 2\pi nt$



Difference between A.C. and D.C.

Direct current (D.C.)		Alternating current (A.C.)	
(1)	Current of constant magnitude	(1)	Current of varying magnitude
(2)	Flows in one direction	(2)	Reverses its direction while flouring in circuit
(3)	Source – Cell (or battery)	(3)	Source – AC generator or mains

Advantages of A.C. over D.C.:

(i) A.C. at any desirable voltage can be obtained using transformer.

(ii) A.C. can be converted to D.C. using rectifier.

(iii) Cost of generation is less.

(iv) A.C. can be controlled without much power loss.

(v) A.C. machines are very durable.

(vi) For long-distance transmission A.C. is preferred as it causes minimum loss of energy.

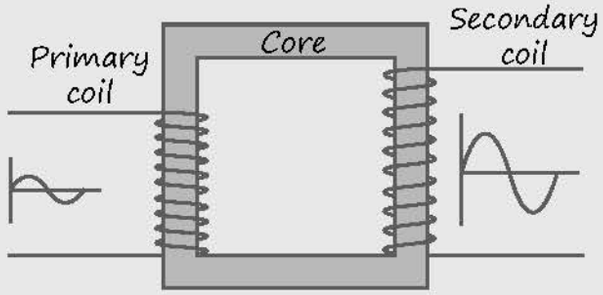
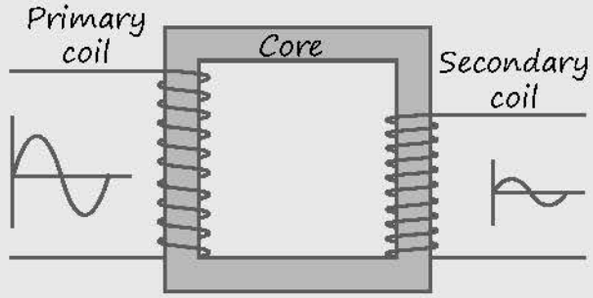
Transformer: Device which is used to increase or decrease the amplitude of an alternating emf.

They are of two types:

(1) Step-up transformer

(2) Step-down transformer

Principle: Transformer is based on the principle of mutual induction.

Step-up transformer	Step-down transformer
Changes low alternating voltage too high alternating voltage. ($E_S > E_P$, $N_S > N_P$)	Changes high alternating voltage to low alternating voltage. ($E_S < E_P$, $N_S < N_P$)
	
Uses: ❑ Transmission of electric power at the power generating station to step up the voltage ❑ Television ❑ Wireless sets ❑ X-ray tubes to provide high accelerating voltage.	**Uses:** ❑ Electric bells, night bulbs, mobile phones, computers. ❑ At the power substation to step down the voltage before its distribution to consumers.

11 CHAPTER

Calorimetry

Heat

Heat is the internal energy of molecules constituting the body. It flows from a hot body to a cold body when they are kept in contact.

It is represented by Q. S.I. unit $\Rightarrow$ joule (J). Other units: 1 calorie = 4.2 joule. 1 kcal = 10^3 cal = 4.2×10^3 joule

Note: The measurement of the quantity of heat is called calorimetry.

Temperature

It is the measure of hotness or coldness of any body. S.I. unit $\Rightarrow$ Kelvin (K)

Other units: Celsius (°C), Fahrenheit (°F)

$$\boxed{\begin{aligned} T(K) &= T(°C) + 273.15 \\ \Delta T(°C) &= \Delta T(K) \end{aligned}}$$

Factors affecting the quantity of heat absorbed to increase the temperature of a body:

(1) mass of body (m)

(2) increase in temperature of body Δt

(3) material of the body

$$Q \propto m \qquad Q \propto \Delta t$$

$$\boxed{Q = cm\Delta t} \qquad c = \text{specific heat capacity}$$

Thermal Capacity or Heat Capacity (c)

Amount of heat energy required to raise the temperature of a body by 1°C (or 1K). It is denoted by C.

$$\boxed{\text{Heat capcity}(C) = \frac{\text{Heat supplied}(Q)}{\text{Rise in temperature}\ (\Delta t)}}$$

S.I. unit $\Rightarrow JK^{-1}$ or $J°C^{-1}$

Specific Heat Capacity (c)

Amount of heat energy required to raise the temperature of unit mass of the substance through 1°C (or 1K)

$$c = \frac{\text{Amount of heat energy supplied}}{\text{mass} \times \text{rise in temperature}} \qquad \boxed{c = \frac{Q}{m\Delta t}}$$

S.I. unit $\Rightarrow J\ Kg^{-1}\ K^{-1}$ or $J\ Kg^{-1}\ °C^{-1}$

Relationship between C and c

$$\boxed{C = m \times c}$$

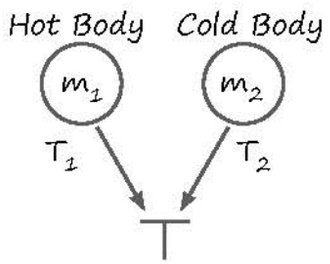
Specific heat capacity (c) → low ⇒ conductor
Specific heat capacity (c) → high ⇒ Poor conductor
Water has an unusually high specific heat capacity (= 4200 J Kg^{-1} K^{-1})

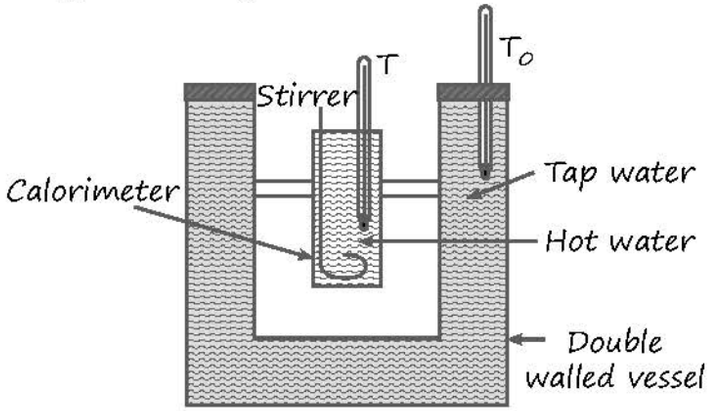
Calorimeter

It is a cylindrical vessel which is used to measure the amount of heat gained (or loss) by body when it is mixed with the other body.

Principle

Heat lost by hotter body = Heat gained by cooler body



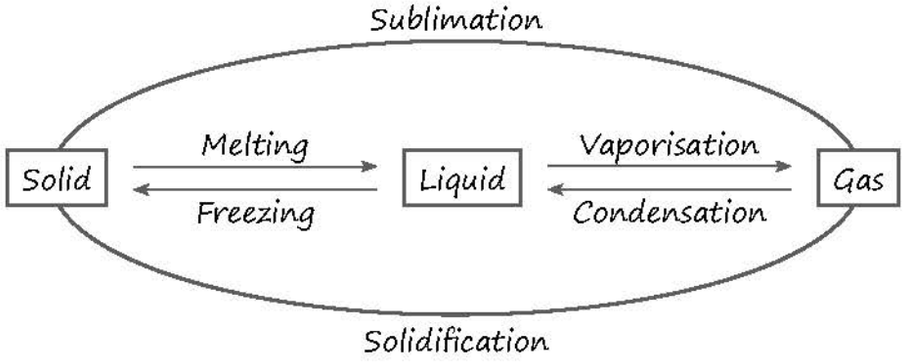


Heat lost (Q_1) = Heat gained (Q_2)

$m_1c_1(T_1 - T) = m_2c_2(T - T_2)$

Changes of Phase (State)

The process of converting one state of substance into another state is known as change of state of substance. It happens by heat absorption or rejection.



Latent Heat

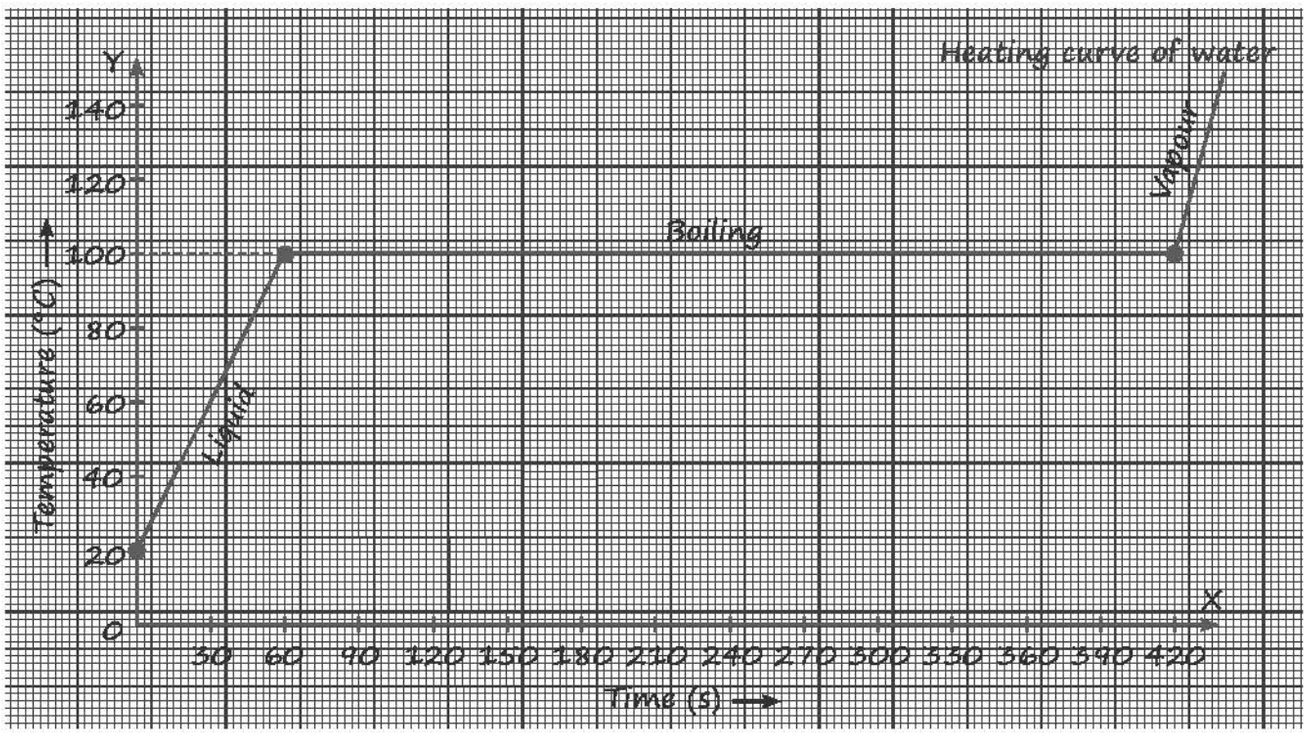
It is the hidden amount of heat that is given out or taken by the unit mass of any substance to change its state without change in temperature. It may also be called as change in potential energy of molecules.

Specific Latent Heat

Quantity of heat absorbed (or liberated) by the unit mass of the substance for the change in its phase at constant temperature. It is denoted by L.

$$\boxed{L = \frac{Q}{m}}$$

S.I. unit $\Rightarrow J\ kg^{-1}$, Other unit $\Rightarrow cal\ g^{-1}$, $kcal\ kg^{-1}$



Formula Sheet

(1) $T(K) = T(°C) + 273.15$

(2) $Q = mc\Delta t$ [Heat energy]

(3) $C = \frac{Q}{\Delta t}$ [Heat capacity]

(4) $c = \frac{Q}{m\Delta t}$ [Specific heat capacity]

(5) $C = m \times c$ [Relation between C and c]

(6) $m_1c_1(T_1 - T) = m_2c_2(T - T_2)$ [Principle of calorimetry]

(7) $L = \frac{Q}{m}$ [specific heat capacity]

12 CHAPTER

Radioactivity

Atoms

Smallest part of any substance which consists of a centralized dense part called nucleus.

Atomic Number (Z)

Number of proton present in nucleus of an atom.

Mass Number (A)

Total number of proton and neutron present in nucleus.

Isotopes

Atoms belonging to the same element having same atomic number (z) but different mass number. E.g., Protium $\left({}_{1}^{1}H\right)$ and deuterium $\left({}_{1}^{2}H\right)$

Isotones

Atoms with same neutrons (N) but different atomic number (Z). E.g., $\Rightarrow {}_{11}^{23}Na$ and ${}_{12}^{24}Mg$

Isobars

Atoms with same mass number (A) but different atomic number. E.g., $\Rightarrow {}_{11}^{23}Na$ and ${}_{12}^{23}Mg$

Radioactivity

Spontaneous process by which a nucleus of an unstable atom decay by emitting radiations such as α, β and γ. Substances which disintegrate by spontaneous emissions are radioactive substances. E.g., Uranium, Radium, Polonium, Thorium, Actinium etc. Radioactivity is a nuclear phenomenon.

Properties of α (alpha) particles

- Consists of Helium nuclei, ${}_{2}He^{4}$.
- It is positively charged particle.
- Least penetration power.
- Highest ionisation power.
- Can't penetrate skin but harmful for humans.
- Effects a photographic plate.
- Causes fluorescence on striking a fluorescence material.
- Affected by electric and magnetic fields.

Properties of β (Beta) particles

- Highly energetic electrons, which release from inside of a nucleus.
- Denoted by $_{-1}\beta^{0}$ and $_{+1}\beta^{0}$.
- They are negatively and positively charged.
- They have negligible mass.
- For emission of negative β particles, a neutron divides into proton and an electron and for +β protons converts into neutron.
- Greater penetration power compared to α particles.
- Less ionization power, still harmful.
- Effects photographic plate
- Causes fluorescence on striking fluorescent material.
- Affected by electric & magnetic field.

Properties of γ particles (Gamma)

- Waves from high frequency end of electromagnetic spectrum which do not have mass.
- Neutral in nature.
- Denoted by $_{0}\gamma^{0}$
- High penetrating power compared by α and β particle.
- Very harmful for living cell & causes mutation.
- Effects photographic plate.
- Cause fluorescence when they strike a fluorescent material.
- Not affected by electric and magnetic fields.

Changes within the nucleus in α, β and γ emission

(1) Alpha emission

$$\underset{\text{Parent nucleus}}{{}^{A}_{Z}X} \longrightarrow \underset{\text{Daughter nucleus}}{{}^{A-4}_{Z-2}Y} + \underset{\alpha\text{-particle}}{{}^{4}_{2}He}$$

(2) Beta emission

$$\underset{\text{Parent nucleus}}{{}^{A}_{Z}P} \longrightarrow \underset{\text{Daughter nucleus}}{{}^{A}_{Z+1}Q} + \underset{\beta\text{-particle}}{{}^{0}_{-1}e}$$

(3) Gamma emission

$$\underset{\text{Parent nucleus}}{{}^{A}_{Z}X} \longrightarrow \underset{\text{Daughter nucleus}}{{}^{A}_{Z}X} + \underset{\gamma\text{-radiation}}{\gamma}$$

Uses of Radioactivity

(1) Medical use (2) Industrial use (3) Scientific use

X-Rays

- These are electromagnetic waves
- Frequency ranges from 10^{16} to 3×10^{20} Hz

Uses:

(i) Surgery to detect fracture, stones in body
(ii) To detect fault, crack on bridges
(iii) For security purpose to detect metal or explosive material.
(iv) Used in scientific research.

Sources of harmful radiation

(1) Radioactive fall out from nuclear power plants
(2) Nuclear waste
(3) Cosmic radiations

Background radiations

All the radiations which are present in the atmosphere, even if there is no radioactive source present are called background radiations. The sources of these radiations are:

(i) cosmic radiations
(ii) natural and artificial radioactive isotopes
(iii) products after nuclear explosion
(iv) traces of radioactive substance present in earth, i.e., rocks.

Safety measures for handling radioactive materials

- Know radiation safely limits to prevent excessive exposure.
- Wear lead lined apron and gloves for protection.
- Store radioactive substances in thick lead containers with narrow opening to prevent radiation spread.

Nuclear energy

Energy released during the transformation of nuclei to another nuclei.
Two distinct ways of obtaining nuclear energy.

Nuclear Fission		Nuclear Fusion	
(1)	Breaking of heavy nucleus into lighted nucleus	(1)	Involves binding of two nucleus
(2)	It is a chain reaction	(2)	It is not a chain reaction
(3)	It is a controlled process	(3)	It is uncontrolled process
(4)	Produces enormous energy	(4)	More energy than fission
(5)	Fission products are hazardous	(5)	It does not cause pollution
	$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3{}_{0}n^{1} + Q$ Heavy nucleus; Thermal nucleus; Fission product; Product neutron; Huge amount of energy (220 MeV)		$_1H^1 + {}_1H^1 \rightarrow {}_1H^2 + {}_1e^+ + \nu + 0.42\ meV$ $_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + {}_0n^1 + 3.27\ meV$ $_1H^2 + {}_1H^2 \rightarrow {}_1H^3 + {}_1H^1 + 4.03\ meV$

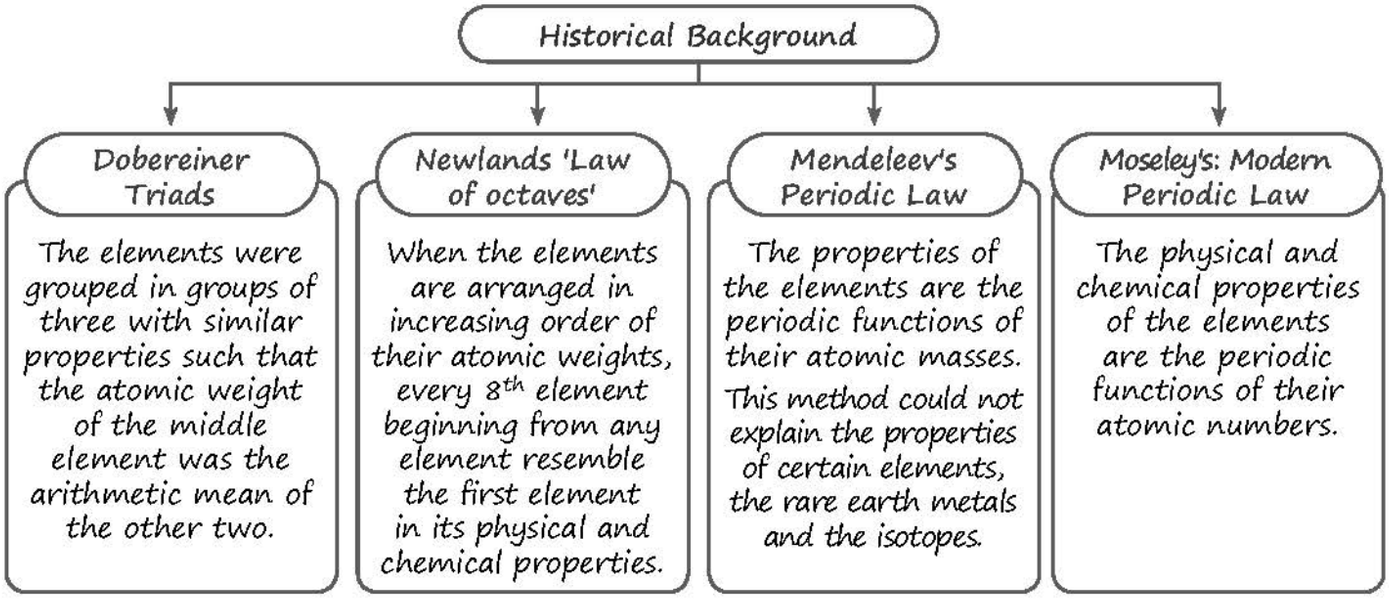
CHEMISTRY

1 CHAPTER

Periodic Table, Periodic Properties and Variation of Properties

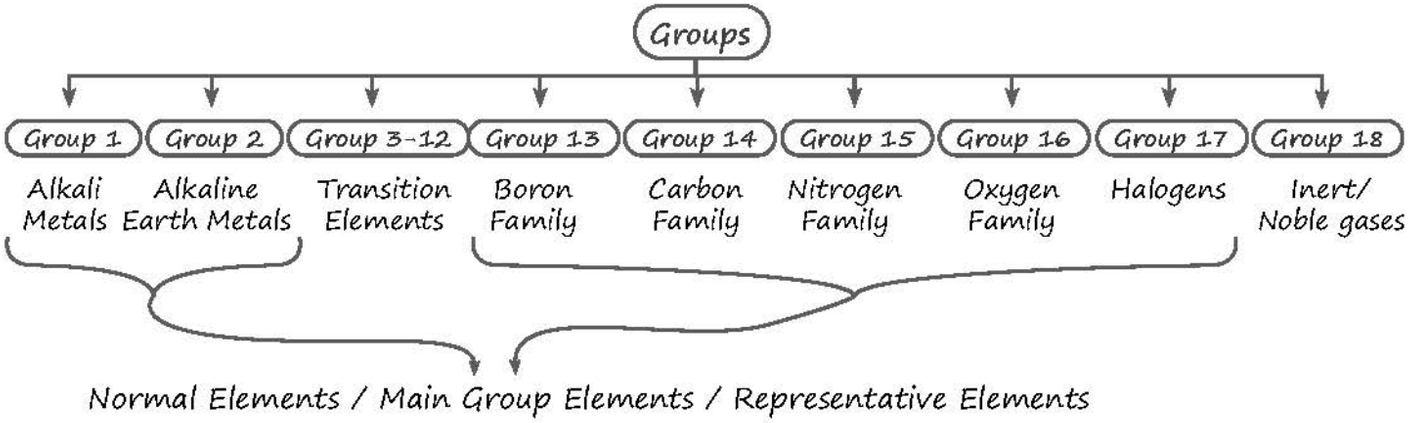
Periodic Table, Periodic Properties and Variation of Properties

- Periodic table may be defined as a stable which classifies all the known elements in accordance with their properties in such a way that elements with similar properties are grouped together in the same vertical columns and dissimilar elements are separated from one another.



Modern Periodic Table:

Modern periodic table has 18 vertical columns. The vertical columns are known as 'Groups' arranged from left to right in the order of increasing atomic numbers.



Periods

There are seven horizontal rows in the modern periodic table. They are known as periods.

- The number of shells present in an atom determines its period.

Period No.	Type	No. of Elements	Atomic Number
1.	Shortest	2	1 and 2
2.	Short	8	3 to 10
3.	Short	8	11 to 18
4.	Long	18	19 to 36
5.	Long	18	37 to 54
6.	Longest	32	55 to 86
7.	Longest		87 to ....

Lanthanides (Rare Earth Elements): Elements with atomic number 55 to 86

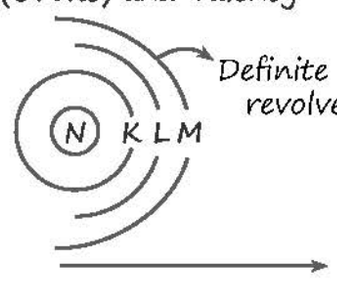
Actinides (Radioactive Elements): Elements with atomic number 87 to ....

Lanthanides and actinides: Inner transition Elements

Periodicity

- The properties that reappear at regular intervals or in which there is a gradual variation at regular intervals are called periodic properties and the phenomenon is called **periodicity of elements.**
- The cause of periodicity is the recurrence of similar electronic configuration.

Shells (Orbits) and Valency



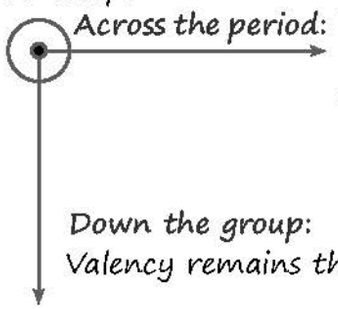
Definite circular paths in which electrons revolve around the nucleus = ORBITS

Along the period/Across the period, the number of shells remains the same.

Down the group no of shells/orbits increases and is equal to the number of the period to which that element belongs

Valency

Valency denotes the combining capacity of an element. It is equal to the number of electrons an atom can donate or accept.

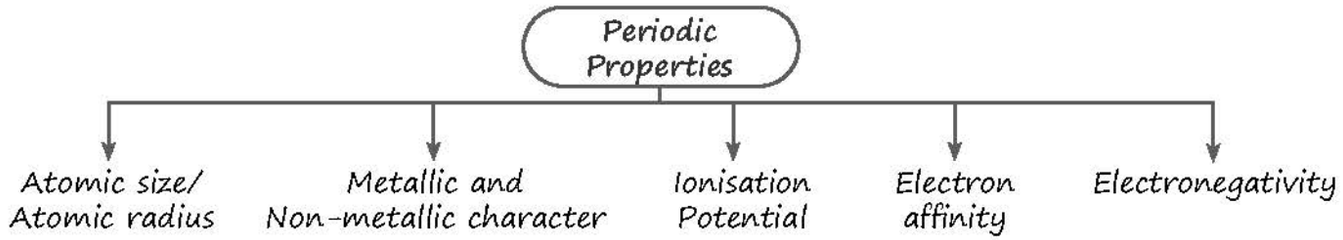


Across the period: Valency increases upto group 14 where it becomes 4 and then it decreases in group 17, it becomes 1.

Down the group:
Valency remains the same

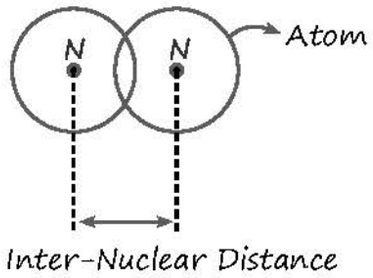
Periodic Properties

The properties which are directly or indirectly related to their electronic configuration and which shows a regular gradation when we move from left to right in a period or from top to bottom in the group are called as 'Periodic properties'.



I. Atomic Radius

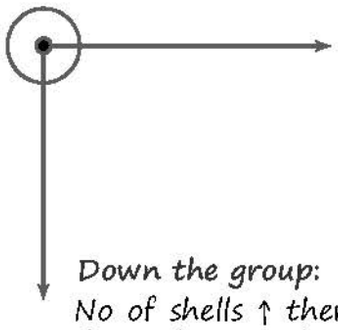
It is the distance between the centre of the nucleus of an atom and its outermost shell.



$$\text{At. radius} = \frac{1}{2}\text{ (Internuclear distance)}$$

Unit: $1A^\circ = 10^{-10}$ m

1 pm = 10^{-12} m



Across the period:
Atomic radius decreases as we move from left to right in a period due to the increase in the effective Nuclear charge.

Down the group:
No of shells ↑ thereby increasing the atomic radius due to increase in the distance of the outermost shell.

Ionic Radius

The effective distance from the centre of the nucleus of the ion upto which it exerts its influence on its electronic cloud.

Anion > Parent atom > Cation

Isoelectronic Species

The ions having the same number of electrons are called isoelectronic species.

$$\text{Size} \propto \frac{1}{\text{Nuclear charge}}.$$

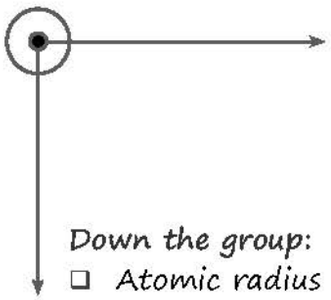
II. Metallic Character

Those elements, which have a tendency to lose their valence electrons and form a positive ion are called metals.

- Metallic character depends on: (i) Atomic size, (ii) Nuclear charge

$$\text{Atomic size} \propto \text{Metallic character}$$

$$\text{Nuclear charge} \propto \frac{1}{\text{Metallic character}}$$



Across the period:
- Atomic radius decreases
- Nuclear pull increases
- Metallic nature decreases

Down the group:
- Atomic radius increases
- Tendency to lose electrons increases
- Metallic character increases

Metals are good Reducing agents as they have a greater tendency to lose e^-

Li Be — Metals; B — Metalloid; C N O F — Non-Metals

Non-Metallic Character

Those elements which have a tendency to gain electrons, in order to attain octet in their outermost orbit are considered as non-metals.

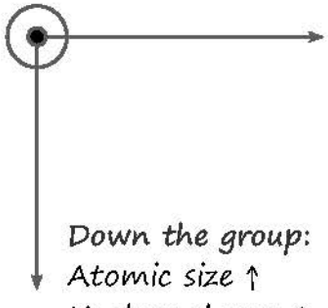
- Non Metallic character Depends on: (i) Atomic size, (ii) Nuclear charge

- Nuclear charge $\propto$ non-metallic character

$$\text{Atomic size} \propto \frac{1}{\text{Non-Metallic character}}$$

C — Non-Metals; Si Ge — Metalloid; Sn Pb — Metals

Non-metals are good oxidising agents as they have a greater tendency to accept electrons



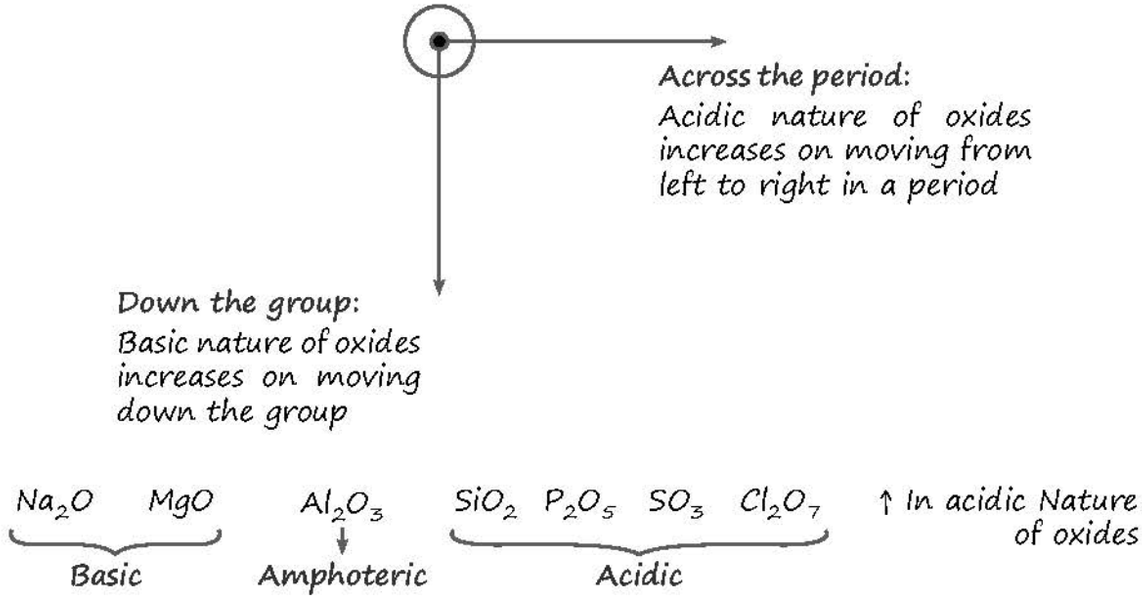
Across the period:
Atomic size ↓
Nuclear pull ↑
Non-metallic character ↑

Down the group:
Atomic size ↑
Nuclear charge ↑
Non-metallic character ↓

Na < Mg < Al < Si < P < S < Cl

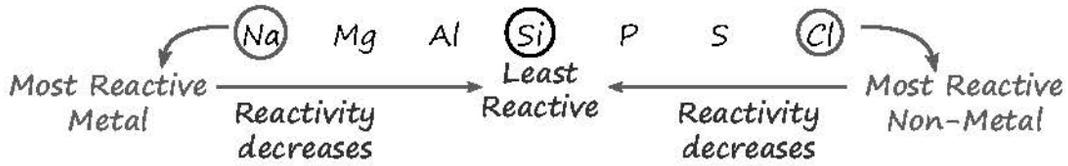
Metals — Metalloid — Non-metals

Nature of Oxides



Chemical Reactivity

Greater the tendency to lose / gain electron ∝ Chemical reactivity



- As we move down the group, tendency to lose electrons increases therefore, reactivity increases on moving down the group. Fr is the most reactive metal
- The chemical reactivity of non-metals decreases on moving down the group. Fluorine is the most reactive non-metal.

Gradation in Physical Properties

Melting and Boiling Points

Metals- mpt and bpt decrease on going down the group

Non-Metals- mpt and bpt increase on going down the group

- Across a period, left to right mpt and bpt usually increase upto group 14 and then decrease.

Density

Down a group: Density ↑

Across a period, density gradually increases to maximum and then decrease slightly.

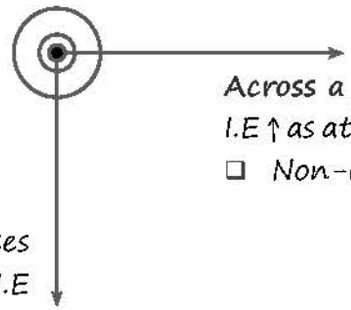
Ionisation Potential or Ionisation Energy or Ionisation Enthalpy

The energy required to remove an electron from a neutral gaseous isolated atom and convert it into a positively charged gaseous ion is called ionisation potential (I.P.)/ ionisation energy (I.E.)

$M(g) + I.E \rightarrow M^+(g) + e^-$

I.E depends on: (i) Atomic size (ii) Nuclear change Unit: eV/atom, kJ/mol

$\text{Atomic size} \propto \frac{1}{I.E}$ Nuclear change $\propto$ I.E



Across a period:
I.E ↑ as atomic size decreases
- Non-metal have high I.E

Down the group:
I.E ↓ as atomic size increases
- Metal usually have low I.E

Electron Affinity (E.A) or Electron Gain Enthalpy

The amount of energy released while converting a neutral gaseous isolated atom into a negatively charged gaseous ion by the addition of electron is called electron affinity (E.A)

$$X_{(g)} + e^- \longrightarrow X^-_{(g)} + E.A$$ Unit: eV/atom, kJ/mol

Electron affinity depends on: (i) Atomic size (ii) Nuclear change

$\text{Atomic size} \propto \frac{1}{E.A}$ Nuclear change $\propto$ E.A

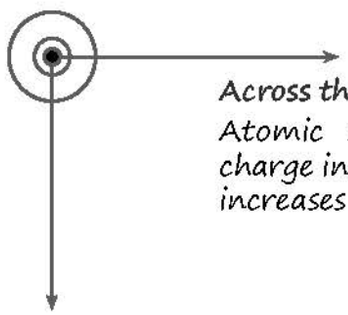
- Exception: In group 17, Fluorine has lower electron affinity than Chlorine.

  $\boxed{Cl > F}$

  In group 16, Oxygen has lower E.A than Sulphur

  $\boxed{S > O}$

- Inert Gases have positive values of E.A/Zero Electron Affinity



Across the period:
Atomic size decreases, nuclear charge increases, electron affinity increases

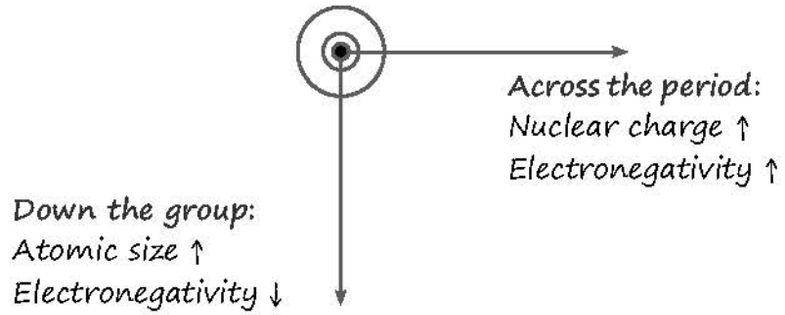
Down the group:
Electron affinity decreases, as the atomic size increases more than the nuclear charge

Electro Negativity

The tendency of an atom in a molecule to attract the shared pair of electrons towards itself is called its electronegativity.

- Dimensionless property
- Electronegativity values depends on: (i) Atomic size (ii) Nuclear charge

$$\text{Atomic size} \propto \frac{1}{\text{Electronegativity}} \qquad \text{Electronetgativity} \propto \text{Nuclear charge}$$



- Metals show lower electronegativity as compared to non-metals.
- Metals are electropositive
- Non-metals are electronegative

Diagonal Relationship

Properties which have increasing trend in a group, generally have decreasing trend in a period.

- Elements of the second period shows resemblance in properties with the elements of the third period, present at right diagonal of the element. This is called as 'Diagonal Relationships'

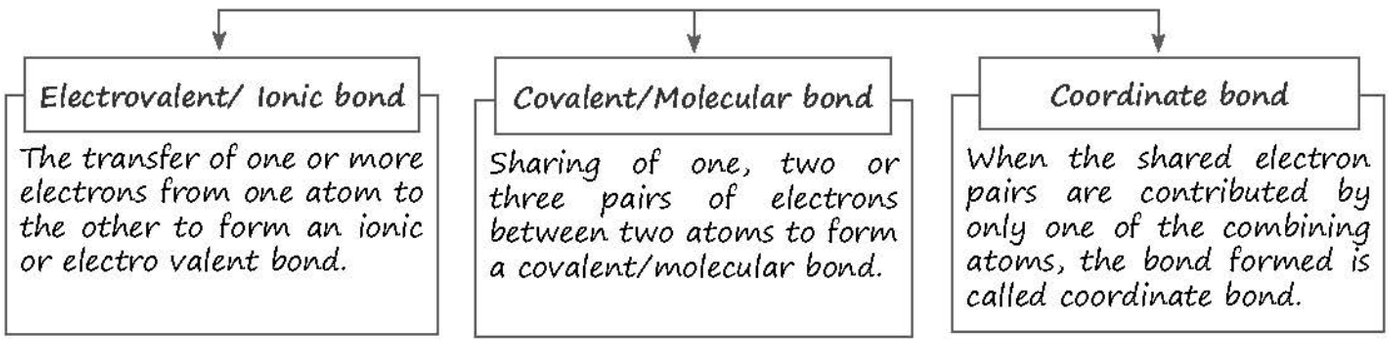
Group →	1	2	13	14	
Period 2 →	Li	Be	B	C	Bridge Element
Period 3 →	Na	Mg	Al	Si	

2 CHAPTER

Chemical Bonding

Chemical Bond

- A chemical bond may be defined as the force of attraction between any two atoms, in a molecule, to maintain stability.
- **Cause of chemical combination:** It is the tendency of elements to acquire the nearest noble gas configuration in their outermost orbit and become stable.
- Three methods in which atoms can achieve a stable configuration:



Electrovalent Bond/Ionic Bond

A chemical bond that is formed due to the electrostatic force of attraction between a cation and an anion is called Electrovalent or an Ionic bond.

- **Electrovalent compounds:** The chemical compounds formed as a result of the transfer of electrons from one atom of an element to one atom of another element are called ionic or electrovalent compounds.
- **Electrovalency:** The number of electrons that an atom of an element loses or gains to form an electrovalent bond is called electrovalency.

Conditions for the Formation of an Electrovalent or Ionic Bond

LOW I.E.	HIGH E.A.	Large Electronegativity Difference
❑ If the ionisation potential of a particular atom is low it will lose electron(s) easily i.e. a cation is formed easily.	❑ If the electron affinity value is high, anion will be formed easily. ❑ A higher electron affinity value favours ionic bonding.	❑ Higher the electronegativity difference more will be the ionic nature of the resulting compound.

Properties of Electrovalent Compounds

1. **Nature:** Their constituent particles are ions. They are hard solids consisting of ions.

   Reason: They have strong electrostatic forces of attraction between their ions, which cannot be separated easily.

2. **Boiling point and Melting point:** These are non-volatile, having high mpt and bpt

   Reason: There exists a strong force of attraction between the oppositely charged ions, so a large amount of energy is required to break the strong bonding force between ions.

3. **Conducting Nature:** They do not conduct electricity in the solid state. They are good conductors of electricity in the fused or in aqueous state.

   Reason: Electrostatic forces of attraction between ions in the solid state are very strong. These forces weaken in fused state or in solution state. Hence, ions become mobile.

4. **Dissociation:** Electrovalent compounds are composed of ions. In solution, these ions become mobile or in molten state these ions dissociate. Their ions dissociate and migrate when an electric current passes through them in their molten or aqueous solution state.

   $$NaCl_{(m)} \rightleftharpoons Na^+ + Cl^-$$

   Reason: Water being a polar covalent compound decreases the electrostatic forces of attraction resulting in free ions in aqueous solution. Ions dissociate in water or in molten state.

   Electrovalent compounds are good conductors of heat.

5. **Solubility:** These are soluble in water but insoluble in organic solvents.

   Reason: Water being a polar covalent compound, decreases the electrostatic forces of attraction, resulting in free ions in aqueous solution.

6. **Rate of Reaction:** They show rapid speed of chemical reactions in aqueous solutions.

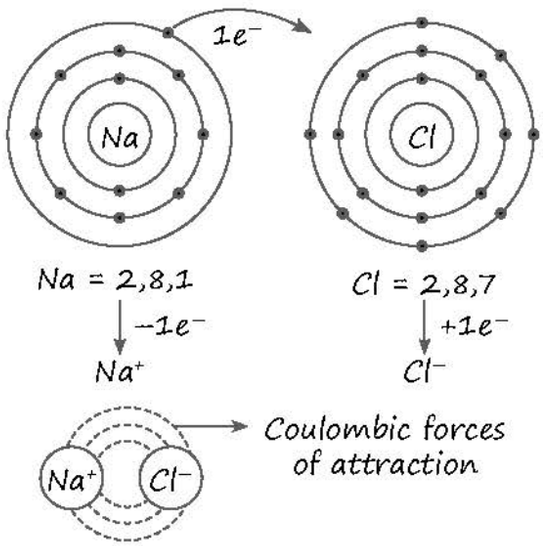
   Reason: Free ions are easily formed in different solutions, they unite very fast forming compounds.

Structures of Some Electrovalent Compounds: Forming Compounds

1. Sodium Chloride (NaCl)

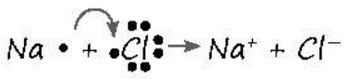
   Properties of sodium atom and sodium ion:

	Sodium Atom (Na)	(Na^+) Sodium ion
1.	Silvery white	Colourless
2.	Poisonous	Non-Poisonous
3.	Very active	Inactive
4.	Incomplete outer-most shell	Complete outermost shell
5.	Neutral	Positively charged
6.	Combined state	Independent Existence



Electron Dot Structure of Sodium Chloride:

Chlorine Atom (Cl)		Chloride Anion (Cl^-)
1.	Yellowish green	Colourless
2.	Poisonous	Non-Poisonous
3.	Suffocating	Odorless
4.	Very active	Inactive
5.	Incomplete outermost shell	Complete outermost shell
6.	Neutral	Negatively charged
7.	Not independent	Independent



or

NaCl/Sodium chloride

Covalent Bond/Molecular Bond

The chemical bond that is formed between two combining atoms by mutual sharing of one or more pairs of electrons is called covalent bond and the compound formed due to this bond is called covalent compound.

Type of covalent Bond

- **Single covalent bond:** Formed by sharing of one pair of electrons between the atom H — H, Cl — Cl, H — O — H, H — Cl
- **Double covalent bond:** Formed by the sharing of two pairs of electrons between two atoms. O = O, O = C = O
- **Triple covalent bond:** Formed by the sharing of three pairs of electrons between two atoms. H — C ≡ C — H, N ≡ N

Covalent Compounds

Non-Polar Covalent Compounds:

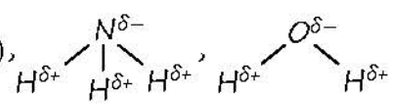
Covalent compounds are non polar when shared pair of electron(s) are equally distributed between the two atoms.

Eg: O_2, Cl_2, H_2

Polar Covalent Compounds:

Covalent compounds are said to be polar when the shared pair of electrons are not at equal distance between the two atoms.

- This develops the fractional positive and negative charges on them.

  Eg: $H-Cl(H^{\delta+}-Cl^{\delta-})$, 

- Since a polar covalent molecule has both positive and negative poles, it is also known as 'DIPOLE MOLECULE'.

Conditions for Formation of a Covalent Bond

1. Both atoms should have 4 or more electrons in their outermost shells
2. Both atoms should have high electronegativity
3. Both atoms should have high electron affinity
4. Both atoms should have high ionisation energies
5. The electronegativity difference between the combining atoms should be either zero or negligible.

Properties of Covalent Compounds

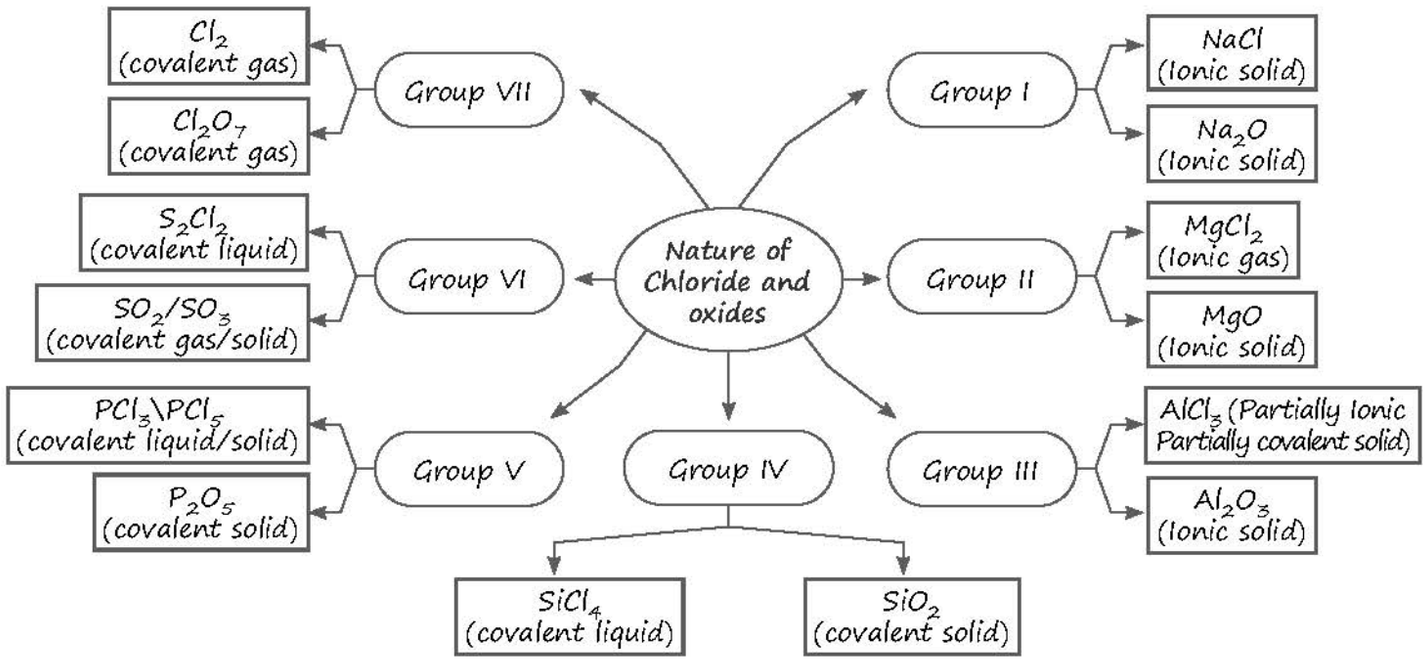
1. **Nature:** Their constituent particles are molecules. These are gases or liquids or soft solids
   Reason: They have weak forces of attraction.
2. **Boiling point and Melting point:** These are volatile, with low mpt and bpt
   Reason: They have weak forces of attraction.
3. **Conducting Nature:** They are non-conductors of electricity in solid, molten or aqueous state.
   Reason: Due to absence of free ions
4. **Ionisation in Solutions:**

   Non Polar covalent compounds $\xrightarrow[\text{Current}]{\text{Electric}}$ No IONS

   Polar Covalent compounds $\xrightarrow[\text{Current}]{\text{Electric}}$ Ionise in their solution

   $HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$

   Covalent compounds are poor conductors of heat.
5. **Solubility:** They are insoluble in water but dissolve in organic solvents
   Reason: As organic solvents are non-polar, hence these dissolve in non-polar covalent compounds.
6. **Rate of Reaction:** In aqueous solutions, their rate of reaction is slow.
   Reason: In covalent molecules; reactions are accompanied by breaking of old bonds and creation of new bonds, thus the reaction in slow.



Coordinate Bond

The bond formed between two atoms by sharing a pair of electrons, provided entirely by one of the combining atoms but shared by both is called is called coordinate bond/Dative bond.

Eg: H_3^+O (Hydronium ion), NH_4^+ (Ammonium ion)

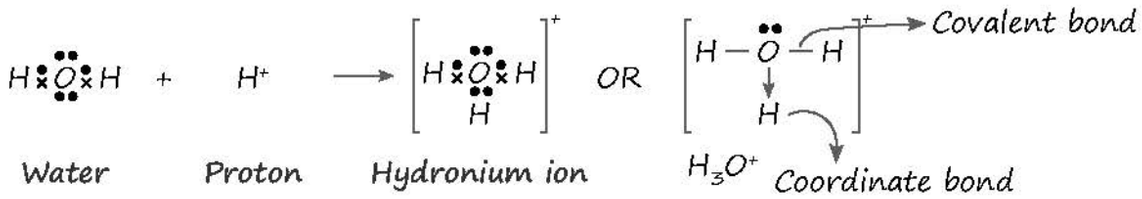
Conditions for the Formation of Coordinate Bond

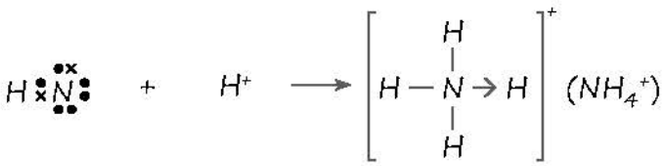
(i) One of the two atoms must have at least one lone pair of electrons.

Eg.: NH_3, H_2O

The lone pair of electrons is provided to the other atom for the formation of coordinate bond.

(ii) Another atom should be short of at least a pair of electrons. Hydrogen (H^+)





Self Ionisation of Water

Self-ionisation of water/auto protolysis of water/auto dissociation of water is an ionisation of water in aqueous medium. Water is amphiprotic, one water molecule can react with another to form an OH^- ion and an H_3^+O in an autoionisation process.

$$2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

3 CHAPTER

Acids, Bases and Salts

Acids

Acids are defined as compounds which contains one or more hydrogen atoms and when dissolved in water, produce H_3O^+ ions the only positively charged ions.

$HCl + H_2O \rightleftharpoons H_3O^+ + Cl^-$

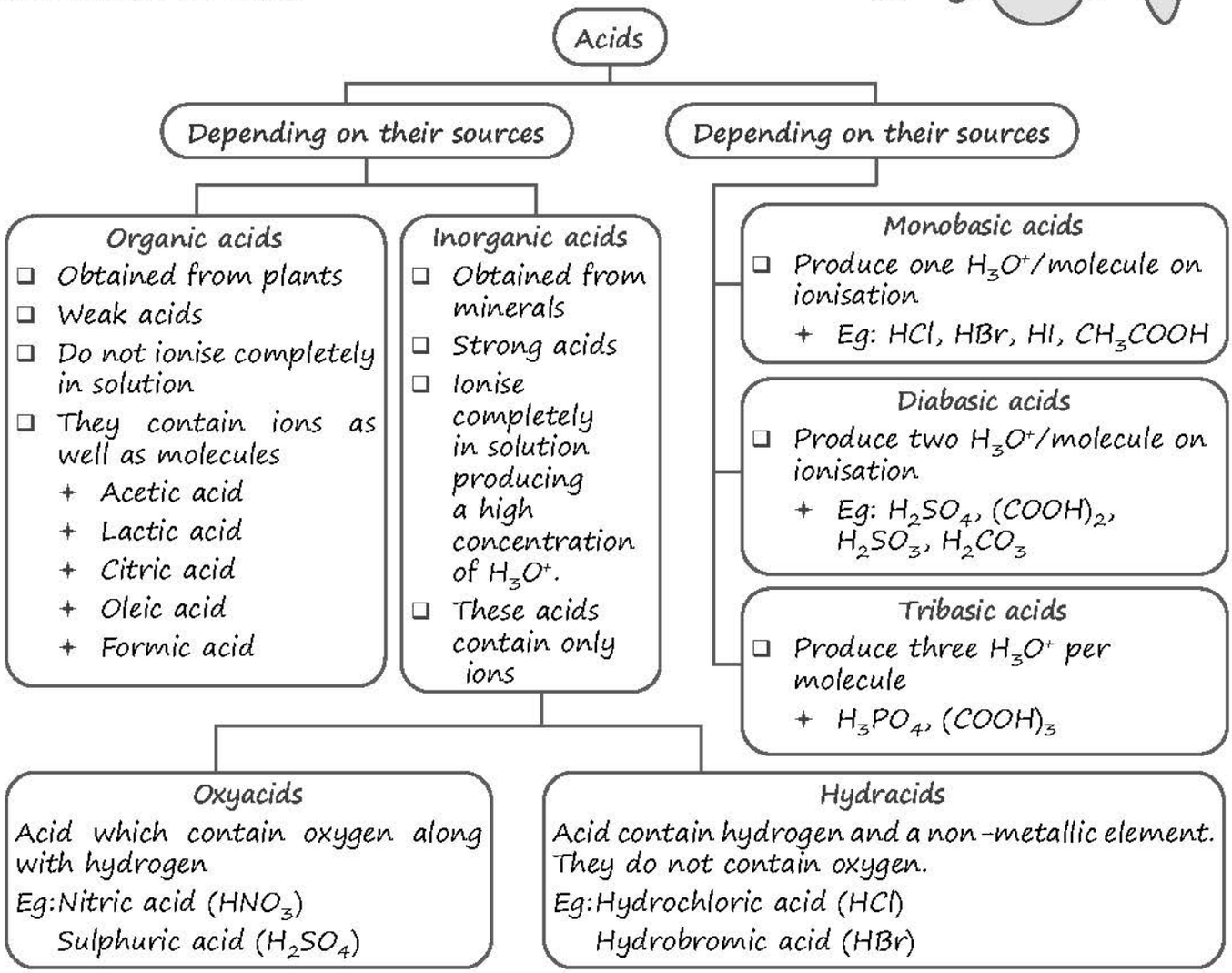
$H_2SO_4 + H_2O \rightleftharpoons 2H_3O^+ + SO_4^{-2}$

$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$

Basicity of an Acid

The number of hydronium ions (H_3O^+) that can be produced by the ionsiation of one molecule of that acid in aqueous solution.

Classification of Acids



Concentration of an Acid

It is the amount of acid present in a definite amount of its aqueous solution.

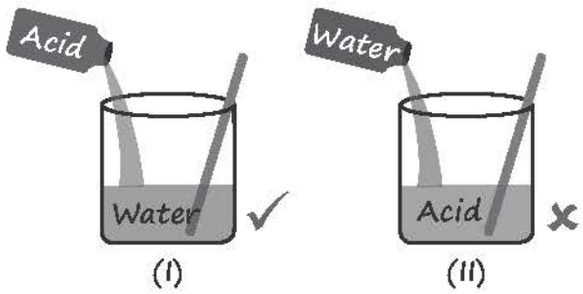
Concentration

- **Dilute acid**: An acid which contains far more amount of water than its own mass is called dilute acid.
- **Concentrated acid**: An acid which contains a very small amount of water / no water is called as concentrated acid.

Dilution of Acid

(I) Correct way of dilution of an acid.

(II) Incorrect way: water is not added to acid as it is an Exothermic process and in this process enormous amount of heat is produced which may break the container.



- **Strength of an Acid**: It is the measure of concentration of hydronium ions it produces in its aqueous solution.
  dil HCl > CH_3COOH ... dil HCl produces more H^+ ion
  (conc)

Preparation of Acids

By Synthesis

Binary acids are prepared by this method.

$$H_2 + Cl_2 \rightarrow 2HCl$$
$$H_2 + Br_2 \rightarrow 2HBr$$

By the action of water on non-metallic or acidic oxides

Acidic oxide/Acidic + Water → Acid anhydride

$$SO_3 + H_2O \rightarrow H_2SO_4$$
$$CO_2 + H_2O \rightarrow H_2CO_3$$
$$N_2O_5 + H_2O \rightarrow 2HNO_3$$

By the oxidation of non-metals

$$S + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$$
$$P + 5HNO_3 \rightarrow H_3PO_4 + H_2O + 5NO_2$$

By displacement

Normal salts of more volatile acids are displaced by less or non-volatile acid.

$$NaCl + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HCl$$
$$NaNO_3 + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HNO_3$$

Physical Properties

Physical State

Some acids are solids and some are liquids at room temperature.

- **Solid Acids:** Boric acid, tartaric acids, citric acid
- **Liquid Acids:** Sulphuric acid, nitric acid, acetic acid
- **Non-Volatile acids:** Sulphuric acid

Effect on Skin

All mineral acids have corrosive action on skin.

- Conc. H_2SO_4: Chars the skin black
- Conc. HNO_3: Yellowing of skin
- Conc. HCl: Amber colour to skin

Carbonic acid and organic acids are non-corrosive

Indicators

Indicators are complex substances that acquire distinct colours in acidic and basic mediums.

- **Litmus:** Blue to red in acidic medium.
- **Methyl orange:** orange to pink in acidic medium.
- **Phenolphthalein:** Remains colourless in acidic medium.

Taste

Acids have sour taste. Mineral acids such as HNO_3, H_2SO_4 are highly corrosive in nature.

Electrolytes

They are electrolytes i.e they conduct electricity in the aqueous state.

Chemical Properties

Decomposition of Sulphites and Bisulphites

Sulphite/Bisulphite + Acid → Salt + H_2O + $SO_2\uparrow$

- $CaSO_3 + 2HCl \rightarrow CaCl_2 + H_2O + SO_2\uparrow$

Decomposition of Sulphides

Metal Sulphide + Acid → Salt + $H_2S\uparrow$

$ZnS + 2HCl \rightarrow ZnCl_2 + H_2S\uparrow$

Reaction with active metals

- Active metal + Acid → Salt + $H_2\uparrow$

  Zn/Mg/Fe + dil HCl or dil H_2SO_4 → metallic salt + $H_2\uparrow$
- HNO_3 is an oxidising agent so it is not used in the preparation of hydrogen. It oxidices H_2 to form water.

  $Mn + 2HNO_3 \rightarrow Mn(NO_3)_2 + H_2\uparrow$

  (dilute)

Reaction with Chlorides and Nitrates

- Few chlorides and nitrates react with dilute H_2SO_4 and dil HCl:

  $BaCl_2 + H_2SO_4 \rightarrow 2HCl + BaSO_4\downarrow$
- Mostly, chlorides and nitrates react only with conc. H_2SO_4 on warming to liberate HCl and HNO_3.

Reaction with Bases (Neutralisation)

Acid + Basic oxide/Hydroxide → Salt + water

Decomposition of Hydrogen Carbonates and Carbonates

Carbonates + Acid → Salt + H_2O + CO_2

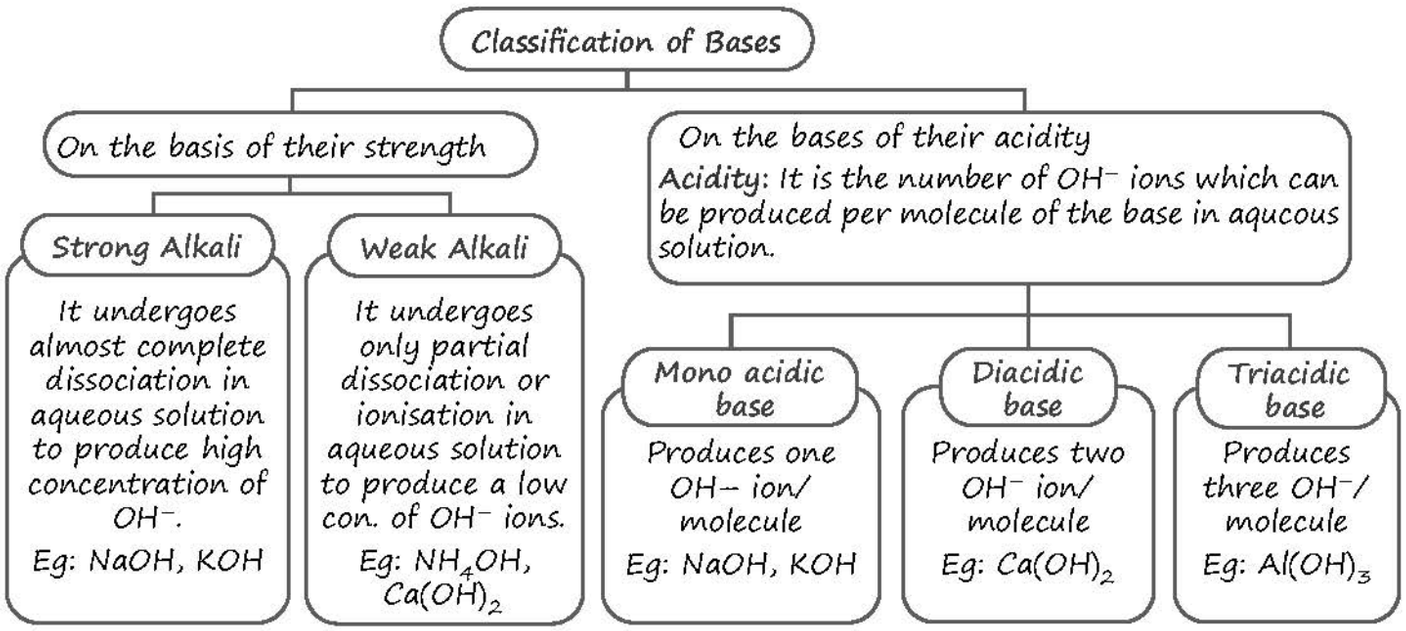
Bicarbonates + Acid → Salt + H_2O + CO_2

- If the salt produced is insoluble, then reaction does not proceed. Lead carbonate and calcium carbonate do not react with H_2SO_4.

Bases

- A base is either a metallic oxide or a metallic hydroxide or ammonium hydroxide which reacts with hydronium ions of an acid to form salt and water only.
  Eg: CuO(metallic oxide), $Mg(OH)_2$
- **Basic oxide:** A basic oxide is a metallic oxide which contains the ion O^{2-} and reacts with an acid to form salt and water only.
- **Basic hydroxide:** It is a metallic hydroxide which contains OH^- and will react with an acid to form salt and water.
  Eg: NaOH, $Al(OH)_3$
- **Alkalis:** An alkali is a basic hydroxide which when dissolved in water produces OH^- ions as the only negatively charged ions.
  + Alkali is a base soluble in water. Eg: NaOH, KOH, $Ca(OH)_2$, NH_4OH

  All alkalis are bases but all bases are not alkalis:
  For Eg: $Fe(OH)_3$ and $Cu(OH)_2$ are bases, but not alkalis because they are insoluble in water.



Properties of Bases/Alkalis

Physical Properties

- They have sharp and bitter taste
- They are soapy to touch
- They change the colour of the indicators
  - **Litmus:** Blue to red in basic medium
  - **Phenolphthalein:** Colourless to pink
  - **Methyl orange:** Orange to yellow.
- They are strong electrolytes and they show a mild corrosive action on skin.

Chemical Properties

- Strong alkali + CO_2 ⟶ Carbonate + Water
  - + $2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$
  - + $2KOH + CO_2 \longrightarrow K_2CO_3 + H_2O$
- Base/Alkali + Acid ⟶ Salt + Water
  - + $Ca(OH)_2 + 2HCl \longrightarrow CaCl_2 + H_2O$
- They precipitate as insoluble metallic hydroxides when added to the solutions of the salts of heavy metals

  Metalic salt + Base/alkali ⟶ salt + Insoluble hydroxides

  (a) $CuSO_4 + 2NH_4OH(aq) \longrightarrow (NH_4)_2SO_4 + Cu(OH)_2\downarrow$ (pale blue)

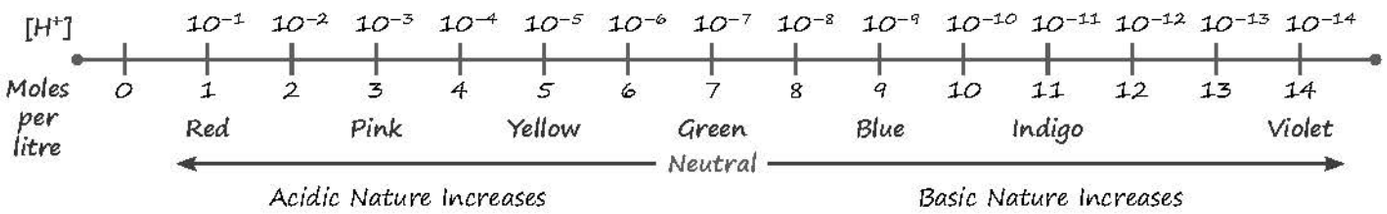
  (b) $ZnSO_4 + 2NaOH \longrightarrow Na_2SO_4 + Zn(OH)_2\downarrow$ (gelatinous white ppt)

  (c) $FeSO_4 + 2NaOH \longrightarrow Na_2SO_4 + Fe(OH)_2\downarrow$ (dirty green)
  - + The hydroxides of Zinc, Aluminium and Lead being amphoteric in nature, dissolve in excess of NaOH or KOH but other hydroxides do not.
- When alkalies are warmed with an ammonium salt, ammonia gas is given out.
  - $NH_4Cl + NaOH \xrightarrow{\Delta} NaCl + H_2O + NH_3\uparrow$
  - $2NH_4Cl + Ca(OH)_2 \xrightarrow{\Delta} CaCl_2 + 2H_2O + 2NH_3\uparrow$

pH Scale: Devised by "Sorensen (1909)"

pH of the solution is the negative logarithm to the base 10 of H^+ conc. expressed in moles/litre

$$pH = \log_{10}\frac{1}{[H^+]} = -\log_{10}[H^+]$$



- Universal Indicator: It is a mixture of indicator dyes that gives a spectrum of colours depending on how acidic or alkaline a solution is Universal indicators give different colours at different concentrations of H^+ ions in a solution.

Importance of pH in Everyday Life

Body pH for normal functioning is 7 to 7·8

In preventing tooth decay

If pH of mouth falls bellow 5.5 tooth decay starts. Tooth paste is used to neutralise the excess acid in the mouth.

In Agriculture

- Rice: Slightly acidic soil
- Sugar cane: Neutral
- Citrus fruits: Alkaline soil

In digestive system

HCl is produced in the stomach which aids in digestion. Excess acid causes acidity which can be relieved by antacids.

In Medicines

Diagnosis of certain diseases are done on the basis of pH value of blood and urine.

pH less than 5.6 of rain water is Acid Rain. Acid rain reduces fertility of soil, destroys marine life and causes damage to monuments.

Salts and Their Preparation

Salt: Salt is the neutralisation product of an acid and a base. Salt is a compound formed by the partial or total replacement of the ionisable hydrogen atoms of an acid by a metallic ion or an ammonium ion.

- $NaNO_3 + \underset{(conc)}{H_2SO_4} \xrightarrow{< 200°C} NaHSO_4 + HNO_3$ **(Partial Replacement)**
- $2NaNO_3 + \underset{(conc)}{H_2SO_4} \xrightarrow{> 200°C} Na_2SO_4 + 2HNO_3$ **(Complete Replacement)**

Classification of Salts

Normal Salts

Normal salts are the salts formed by the complete replacement of the ionisable hydrogen atoms of an acid by a metallic or an NH_4^+ ion.

Eg: $HCl + NaOH \longrightarrow NaCl + H_2O$

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

Double Salts

- Formed by the union of two simple salts that dissolve in water and crystallise together.
  - Potash alum: $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
  - Mohr salt: $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
  - Dolomite: $CaCO_3 \cdot MgCO_3$

Acid Salts

Formed by the partial replacement of the ionisable hydrogen atoms of a polybasic acid by a metal or an NH_4^+ ion.

$H_2SO_4 + NaOH \longrightarrow \underset{\text{Acidic salt}}{NaHSO_4} + H_2O$

Mixed Salts

Those salts that contain more than one basic or acid radical.

Eg: $CaOCl_2$, $NaKCO_3$

Basic Salts

Formed by a partial replacement of the OH group of a ditri-acidic base by an acid radical. $Pb(OH)Cl$, $Mg(OH)Cl$, $Cu(OH)Cl$.

Complex Salts

Those salts which on dissociation give one simple ion and one complex ion. Eg:

- $[Ag(NH_3)_2]Cl$
- $K_2[HgI_4]$

Solubility of Salts

1. All the compounds of ammonium, sodium and potassium are Soluble in water.
2. All nitrates and nitrites are soluble.
3. All chlorides, bromides and Iodides are soluble in water.
   [Hg_2Cl_2, AgCl, $PbCl_2$] [$PbCl_2$ is soluble in hot water, not in cold water]
4. All sulphates are soluble ($CaSO_4$, $PbSO_4$, $BaSO_4$)
5. All carbonates, sulphides, sulphites and phosphates are insoluble. (except of ammonium, sodium and potassium)

All metallic oxides and hydroxides are insoluble except of sodium, potassium and ammonium.
$Ca(OH)_2$: Slightly soluble

- $NaHCO_3$ and $KHCO_3$: Sparingly soluble.

Methods of Preparation of Salts (Soluble Salts)

Direct Combination

Metal + Non-metal ⟶ Salt

$2Na + Cl_2 \longrightarrow 2NaCl$

$2Al + 3Cl_2 \longrightarrow 2AlCl_3$

Neutralisation

- Acid + Soluble base ⟶ Salt + Water
  $HNO_3 + NaOH \longrightarrow NaNO_3 + H_2O$
- Acid + Insoluble base ⟶ Salt + Water
  $2HCl + MgO \longrightarrow MgCl_2 + H_2O$

Decomposition

1. Decomposition of carbonates/bicarbonates by acids
   $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$
2. Decomposition of Sulphites/hydrogen sulphites by acids
   $Na_2SO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2$
3. Decomposition of Sulphides by acids
   $K_2S + H_2SO_4 \longrightarrow K_2SO_4 + H_2S$

By the action of Alkalis on Certain metals, their Hydroxides and their Oxides

1. $2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2$
2. $Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$
3. $Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$

Simple Displacement

Active Metal + Acid(dil) ⟶ Salt + Hydrogen

Methods of Preparation of Insoluble Salts

- **By Direct Combination**
  Metal + Non-metal ⟶ Salt
  + $Pb + S \xrightarrow{\Delta} PbS$

- **By the Combination of an acidic oxide with a basic oxide**
  Acidic oxide + Basic oxide ⟶ Salt
  + $SO_2 + CaO \xrightarrow{\Delta} CaSO_3$

- **By Precipitation reaction**
  Soluble Slat + Acid or Soluble salt ⟶ Insoluble salt + Acid or salt
  + $CuSO_4 + H_2S \longrightarrow CuS\downarrow + H_2SO_4$
  + $BaCl_2 + H_2SO_4 \longrightarrow 2HCl + BaSO_4\downarrow$

4 CHAPTER Analytical Chemistry

Action of Sodium Hydroxide Solution on Certain Metallic Salt Solution

SALT SOLUTION + ALKALI ⟶ METAL HYDROXIDE [PPT.] + SALT FORMED IN SOLUTION	COLOUR OF PPT AND ITS SOLUBILITY IN EXCESS OF ALKALI
1. CALCIUM SALTS [Ca^{2+} ion] $Ca(NO_3)_2$ + $2NaOH$ ⟶ $Ca(OH)_2\downarrow$ + $2NaNO_3$ Calcium nitrate (colourless); Caustic soda (Colourless); Calcium hydroxide (white precipitate); Sodium nitrate (colourless)	WHITE SPARINGLY SOLUBLE
2. IRON: (A) FERROUS SALTS [Fe^{2+} ion] $FeSO_4$ + $2NaOH$ ⟶ $Fe(OH)_2\downarrow$ + Na_2SO_4 Ferrous sulphate (pale green); Caustic soda (colourless); Ferrous hydroxide (dirty green, gelatinous ppt.); Sodium sulphate (colourless)	DIRTY GREEN INSOLUBLE
(B) FERRIC SALT [Fe^{3+} ion] $FeCl_3$ + $3NaOH$ ⟶ $Fe(OH)_3\downarrow$ + $3NaCl$ Ferric chloride (yellow); Caustic soda (colourless); Ferric hydroxide (reddish brown ppt.); Sodium chloride (colourless)	REDDISH BROWN INSOLUBLE
3. COPPER SALTS [Cu^{2+} ion] $CuSO_4$ + $2NaOH$ ⟶ $Cu(OH)_2\downarrow$ + Na_2SO_4 (blue); (colourless); (pale blue ppt.); (colourless)	PALE BLUE INSOLUBLE
4. ZINC SALTS [Zn^{2+} ion] $ZnSO_4$ + $2NaOH$ ⟶ $Zn(OH)_2\downarrow$ + Na_2SO_4 (colourless); (colourless); (white, gelatinous ppt.); (colourless) (With excess of NaOH ppt dissolves) $Zn(OH)_2$ + $2NaOH$ ⟶ Na_2ZnO_2 + $2H_2O$ (excess); Sodium zincate (colourless)	GELATINOUS WHITE SOLUBLE

5. LEAD SALTS [Pb^{2+} ion] $Pb(NO_3)_2$ + $2NaOH$ ⟶ $Pb(OH)_2\downarrow$ + $2NaNO_3$ (colourless) (colourless) (white ppt.) (colourless) *(With excess of NaOH ppt dissolves)* $Pb(OH)_2 + 2NaOH \longrightarrow \underset{\text{Sodium plumbite (colourless)}}{Na_2PbO_2} + 2H_2O$ (excess) *(white precipitate of lead hydroxide is readily soluble in acetic acid)*	CHALKY WHITE SOLUBLE
6. AMMONIUM SALTS [NH_4^+ ion] When sodium hydroxide (or any water soluble base) is heated with ammonium salts, ammonia gas is evolved. $NH_4Cl + NaOH \xrightarrow{\Delta} NaCl + H_2O + NH_3$ $(NH_4)_2SO_4 + 2NaOH \xrightarrow{\Delta} Na_2SO_4 + 2H_2O + 2NH_3$	

Note: Potassium hydroxide (caustic potash) solution also shows similar behaviour.

Action of Ammonium Hydroxide on Certain Salt Solutions

SALT SOLUTION + AMMONIUM HYDROXIDE ⟶ METAL HYDROXIDE + SALT FORMED IN SOLUTION	COLOUR OF PPT AND ITS SOLUBILITY IN EXCESS OF AMMONIUM HYDROXIDE
1. CALCIUM SALTS No precipitation occurs even with addition of excess of NH_4OH. This is because the concentration of OH^- ions from the ionisation of NH_4OH is so low that it cannot precipitate the hydroxide of calcium.	
2. IRON: (A) IRON (II) SALTS [Fe^{2+} ion] $FeSO_4$ + $2NH_4OH$ ⟶ $Fe(OH)_2\downarrow$ + $(NH_4)_2SO_4$ (green) (dirty green ppt.) (colourless in solution)	DIRTY GREEN PPT INSOLUBLE
(B) IRON (III) SALTS [Fe^{3+} ion] (i) $FeCl_3$ + $3NH_4OH$ ⟶ $Fe(OH)_3\downarrow$ + $3NH_4Cl$ (yellow solution) (reddish brown ppt.) (colourless in solution) (ii) $Fe_2(SO_4)_3$ + $6NH_4OH$ ⟶ $2Fe(OH)_3\downarrow$ + $3(NH_4)_2SO_4$ (yellow solution) (reddish brown ppt.) (colourless in solution)	REDDISH BROWN PPT INSOLUBLE

3. COPPER (II) SALTS [Cu^{2+} ion]

$$\underset{\text{(blue)}}{CuSO_4} + 2NH_4OH \longrightarrow \underset{\text{(pale blue ppt.)}}{Cu(OH)_2}\downarrow + \underset{\text{(colourless in solution)}}{(NH_4)_2SO_4}$$

with excess of NH_4OH ppt. dissolves

$$Cu(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \longrightarrow \underset{\substack{\text{Tetraammine}\\ \text{copper (II) sulphate}}}{[Cu(NH_3)_4]SO_4} + 4H_2O$$

OR

$$Cu(OH)_2 + 4NH_4OH \longrightarrow \underset{\substack{\text{Tetraammine}\\ \text{copper hydroxide}}}{[Cu(NH_3)_4](OH)_2} + 4H_2O$$

This reaction is a characteristic property of Cu^{2+} ion and is used for its detection in qualitative analysis.

PALE BLUE PPT
SOLUBLE in excess of NH_4OH and forms deep blue solution

4. ZINC SALTS [Zn^{2+} ion]

$$\underset{\text{(colourless solution)}}{ZnSO_4} + 2NH_4OH \longrightarrow \underset{\text{(white, gelatinous ppt.)}}{Zn(OH_2)}\downarrow + \underset{\text{(colourless in solution)}}{(NH_4)_2SO_4}$$

With excess of NH_4OH ppt. dissolves.

$$\underset{\text{zinc hydroxide}}{Zn(OH)_2} + (NH_4)_2SO_4 + \underset{\text{(excess)}}{2NH_4OH} \longrightarrow \underset{\substack{\text{Tetraammine zinc (II)}\\ \text{sulphate}\\ \text{(colourless solution)}}}{[Zn(NH_3)_4]SO_4} + 4H_2O$$

OR

$$\underset{\text{zinc hydroxide}}{Zn(OH)_2} + 4NH_4OH \longrightarrow \underset{\substack{\text{Tetraammine zinc}\\ \text{hydroxide}}}{[Zn(NH_3)_4](OH)_2} + 4H_2O$$

GELATINOUS WHITE PPT
SOLUBLE in excess of NH_4OH

5. LEAD SALTS [Pb^{2+} ion]

$$\underset{\text{(colourless)}}{Pb(NO_3)_2} + 2NH_4OH \longrightarrow \underset{\text{(white ppt.)}}{Pb(OH)_2}\downarrow + 2NH_4NO_3$$

CHALKY WHITE PPT
INSOLUBLE

Action of Alkalis on Certain Metals

METAL + ALKALI ⟶ SALT + HYDROGEN

1. ZINC

(i) $Zn + \underset{\text{(Hot and Conc.)}}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium zincate}\\ \text{(colourless)}}}{Na_2ZnO_2} + H_2$

(ii) $Zn + \underset{\text{(Hot and Conc.)}}{2KOH} \longrightarrow \underset{\substack{\text{Potassium zincate}\\ \text{(colourless)}}}{K_2ZnO_2} + H_2$

2. ALUMINIUM

Aluminium reacts with boiling caustic alkalis solution

(i) $2Al + NaOH + 2H_2O \longrightarrow \underset{\text{Sodium meta aluminate (colourless)}}{2NaAlO_2} + 3H_2$

(ii) $2Al + 2KOH + 2H_2O \longrightarrow \underset{\text{Potassium meta aluminate (colourless)}}{2KAlO_2} + 3H_2$

Aluminium reacts with fused alkali to produce sodium aluminate

$2Al + 6NaOH \longrightarrow 2Na_3AlO_3 + 3H_2$

3. LEAD

(i) $Pb + 2NaOH \longrightarrow \underset{\text{Sodium plumbite (colourless)}}{Na_2PbO_2} + H_2$

(ii) $Pb + 2KOH \longrightarrow \underset{\text{Potassium plumbite (colourless)}}{K_2PbO_2} + H_2$

Note: Sodium meta aluminate or potassium meta aluminate are generally known as sodium aluminate and potassium aluminate.

Action of Alkalis on Metal Oxides

AMPHOTERIC OXIDE/ HYDROXIDE + ALKALI ⟶ SALT + WATER

1. Zinc oxide/Zinc hydroxide

(i) $\underset{\text{(white)}}{ZnO} + 2NaOH \longrightarrow \underset{\text{Sodium zincate (colourless, soluble)}}{Na_2ZnO_2} + H_2O$

$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$

(ii) $\underset{\text{(white)}}{ZnO} + 2KOH \longrightarrow \underset{\text{Potassium zincate (colourless, soluble)}}{K_2ZnO_2} + H_2O$

$Zn(OH)_2 + 2KOH \longrightarrow K_2ZnO_2 + 2H_2O$

2. Aluminium oxide/Aluminium hdyroxide

(i) $\underset{\text{(white)}}{Al_2O_3} + 2NaOH \longrightarrow \underset{\text{Sodium meta aluminate}}{2NaAlO_2} + H_2O$

$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$

(ii) $\underset{\text{(white)}}{Al_2O_3} + 2KOH \longrightarrow \underset{\text{Potassium meta aluminate}}{2KAlO_2} + H_2O$

$Al(OH)_3 + KOH \longrightarrow KAlO_2 + 2H_2O$

3. Lead oxide/Lead hydroxide

(i) $\underset{\text{(yellow)}}{PbO} + 2NaOH \longrightarrow \underset{\substack{\text{Sodium plumbite}\\\text{(colourless, soluble)}}}{Na_2PbO_2} + H_2O$

$Pb(OH)_2 + NaOH \longrightarrow Na_2PbO_2 + 2H_2O$

(ii) $\underset{\text{(yellow)}}{PbO} + 2KOH \longrightarrow \underset{\substack{\text{Potassium plumbite}\\\text{(colourless, soluble)}}}{K_2PbO_2} + H_2O$

$Pb(OH)_2 + 2KOH \longrightarrow K_2PbO_2 + 2H_2O$

5 CHAPTER

Mole Concept and Stoichiometry

- Pressure-Volume relationship: BOYLE'S LAW

  The volume of a given mass of dry gas is inversely proportional to its pressure at a constant temperature.

  $P \propto \frac{1}{V}$ [Temp = constant]

  PV = constant $\boxed{P_1V_1 = P_2V_2 = K}$

- Temperature – Volume relationship: CHARLE'S LAW

  The volume of a given mass of a dry gas is directly proportional to its absolute (Kelvin) temperature, if the pressure is kept constant.

  The pressure remaining constant, the volume of a given mass of a dry gas increases or decreases by 1/273 of its volume for each 1°C increase or decrease in temperature respectively.

  $V \propto T$ [P = constant]

$$\frac{V}{T} = \text{constant} \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Gas Equation:

$$V \propto \frac{1}{P} \propto T$$

$$\Rightarrow V = \frac{T}{P} \times \text{Constant} \Rightarrow \frac{PV}{T} \times \text{Constant} \Rightarrow \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Standard Temperature Pressure: (STP)

T = 0°C or 273K

P = 1 atm or 760 mm or 760 mm of Hg

Temperature Scales:

Absolute Scale/Kelvin Scale: A temperature scale with absolute zero as the starting point is called the absolute scale or the Kelvin scale.

$\boxed{\text{Absolute zero} = 0K = -273°C}$

- **Stoichiometry:** It measures the quantitative relationships, and is used to determine the amount of products/reactants that are produced/needed in a given reaction.
- Describing the quantitative relationships among substances as they participate in chemical reactions in known as **Reaction stoichiometry.**

Gay-Lussac's Law of Combining Volumes

When gases react, they do so in volumes which bear a simple ratio to one another, and to the volume of the gaseous product, provided that all the volumes are measured at the same temperature and pressure.

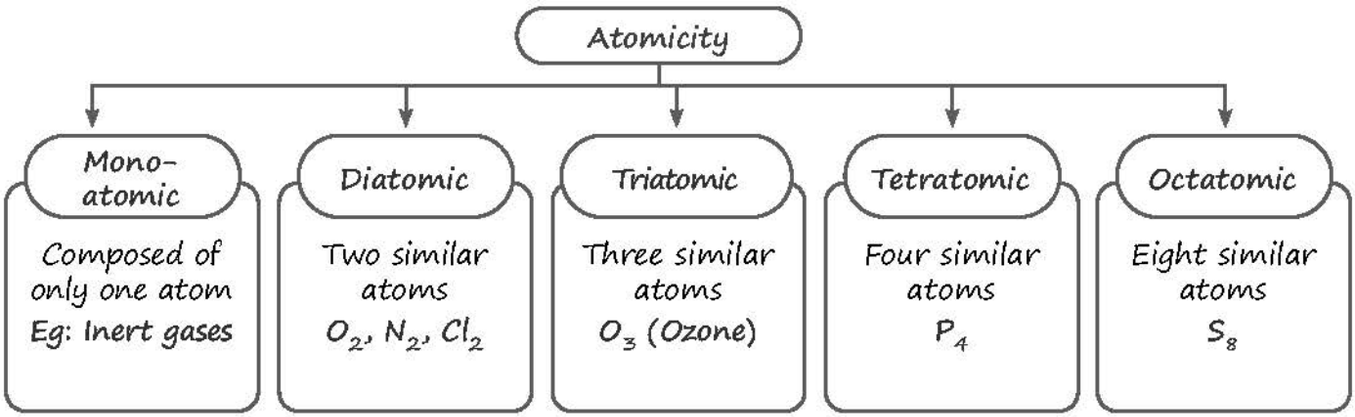
- Valid only for gases. The volumes of solid and liquids are considered to be zero.

Eg.: (i) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
1 : 3 : 2 (combining volume ratio)

(ii) $2CO_{(g)} + O_{2(g)} \longrightarrow 2CO_{2(g)}$
2 : ⊥ : 2

Avogadro's Law: It states that, "equal volumes of all gases under similar conditions of temperature and pressure contain the same number of molecules".

Atomicity: The number of atoms in a molecule of an element is called its atomicity.



- **Relative atomic mass:**

The relative atomic mass or atomic weight of an element is the number of times one atom of the element is heavier than $\frac{1}{12}$ times of the mass of an atom of carbon –12.

$$\text{Relative atomic mass} = \frac{\text{Mass of 1 atom of an element}}{1/12^{th} \text{ the mass of one atom of C-12 atom}}$$

- **Expressed as a.m.u.**
- Atomic mass unit is defined as $\frac{1}{12}^{th}$ the mass of C-12 atom.

Fractional atomic masses

It is the weighted average of the relative atomic masses of its natural isotopes.

Eg: Chlorine consists of a mixture of two isotopes of masses 35 and 37 in the ratio 3 : 1.

$$\therefore \text{ Av. relative mass of chlorine } = \frac{35 \times 3 + 37 \times 1}{4} = 35.5$$

- **Relative molecular mass: (Molecular weight)**

  The relative molecular mass of an element or compound is the number that represents how many times one molecule of the substance is heavier than $\frac{1}{12}^{th}$ of the mass of an atom of carbon-12

  Eg: Relative molecular mass of H_2SO_4: $2 \times 1 + 32 + 16 \times 4 = 98u$.

- **Gram atomic mass:** The atomic mass of an element expressed in grams is called gram atomic mass.

  Eg. 1g atom of sodium = 23g of sodium

- **Gram molecular mass:** The molecular mass of a substance expressed in grams is called gram molecular mass or molar mass.

  Eg. Molecular mass of H_2O = 18 a.m.u.

  Gram molecular mass of water = 18g.

Mole Concept

A mole is the amount of pure substance containing the same number of chemical units as there are atoms in exactly 12 grams of carbon-12.

- **Avogadro's Number:** It is defined as the number of atoms present in 12g of C-12 isotope ie: 6.022×10^{23} atoms.
  - **Mole of Atoms:** One mole of atoms contain 6.022×10^{23} atoms having mass equal to its gram atomic mass.

    1mole of oxygen atom $= 6.022 \times 10^{23}$ atoms of oxygen = 16 g

  - **Mole of Molecules:** One mole contain 6.022×10^{23} molecules and is equivalent to the gram molecular mass of any given substance.

    1mole of oxygen molecule $= 6.022 \times 10^{23}$ molecules = 32 g

- **Molar Volume:** The molar volume of a gas can be defined as the volume occupied by one mole of the gas at S.T.P.

  1 mole of any gas = 22.4 dm^3 (litre) or 22400 cm^3(ml)

- **Vapour density and its relation to relative molecular mass:**

$$\text{Relative VD} = \frac{\text{Mass of volume 'v' of the gas under similar conditions}}{\text{Mass of volume 'v' of } H_2 \text{ gas under similar conditions}}$$

$$\text{Relative V.D} = \frac{\text{Mass of 1 molecule of gas or vapour}}{\text{Mass of 2 atoms of hydrogen}}$$

$$2 \times \text{Relative V.D} = \frac{\text{Mass of 1 molecule of gas or vapour}}{\text{Mass of 1 atom of hydrogen}}$$

$$2 \times \text{Relative V.D} = \text{Relative molecular mass of a gas or vapour}$$

- **Percentage composition:** It is the percentage by weight of each element present in it.

$$\% \text{ of an element in a compound} = \frac{\text{Total wt. of the element in one molecule}}{\text{Gram molecular wt of the compound}} \times 100$$

- **Empirical formula of a compound:**

  The empirical formula of a compound is the simplest formula, which gives the simplest ratio in whole numbers of atoms of different elements present in one molecule of the compound.

  Example: Empirical formula of glucose: ($C_6H_{12}O_6$) is CH_2O

- **Relation between molecular weight and Empirical formula weight:**

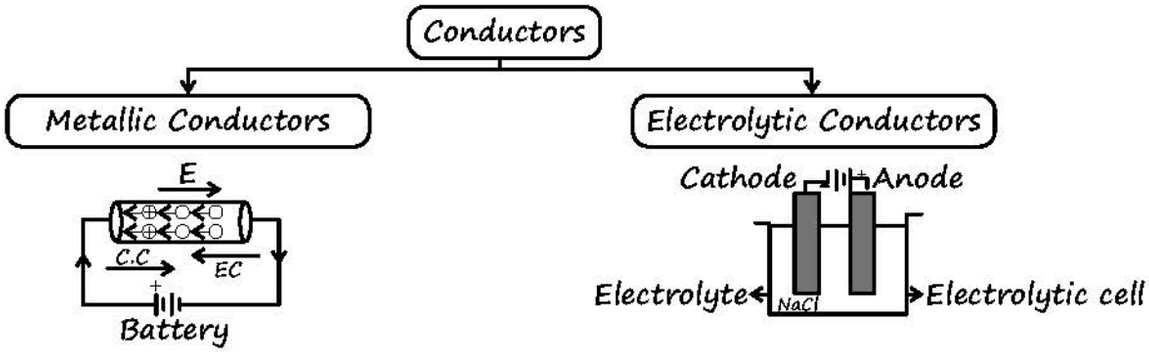
$$n = \frac{\text{Molecule weight}}{\text{Empirical Formula Weight}}$$

6 CHAPTER

Electrolysis

Conductors and Non-Conductors

- Conductors are materials or substances which allow the passage of electric current. Most metallic elements like copper, aluminium, iron, some alloys whether in solid state or in molten state, allow an electric current to pass through them without undergoing any chemical change.
- Non-conductors are materials or substances which do not allow the passage of electric current.
  Eg: Non-metals.



Metallic	Electrolytic conductors
❑ Flow of current consists of a single stream of electrons flowing from the -ve pole to the +ve pole.	❑ Flow of current consists of two streams, one of the +ve cations flowing towards the -ve cathode, and -ve anions flowing towards the +ve anode.
❑ Current carrying particles are electrons which are extremely small.	❑ Current carrying particles are ions.
❑ This is a physical process as no change of form or composition occurs during conduction of electricity.	❑ New products are formed at the electrodes. This involves chemical changes.
❑ Metals, both in solid and liquid state, are good conductors of electricity.	❑ Electrolytes are good conductors only in aqueous solution or in molten state.

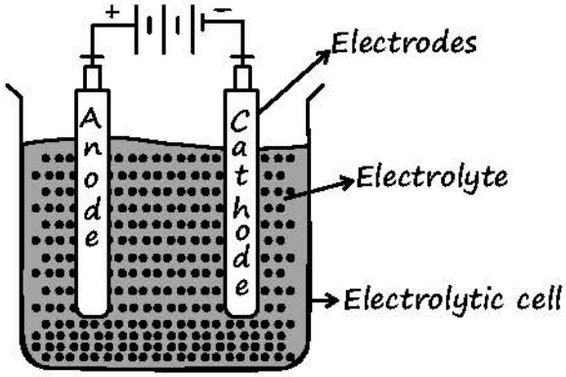
- **Electrolysis:** It is a process of decomposition of a chemical compound in aqueous solution or in molten state accompanied by a chemical change using direct electric current.
- **Electrolytes:** They are compounds which either in aqueous solutions or in molten state allow electric current to pass through them.

Electrolytes	
Strong Electrolytes	**Weak Electrolytes**
❑ Electrolytes which allow a large amount of electricity to flow through them.	❑ Electrolytes which allow small amounts of electricity to flow through them.
❑ They are almost completely dissociated in fused or in aqueous state. These solutions contain only free mobile ions.	❑ These are partially dissociated in fused or aqueous solution states. These solutions contain ions as well as molecules.
Eg: ❑ Acids – HCl, H_2SO_4, HNO_3 ❑ Bases – NaOH, KOH (aqueous or molten state) ❑ Salts – NaCl (molten or aqueous), $PbBr_2$ (molten), $CuCl_2(aq)$, $CuSO_4(aq)$	Eg: ❑ Acids – H_2CO_3, CH_3COOH, Oxalic acid ❑ Bases – NH_4OH, $Cu(OH)_2$ ❑ Salts – $(CH_3COO)_2Pb$, $(NH_4)_2CO_3$

❑ **Non-Electrolytes:** It is a compound which neither in solution nor in molten state allows an electric current to pass through it.

Eg: Distilled water, alcohol, kerosene, glucose, urea, etc.

❑ **Electrolytic Cell (Voltameter):** A non-conducting vessel containing two electrodes immersed in a solution of electrolytes used to bring about a chemical reaction.



❑ **Electrochemical Cell:** It is a device used to convert chemical energy into electrical energy.

Eg: Daniel cell, Simple voltaic cell.

❑ **Electrodes:** Two metal plates or wires or graphite rods or gas carbon rods immersed in the electrolyte through which the current enters and leaves the electrolytic cell are called electrodes.

Electrodes

Anode
- ❑ Positive electrode
- ❑ Anions migrate to the anode
- ❑ It is an oxidising electrode. Oxidation takes place at the anode
- ❑ Anions → negatively charged ions

Cathode
- ❑ Negative electrode
- ❑ Cations migrate to the cathode
- ❑ It is a reducing electrode. Reduction takes place at the cathode
- ❑ Cations → positively charged ions

- **Oxidation:** A chemical process which involves addition of oxygen or removal of hydrogen or loss of electron from an atom or ion.

  Examples:

  - $C + O_2 \rightarrow CO_2$
  - $Mg + O_2 \rightarrow MgO$
  - $H_2S + Cl_2 \rightarrow 2HCl + S$
  - $Zn \rightarrow Zn^{+2} + 2e^-$

- **Oxidising Agent:** An oxidising agent is one that oxidises other substances by either accepting electrons or by providing oxygen or an electronegative ion.

  Examples:

  - MnO_2
  - PbO
  - H_2O_2
  - conc. HNO_3
  - H_2SO_4
  - O_2
  - O_3
  - Cl_2

- **Reduction:** Reduction is defined as a chemical process which involves removal of oxygen or addition of hydrogen or gain of electrons by an atom or ion.

  Examples

  - $CuO + H_2 \rightarrow Cu + H_2O$
  - $Cu^{+2} + 2e^- \rightarrow Cu$
  - $Cl_2 + H_2S \rightarrow 2HCl + S$
  - $Fe^{+3} + e^- \rightarrow Fe^{+2}$

- **Reducing Agents:** A reducing agent is one that reduces other substances by providing electrons, or by providing hydrogen or an electropositive ion.

- **Examples:**

  Carbon, stannous chloride, H_2O_2, HI, H_2S, CO, SO_2

- **Theory of Electrolytic Dissociation:** (Arrhenius)

  1. $\text{Electrolyte} \underset{\text{Current}}{\overset{\text{Electric}}{\rightleftharpoons}} \underset{(+ve)}{\text{Cation}} + \underset{(-ve)}{\text{Anion}}$ They allow the flow of an electric current through it.
  2. Conductivity $\propto$ concentration of ions in the solution.
  3. Number of +ve charges = Number of -ve charges.
  4. In non-electrolytes (such as sugar solution, benzene, etc.), there is no ionization, and therefore only molecules are present in solution.
  5. The degree of dissociation is the extent to which an electrolyte dissociates or breaks up into ions.

- **Conduction In Compounds:**
  - Electrovalent compounds are aggregates of oppositely charged ions held together by strong electrostatic forces of attraction.

When these compounds are melted or dissolved in water, the forces of attraction between their ions are broken, making the ions free to conduct electricity.

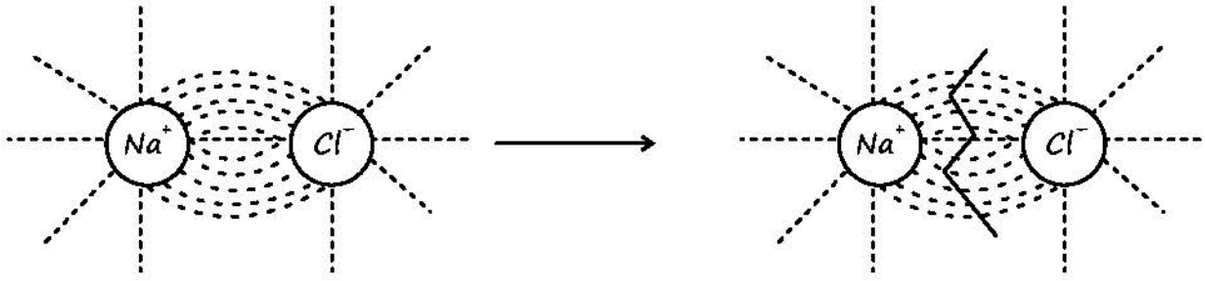
Characteristics of Electrolysis

- The mass of a substane produc ed at an electrode during electrolysis is proportional to the quantity of electricity passess through the electrolyte
- Only non-metals are liberated at anode, hence are called electronegative elements.
- Electrolysis caused by DC
- H_2 gas and metals are liberated at cathode, hence c/a electropositive elements.
- Redox process
- The passage of electricity through an electrolyte causes the movement of ions. i.e. +ve ions to migrate towards cathode and -ve ions to migrate towards anode.
- No. of e^- gained by the anode = No. of e^- lost by the cathode
- The products of electrolysis are formed at the anode and cathode itself since the exchange of electrons takes place only at the surface of the electrodes

- **Electrolytic Dissociation And Ionisation**
  - Dissociation → It is the separation of ions which are already present in an ionic compound!
  - Electrovalent compounds show dissociation.

Examples

NaCl, KCl



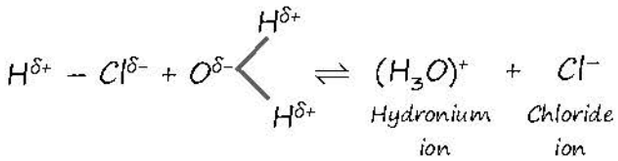
In solid state, ions are held by strong electrostatic forces of attraction

In aqueous solution or is molten state, the force of attraction weakens between the particles and the ions are set free.

- **Ionisation:** Formation of positively or negatively charged ions from molecules which are not initially in the ionic state. Polar covalent compounds show ionisation.

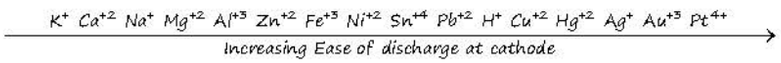
  Examples:

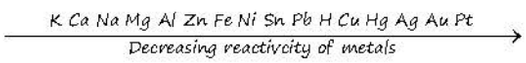
  NH_4OH, HCl



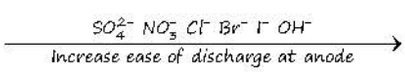
- **Electrochemical Series:** Based on the ease with which atoms of metals lose electrons to form positively charged ions, the metals are arranged in a series known as

  Electrochemical series.

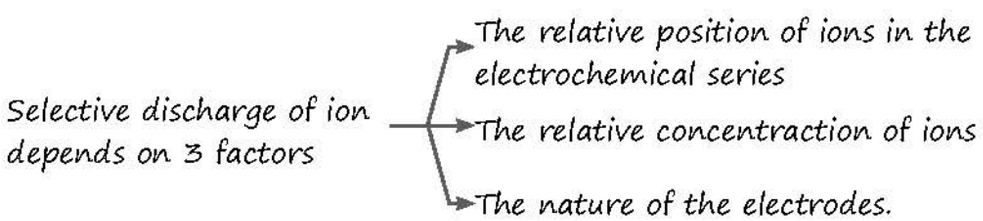




- **About Electrochemical Series of Metals**
  1. A metal with a higher position in the series displaces the one with a lower position.
  2. Metals below hydrogen cannot displace hydrogen from acids.
  3. Elements lower in the series get discharged more easily at the cathode during electrolysis because their cations can easily gain electrons.
  4. Metal atoms that form their ions most easily will accept the electrons back to form the atoms with greatest difficulty.
- **Electrochemical Series of Anions**



  + Lower the position of the anion in the series, more easily it gets discharged at the anode.
  + Higher the position of an anion in the series, more difficult for it to lose electrons or get oxidised.
- **Potential or Selective Discharge of Ions at Electrodes**
  + When two or more ions of the same charge are present in a solution of an electrolyte, under identical conditions, and are competing for discharge at the same electrode, one of them gets preferentially discharged.



- **Electrolysis of Aqueous Copper Sulphate Using Copper Electrodes**
  - **Electrolyte:** Aq. $CuSO_4$ + dil. H_2SO_4
  - **Electrode:** Copper

    $CuSO_4 \rightleftharpoons Cu^{2+} + SO_4^{2-}$

    $H_2O \rightleftharpoons H^+ + OH^-$

  - **At Anode**

    $Cu - 2e^- \rightarrow Cu^{2+}$ (SO_4^{2-} and OH^- ions migrate to the anode, but neither gets discharged due to the nature of the electrode.)
  - **At Cathode**

    $Cu^{2+} + 2e^- \rightarrow Cu$ (Both Cu^{2+} and H^+ ions migrate to the cathode, but Cu^{2+} is discharged as neutral copper atoms.)

    Products at Anode: Copper anode keeps dissolving during the reaction.

    Products At Cathode: Pinkish or reddish-brown copper is deposited.
- **Observations & Conclusion**
  - The blue color of aqueous copper(II) sulphate solution remains unchanged during electrolysis because for every Cu^{2+} discharged at the cathode, another Cu^{2+} ion is released into the solution by the anode, keeping the total Cu^{2+} ion concentration the same.
  - The cathode becomes thicker and the anode becomes thinner over time.
- Since, SO_4^{-2} ions migrate to the anode and H^+ ions have been discharged, the concentration of H_2SO_4 will decrease at the cathode.
- **At the anode:** The discharge of OH^- disturbs the ionic equilibrium of H_2O and to maintain it, more H_2O ionizes:
- $H_2O \rightarrow H^+ + OH^-$.....The excess of H^+ ions, thus produced, and the SO_4^{-2} ions present increase the conc of H_2SO_4 at anode.
- Electrolysis of water gives 2 volumes of H_2 and 1 volume of O_2.

3. Electrolysis of $CuSo_4$ solution using platinum anode and copper or platinum cathode:

Platinum Cathode

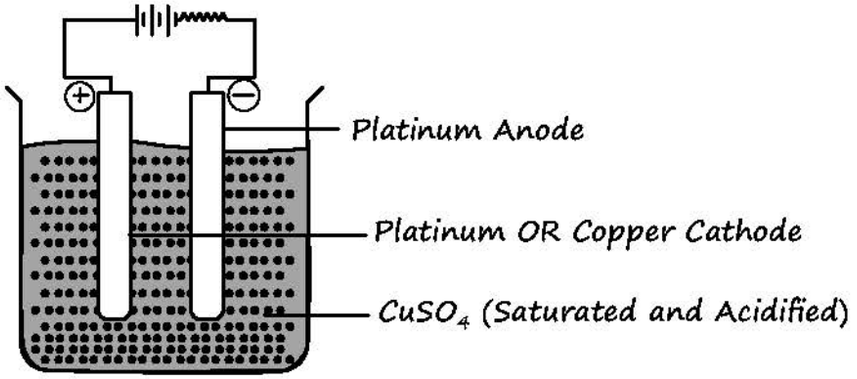
$CuSO_4 \rightleftharpoons Cu^{+2} + SO_4^{2}$

$H_2O \rightleftharpoons H^+ + OH$

At Cathode:

$Cu^{+2} + 2e \rightarrow Cu$

Copper atoms deposit on the cathode (pink or reddish brown copper)



- At anode: SO_4^{-2} and OH^- ions both migrate to the anode. OH^- ions, being lower in the electrochemical series as compared to SO_4^{-2} ions,

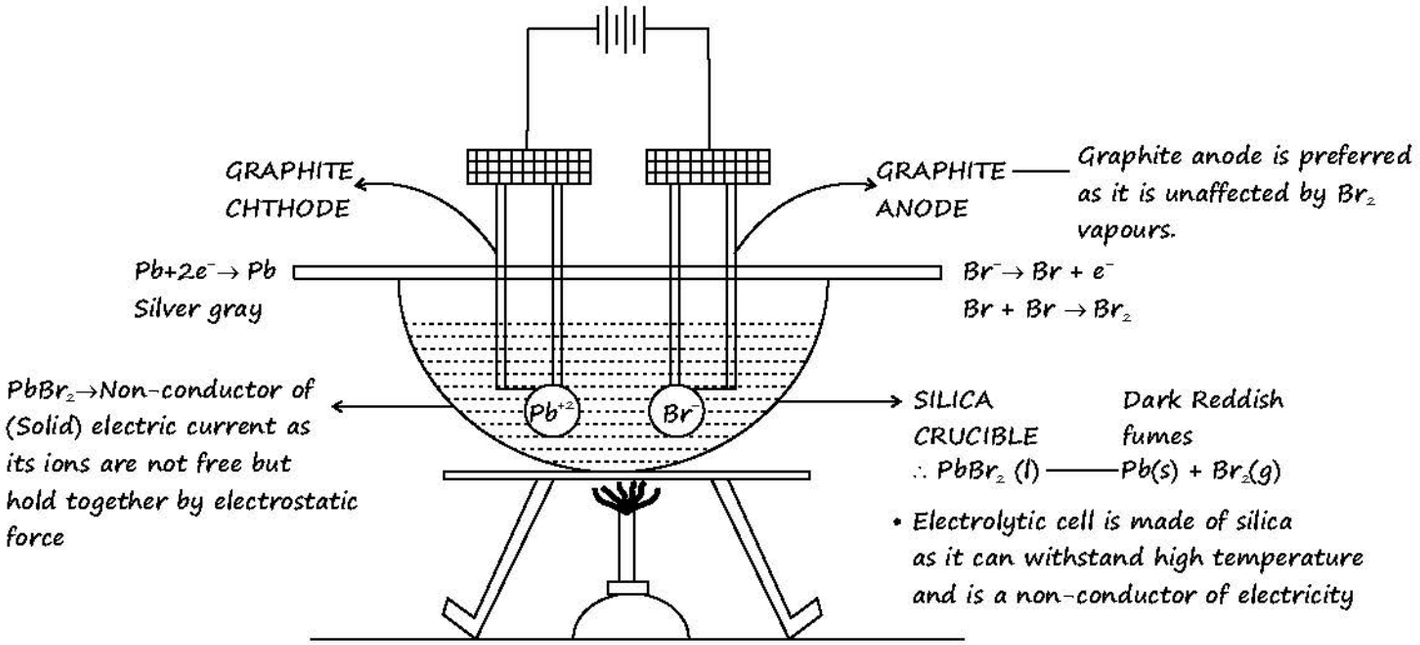
  $4OH^- \rightarrow 4e^- \rightarrow 4OH$

  $2OH^+ 2OH \rightarrow 2H_2O + O_2$

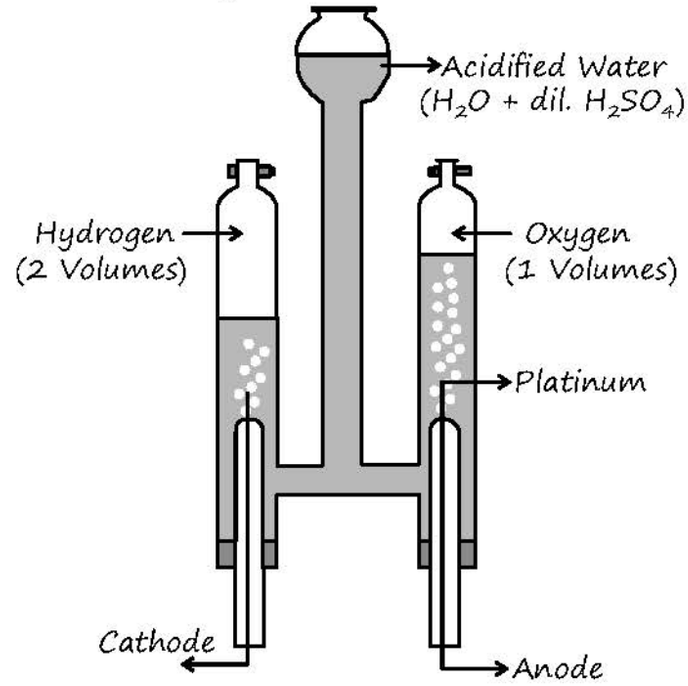
  At Anode

- The blue colour of the Cu^{+2} ions in the electrolyte solution fades due to the decrease in Cu^{+2} ions, and finally, the solution becomes colourless.

1. Electrolysis of Molten lead Bromide



2. Electrolysis of Acidified Water Using Platinum Electrodes



$$H_2SO_4 \rightleftharpoons \overset{\text{At cathode}}{2H^+} + \overset{\text{Anode}}{SO_4^{-2}}$$

$$H_2O \rightleftharpoons H^+ + OH^-$$

At cathode

$H^+ + e^- \rightarrow H$ At Anode: SO_4^{-2} and OH^- both migrate but OH^- gest preferentially discharged

$H+H \rightarrow H_2$

$OH^- \rightarrow H_2O + O \quad \therefore O + O \rightarrow O2\uparrow$

Electrolyte	Dissociation of Electrolytes	Ions Migrating to Cathode	Reaction at Cathode	Ions Migrating to Anode	Reaction at Anode
Dilute $CuSO_4$ Solution Using Inert Electrodes	$CuSO_4 \rightarrow Cu^{+2} + SO_4^{-2}$ $H_2O \rightleftharpoons H^+ OH^-$	Cu^{+2}, H^+	$Cu^{+2} + 2e^- \rightarrow Cu$	SO_4^{-2}, OH^-	$4OH^- - 4e^- \rightarrow 4OH$ $2OH + 2OH \rightarrow 2H_2O + O_2$
Dilute $CuSO_4$ Solution Using Copper Electrode	$CuSO_4 \rightarrow Cu^{+2} + SO_4^{-2}$ $H_2O \rightleftharpoons H^+ + OH^-$	Cu^{+2}, H^+	$Cu^{+2} + 2e^- \rightarrow Cu$	SO_4^{-2}, OH^-	$Cu - 2e^- \rightarrow Cu^{+2}$
Dilute NaCl Solution	$NaCl \rightarrow Na^+ + Cl^-$ $H_2O \rightleftharpoons H^+ + OH^-$	Na^+, H^+	$H^+ + e^- \rightarrow H$ $H + H \rightarrow H_2$	Cl^-, OH^-	$4OH^- - 4e^- \rightarrow 4OH$ $2OH + 2OH \rightarrow 2H_2O + O_2$

Nature of Electrodes

- Electrolysis can be done by use of either an inert electrode or an active electrode.
- If inert electrodes like graphite and platinum are used, they will not take part in the electrolytic reaction.

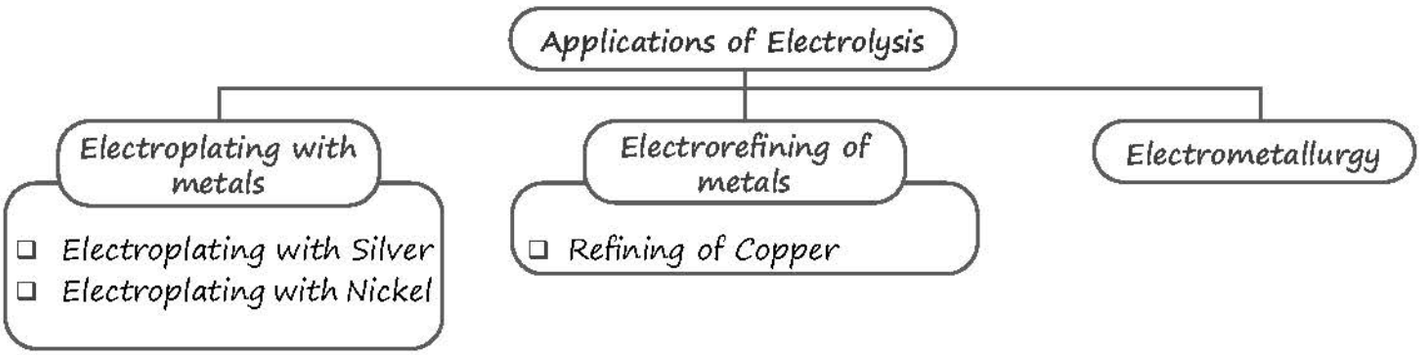
Electrolysis will depend on

1. Position in the electrochemical series
2. Concentration of ions

- Active electrodes like copper, Nickel, Silver, if used, take part in the electrolytic reaction.

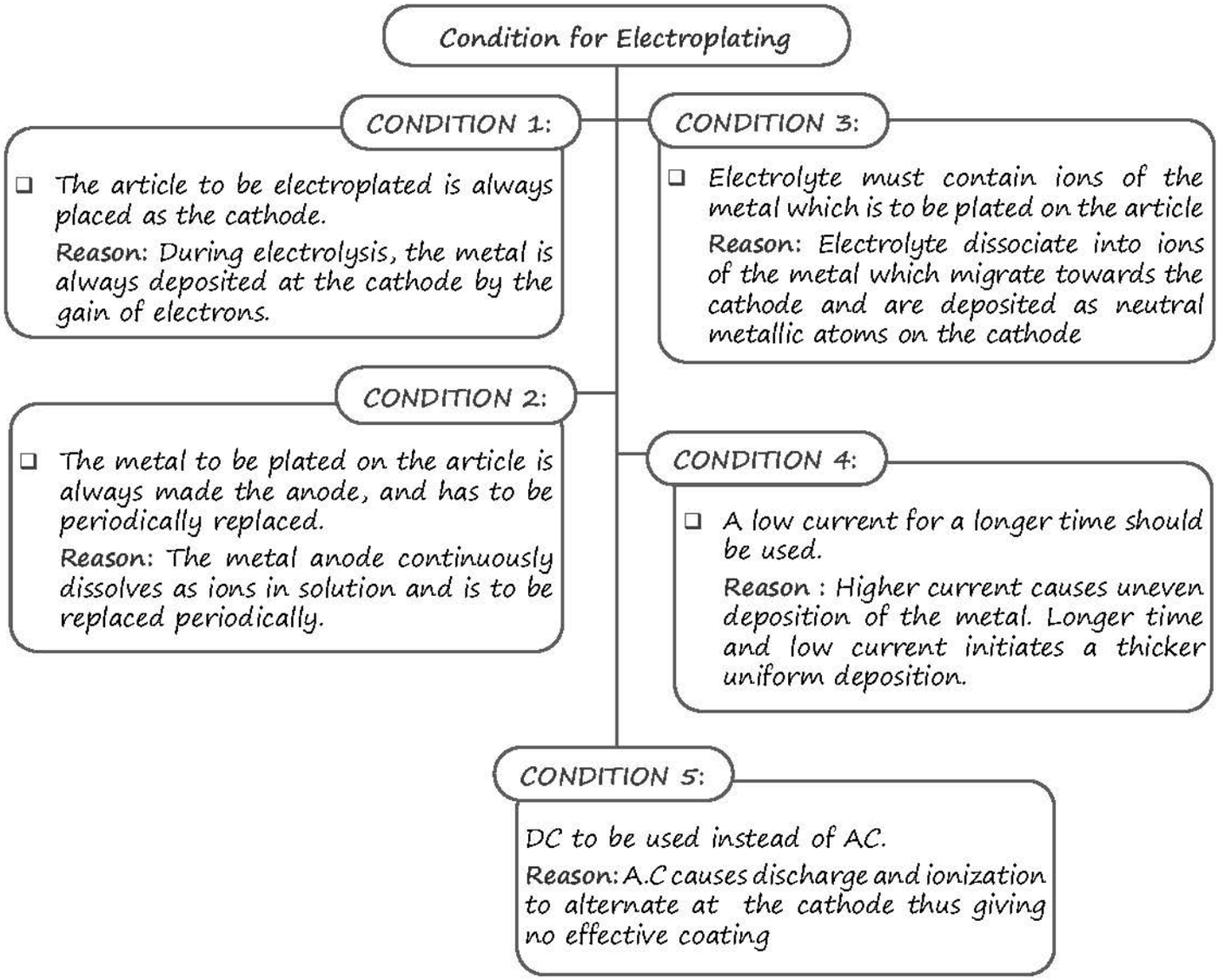
  Eg: In Electrolysis of $CuSO_4$:

Active electrode: Copper	Inert electrode : Platinum
At Cathode: Copper atom	At cathode : Copper
At Anode: Copper ions	At Anode : $O_2\uparrow$

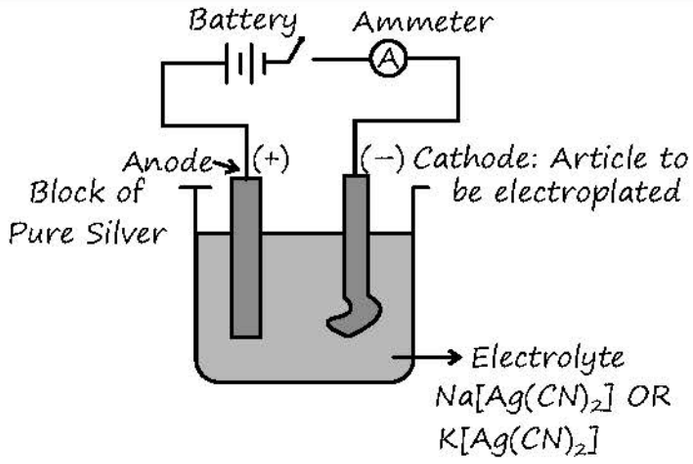


1. **Electroplating** : A process in which a thin film of a metal gets deposited on another metallic article with the help of electricity.

   Reasons: Decoration, to protect from corrosion



Electroplating an Article with Silver



Preparation of electrolyse

$AgNO_3 + NaCN \rightarrow AgCN\downarrow + NaNO_3$

$AgCN + NaNO_3 \rightarrow Na[Ag(CN)_2]$

Sodium argento cyanide

At Cathode: Article to be electroplated is duly cleaned.

$Ag^+ + e^- \rightarrow Ag$

(from electrolyte)

At Anode:

$$Ag - e^- \rightarrow Ag^+$$

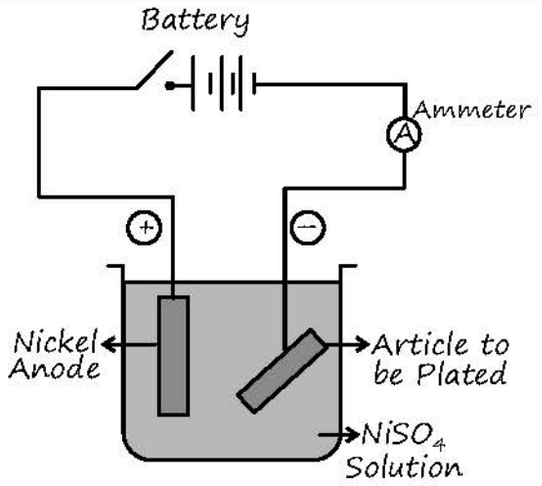
(From anode) (goes into electrolyte)

(Due to the use of active electrode)

Key Pointers

1. The thickness of the coating will depend on the duration of the current passed.
2. In the process the anode dissolves
3. The electrolyte solution maintains the some concentration, thus if it is coloured, the intensity of the colour remains the same
4. If $AgNO_3$ is used directly instead of $Na[Ag(CN)_2]$ the deposition of silver will be very fast and hence not very smooth and uniform.

Electroplating an Article with Nickel



Electrolyte: Aqueous solution of $NiSO_4$

$NiSO_4 \rightleftharpoons Ni^{+2} + SO_4^{-2}$

$H_2O \rightleftharpoons H^+ + OH^-$

At cathode:

$Ni^{+2} + 2e^- \rightarrow Ni$(deposited)

At Anode:

$Ni - 2e^- \rightarrow Ni^{+2}$ (due to the use of active electrode)

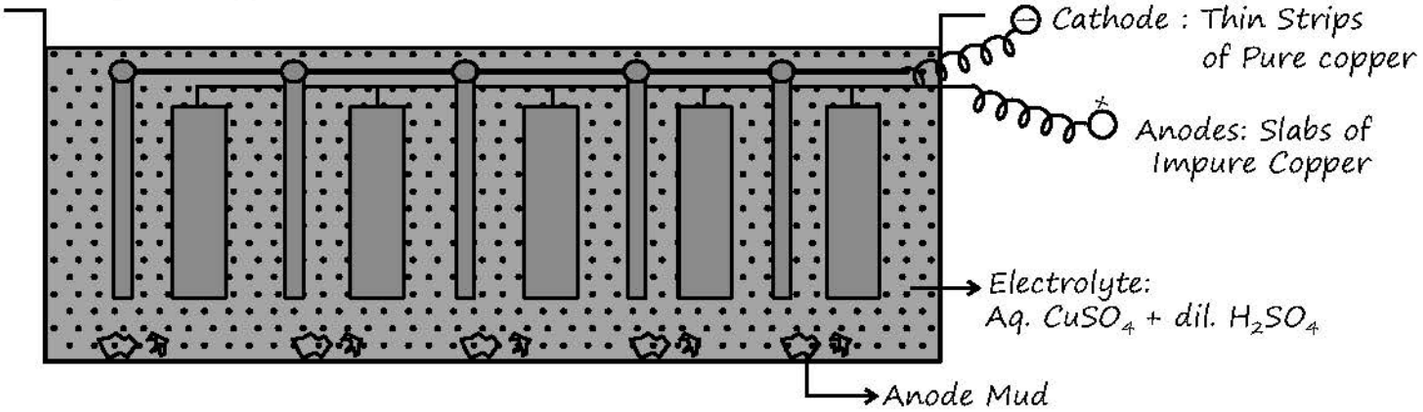
Key Pointers

1. Article to be plated is always kept at the cathode
2. The electrolyte must be a solution of a salt of metal to be electroplated
3. The anode is mode of a strip of metal to be plated.

Electrolytic Refining of Metals

It is a process by which metals containing impurities are purified electrolytically to give a pure metal

Refining of Copper



Reaction

At cathode: $Cu^{+2} + 2e^- \rightarrow Cu$

At cathode: $Cu - 2e^- \rightarrow Cu^{+2}$

- These coper ions migrate to the cathode and gain electrons and then gets deposited at cathode. The net result is that, gradually the impure slab of copper gets finished and this strips of pure copper build up.
- Some impurities gets dissolved is the acid while others namely silver and gold which are insoluble gets deposited below anode as 'ANODE MUD'

Electro Metallurgy

- It is the process of extraction of metals by electrolysis
- Metals which are higher is electrochemical series such as potassium, calcium, sodium etc. are extracted by electrolysis
  - These metals have high affinity for oxygen, their oxides are stable and cannot be reduced by using common reducing agents such as CO and H_2.
  - Reactive metals are extracted from their halides by electrolytic reduction using inert electrodes.

Eg. $KBr \rightleftharpoons K^+ + Br^-$ (Fused)

At cathode: $K^+ + e^- \rightarrow K$

At cathode: $Br^- \rightarrow Br + e^-$

$Br + Br \rightarrow Br_2$

$Al_2O_3 \rightleftharpoons 2Al^{+3} + 3O^{-2}$

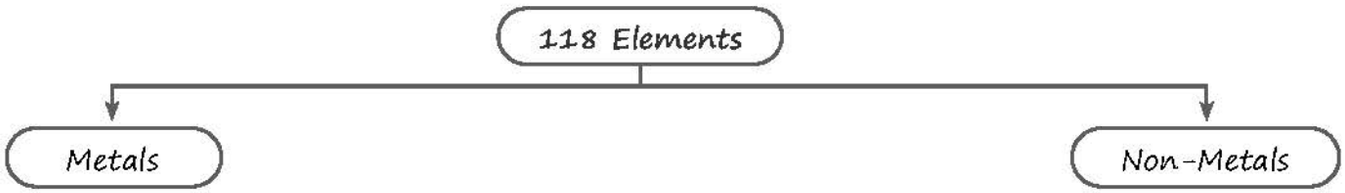
At cathode: $2Al^{+3} + 6e^- \rightarrow 2Al$

At anode: $3O^{-2} - 6e^- \rightarrow 3[O]$

$3[O] + 3[O] \rightarrow 3O_2$

7 CHAPTER

Metallurgy



- Mostly solid (except Hg), brilliant lustre, high density, good conductor of heat and electricity, malleable, ductile, tenacious.
- They usually form basic oxides, are non-volatile, have electropositive nature, form +ve ions by loss of electrons.
- They react with dilute acids to liberate H_2.
- Non-metals are found in all three states. Non-lusturous, bad conductors of heat and electricity, non-malleable, non-ductile, low density, low tensile strength.
- They usually form acidic oxides, volatile, are electronegative in nature and form –ve ions by gain of electrons.
- They do not react with dilute acids to liberate H_2.

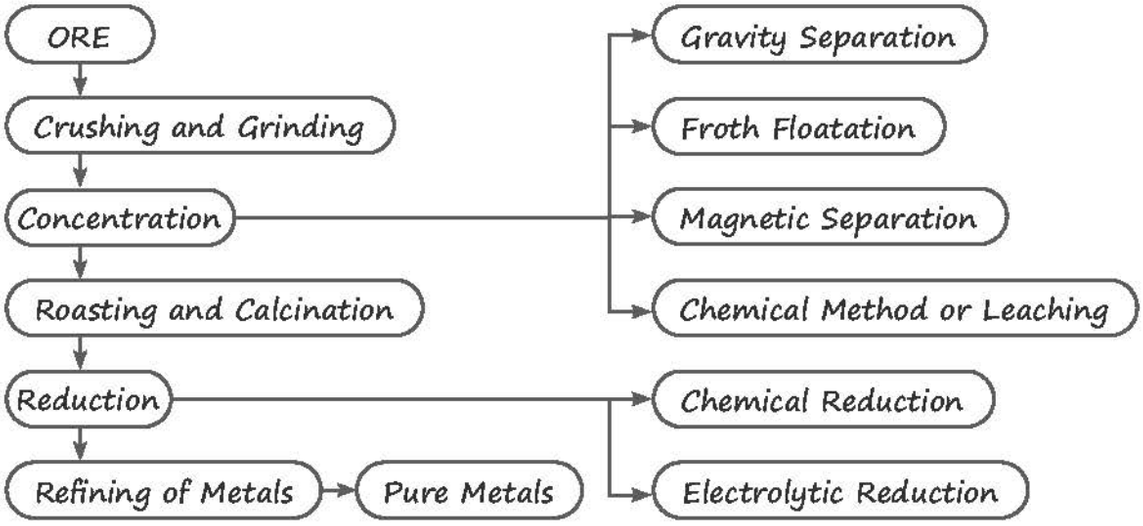
Minerals: The elementary state or the compounds in the form of which the metals occur in nature.

ORE: The mineral from which the metal can be extracted conveniently and economically.

I. All Oare Minerals But All Minerals are not Ores

Metallurgy: The various steps involved in the extraction of metal from its ores followed by refining of metal.

Stages Involved in The Extraction of Metals

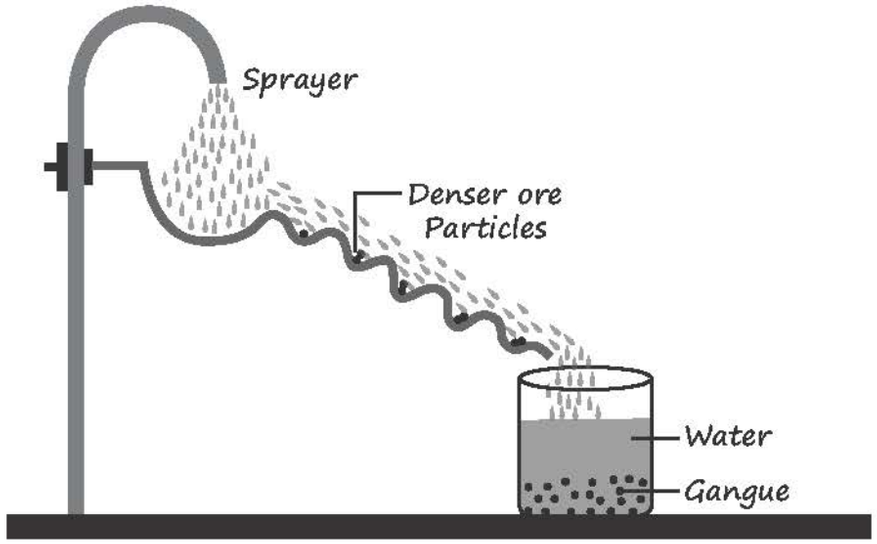


1. *Crushing and Grinding of the Ore*: Ores are pulverised that is crushed into small pieces using crushers and ball mills.
2. *Concentration of ORE/Enrichment of Ore*: The separation or removal of earthy impurities called "gangue" from the ore so that concentration of the metal in the ore increases.

Hydraulic Washing or Gravity Separation or Levigation

Principle: The difference in the densities of the ore and the gangue

- Oxides of iron and tin are separated by this method

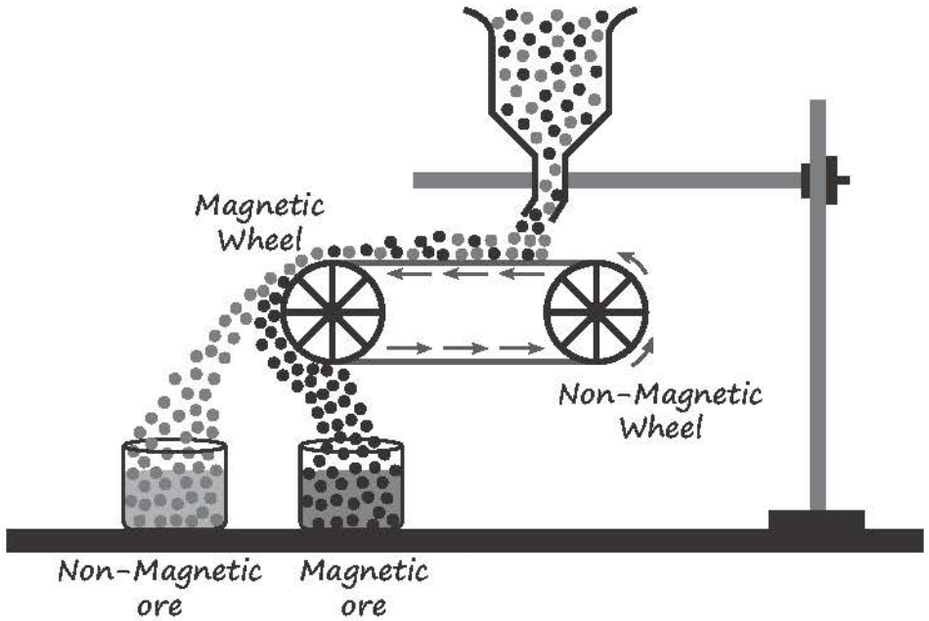


Gravity separation using water and vibrating corrugated table.

Magnetic Separation

Principle: Magnetic properties of the ore.

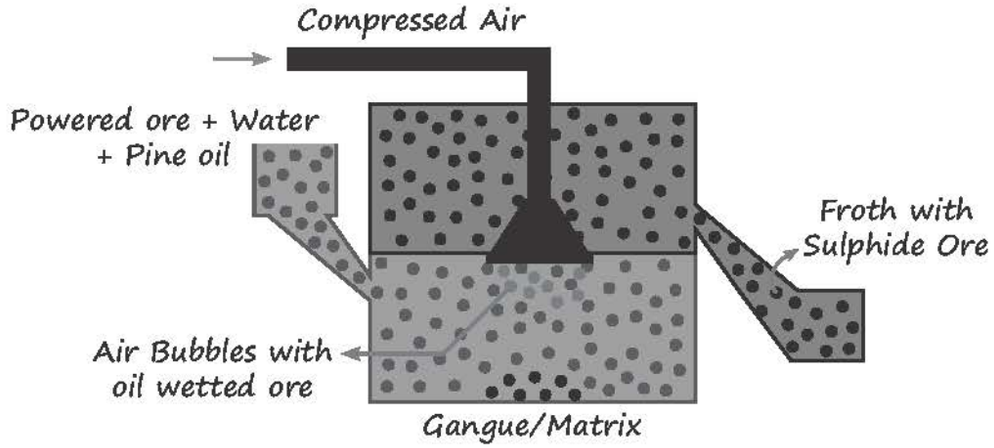
- Used to concentrate tin stone (SnO_2). It contains magnetic oxide of iron (Fe_3O_4) as impurity.



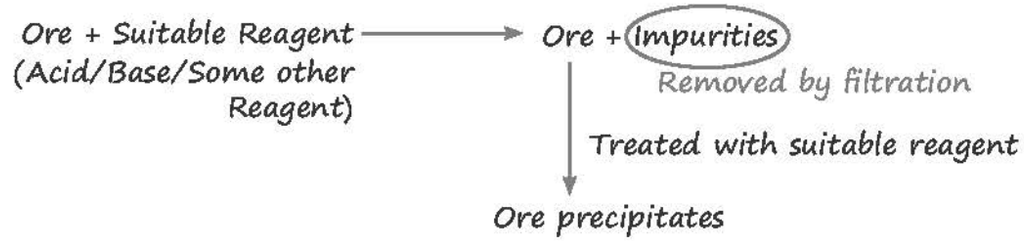
Froth Floatation Method

Principle: This process depends on the preferential wettability of the ore with oil and the gangue particles by water.

- ZnS and PbS are higher than the impurities present. They are concentrated by this method.



Chemical Method or Leaching



- Used for the concentration of metals like silver, gold, aluminium.

II. Conversion of Concentration Ore to Its Oxids

Conversion of Concentration Ore to its Oxids

Roasting

- Process of heating the ore to a high temperature in presence of air.
- Generally, sulphide ores are roasted

$$2ZnS + 3O_2 \xrightarrow{800-900^\circ C} 2ZnO + 2SO_2\uparrow$$

- Volatile impurities are removed as oxides and the ore becomes porous and more reactive

Calcination

- Process of heating the ore in the absence of air to a temperature that is high but insufficient to melt the ore.
- Generally, carbonate and hydrated ores are calcined.

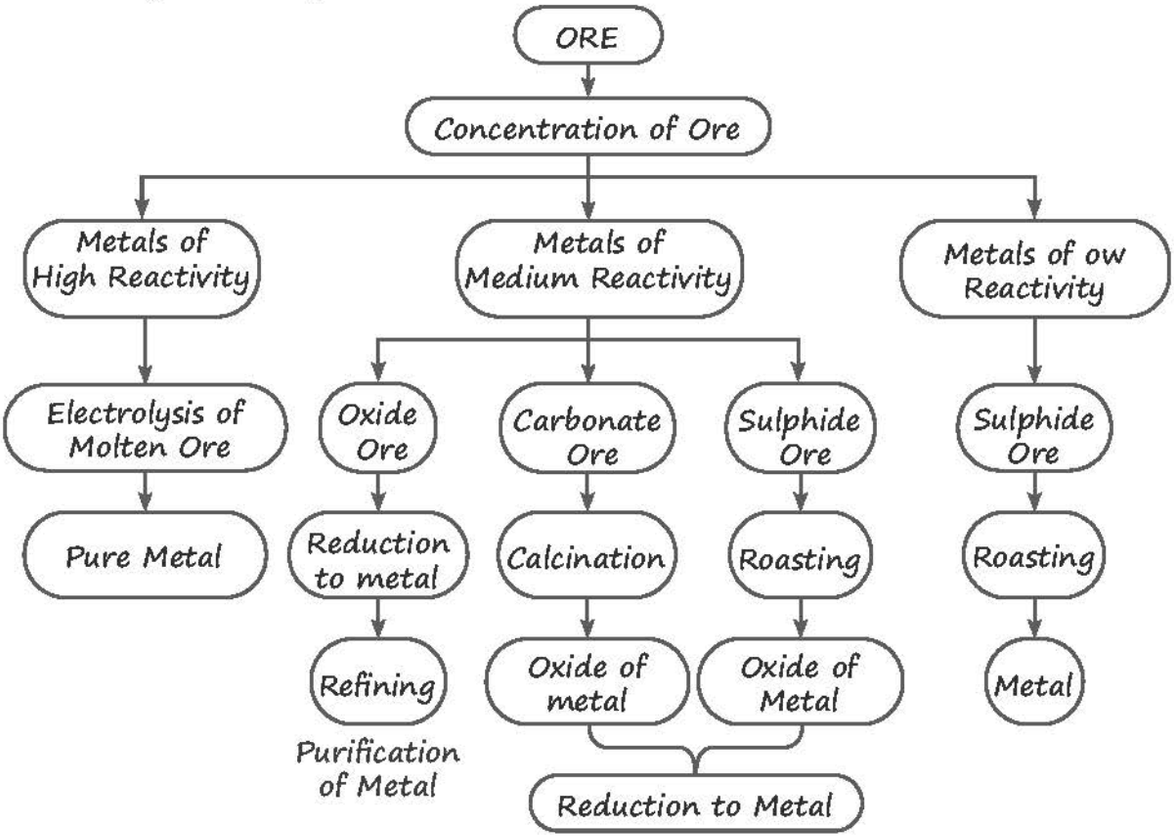
$$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$$

- Moisture and organic impurities are removed and the ore becomes porous and more reactive.

III. Reduction of the Metal Oxides to Metals

After roasting or calcination, the metal oxide is reduced by heating it with reducing agents. The selection of reducing agents depends upon the position of the metal in the reactivity series.

- The arrangement of metals in the decreasing order of their reactivity is called the activity series or reactivity series of the metals.
- Based on their reactivity, metals are categorised as:
  - Metals of low reactivity
  - Metals of medium reactivity
  - Metals of high reactivity



Reduction of Metal Oxide to Metal

Reduction of ZnO, Fe_2O_3 etc is done by heating with Carbon (coke). Reduction of MnO_2, Cr_2O_3 etc is done by heating with more active metals like Na, K, Al. Reduction of Fe_2O_3 with Al is called thermite reaction. The reduction with Al is called **Aluminothermy.**

$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + heat$

- **Refining of the metal:**

Refining is the process by which crude metal is purified. The method of purification used depends upon the:

(i) nature of the metal

(ii) nature of the impurities

(iii) purpose for which the metal is to be used

Refining

Distillation

Volatile metal like Zn, Cd, Hg are refined by distillation

Liquation

Easily fusible metals (having low mpts) like lead and tin can be refined by liquation.

Electro refining

Many metals like Cu, Al, Zn, Ag, Au may be refined using the impure metals as anodes, pure metal strips as cathode and metal salt solution as electrolyte.

Extraction of Aluminium

- Aluminium: Al, silvery white metal, Atomic number : 13, Atomic mass : 27, Valency : 3, Electronic configuration: 2,8,3
- Most abundant metal in the earth's crust. Discovered by Oersted.

Extraction of Aluminium

- Purification of bauxite ore Bayer's process

Step 1: Impure bauxite $\longrightarrow$ Sodium meta aluminate

$$Al_2O_3.2H_2O + 2NaOH \xrightarrow{150^\circ C} 2NaAlO_2 + 3H_2O$$

(Bauxite) (conc) Sodium meta aluminate

Step 2: Sodium meta aluminate $\longrightarrow$ Aluminium hydroxide

$$NaAlO_2 + 2H_2O \xrightarrow{50^\circ C} Al(OH)_3\downarrow + NaOH$$

Step 3: Aluminium hydroxide $\longrightarrow$ Alumina

$$2Al(OH)_3 \xrightarrow{1000^\circ C} Al_2O_3\downarrow + 3H_2O$$

Refining of Aluminium

- Hoope's electrolytic process
- It is done in an electrolytic cell which comists of three layers of molten substances differing in specific gravity.
- Molten impure aluminium forms the bottom layer and acts as anode, pure molten aluminium forms the upper layer and acts as cathode. The middle layer is of fluorides.

At Anode:

$Al \longrightarrow Al^{+3} + 3e^-$

At Cathode:

$Al^{+3} + 3e^- \longrightarrow Al$

Electrolytic Reduction of Fused Alumina

- Hall-Heroult's process
- Electrolyte:

Al_2O_3 : Cryolite : Fluorspar

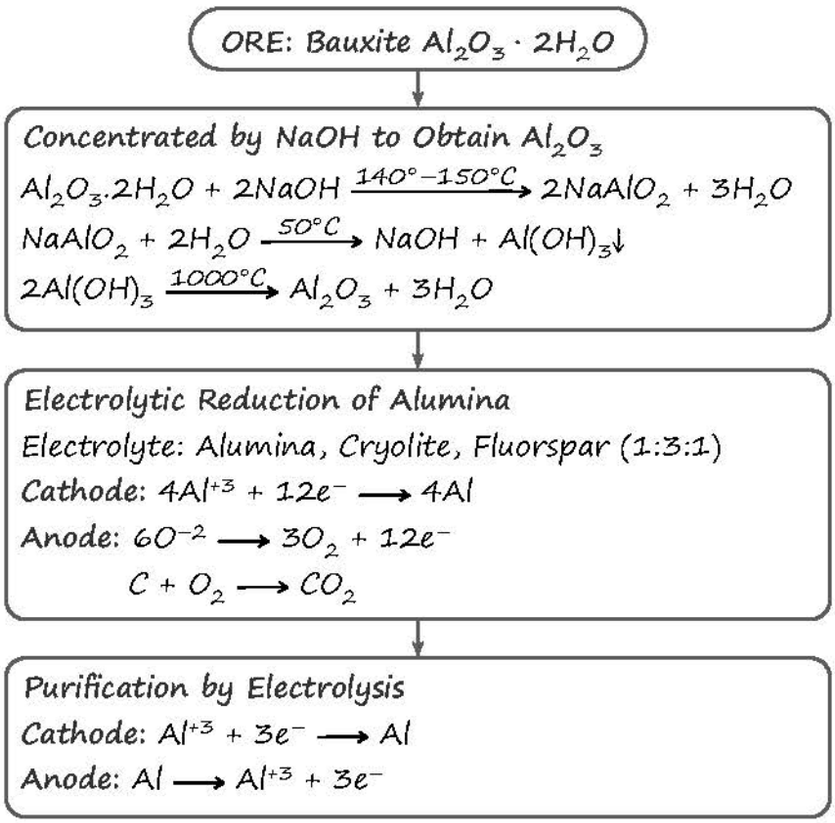
1 : 3 : 1

- Electrode: Cathode: Carbon lining, Anode: Graphite rods
- Temperature: 950°C
- Products formed: Cathode: Pure Al, Anode: $O_2(g)$

Role of Cryolite:

(i) Mpt of Al_2O_3 gets lowered from 2050°C to 950°C

(ii) Increases electrical conductivity of the mixture.



Alloys

An alloy is a homogeneous mixture of two or more metals or of one or more metals with certain non-metallic elements.

Purpose of Making Alloys

The purpose of an alloy is to improve the specific usefulness of the primary component and not to adulterate or degrade it.

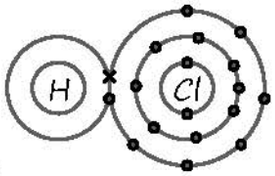
Reason for Alloying

- To modify appearance and colour
- To modify chemical activity
- To modify casting ability
- To lower the melting point
- To increase the hardness and tensile strength
- To increase resistance to electricity

8 CHAPTER

Hydrogen Chloride

Molecular Formula : HCl
Molecular Mass : 36.5
Bond : Covalent
Historical Background : Glauber: Prepared by heating common salt with conc. H_2SO_4.
Lavoisier: Named it Muriatic acid.
Davy: Named it Hydrochloric acid.



or H — Cl

Preparation of Hydrogen Chloride Gas

By Synthesis (Direct Combination)

Reaction in direct sunlight is explosive but not in dark

$$H_2(g) + Cl_2(g) \xrightarrow[\text{Sunlight}]{\text{Diffused}} 2HCl(g)$$

Activated Carbon Absorbs Hydrogen which increases reactivity allowing Reaction to occur in dark.

By Heating Metallic Chloride with Concentrated Sulphuric Acid

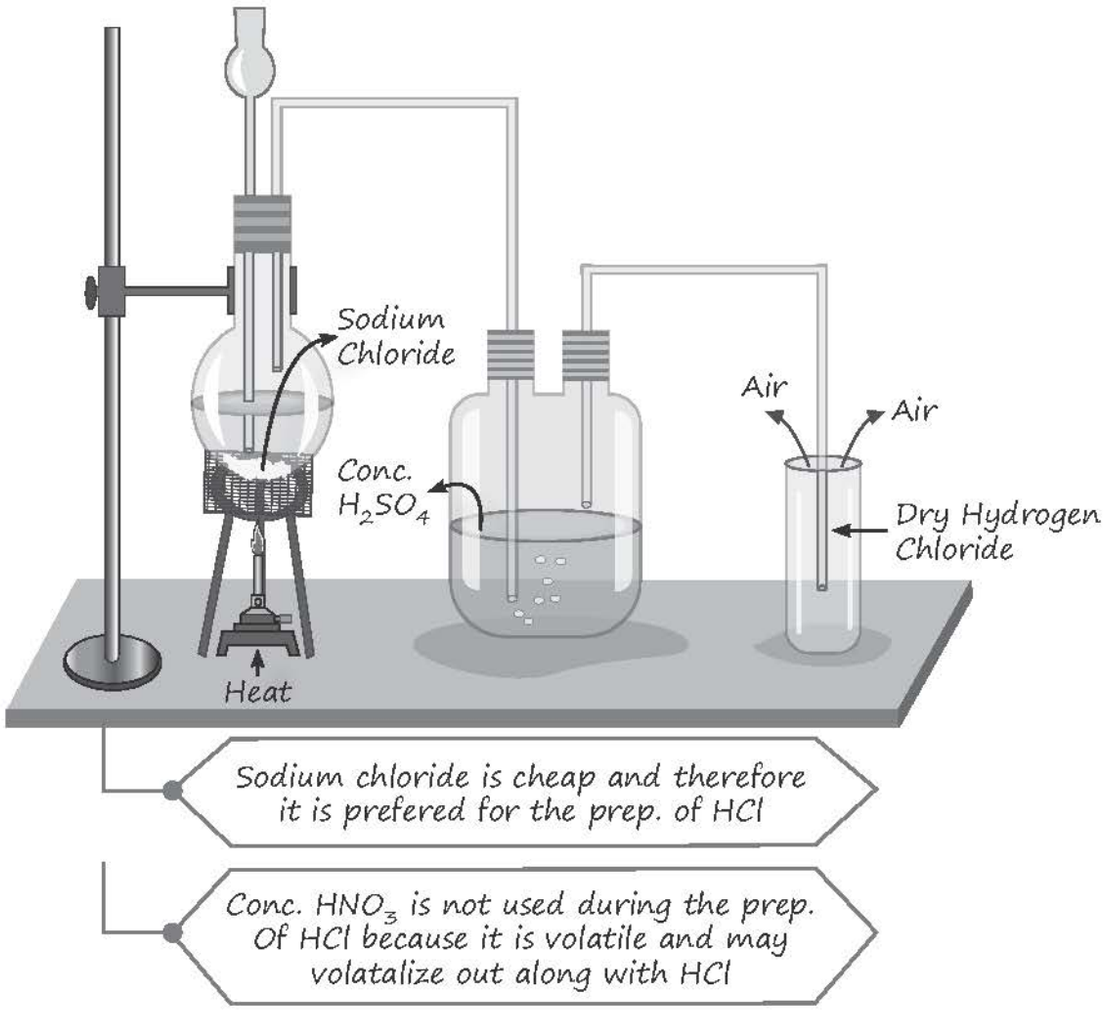
Metal Chloride + Conc. H_2SO_4 ⟶ Salt + Hydrogen Chloride

Laboratory Preparation of Hydrogen Gas

Reaction Involved:

$$NaCl + H_2SO_4 \xrightarrow{< 200°C} NaHSO_4 + HCl\uparrow$$

- The reaction can occur up to the stage of formation of Na_2SO_4 on heating above 200°C.
- $NaHSO_4 + NaCl \xrightarrow{> 200°C} Na_2SO_4 + HCl\uparrow$
- $2NaCl + \underset{\text{(conc.)}}{H_2SO_4} \xrightarrow{> 200°C} Na_2SO_4 + 2HCl$



Purification of HCl Gas

- Dried by using concentrated H_2SO_4
- Other drying agents like P_2O_5, CaO cannot be used as they react with HCl

$$2P_2O_5 + 3HCl \rightarrow POCl_3 + 3HPO_3$$
$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

Collection of HCl

- HCl is collected by downward delivery (upward displacement of air) as it is 1.28 times heavier than air.
- It is not collected over water, since it is highly soluble in water.

Identification

- HCl gas on exposure to air, gives white fumes due to the formation of HCl acid on dissolving in atmospheric water vapour.

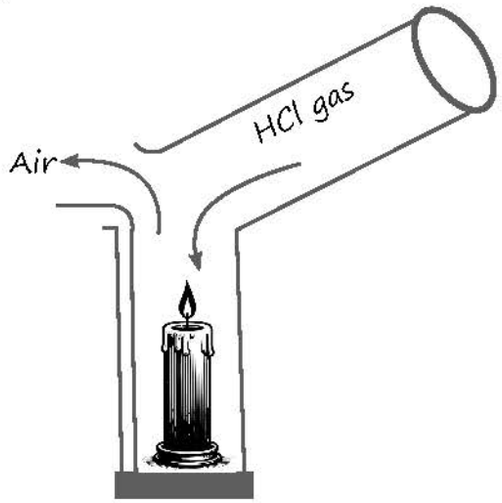
Test: Bring a rod dipped in NH_4OH near the mouth of the test tube, dense white fumes of NH_4Cl will be produced.

$$HCl + NH_4OH \rightarrow NH_4Cl + H_2O$$

Physical Properties of Hydrogen Chloride Gas

- **Colour** — Colourless Gas
- **Smell** — Pungent Choking Smell
- **Taste** — Sour (acidic) in taste
- **Physiological Nature** — It is corrosive in nature. It irritates nose, throat and lungs.
- **Density** — V.D of HCl is 18.25
- **Melting Point** — –113°C
- **Boiling Point** — –83°C
- **Liquefaction & Solidification** — Liquefaction occurs at high pressure and low temperature
- **Solubility** — Highly soluble in H_2O. HCl being a polar covalent compound is soluble in organic non-polar solvents such as acetone and toluene

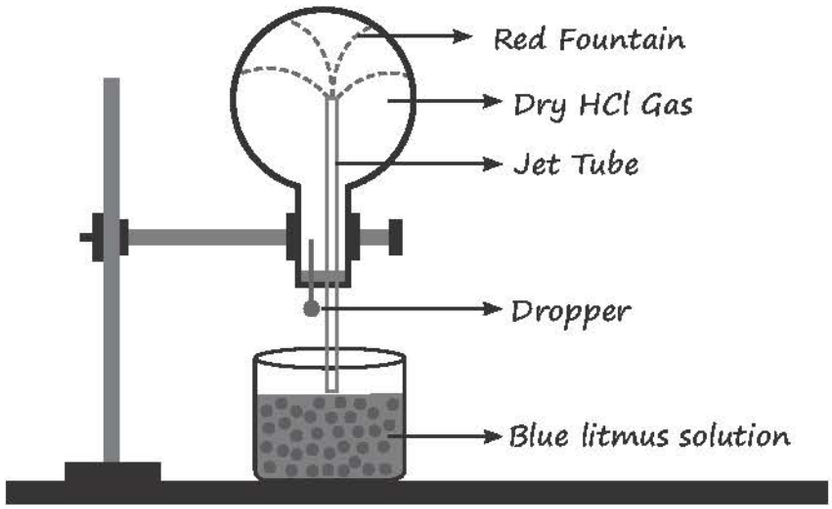
Experiment: To show that HCl gas is heavier than air



This experiment also proves that HCl gas is neither combustible nor a supporter of combustion.

Experiment: To show that HCl gas is highly soluble

As the water goes in the flas from the dropper, HCl gas present in the flask dissolves due to its high solubility, thereby lowering the pressure inside the outside pressure being higher pushes the blue litmus solution inside, through the jet tube.



- The blue litmus solution turns red due to the acidic nature of HCl gas.

Chemical Properties of Hydrogen Chloride Gas

- **Combustibility** — Gas is neither combustible nor a supporter of combustion. It does not burn, rather it extinguishes a burning splint.
- **Thermal Dissociation** — On heating above 500°C, it dissociates into hydrogen and chlorine

$$2HCl \xrightarrow{> 500°C} H_2 + Cl_2$$

- **Reaction with Metals** — $Metal + HCl(g) \longrightarrow MCl_2 + H_2\uparrow$
- **Reaction with Ammonia** — It combines with NH_3 to form dense white fumes of ammonium chloride.

$$NH_3 + HCl \longrightarrow NH_4Cl_{(s)}$$

Hydrochloric Acid

Dissolution of HCl gas in water yields HCl acid. It is a polar covalent compound.

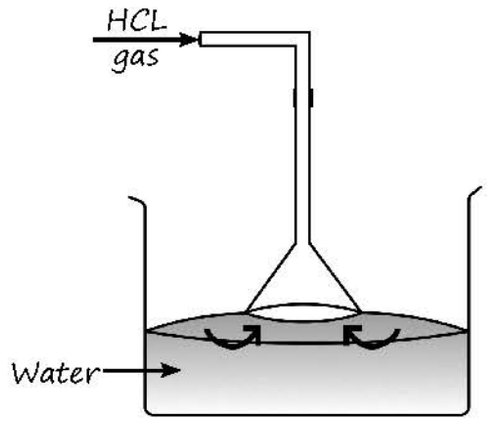
$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

Dry HCl gas and liquefied HCl do not produce H^+ ions. Hence, they do not conduct electricity and they do not turn blue litmus paper red. It shows non-acidic character and covalent nature of HCl gas.

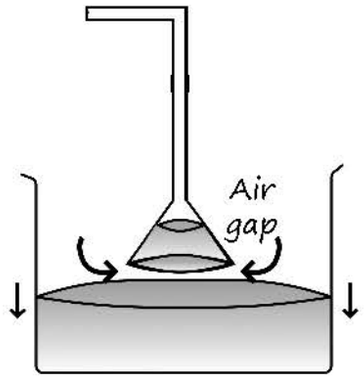
Laboratory Method of Preparation of Hydrochloric Acid

- It is prepared by dissolving HCl gas in water. The gas is passed into H_2O until no more gas is absorbed. The product is concentrated and contains about 36% of HCl by maass.

(I) Since HCl gas is highly soluble in water, water rises up in the funnel, back suction occurs and in turn, the level outside the funnel falls creating an air gap between the rim of the funnel.



(II) The pressure inside and outside becomes equal and the water which have rinse falls, gradually saturating the water in the beaker.



- The mechnism by which back suction is avoided on minimised: The funnel Arrangement
  (a) Prevents or minimizes the back suction of water.
  (b) Provides a large surface area for absorption of HCl gas.
- A constant boiling mixture or azeotrope is a solution which boils without any change in its composition. HCl acids forms constant boiling mixture at 110°C.

Physical Properties of Hydrochloric Acid

- **Colour** — Colourless Gas
- **Smell** — Pungent Choking Smell
- **Taste** — Sour (acidic)
- **Effect on Skin** — Concentrated acid is corrosive and causes blisters on skin.
- **Solubility** — Readily soluble in water in all proportions
- **Boiling Point** — 110°C

Chemical Properties of Hydrochloric Acid

Nature

Aqueous Solution is strongly acidic.

Action on litmus:	Moist litmus:	Blue → Red
	Methyl orange:	Orange → Pink
	Phenolpthalein:	Colourless → Colourless

Action on Metals

Metal + Hydrochloric acid → Metallic Chloride + Hydrogen

Action on Oxides and Hydroxides

Metallic oxide or Metallic Hydroxide + Hydrochloric acid → Metallic Chloride + Water

With Salts of Weaker Acids

It decomposes salts of weaker acids i.e., carbonates, hydrogen carbonates, sulphites and sulphides.

- Carbonates/ Hydrogen carbonates + Hydrochloric acid (dilute) → Metal Chloride + Water + $CO_2\uparrow$
- Sulphites/ Hydrogen sulphites + Hydrochloric acid (dilute) → Metal Chloride + Water + $SO_2\uparrow$
- Metallic Sulphide + Hydrochloric acid (dilute) → Metal Chloride + $H_2S\uparrow$
- Thiosulphate ($Na_2S_2O_3$) + Hydrochloric acid (dilute) → Metal Chloride + $SO_2\uparrow$ + $S\downarrow$ + H_2O

$$Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2 + S\downarrow$$

Precipitation Reaction

Dilute HCl does not normally react with nitrates.

- Lead nitrate: $Pb(NO_3)_2 + 2HCl \longrightarrow PbCl_2\downarrow + 2HNO_3$
- $Hg_2(NO_3)_2 + 2HCl \longrightarrow Hg_2Cl_2\downarrow + 2HNO_3$

* White ppt of AgCl is insoluble in HNO_3 but soluble in NH_4OH Solution, forms a complex salt c/a diammine silver (I) chloride $[Ag(NH_3)_2]^+Cl^-$

$$AgCl + 2NH_4OH \longrightarrow [Ag(NH_3)_2]^+Cl^- + 2H_2O$$

$$\downarrow + HNO_3$$

$$2AgCl\downarrow \xrightarrow[\text{to Sunlight}]{\text{on exposure}} Ag + Cl_2$$

White ppt.

Concentrated Hydrochloric Acid

Conc. HCl can be readily oxidised to chlorine by strong oxidising agents such as MnO_2, PbO_2 and Pb_3O_4

$$\text{Oxidising agent} + \text{Hydrochloric acid} \longrightarrow \text{Metallic chlordie} + H_2O + Cl_2$$

- $MnO_2 + \underset{\text{(conc.)}}{4HCl} \xrightarrow{\Delta} MnCl_2 + 2H_2O + Cl_2\uparrow$
- $PbO_2 + \underset{\text{(conc.)}}{4HCl} \xrightarrow{\Delta} PbCl_2 + 2H_2O + Cl_2\uparrow$
- $Pb_3O_4 + \underset{\text{(conc.)}}{8HCl} \xrightarrow{\Delta} 3PbCl_2 + 4H_2O + Cl_2\uparrow$
- $\underset{\text{(Bleaching Powder)}}{CaOCl_2} + \underset{\text{(dil.)}}{2HCl} \xrightarrow{\Delta} CaCl_2 + H_2O + Cl_2\uparrow$

Uses of Hydrochloric Acid

General Uses:

As a laboratory reagent and for the preparation of aquaregia.

Medicines

Dilute HCl is prescribed to patients with decreased activity of their gastric juices.

Industrial Uses

- In the manufacture of chlorine and chlorides, dyes, drugs, glucose from starch.
- For purifying bone black, because HCl dissolves the calcium phosphate present in bones.
- To pickle steel as HCl dissolves the oxides.
- To remove rust from iron sheets.
- For cleaning metal surfaces before painting, electroplating, galvanisation.
- In extraction of glue from bones.

Tests for Hydrogen Chloride Gas and Hydrochloric acid

1. HCl possesses a characteristic irritating smell
2. $NH_3 + HCl \rightarrow NH_4Cl$: HCl gives thick white fumes of NH_4Cl, when a glass rod dipped in ammonia solution is held near the vapours of the acid.
3. $AgNO_3 + HCl \rightarrow \underset{\text{White ppt.}}{AgCl\downarrow} + HNO_3$

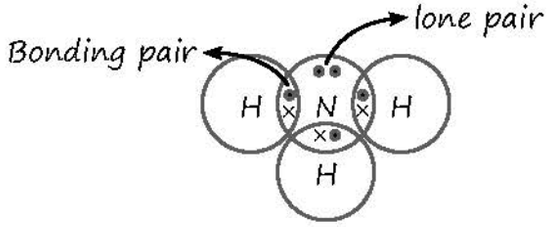
   $\rightarrow$Insoluble in nitric acid but soluble in NH_4OH
4. $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$ (Greenish yellow gas)

   The gas liberated turns moist starch iodide paper blue-black.

9 CHAPTER

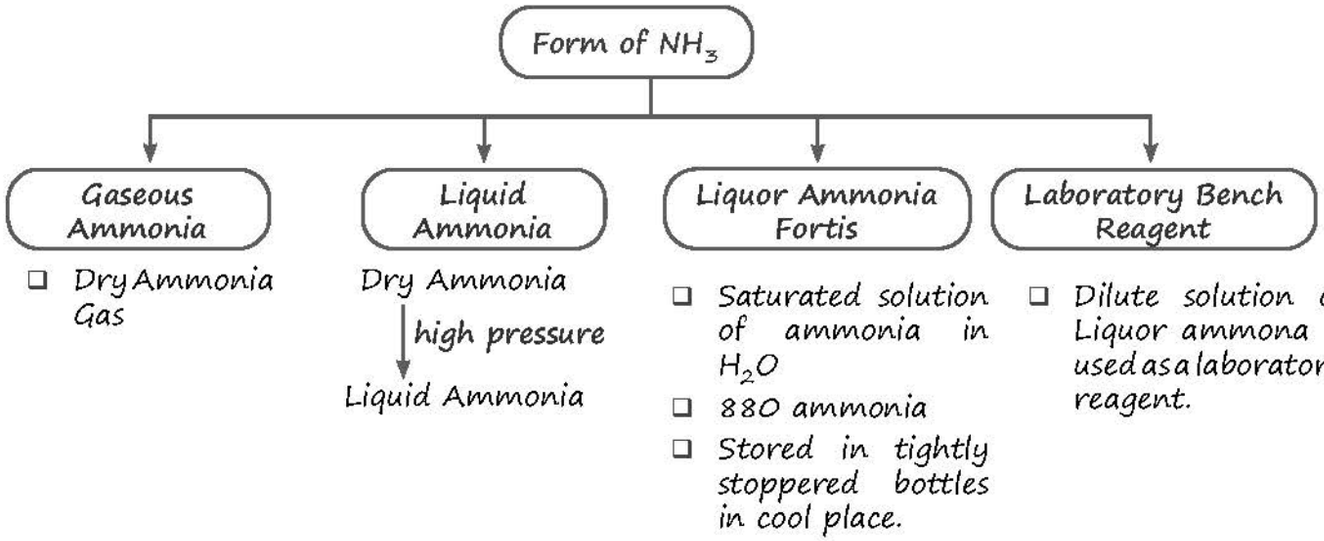
Ammonia

Molecular Formula : NH_3
Relative Molecular Mass : 17
Bonding : Covalent



- NH_3 and ammonium compounds being highly soluble in water, do not occur as minerals.
- The bacterial decomposition of urea yields ammonia which gives a pungent odour.

$$\underset{\text{Urea}}{CO(NH_2)_2} + 2H_2O \rightarrow 2NH_3 + H_2O + CO_2\uparrow$$



I. Preparation of Ammonia Gas

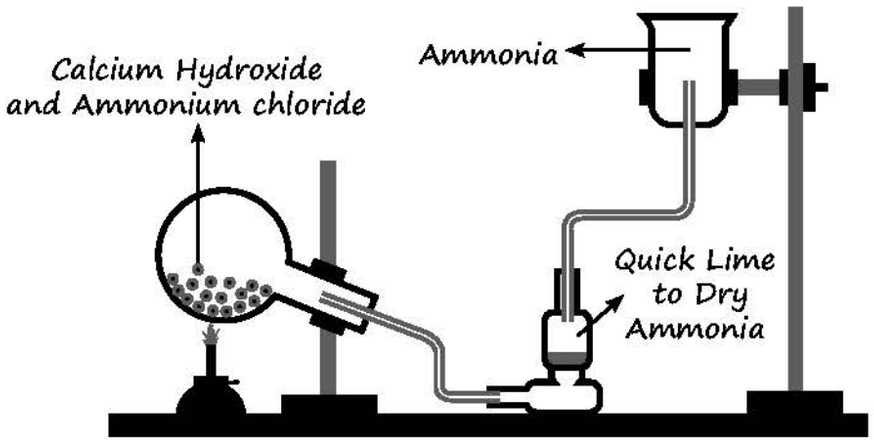
General Methods of Preparation

$$\text{Ammonium Salt} + \underset{\text{(Slaked lime/ Caustic soda/ Caustic potash)}}{\text{Alkali}} \xrightarrow{\Delta} \text{Salt} + \text{Water} + \text{Ammonia}$$

- $2NH_4Cl(s) + Ca(OH)_2 \xrightarrow{\Delta} CaCl_2 + 2H_2O + 2NH_3\uparrow$
- $(NH_4)_2SO_4 + Ca(OH)_2 \xrightarrow{\Delta} CaSO_4 + 2H_2O + 2NH_3\uparrow$

- $(NH_4)_2SO_4 + 2NaOH \xrightarrow{\Delta} Na_2SO_4 + 2H_2O + 2NH_3\uparrow$
  Caustic Soda
- $(NH_4)_2SO_4 + 2KOH \xrightarrow{\Delta} K_2SO_4 + 2H_2O + 2NH_3\uparrow$
  Cautic potash

Laboratory Preparation of Ammonia Gas



I. From Ammonium Chloride

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3\uparrow$$

In order to get dry NH_3, the gas is passed through a drying tower containing lumps of quick lime (CaO).

Unsuitability of other drying agents

Other drying agents like Conc. H_2SO_4, P_2O_5, Anhydrous $CaCl_2$ are not used as ammonia being basic reacts with them.

- $2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$
- $6NH_3 + P_2O_5 + 3H_2O \longrightarrow 2(NH_4)_3PO_4$
  Ammonium Phosphate
- $CaCl_2 + 4NH_3 \longrightarrow CaCl_2.4NH_3$
  Addition Compound

PTR:

1. Higher ratio by weight of alkali is used as it may counteract the loss by sublimation of NH_4Cl.
2. $Ca(OH)_2$ is used as it is cheap and not deliquescent like other alkalies.

3. Though all ammounium salts on heating with alkalis give NH_3 Except: NH_4NO_3, NH_4NO_2

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

Collection

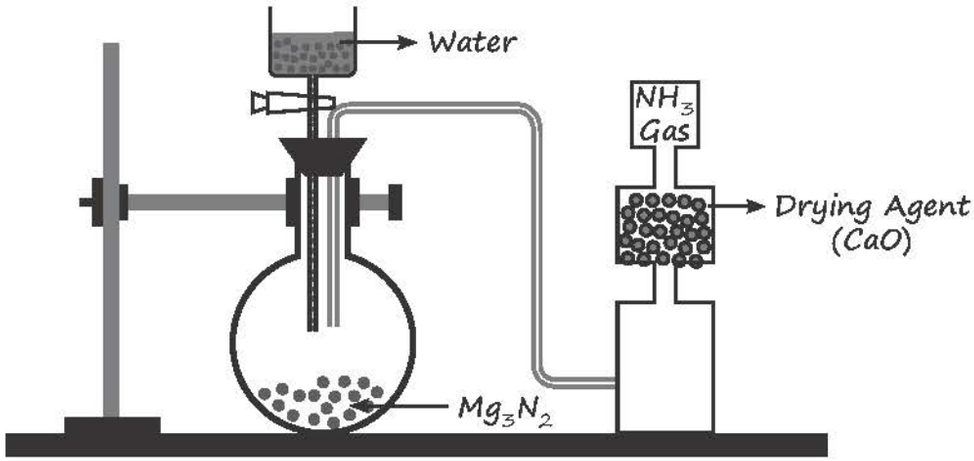
NH_3 gas is collected in inverted gas jars by the downward displacement of air because it is:

(i) Lighter than air (V.D. of NH_3 8.5, V.D. of air 14.4)

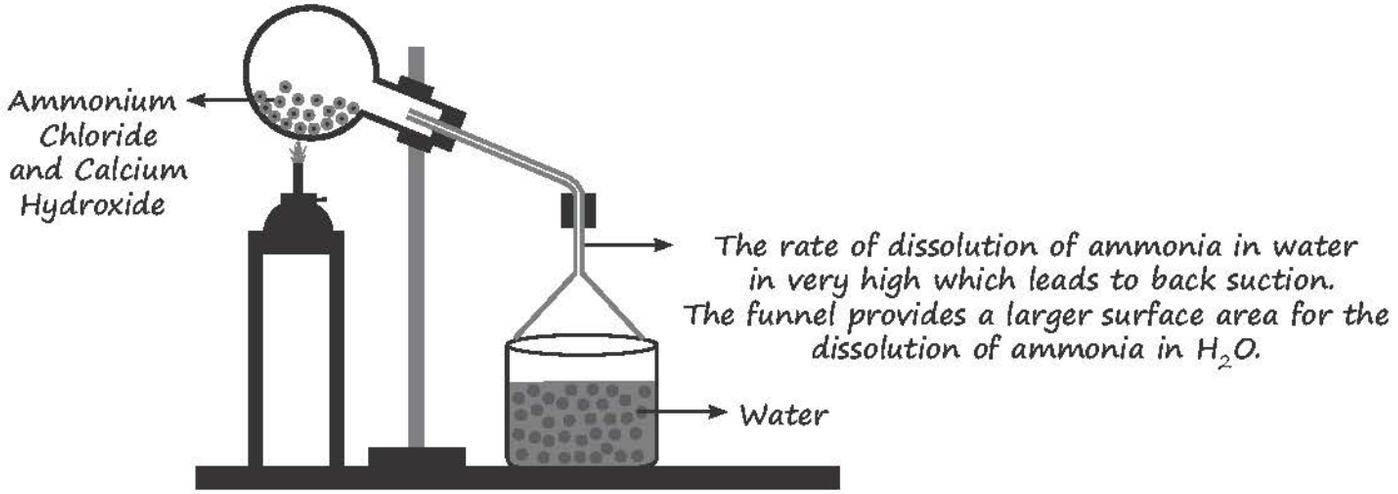
(ii) Highly soluble in H_2O and therefore cannot be collected over water.

II. From Metal Nitrides

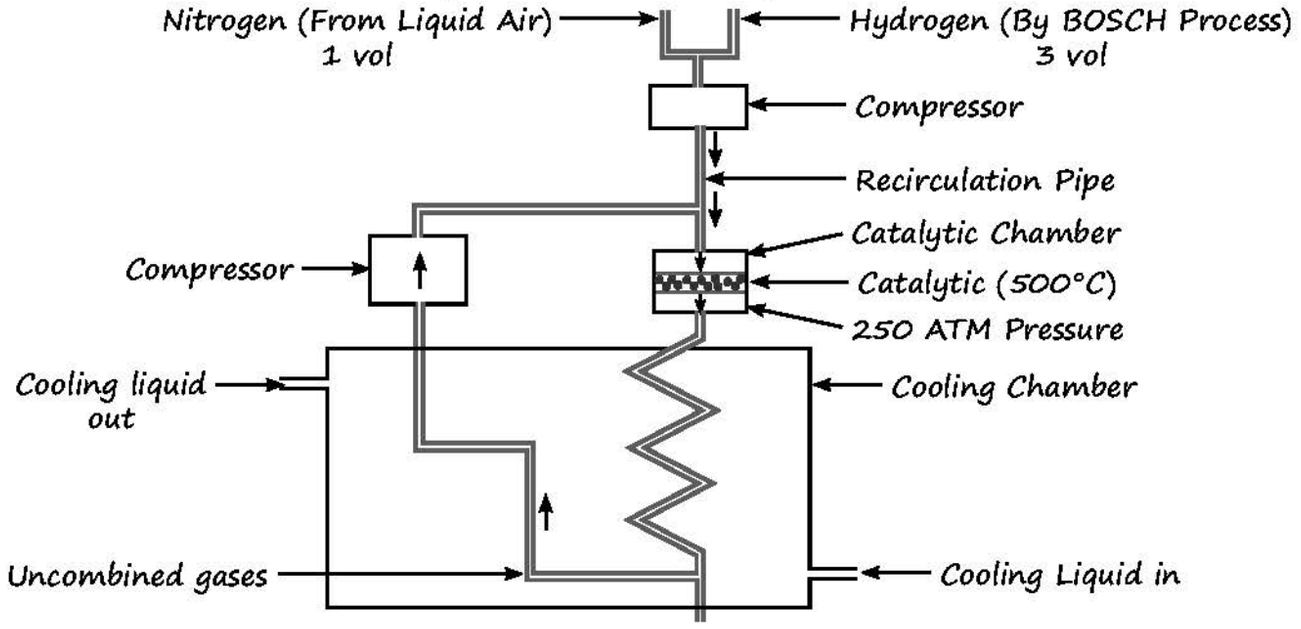
- $Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3\uparrow$
- $AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3\uparrow$



Preparation of Aqueous Ammonia



Manufacture of Ammonia (Haber's Process)



Reactants

$$N_2 : H_2 :: 1 : 3 \text{ (By volume)}$$

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + \text{Heat}$$

Reaction is reversible, exothermic and proceeds with decrease in volume.

Temperature: 450 - 500°C (Optimum temperature)

Pressure: Above 200 atm

Catalyst: Finely divided Iron

Promoter: Traces of Mo or Al_2O_3

NH_3 is separated from the unreacted N_2 and H_2 by:

(i) Liquefaction

(ii) Absorbing in water

Favourable conditions

(1) The reaction is exothermic, hence low temperature will favour the synthesis.

(2) High pressure favours the formation of NH_3.

(3) The speed of the reaction can be increased by using a catalyst, which is finely divided iron. A promoter Molybdenum or Al_2O_3 is used to increase the efficiency of catalyst.

Physical Properties of Ammonia

Colour: Colourless

Odour: Strong Pungent choking smell

Taste: Slightly bitter (alkaline) taste

Physiological action: Non-poisonous

Density: Vapour denistry = 8.5

Nature: Alkaline nature

Liquefaction: Easily liquefied at 10°C by compressing it at 6 atm. pressure.

Boiling point: –33.5°C

Freezing point: –77.7°C

Solubility: Highly soluble in water

- To demonstrate the high solubility of NH_3 gas in water: Fountain experiment

Chemical Properties of Ammonia

Thermal Dissociation

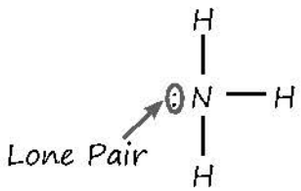
$$2NH_3 \rightleftharpoons N_2 + 3H_2 \text{ (at high temp.)}$$

Basic Properties

Dry NH_3: Covalent molecule, neutral in liquefied form

Aq. Solution: Weak base, basic nature is due to the presence of lone pair.

$$\underset{\text{Weak base}}{NH_4OH} \rightleftharpoons NH_4^+ + OH^-$$



Burning of NH_3 in Oxygen

Ammonia is not a supporter of combustion It burn in oxygen with greenish yellow flame.

Catalytic Oxidation of Ammonia

$$4NH_3 + 5O_2 \xrightarrow{Pt,\ 800°C} 6H_2O + 4NO\uparrow + Heat$$

$$2NO + O_2 \longrightarrow 2NO_2 \text{ (Reddish brown gas)}$$

Ammonia as a Reducing Agent

(i) Reduction of Metallic oxides:

$$\underset{\text{Black Solid}}{2NH_3 + 3CuO} \longrightarrow \underset{\text{Reddish Brown solid}}{3Cu} + 3H_2O + N_2\uparrow$$

(ii) Reducing of Cl_2 to HCl:

(a) When NH_3 is in excess

$$\underset{\text{(excess)}}{8NH_3} + 3Cl_2 \longrightarrow N_2 + 6NH_4Cl$$

(b) When Cl_2 is in excess

$$NH_3 + 3Cl_2 \longrightarrow 3HCl + \underset{\text{(yellow liq.)}}{NCl_3}$$

Reaction with Acids

$$\text{Ammonia gas} + \text{Acid} \longrightarrow \text{Ammonia Salt}$$

$$NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$$

$$NH_3(g) + HNO_3 \longrightarrow NH_4NO_3$$

$$2NH_3(g) + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

Aqueous Solution of NH_3

$$NH_3 + H_2O \longrightarrow NH_4OH$$

$$\text{Aq. } NH_3 + \text{Acid} \longrightarrow \text{Ammonium salt} + \text{water}$$

$$NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$$

Reaction of Aqueous Solution of Ammonia with Soluble Metal Salts

$$\text{Metallic Salt Solution} + NH_4OH \longrightarrow \text{Ammonium Salt} + \text{Metallic hydroxide (ppt)}$$

- $FeSO_4 + 2NH_4OH \longrightarrow (NH_4)_2SO_4 + Fe(OH)_2\downarrow$ (Insoluble in excess of NH_4OH)
  (dirty green)
- $FeCl_3 + 3NH_4OH \longrightarrow 3NH_4Cl + Fe(OH)_3\downarrow$ (Insoluble in excess of NH_4OH)
  (Reddish Brown)
- $Pb(NO_3)_2 + 2NH_4OH \longrightarrow 2NH_4NO_3 + Pb(OH)_2\downarrow$ (Insoluble in excess of NH_4OH)
  (White)
- $Zn(NO_3)_2 + 2NH_4OH \longrightarrow 2NH_4NO_3 + Zn(OH)_2\downarrow$ (Soluble in excess of NH_4OH)
  (White Gelatinous)
- $CuSO_4 + 2NH_4OH \longrightarrow (NH_4)_2SO_4 + Cu(OH)_2\downarrow$ (Soluble in excess of NH_4OH)
  (Pale Blue)
- Ammonium hydroxide is used in qualitative analysis for identifying cations.

Test for Ammonia Gas and Ammonium Ion

1. It has a sharp characteristic odour
2. It turns:
   (i) moist red litmus blue
   (ii) moist turmeric brown
   (iii) phenolphthalein solution pink
3. It gives dense white fumes with conc. HCl
   $$NH_3 + HCl \longrightarrow NH_4Cl$$
4. $NH_3(g) + CuSO_4 \longrightarrow$ deep blue ppt $\xrightarrow[\text{In excess}]{NH_3(g)}$ deep blue solution
5. Ammonium salts gives brown colour ppt with Nessler's solution (K_2HgI_4)

Uses of Ammonia

1. Liquid NH_3: used as a refrigerant in ice plants.
2. Anhydrous NH_3: It is a clear, colourless liquid under pressure. It evaporates rapidly and produces cooling effect. This makes ammonia a good refrigerant.
2. Ammonia is environmentally compatible. It does not deplete O_3 layer and does not contribute towards the global warming.
3. Ammonia being lighter than air goes up in the atmosphere not affecting the life too much on earth.
4. NH_3 Solution is used as an important laboratory reagent in qualitative analysis as it gives characteristic coloured metallic hydroxide precipitates.
5. Aqueous NH_3 emulsifies or dissolves fats, grease etc.
6. Ammonia is used in the manufacture of nitrogenous fertilizers, explosives, other ammonium salts such as $(NH_4)_2CO_3$, NH_4Cl, Nylon and Rayon.

10 CHAPTER

Study of Compounds - Nitric Acid

Study of Compounds: (c) Nitric Acid

- MOLECULAR FORMULA: HNO_3
- RELATIVE MOLECULAR MASS: 63u
- STRUCTURE: $H - O - N \begin{matrix} = O \\ - O \end{matrix}$
- OCCURRENCE: (i) Free State: Nitric acid is found in rainwater, where it occurs in traces after lightning.
  (ii) Combined State: Chile saltpetre ($NaNO_3$), Nitre (KNO_3), Calcium nitrate [$Ca(NO_3)_2$]

Formation of Nitric Acid in the Atmosphere

$N_2 + O_2 \underset{}{\overset{(3000°C)}{\rightleftharpoons}} 2NO$ .... during lightning discharge

$2NO + O_2 \rightleftharpoons 2NO_2$ .... oxidation of nitric oxide

$4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$ .... Nitrogen dioxide dissolves in atmospheric moisture or rainwater in the presence of oxygen and forms nitric acid in the free state.

The conversion of free atmospheric nitrogen into useful nitrogenous compounds in the soil is known as the fixation of atmospheric nitrogen.

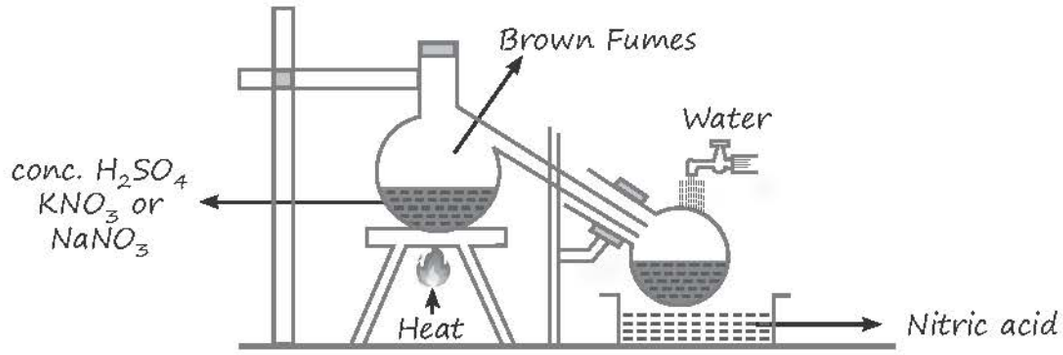
Laboratory Preparation of Nitric Acid

$KNO_3 + \underset{(conc.)}{H_2SO_4} \xrightarrow{<200°C} KHSO_4 + HNO_3$ $\qquad$ $NaNO_3 + \underset{(conc.)}{H_2SO_4} \xrightarrow{<200°C} NaHSO_4 + HNO_3$

The vapors of nitric acid are condensed to a light yellow liquid by chilling.

Pure nitric acid is colorless, but the acid obtained in the lab is slightly yellow.

The yellow color is due to the dissolution of reddish-brown colored NO_2 gas in the acid.



Removal of Yellow Colour From Acid

(i) If dry air or CO_2 is bubbled through yellow acid, acid turns colourless because it removes NO_2 from warm acid which is further oxidised to HNO_3.

(ii) $NO_2 + H_2O \longrightarrow$ colourless
(excess)

PTR (Points to Remember)

(i) In lab method all glass apparatus is used because HNO_3 vapours attack rubber and cork.

(ii) Conc. HCl is not used in place of conc. H_2SO_4 because HCl is volatile and hence HNO_3 vapours will carry HCl vapours.

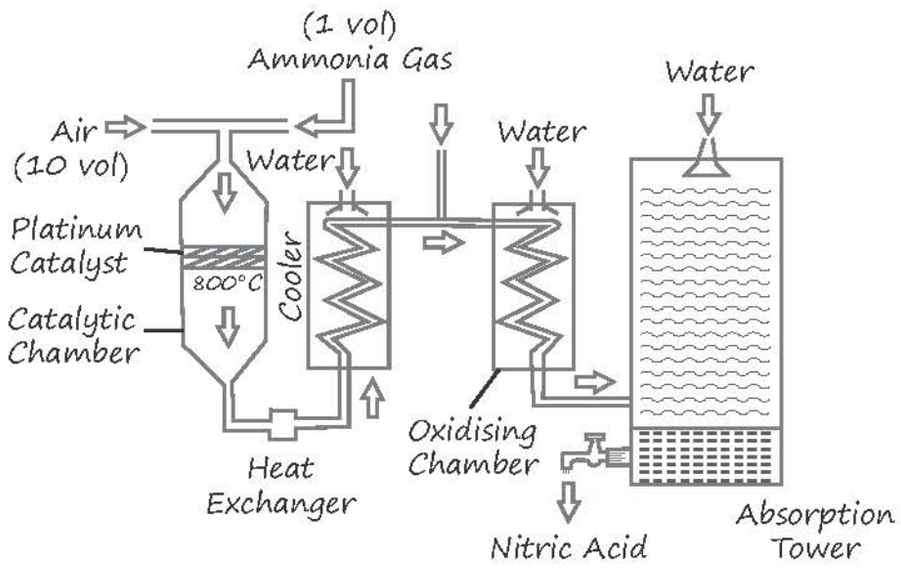
(iii) The temperature of the reaction should not exceed 200°C; Na_2SO_4 formed at higher temperature forms a hard crust which sticks to the wall of the retort and is difficult to remove.

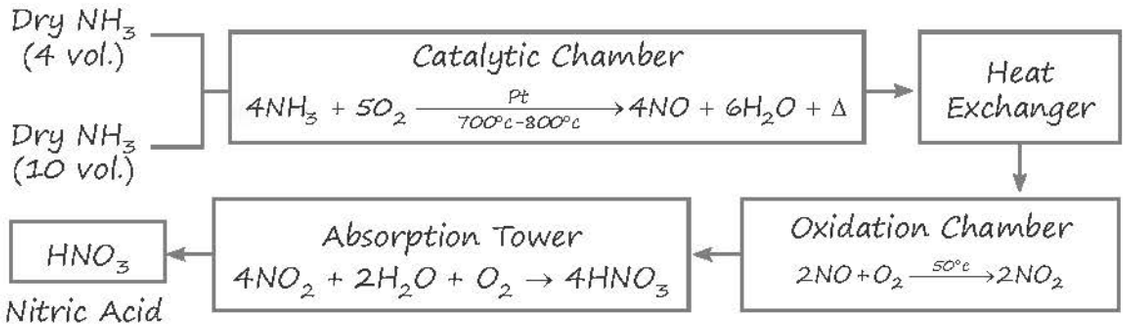
$$NaNO_3 + NaHSO_4 \xrightarrow{>200°C} Na_2SO_4 + HNO_3$$

$$2NaNO_3 + NaHSO_4 \xrightarrow{>200°C} Na_2SO_4 + 2HNO_3$$

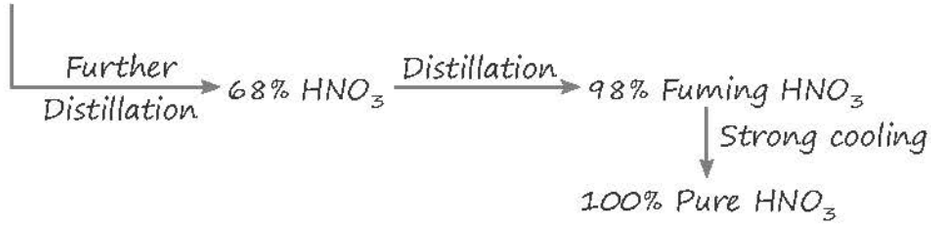
Higher temp may lead to → damage of glass apparatus
→ decomposition of HNO_3
→ wastage of fuel.

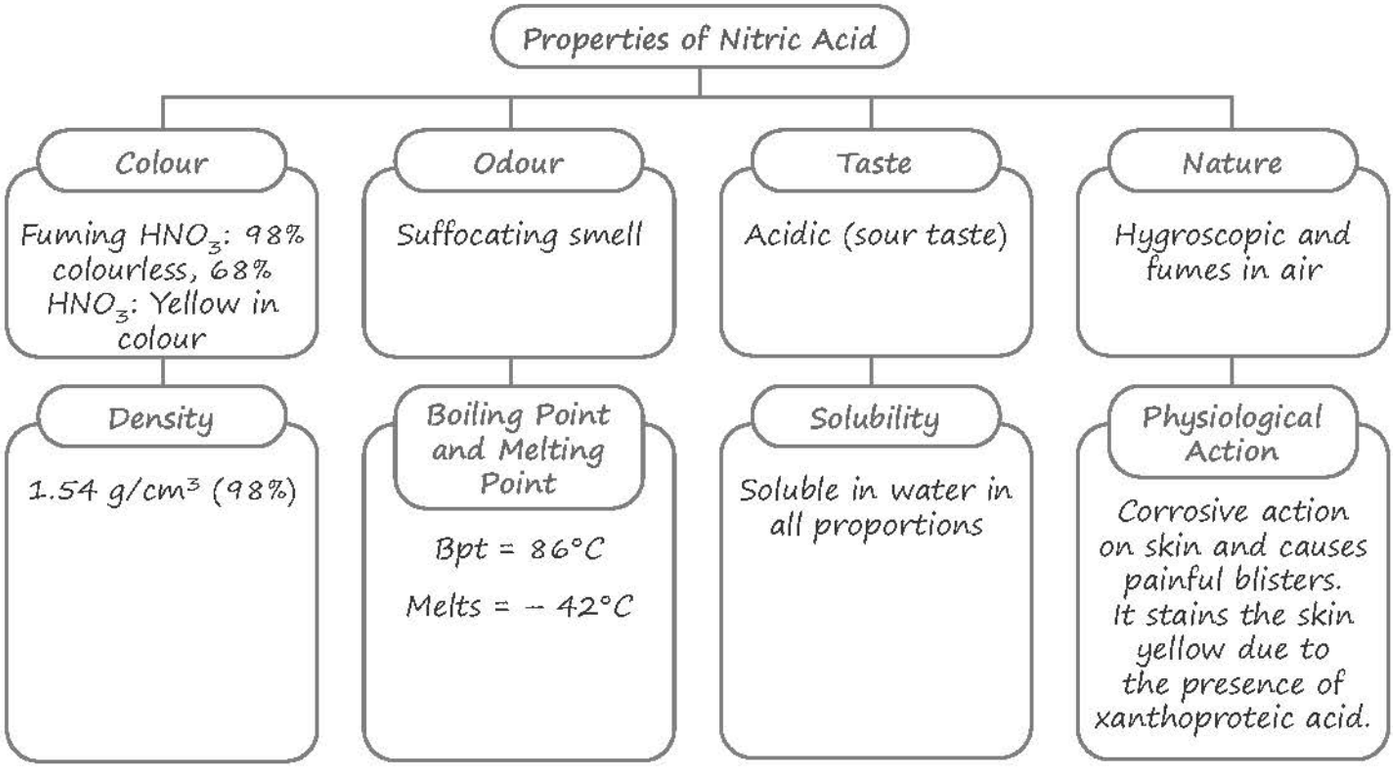
Manufacture of Nitric Acid "Ostwald Process"





- Higher ratio of air is used b'coz oxygen is only $1/5^{th}$ in air and it is requried in all three chambers
- Acid obtained at the bottom of the tower is concentrated above 50%





Chemical Properties of Nitric Acid

Stability

Pure HNO_3 is unstable to heat or sunlight.

$$4\,HNO_3 \xrightarrow[\text{Sunlight}]{\Delta} 4\,NO_2 + 2H_2O + O_2 \ldots.$$

The nitric acid stored in a bottle turns yellow. The colour is due to dissolved NO_2 in HNO_3. To avoid this decomposition HNO_3 is stored in coloured bottles.

Acidic Properties

Strong monobasic acid $HNO_3 \rightleftharpoons H^+ + NO_3^-$

Litmus test: Blue → Red

Phenolphthalein: Colourless

Methyl orange: Pink

- Reactions: with alkalis:
  Base + HNO_3 (dil) → Salt + Water
- Reaction with carbonates and bicarbonates:
  Carbonate + HNO_3 → Salt + Water + HO_2
- Reaction with metallic sulphites and bisulphites:
  Metallic Sulphite + HNO_3 (dil) → Metalic Nitrate + H_2O + $SO_2(g)$

Oxidising Properties

HNO_3: Powerful oxidising agent

$2HNO_3$ (conc) → $2NO_2 + H_2O + [O]$

$2HNO_3$ (dil) → $2NO + H_2O + 3\,[O]$

(i) Action on non-metals

Non-metal + Acid (conc.) → oxidised product + water + NO_2

$C + 4HNO_3$ (conc.) → $CO_2 + H_2O + NO_2$

(ii) Action on metals

(Metals like Iron, cobalt, aluminium, Nickel show passivity when treated with pure conc. HNO_3)

(a) **Cold and dilute HNO_3**

Metal + HNO_3 → oxidised product + water + $2NO$
(cold and dilute)

Metals like iron, copper, cobalt, aluminium, chromium, etc., are rendered passive when treated with pure conc. HNO_3

(b) **Conc. HNO_3 or hot dilute HNO_3:**

Conc. HNO_3 → oxidised + H_2O + $2NO_2$
(conc./ hot dilute)

(c) **Very dilute (1% HNO_3) (only Mg and Mn)**

$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$
(v-dilute)

Reaction as Aqua Regia

Conc. HNO_3 : Conc. HCl ... aqua regia /Royal water

1 : 3

$HNO_3 + 3HCl \rightarrow NOCl + 2H_2O + 2[Cl]$

(Nitrosyl chloride)

$Pt + 4[Cl] \rightarrow PtCl_4$ (soluble); $Au + 3[Cl] \rightarrow AuCl_3$ (soluble)

(Platinum (IV) chloride) Gold (III) chloride

Uses of Nitric Acid

1. To etch designs on copper and brassware
   - Acts as a solvent for a large number of metals
2. To purify gold
3. It acts as a rocket fuel oxidant
4. In the preparation of fertilizers:
   - $Ca(NO_3)_2$
   - NH_4NO_3
   - $NH_4NO_3 + CaCO_3$
   - $CaO \cdot Ca(NO_3)_2$
5. In the preparation of aqua regia, which dissolves noble metals
6. In the manufacture of explosives like T.N.T.
7. In the manufacture of synthetic fibers like artificial silk, nylon
8. In the manufacture of imp. compounds like nitrates of potassium, silver, etc.

Tests for Nitric Acid and Nitrates

1. Conc. HNO_3 gives brown fumes on heating:

   $4HNO_3 \xrightarrow{\Delta} 2H_2O + 4NO_2 + O_2$

   (Brown fumes)

2. Nitrates (other than potassium, sodium, ammonium) produce reddish-brown fumes:

   Metallic nitrate $\xrightarrow{\Delta}$ Metallic oxide + $NO_2 + O_2$

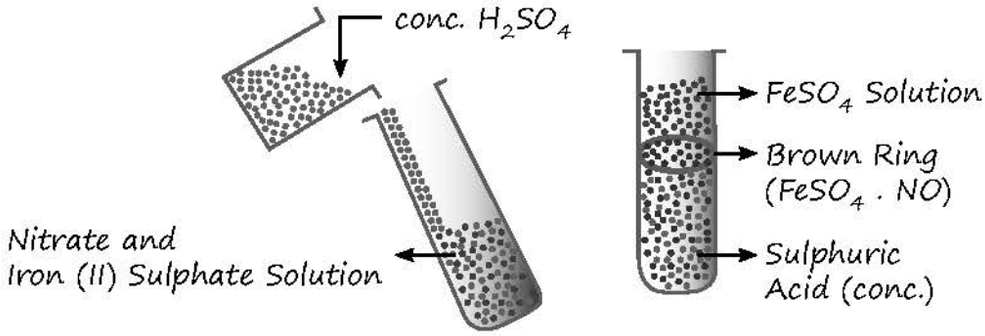
3. On adding copper to HNO_3 or acidified nitrates:

   Dense reddish brown fumes of NO_2 are evolved

   $Cu + 4HNO_3 \xrightarrow{\Delta} Cu(NO_3)_2 + 2H_2O + 2NO_2$

   $Cu + 4NaNO_3 + 4H_2SO_4 \xrightarrow{\Delta} 4NaHSO_4 + Cu(NO_3)_2 + 2H_2O + 2NO_2$

Brown Ring Test



Reaction: $6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$

$FeSO_4 + NO \rightarrow FeSO_4 . NO$ [Nitrosofernous sulphate "Brown ring"]

Points to Remember

(i) A freshly prepared ferrous sulphate solution is used because, on exposure to the atmosphere, it is oxidized to ferric sulphate, which will not give the ring test.

(ii) The brown ring of nitrosoferrous sulphate is formed at the junction of two liquids.

(iii) The brown ring decomposes on disturbing the test tube. The heat evolved decomposes the brown ring.

Effects of Heat on Nitrates

I. Alkali Metal Nitrates:

On heating, melts into colorless liquid, which decomposes on heating to give O_2 gas.

$2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2 \uparrow$

$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2 \uparrow$

As O_2 is formed glowing splinter bursts into flames

II. All other Nitrates (except silver and mercury)

$2Ca(NO_3)_2 \xrightarrow{\Delta} 2CaO + 4NO_2(g) + O_2 \uparrow$
White

$2Zn(NO_3)_2 \xrightarrow{\Delta} 2ZnO + 4NO_2 + O_2$
Yellow when hot
White when cold

III. Silver and Mercuric Nitrate:

$2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + O_2$
White

$Hg(NO_3)_2 \xrightarrow{\Delta} Hg + 2NO_2 + O_2$

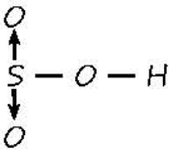
IV. $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O + O_2$

11 CHAPTER

Sulphuric Acid

Study of Compounds

- Molecular Formula: H_2SO_4
- Relative Molecular Mass: 98
- Structure:



Preparation of Sulphuric Acid

BY OXIDATION OF AN AQUEOUS SOLUTION OF SULPHUR DIOXIDE:

- $2SO_2 + 2H_2O + O_2 \rightarrow 2H_2SO_4$
- $SO_2 + 2H_2O + Cl_2 \rightarrow H_2SO_4 + 2HCl$ ..... Greenish yellow colour of chlorine fades
- $SO_2 + 2H_2O + Br_2 \rightarrow H_2SO_4 + 2HBr$ ..... Reddish brown colour of bromine fades
- $S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$

Manufacture of Sulphuric Acid: Contact Process

STEP – I

Production of SO_2

$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

STEP – II

Purification of Gases Using Electrical Precipitator, Water Scrubber and Arsenic Purifiers

STEP – III

Catalytic Oxidation of SO_2

$2SO_2 + O_2 \underset{450°C}{\overset{V_2O_5/Pt}{\rightleftharpoons}} 2SO_3 + \text{heat}$

STEP – IV

Absorption of SO_3 in H_2SO_4

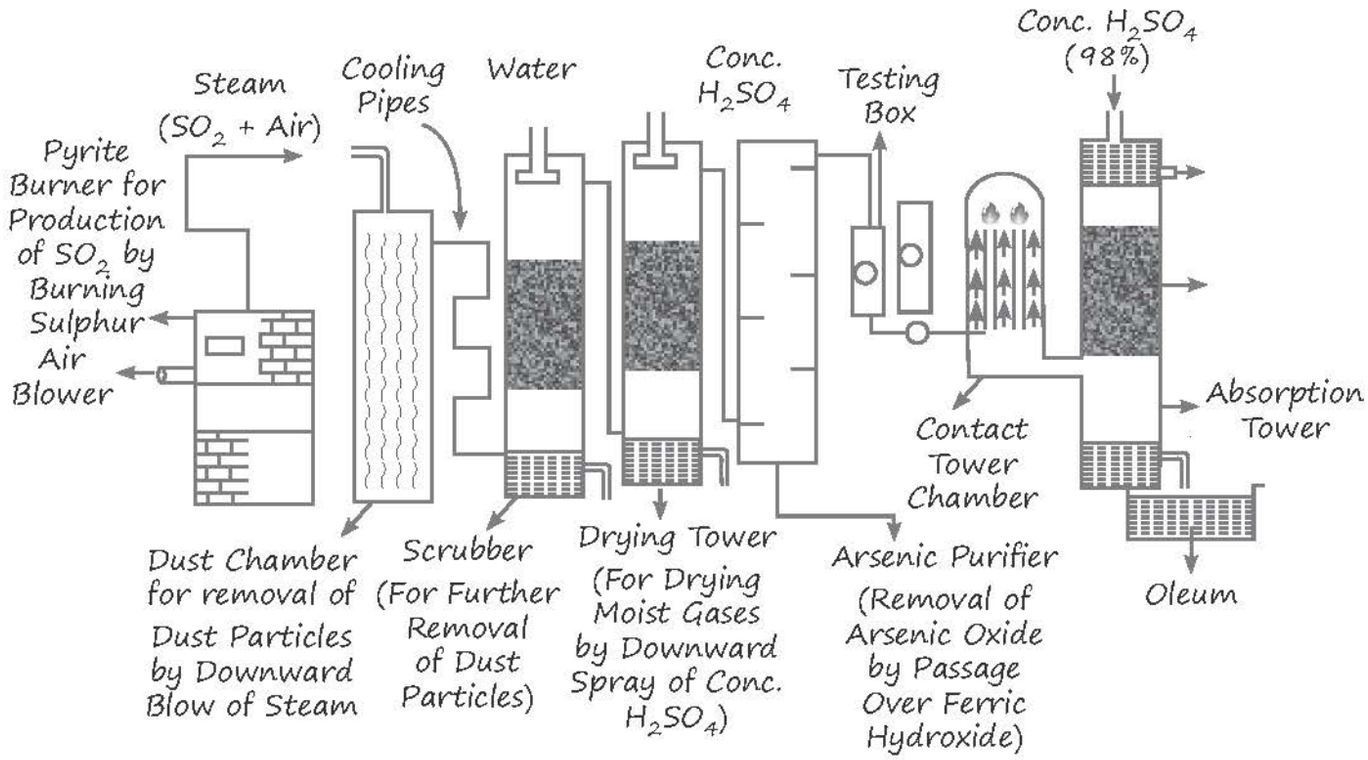
$SO_3 + H_2SO_4 \rightarrow H_2SO_4$ (oleum/pyrosulphuric acid)

STEP – V

Dilution of Oleum to Obtain H2SO4

$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

Contact Process For Manufacture of H_2SO_4



Favourable Conditions for the Conversion of SO_2 to SO_3

Contact Process

1. **Exothermic Reactions Are Favoured at Low Temperature:** The reactions are favoured at low temperature. optimum temperature: 410–450°c
2. **High Pressure:** 1–2 atm. pressure
3. **Excess of Oxygen:** This increases the production of SO_3
4. **Catalyst:** Platinum is more efficient as a catalyst than V_2O_5 but platinum is expensive and also it gets poisoned by impurities like arsenic (III) oxide.

   * V_2O_5: catalyst.

Properties of Sulphuric Acid

Physical Properties of Sulphuric Acid

- **Colour:** Colourless
- **Odour:** Odourless
- **Taste:** Slightly sour in taste
- **Nature:** Dense, oily, hygroscopic liquid. It absorbs moisture, so H_2SO_4 should always be kept stoppered.
- **Density:** Pure acid: 1.85 g/cc
- **Boiling Point:** 338°C
- **Melting Point:** 10.4°C
- **Solubility:** Soluble in water in all proportion.
- **Conductivity:** Pure acid: Almost a non conductor of electricity Dilute acid: Good conductor of electricity
- **Constant Boiling Mixture:** It forms a constant boiling mixture at 338°C containing 98.5% of the acid
- **Physiological Nature:** Concentrated acid is highly corrosive in nature and chars the skin black.

Chemical Properties

- H_2SO_4 when dissolved in water form hydronium ion and shows acidic properties

$$H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^- \\ HSO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-}$$ } dil. basic Acid

- It can form two types of salt:

(i) Acid salt: $NaCl + H_2SO_4 \xrightarrow{\text{Below } 200^\circ C} \underset{\text{(acid salt)}}{NaHSO_4} + HCl$

(ii) Normal salt: $2NaCl + H_2SO_4 \xrightarrow{\text{below } 200^\circ C} \underset{\text{normal salt}}{Na_2SO_4} + 2HCl$

Acidic Property

Properties of dilute H_2SO_4

With Metals Above Hydrogen	Active metal + Acid (dil) ⟶ metal Sulphate+ hydrogen $Mg + H_2SO_4$ (dil) ⟶ $MgSO_4 + H_2\uparrow$
Bases: Neutralisation	Base/alkali + Acid ⟶ salt + water $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O$
With Carbonates and Bi carbonates	Carbonate/bicarbonate + Acid → salt + CO_2 + H_2O $CuCO_3 + H_2SO_4 \longrightarrow CuSO_4 + H_2O + CO_2\uparrow$ $2NaHCO_3 + H_2SO_4 \longrightarrow Na_2So_4 + CO_2\uparrow + H_2O$
With sulphites and Bisulphites	Sulphites/bisulphites + Acid → salt + H_2O + $SO_2\uparrow$ $Na_2SO_4 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2\uparrow$
With metal sulphides	Metal sulphide + Acid ⟶ Salt + $H_2S\uparrow$ $Na_2S + H_2SO_4 \longrightarrow Na_2SO_4 + H_2S\uparrow$

Properties of conc. Sulphuric Acid

(1) Non volatile Nature

Due to its high boiling point it is used for the prep. of volatile acids from their salts by double decomposition.

Salt of volatile acid + Acid (conc) $\longrightarrow$ Acid salt + volatile acid

(2) Oxidising agent

Thermal decomposition of H_2SO_4 yields nascent oxygen

$H_2SO_4 \longrightarrow H_2O + SO_2 + [O]$

(i) **Non metals:** $C + H_2SO_4 \longrightarrow CO_2 + 2SO_2\uparrow + H_2O$

(ii) **Metals:** $Cu + 2H_2SO_4 \longrightarrow CuSO_4 + 2H_2O + SO_2\uparrow$

(iii) **Inorganic compounds:** $2HBr + H_2SO_4\,(conc) \longrightarrow Br_2 + 2H_2O + SO_2\uparrow$

(3) Dehydrating Agent

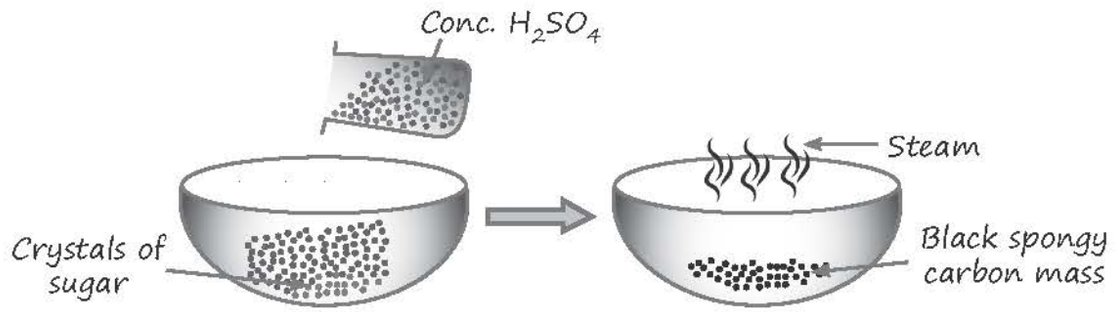
(i) Conc H_2SO_4 removes water of crystallisation from salts

$CuSO_4 \cdot 5H_2O \xrightarrow{Conc.H_2SO_4} CuSO_{4(s)} + 5H_2O$

(Blue) (White)

(ii) All carbohydrates such as glucose, sugar and cellulose react immediately to give a black spongy mass of carbon the reaction is called 'charring'.

$C_6H_{12}O_6 \xrightarrow{Conc.H_2SO_4} 6C + 6H_2O \qquad C_{12}H_{22}O_{11} \xrightarrow{Conc.H_2SO4} 12C + 11H_2O$



Organic compounds and organic acids are dehydrated by conc. H_2SO_4.

(i) $HCOOH \xrightarrow{conc.H_2SO_4} CO + H_2O$

(ii) $COOH \xrightarrow{conc\ H_2SO_4} Co_2 + H_2O + CO$

$|$

$COOH$

(iii) $C_2H_5OH \xrightarrow[170°C]{conc.\ H_2SO_4} C_2H_4\uparrow + H_2O$

Preparation of Insoluble sulphates

H_2SO_4 Precipitates the insoluble sulphates of lead, barium and calcium from the solutions of their salts

soluble salt + Acid $\longrightarrow$ Insoluble salt + Acid

$Pb(NO_3)_2 + H_2SO_4 \longrightarrow PbSO_4\downarrow + 2HNO_3$

$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4\downarrow + 2HCl$

Tests for Sulphuric Acid and Sulphates

Test1: When barium chloride solution is added to H_2SO_4 or any soluble metal sulphate, white ppt of barium sulphate is formed.

$$BaCl_2 + H_2SO_4 \rightarrow \underset{\text{White ppt (Insoluble)}}{BaSO_4\downarrow} + 2HCl$$

Test 2: conc. H_2SO_4 on heating with Cu. evolves SO_2.SO_2 turns acidified $K_2Cr_2O_7$ green

$$Cu + \underset{\text{(conc)}}{2H_2SO_4} \longrightarrow CuSO_4 + 2H_2O + SO_2\uparrow$$

$$\underset{\text{(Orange)}}{K_2Cr_2O_7 + H_2SO_4} \xrightarrow{+SO_2} \underset{\text{(Green)}}{Cr_2(SO_4)_3 + K_2SO_4 + H_2O}$$

Test 3: Conc. H_2SO_4 on heating with NaCl evolves pungent fumes of HCl gas. HCl gives dense white funes of NH_4Cl With a glass rod dipped in NH_4OH

$$NaCl + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HCl\uparrow$$

$$HCl + NH_4OH \rightleftharpoons \underset{\text{(dense white funes)}}{NH_4Cl} + H_2O$$

Uses of Sulphuric Acid

- **It is used in the preparation of:** (a) Halogens (b) Carbon Monoxide (c) Hydrogen (d) Sulphur dioxideconc. conc. H_2SO_4 is used to dry HCl gas.
- **Metalurgy:**
  Extraction- It reaction with metallic compounds given sulphates which on electrolysis give the metal in the pure form.
  Picling Metals- Removes metallic impurities like oxides and carbonates from the surface of metals before galvanising.
- **IN Lead Accumulators:** It undergoes electrolysis in the aqueous state
- **Oil Refining:** Sulphuric acid is used to remove harmful empurities in purification of oil products.
- **Industrial Uses: In the Manufacture of**
  Fertilizers- Ammonium sulphate $[(NH_4)_2SO_4]$
  Dyes, Drugs- Superphosphate of lime $[Ca(H_2PO_4)_2 + CaSO_4]$ Rayon, Nylon
  Explosives: TNT (Tri Nitro Toluene)- Picric acid, Trinitroglycerine
  Acids: HNO_3, HCl, H_3PO_4, CH_3COOH
  Compounds: Sodium sulphate, Ferrous sulphate

12 CHAPTER

Organic Compound

Organic Compounds

Organic compounds are the compounds of carbon and organic chemistry is the study of carbon compounds.

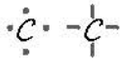
Sources of Organic Compounds

- Plants
- Animals
- Coal
- Petroleum
- Fermentation
- Wood
- Synthetic Methods

Unique Nature of Carbon Atoms

Tetra valency of the Carbon Atom

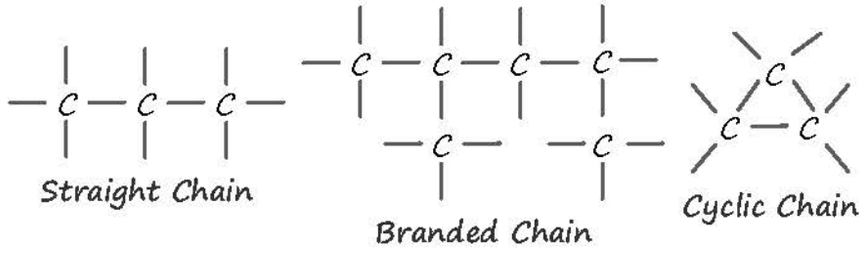
The characteristic of the carbon atom by virtue of which it forms covalent bonds is called the tetravalency of carbon.



Catenation

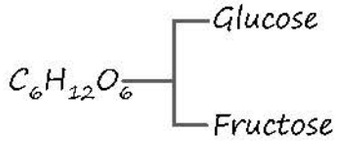
The property of self linking of atoms of an element through covalent bonds in order to form straight chains, branched chains and cyclic chains of different sizes is known as catenation.

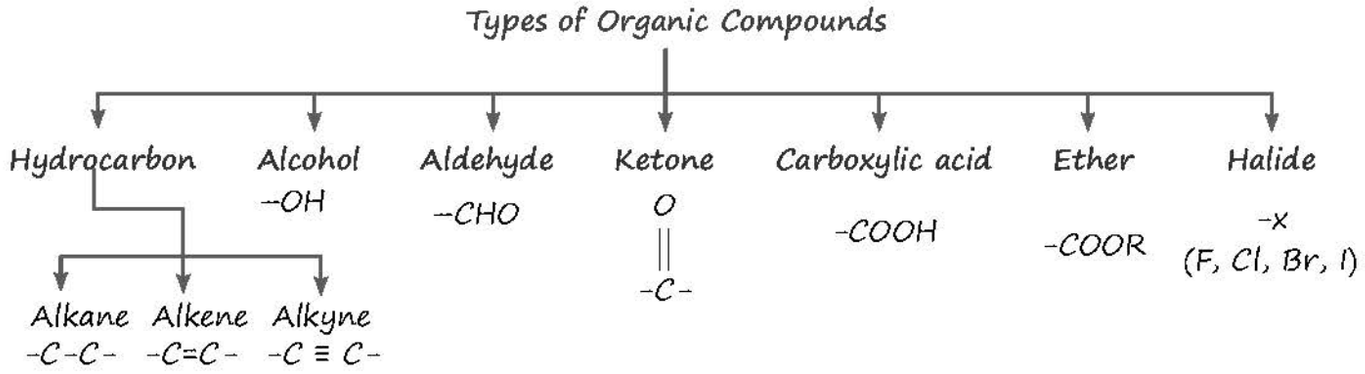
- This is due to
  - Greater strength of carbon-carbon bond
  - Tetra- Covalency of carbon



Isomerism

The phenomenon in which the compounds have the same molecular formula but differ in the structural formula is known as Isomerism and the compounds are refered as Isomers.

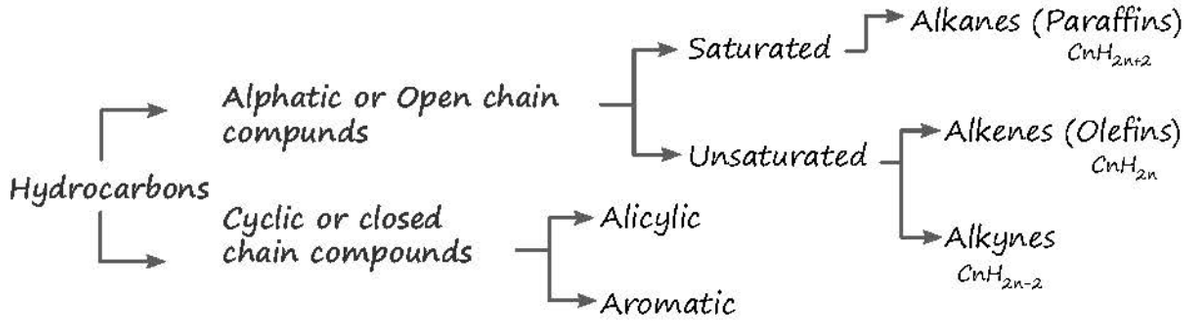




Hydrocarbons

Hydrocarbons are compounds that are made up only of carbon and hydrogen atoms.

Classification of Hydrocarbons



Aliphatic or Open Chain Compounds

Saturated Organic Compounds

- All four Valencies of each carbon atom are satisfied by forming single covalent bonds with carbon and hydrogen atoms. Carbon atoms are joined by single covalent bond. They are less reactive due to the non-availability of electrons.
- They Undergo substitution reaction.

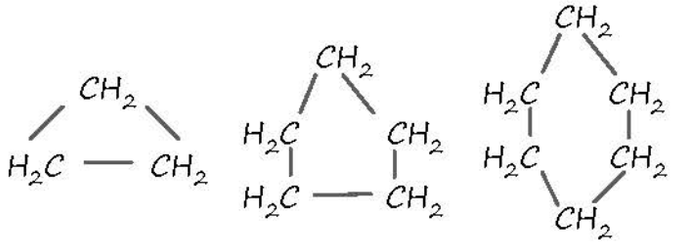
Unsaturated Organic Compounds

- The valencies of at least two carbon atoms are not fully satisfied by the hydrogen atoms.
- Carbon atoms are joined by double or triple covalent bond.
- They are more reactive due to the presence of electrons.
- They undergo addition reactions

Cyclic or Closed Chain or Ring Chain Compounds

Alicyclic Compounds

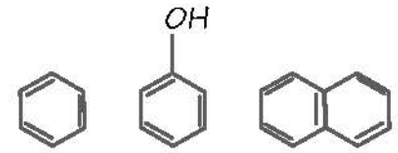
- These are ringed carbon compounds containing three or more atoms of carbon arranged in a closed structure.



Cyclopropane Cyclopentane Cyclohexane

Aromatic Compounds

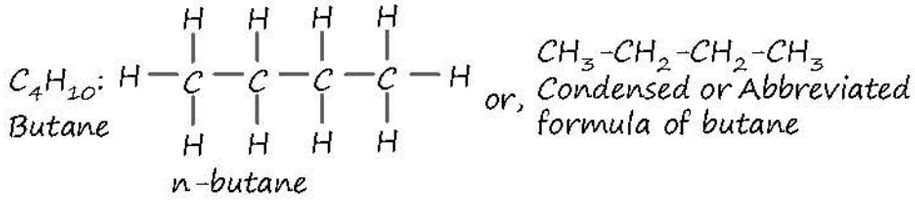
- Some hydrocarbons contain at least one benzene ring in their molecules. These compounds have a pleasant smell. Hence, they are known as aromatic compounds.



Benzene Phenol Naphthalene

Structural Formula

- The formula that shows how atoms of different elements are linked together in a molecule is known as structural formula.



C_4H_{10}: Butane

n-butane

or, $CH_3-CH_2-CH_2-CH_3$ Condensed or Abbreviated formula of butane

Alkyl Group

- An alkyl group is formed by removing one atom of hydrogen atom from an Alkane

$$\underset{\text{Alkane}}{C_nH_{2n+2}} \xrightarrow{-H} \underset{\text{Alkyl}}{C_nH_{2n+1}}$$

Functional Group

- An atom or group of atoms joined in a specific manner which is responsible for the characteristic chemical properties of the organic compounds.

Characteristics of Functional Group

- Chemical properties of the compounds containing the same functional group are similar.
- The physical and chemical properties of the compounds of diff. functional groups are different.
- There exists a homologous series of compounds containing a particular type of functional group.

Homologous Series

It is a group of organic compounds represented by the same general formula and having a similar structure and similar chemical properties in which the successive compounds differ by a $-CH_2$ group.

Charateristics of a Homologous Series

- All member share the same general formula. (G.F of Alkane: C_nH_{2n+2})
- Each member of the series differs from the preceding one by the addition of $-CH_2$ group or 14 a.m.u.
- The physical properties of the members show gradation as the molecular mass increases.
- Members of same homologous series have similar chemical properties.
- They can be prepared by same general method of preparation.

Significance of the Homologous Series

- If the properties of the first few members are known, the nature of any member of that family can be ascertained.
- It helps to predict the properties of the members of the series which are yet to the prepared.
- Systematic study of organic chemistry becomes easy.

Nomenclature

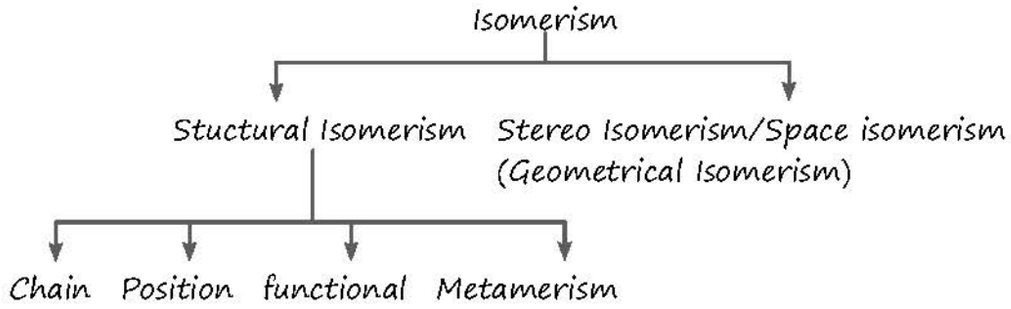
Naming of organic compounds follows Trivial system and IUPAC system.

Name of the compound following IUPAC:

Prefix	Word root	Primary suffix	Secondary suffix
Additional substituent alkyl or functional group and its position in the carbon chain in alphabetical order.	Depends on the number of carbon atoms	Nature of the bond in carbon- carbon atom i.e. C-C: -ane C=C: -ene C≡C: -yne	Particular functional group linked to the carbon chain.

Isomerism

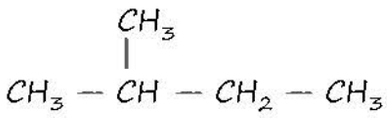
Compounds having the same molecular formula but different structural formula are known as 'Isomers' and the phenomenon is known as Isomerism. Main cause of structural isomerism is difference in the mode of linking of atoms.



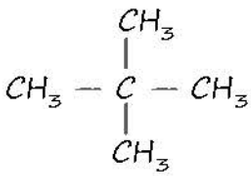
Chain Isomerism

Exhibited by compounds having similar molecular formula but differ in the arrangement of carbon atoms in straight or branched chains.

C_5H_{12} $CH_3CH_2CH_2CH_2CH_3$
n-pentane



Iso. pentane



neo-propane

Position Isomerism

When two or more compounds with same molecular formula differ in the position of substituent atom or functional group on the carbon atom they are called position isomers.

C_4H_6 (Butyne)
- $CH_3CH_2 - C \equiv C - H$ (But-1-yne)
- $CH_3 - C \equiv C - CH_3$ (But-2-yne)

Functional Isomerism

Two or more compounds with the same molecular formula but different functional groups are called functional Isomers.

C_2H_6O
- CH_3CH_2OH (Ethanol)
- CH_3-O-CH_3 (Methyl ether)

Metamerism

- It arises due to unequal distribution of alkyl groups on either side of the functional groups in the molecule

$CH_3-O-C_3H_7$ Methoxy propane $C_2H_5-O-C_2H_5$ Ethoxy ethane

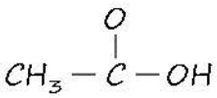
13 CHAPTER

Carboxylic Acids

Carboxylic Acids

- An organic compound containing the Carboxyl group (–COOH).
- General formula: $C_nH_{2n+1}COOH$ or RCOOH known as "Alkanoic Acids"

Ethanolic Acid/Acetic Acid: CH_3COOH



- 4–5% solution of ethanoic acid is called vinegar.

Physical Properties of Ethanoic Acid

- Pure Ethanoic acid is a colourless liquid
- Characteristic pungent smell
- The anhydrous acid on cooling forms a crystalline mass resembling ice; m.pt is 17°C and for this reason it is called 'glacial acetic acid'.
- It boils at 118°C
- It is a hygroscopic liquid, sp. gravity at 0°C is 1.08
- It is miscible with water, alcohol and ether in all the proportions.

Chemical Properties of Ethanoic Acid

Acidity

- It is a weak acid, hence
  - Turns moist blue litmus red.
  - reacts with active metals (i.e. Zn and Mg) evolving hydrogen

$$2\ CH_3COOH + Zn \rightarrow (CH_3COO)_2Zn + H_2\uparrow$$

$$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2\uparrow$$

  - Reacts with alkalies to form salt and water.

$$Acid + Base \rightarrow salt + Water$$

+ Liberates brisk effervescence of CO_2 with carbonates.

$$2CH_3COOH + Na_2Co_3 \rightarrow 2CH_3COONa + CO_2\uparrow + H_2O$$

$$CH_3COOH + NaHCo_3 \rightarrow CH_3COONa + CO_2\uparrow + H_2O$$

+ Esterification: When an acid reacts with alcohol in the presenceof conc. H_2SO_4

$$\underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Ethanol}}{C_2H_5OH} \xrightarrow{\text{Conc. } H_2SO_4} \underset{\text{Ethyl ethanoate}}{CH_3COOC_2H_5} + \underset{\text{water}}{H_2O}$$

Dehydration

It forms acetic anhydride an heating with P_2O_5

$$2CH_3COOH \xrightarrow[P_2O_5]{\Delta} (CH_3CO)_2O + H_2O$$

Reduction

A strong reducing agent such as LiAlH4 will reduce ethanoic acid to ethanol.

$$CH_3COOH + 4[H] \longrightarrow C_2H_5O_4 + H_2O$$

14 CHAPTER

Alkynes

Alkynes

Aliphatic hydrocarbons that contain triple bond (–c ≡ c–) i.e. acetylenic bond.

- They are unsaturated hydrocarbons General formula: c_nH_{2n-2}

Sources of Alkyne

Natural Gas and Petroleum

IsoMERISM

- Position Isomerism
- Chain Isomerism

1. Butyne: shows position isomerism

$CH_3 - CH_2 - C \equiv CH$ (But–1–yne) $CH_3 - C \equiv C - CH_3$ (But–2–yne)

2 Pentyne

Shows chain and Position. isomerism

$CH_3–CH_2–CH_2–C \equiv CH$: Pent–1–yne

$CH_3–CH_2–C \equiv C–CH_3$: Pent–2–yne

$CH_3–CH(CH_3)–C \equiv CH$: 3–Methyl but–1–yne

Ethyne: C_2H_2

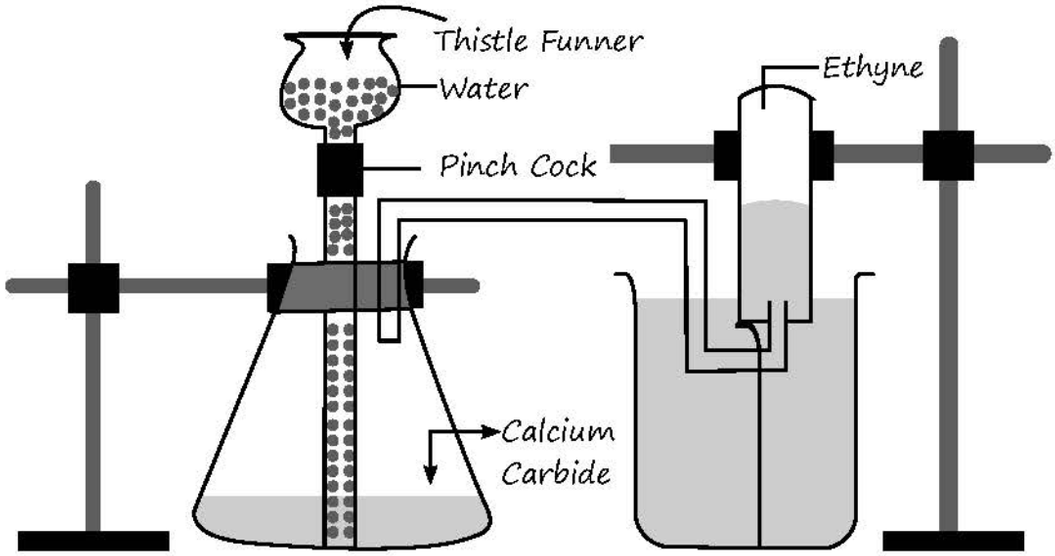
Common Name Acetylene: $H–C \equiv C–H$

Linear

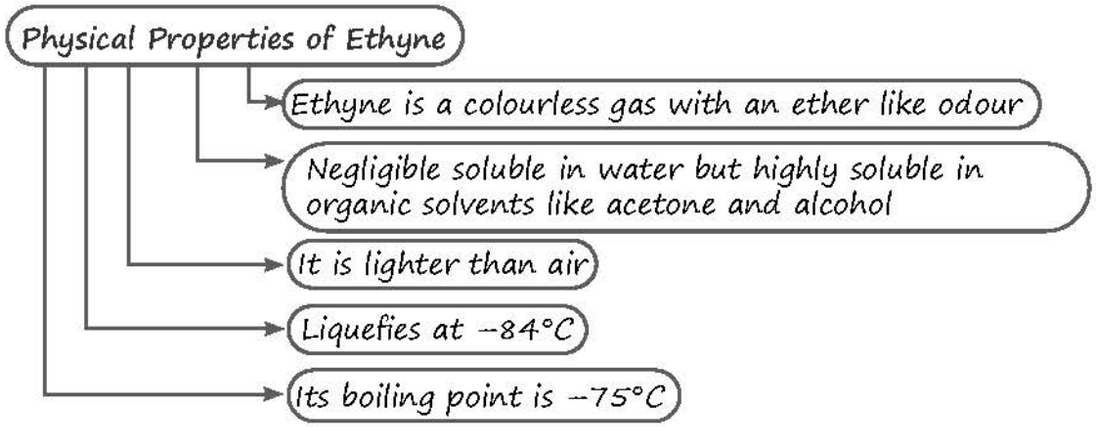
Laboratory Preparation of Ethyne:

$$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2\uparrow$$

Calcium carbide + Water → Calcium hydroxide + Ethyne



- The gas is collected by the downward displacement of water, since it is insoluble in water.
- The flask should not contain air as ethyne forms an explosive mixture with air. Therefore, the air of the flask is displaced by oil or some inert gas.



Chemical Properties of Ethyne:

Ethyne is a highly reactive compound due to the presence of triple bond.

1. Oxidation of Ethyne: Combustion

- Ethyne burns in excess air with a brillliant white flame to produce CO_2, water vapour and large amt. of heat.

$$2CH \equiv CH + 5O_2 \rightarrow 4CO_2 + 2H_2O + Heat$$

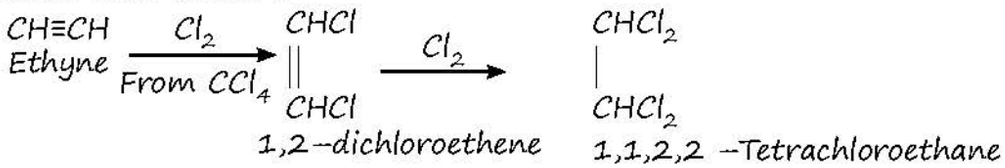
2. Addition Reaction

- Alkynes are unsaturated compounds, so they are associated with addition reactions, since triple bonds breaks easily.
  - + Addition of Hydrogen: Catalytic Hydrogenation

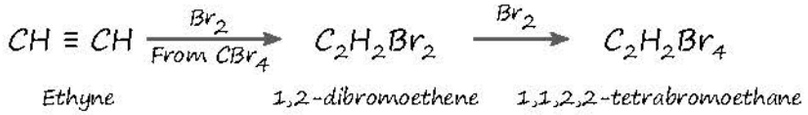
$$HC \equiv CH \xrightarrow[Ni]{+H_2} \underset{\text{Ethene}}{CH_2=CH_2} \xrightarrow[Ni]{+H_2} \underset{\text{Ethane}}{CH_3-CH_3}$$

Addition with Halogen:

Reaction with Chlorine

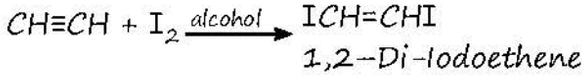


Reaction with Bromine

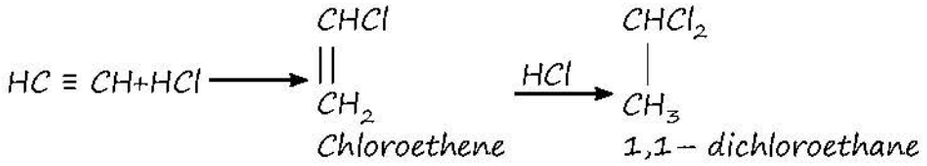


- Reddish brown colour disappears due to the formation of an addition product.

Reaction with Iodine



Reaction with HCl



Chemical Tests to Distinguish Between Alkanes, Alkenes, Alkynes.

	Test	Alkanes (Saturated HC)	Alkenes (Unsaturated Hydrocarbon)	Alkynes (Unsaturated Hydrocarbon)
1.	Hydrocarbon + Br_2/ CCl_4(Reddish brown)	No change	Reddish brown colour gets decolourised	Reddish brown colour gets decolourised
2.	Hydrocarbon + Alk. $KMnO_4$ (Purple colour)	No change	Purple colour fades	Purple colour fades" BAYER'S TEST" $CH{\equiv}CH+4[O] \xrightarrow{Alk.KMNO_4} \begin{matrix}COOH\\ \vert \\ COOH\end{matrix}$ (Oxalic acid)
3.	Hydrocarbon + Ammonical cuprous chloride	No change	No change	Red ppt of copper acetylide is formed
4.	Hydrocarbon + Ammonical $AgNo_3$	No change	No change	White ppt of Silver acetylide is formed.

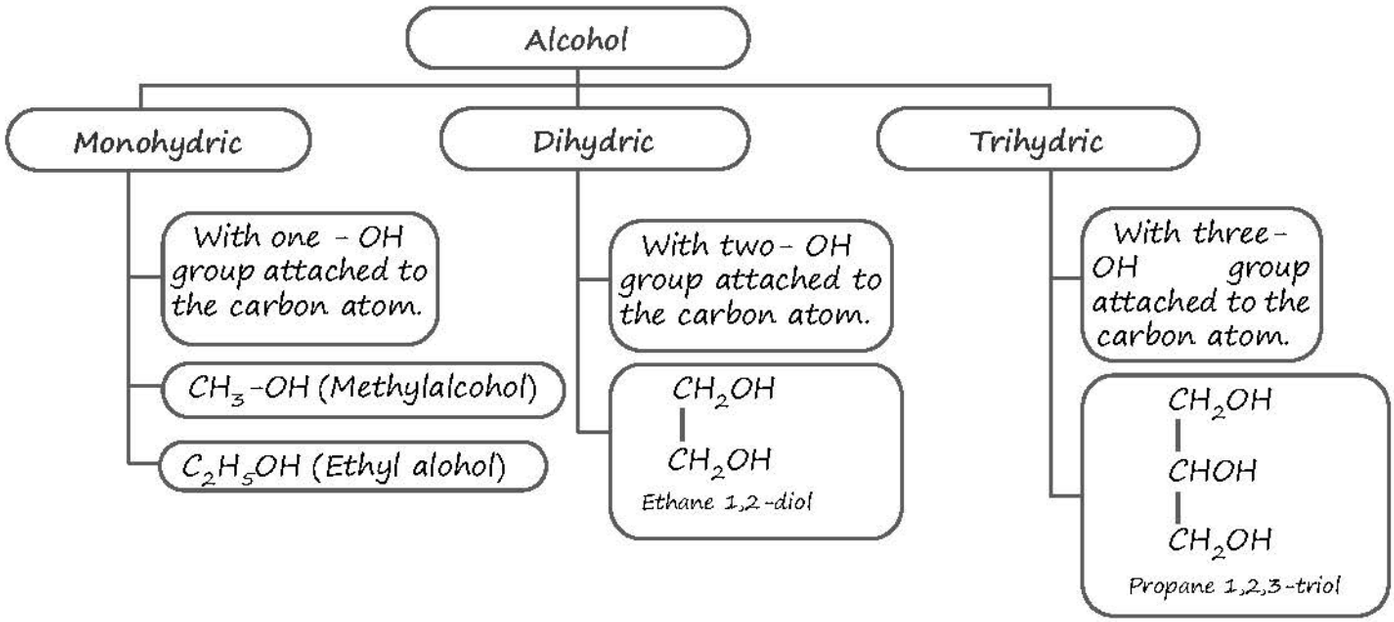
15 CHAPTER

Alcohol

Alcohol

- Hydroxy derivatives of alkanes

General formula: $C_nH_{2n+1}OH$



Ethanol: C_2H_5OH

Isomers in Alcohols: Alcohol with 3 or more carbon atom shows isomerism.

E.g: $CH_3\,CH_2\,CH_2\,OH$ (Propan-1-ol) $CH_3\,CH(OH)\,CH_3$ (Propan-2-ol)

Laboratory Preperation

Alkylhalide + aq. KOH $\xrightarrow{\text{boil}}$ Alcohol + KX

Alkylhalide + aq. NaOH $\xrightarrow{\text{boil}}$ Alcohol + NaX

Hydrolysis of alkylhalide

E.g: $C_2H_5Cl + aq.KOH \xrightarrow{\text{boil}} C_2H_5OH + KCl$

(Ethyl chloride → Ethanol)

Industrial Method: (Large Scale Method)

(i) $C_2H_4 + H_2SO_4 \xrightarrow[30\ atm]{80°C} C_2H_5HSO_4 \xrightarrow[(Boiling)]{H_2O} C_2H_5OH + H_2SO_4$
Ethene — Ethene

(ii) $C_2H_4 + H_2O \xrightarrow[300°C,\ 60\ atm]{H_3PO_4} C_2H_5OH$
Ethanol

Physical Properties

- They are inflammable volatile liquids
- B pt increases with an increase in molecular wt.
- They are soluble in water as well as in organic solvents
- Ethanol is lighter than water
- They are colourless, have a faint odour and a burning taste.
- Some are toxic

Chemical Properties

Combustion (Burning)

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ + heat Ethanol burns with a blue flame.

Oxidation With Acidified $K_2Cr_2O_7$

Alcohol $\xrightarrow{[O]}$ Aldehyde $\xrightarrow{[O]}$ Acid $\xrightarrow{[O]}$ $CO_2 + H_2O$

- $CH_3OH \xrightarrow{[O]} HCHO + H_2O \xrightarrow{[O]} HCOOH$
  Methanol — Formaldehyde — Formic Acid
- $C_2H_5OH \xrightarrow{[O]} CH_3CHO + H_2O \xrightarrow{[O]} CH_3COOH$
  Ethanol — Ethanal — Ethanoic acid

Reaction with Sodium

$2CH_3OH + 2Na \rightarrow 2CH_3ONa + H_2\uparrow$
Methanol — Sodium methoxide

Effervescence of hydrogen
Serves as a test for alcohols

$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2\uparrow$
Ethanol — Sodium ethoxide

Reaction With Acetic Acid: Esterification

- $CH_3OH + CH_3COOH \xrightarrow{Conc.H_2SO_4/\Delta} \underset{\text{Methylacetate}}{CH_3COOCH_3} + H_2O$
- $C_2H_5OH + CH_3COOH \xrightarrow{Conc.H_2SO_4/\Delta} \underset{\text{Ethyl Acetate}}{CH_3COOC_2H_5} + H_2O$

Dehydration With Conc Sulphuric Acid

- $\underset{\text{Ethylalcohol}}{C_2H_5OH} \xrightarrow[170°C]{Conc.\ H_2SO_4} \underset{\text{Ethene}}{CH_2=CH_2} + H_2O$
- $\underset{\text{Ethanol}}{C_2H_5OH} \xrightarrow[350°C]{Al_2O_3} \underset{\text{Ethene}}{C_2H_4} + H_2O$
- $\underset{\text{Methanol}}{2CH_3OH} \xrightarrow[140°C]{Conc.\ H_2SO_4} \underset{\text{Dimethylether}}{CH_3-O-CH_3} + H_2O$
- $\underset{\text{Ethanol}}{2C_2H_5OH} \xrightarrow[140°C]{Conc.\ H_2SO_4} \underset{\text{Diethylether}}{C_2H_5-O-C_2H_5} + H_2O$

Commercial Forms of Ethanol

- **Denatured Alcohol Or Methylated Spirit:** Methylated spirit or denatured alcohol is ethyl alcohol with 5% methyl alcohol, a coloured dye and some pyridine.
- **Spurious Alcohol:** Illicit liquor made by improper distillation. It contains large proportions of methanol in a mixture of alcohols.

MATHEMATICS

1 CHAPTER

Commercial Mathematics

Goods and Services Tax

- Selling Price = List Price – Discount
- Tax = Rate of tax × SP.
- Intra state (Same state)
  - CGST
  - SGST
- Inter State (Different State)
  - IGST
- Input tax Credit (Business purpose only)
  - Output tax – Input tax
  - Tax on SP – Tax on CP (Tax on profit)

Banking

- Recurring Deposit account (R.D. Account)
  - Interest = $\boxed{P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}}$
  - Maturity Value = $\boxed{P \times n + \text{Interest}}$

Shares & Dividends

- The original value of a share is called NOMINAL VALUE.
- The price of a share at any time is called its MARKET VALUE.
- If the market valve of a share is the same as its NOMINAL VALUE, the share is called at PAR.
- If the market value of a share is more than its NOMINAL VALUE, then the share is called at PREMIUM or ABOVE PAR.
- If the market value of a share is less than its NOMINAL VALUE, then the share is called at DISCOUNT or BELOW PAR.

$$\text{Sum Invested} = \text{Number of shares} \times \text{Market value of share}$$

$$\Rightarrow \text{No. of shares} = \frac{\text{Sum Invested}}{\text{Market Value of Share}}$$

$$\Rightarrow \text{No. of shares} = \frac{\text{Total Dividend}}{\text{Dividend of 1 share}}$$

$$\Rightarrow \frac{\text{Total Income (Profit)}}{\text{Income (Profit) on 1 share}}$$

$$\text{Total Dividend} = \text{No. of share} \times \text{Rate of dividend} \times \text{N.V. of share}$$

$$\text{Rate of dividend} \times \text{N.V.} = \text{Profit (return)}\% \times \text{M.V.}$$

2 CHAPTER

Linear Inequations

Linear Inequations

Signs of inequality are $>$, $<$, $\geq$ and $\leq$.

RULE 1: Positive term transferred to other side become negative.

Ex.: $3x + 2 < 8 \Rightarrow 3x < 8 - 2$

RULE 2: Negative term transferred to other side become positive.

Ex.: $x - 3 > 5 \Rightarrow x > 5 + 3$

RULE 3: If each term is multiplied or divided by same positive number, sign of inequality remains same.

Ex.: $3x > 6 \Rightarrow \frac{3x}{3} > \frac{6}{3} \Rightarrow x > 2$ divided by 3

$\frac{x}{2} < 5 \Rightarrow 2 \times \frac{x}{2} < 5 \times 2 \Rightarrow x < 10$ multiplied by 2

RULE 4: Taking reciprocal on both sides reverses inequality sign.

Ex.: $x > 3 \Rightarrow \frac{1}{x} < \frac{1}{3}$

RULE 5: If each term is multiplied or divided by same negative number, sign of inequality reverses.

Ex.: $-2x < 5 \Rightarrow \frac{-2x}{-2} < \frac{5}{-2} \Rightarrow x > \frac{-5}{2}$ divided by -2

RULE 6: If sign on both side changes, inequality reverses.

Ex.: $x < -2$, $-x > 2$

Replacement & Solution Set

- If $x > 4$

  Replacement set = $x \in N$ (x belongs to natural number)

  Solution set = {1, 2, 3}

- Number line – ○ Hollow represent $>$, $<$

  ● Filled represents $\geq$, $\leq$.

3 CHAPTER

Quadratic Equations

Quadratic Equation

An equation with one variable, in which the highest power of variable is two, is known as quadratic equation.

Standard form: $ax^2 + bx + c$, where a, b & c are all real numbers and $a \neq 0$.

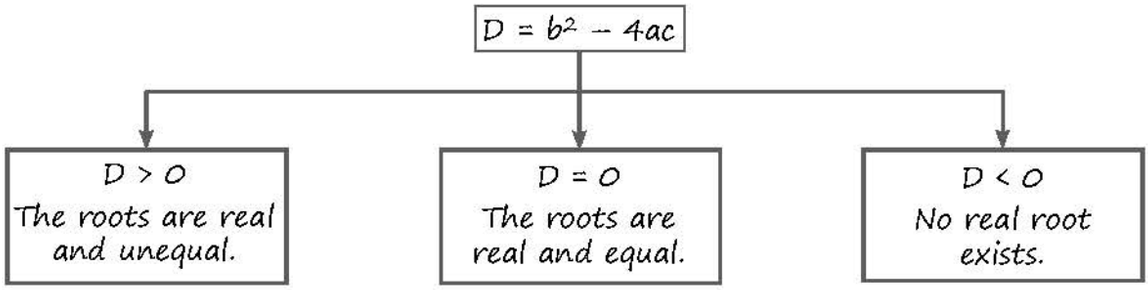
- Every quadratic equation gives two values of the unknown variable used in it & these valves are called ROOTS/SOLUTION of the quadratic equation.
- For the quadratic equation $ax^2 + bx + c = 0, a \neq 0$; the expression $b^2 - 4ac$ is called DISCRIMINANT & is denoted by D.
- The roots of the quadratic equation $ax^2 + bx + c = 0$; where $a \neq 0$ can be obtained by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Also expressed as $x = \frac{-b \pm \sqrt{D}}{2a}$, where $D = b^2 - 4ac$

Nature of Roots

Nature of roots of a quadratic equation depends on DISCRIMINANT.



4 CHAPTER

Ratio and Proportion

Ratio

If a and b are two quantities of the same kind and with the same units such that $b \neq 0$; then the quotient $\frac{a}{b}$ is called the ratio between a and b.

Commensurable and incommensurable quantities

If the ratio between any two quantities of the same kind and having the same unit can be expressed exactly by the ratio between two integers; the quantities are said to be commensurable; otherwise incommensurable.

Composition of ratios:

(i) Compound Ratio

For two or more ratios, the ratio between the product of their antecedents to the product of their consequents is called compound ratio.

e.g. for ratios $a : b$ and $c : d$; the compound ratio is $(a \times c) : (b \times d)$.

(ii) Duplicate ratio

It is the compound ratio of two equal ratios.

e.g. Duplicate ratio of $a : b$ = compound ratio of $a : b$ and $a : b$

$= (a \times a) : (b \times b) = a^2 : b^2$

(iii) Triplicate ratio

It is the compound ratio of three equal ratios.

e.g. Triplicate ratio of $a : b$ = Compound ratio of $a : b$, $a : b$ and $a : b$

$= (a \times a \times a) : (b \times b \times b) = a^3 : b^3$

(iv) Sub-duplicate ratio

For any ratio $a : b$, its sub-duplicate ratio is $\sqrt{a} : \sqrt{b}$

(v) Sub-triplicate ratio

For any ratio $a : b$ its sub-triplicate ratio is $\sqrt[3]{a} : \sqrt[3]{b}$

(vi) Reciprocal ratio

For any ratio $a : b$; where $a, b \neq 0$, its reciprocal ratio $= \frac{1}{a} : \frac{1}{b} = b : a$.

Proportion

Four non-zero quantities a, b, c and d are said to be proportion, if a : b :: c : d and is read as "a is to b as c is to d".

Some Properties of Ratios & Proportion

- Invertendo: If $a : b :: c : d$ then $b : a : d : c = \frac{a}{b}=\frac{c}{d}\Rightarrow\frac{b}{a}=\frac{d}{c}$
- Alternendo: If $a : b :: c : d$ then $a : c :: b : d = \frac{a}{b}=\frac{c}{d}\Rightarrow\frac{a}{c}=\frac{b}{d}$
- Componendo: If $a : b :: c : d$ then $(a + b) : b :: (c + d) : d = \frac{a}{b}=\frac{c}{d}\Rightarrow\frac{a}{b}+1=\frac{c}{d}+1\Rightarrow\frac{a+b}{b}=\frac{c+d}{d}$
- Dividendo: If $a : b :: c : d$ then $(a - b) : b :: (c - d) : d = \frac{a}{b}=\frac{c}{d}\Rightarrow\frac{a}{b}-1=\frac{c}{d}-1\Rightarrow\frac{a-b}{b}=\frac{c-d}{d}$
- Componendo & Dividendo: If $a : b :: c : d$ then $(a + b) : (a - b) :: (c + d) : (c - d)$

  On dividing the results of Componendo by Divendo, we get $= \frac{a+b}{b}\times\frac{b}{a-b}=\frac{c+d}{d}\times\frac{d}{c-d}\Rightarrow\frac{a+b}{a-b}=\frac{c+d}{c-d}$
- Convertendo: If $a : b :: c : d$ then $a : (a - b) :: c : (c - d)$ $\frac{a}{b}=\frac{c}{d}$ ....(i)

  By dividendo, $\frac{a-b}{b}=\frac{c-d}{d}$ ...(ii)

  Diving (i) by (ii), we get, $\frac{a}{b}\times\frac{b}{a-b}=\frac{c}{d}\times\frac{d}{c-d}\Rightarrow\frac{a}{a-b}=\frac{c}{c-d}$
- If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}$, then each ratio $= \frac{a+c+e}{b+d+f}$

5 CHAPTER

Remainder and Factor Theorems

Remainder Theorem

When a polynomial $f(x)$ (whose degree is greater than or equal to 1) is divided by a linear polynomial $(x - a)$, the remainder is given by $R = f(a)$.

Factor Theorem

When a polynomial $f(x)$ is divided by $x - a$, the remainder = $f(a)$. And if $f(a) = 0$;

$\Rightarrow (x - a)$ is a factor of the polynomial $f(x)$.

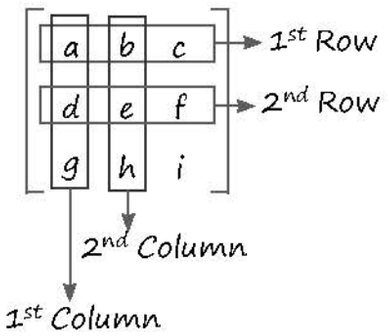
6 CHAPTER

Matrices

Matrices

A rectangular arrangement of numbers in the form of horizontal & vertical lines is called a matrix.

- Horizontal lines are called ROWS & Vertical lines are COLUMNS.
- If a matrix contains m rows & n column then it is called a matrix of order mxn.
- A matrix of order mxn has mn elements.
- An element appearing in the i^{th} row & j^{th} column of a matrix is called its $(i, j)^{th}$ element or the $(i, j)^{th}$ entry.



Equal Matrices

Two matrices A & B are called equl, written as A = B, if and only if

(i) A and B are of same order i.e. No. of rows in A = No. of rows in B & No. of columns in A = No. of columns in B.

(ii) their corresponding elements are equal i.e. the entries of A & B in the same position are equal.

Special Types of Matrices

Row Matrix:

A matrix having only one row is called row matrix.

Column Matrix:

A matrix having only one column is called column matrix.

Square Matrix:

A matrix in which number of rows equals the number of columns is called Square Matrix.

Zero or Null Matrix:

A matrix whose each element is zero is called a zero or null matrix.

Identity or Unit Matrix:

A square matrix in which each diagonal element is 1 and all other elements are zero is called an identity or unit matrix.

Addition of Matrices

If A & B are two matrices of the same order, we say that these are compatible for addition, and their sum A + B is the matrix obtained by adding the corresponding elements of A & B.

Subtraction of Matrices

If A & B are two matrices of the same order, we say that these are compatible for subtraction, and their difference A – B is the matrix obtained by subtracting the corresponding elements of B from A.

Transpose of a Matrix

Transpose of a matrix is the matrix obtained on interchanging its rows & columns. If A is matrix then its transpose is denoted by A^t.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 7 \end{bmatrix}., \text{ then } A^t = \begin{bmatrix} 2 & 0 \\ 3 & 4 \\ 1 & 7 \end{bmatrix}$$

Additive Identity

If any number is added to zero or zero is added to the number, the number remains the same. 0 is said to be additive identity in numbers.

If any matrix is added to null matrix of same order then the matrix remains the same. So, here null matrix is said to be additive identity in matrices.

Additive Inverse

If A & B are two matrices of the same order such that:

A + B = B + A = a null matrix; then A is said to be additive inverse of B & B is said to be additive inverse of A.

Multiplication of Matrices

Two matrices A & B can be multiplied together to get the product matrix AB. If & only if, the numbers of columns in A (the left hand matrix) is equal to the number of rows in B (the right hand matrix).

$$A_{m\times n} \times B_{n\times p} = C_{m\times p} = AB$$

Ex: $A = \begin{bmatrix} 3 & 4 \\ 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $\quad AB = \begin{bmatrix} (3\times1)+(4\times3) & (3\times2)+(4\times4) \\ (5\times1)+(0\times3) & (5\times2)+(0\times4) \end{bmatrix}$

Identity matrix for multiplication:

On multiplying any matrix with the identity matrix of the same order, the matrix remains unaltered.

$$A \times I = A = I \times A$$

Points to remember:

1. For any three matrices A,B, and C which are compatible for multiplication; ABC = A(BC). Similarly, BAC = B(AC); CAB = C(AB) and so on.
2. In general $AB \neq BA$ i.e. product of matrices is not commutative.
3. (AB) C = A(BC) i.e. product of matrices is associative.
4. If $A \neq 0$ and AB = AC. then it is not necessary that B = C.
5. If AB = 0, then it is not necessary that A = 0 or B = 0.
6. If A = 0 or B = 0, then AB = 0 = AB.
7. (i) A(B + C) = AB + AC i.e. in matrices, multiplication is distributive over addition.
   (ii) (A+B) C = AC + BC.
   In the same way, A (B – C) = AB – AC and (A – B) C = AC – BC.

7 CHAPTER Arithmetic and Geometric Progression

Arithmetic & Geometric Progressions

A list of number is called an ARITHMETIC PROGRESSION if and only if the difference of any term from its preceding term is constant.

- An A.P. is called a finite A.P. if and only if it contains finite number of terms.
- An A.P. is called an infite A.P. if and only if it contains infinite number of terms.
- Let a_1, a_2, a_3.....l is an A.P. then
  $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = d$, where d = common difference and ℓ is last term.
- $\boxed{n^{th} \text{ term of an A.P.} = a + (n-1)d}$, where a = first term & d = common difference
- $\boxed{n^{th} \text{ term of an A.P. from last} = \ell + (n-1)(-d)}$

Sum of first n terms of an A.P. = $\boxed{S_n = \frac{n}{2}[2a+(n-1)d]}$

or $\boxed{S_n = \frac{n}{2}[a+l]}$

- Three terms in A.P. = $(a-d), a, (a+d)$
- Four terms in A.P. = $(a-3d), (a-d), (a+d), (a+3d)$; where common difference = $2d$.
- Five terms in A.P. = $(a-2d), (a-d), a, (a+d), (a+2d)$
- Let say, $S_n = n^2 - n$

  $\Rightarrow S_1 = a_1$

  $\Rightarrow S_2 - S_1 = a_2$

  $\Rightarrow a_2 - a_1 = d$

Geometric Progression

A list of non-zero numbers is called a geometric progression (G.P.) if and only if the ratio of its any term to its preceding term is constant. i.e., fixed number a_1, a_2, a_3,.....is a G.P. if and only if $\boxed{\frac{a_{n+1}}{a_n} = r}$, where r is constant & independent of n.

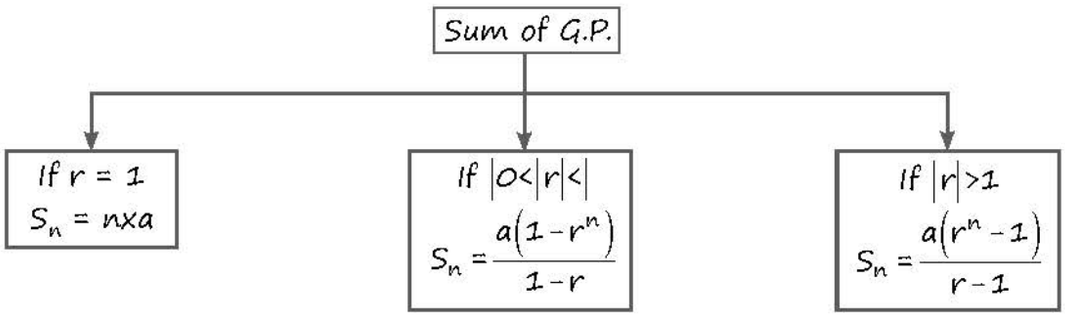
General term of a Geometric Progression

Let a be the first term and L be last term and r ($r \neq 0$) be the common ratio of G.P. Then n^{th} form of GP is $\boxed{a_n = ar^{n-1}}$ and k^{th} term from end of GP (finds GP) $T_k = ar^{n-k}$ where T_k is k^{th} term from the end, and n is total number terms.

Geometric Mean

If a & b be two positive real numbers, then $\sqrt{ab}$ is called Geometric mean between a & b.

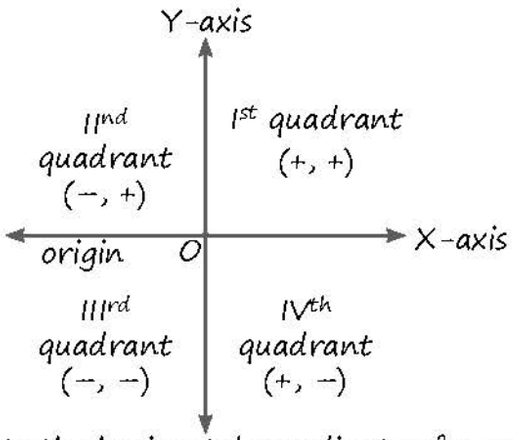
Sum of a geometric progression



8 CHAPTER

Co-ordinate Geometry

Coordinate Geometry



Abscissa: The abscissa represents the horizontal coordinate of a point, typically denoted as X in a Cartesian coordinate system.

Ordinate: The ordinate represents the vertical coordinate of point, usually denoted as y in a Cartesian coordinate system.

Coordinates of Origin $\rightarrow (0, 0)$
Coordinates of a point on X-axis $\rightarrow (X, 0)$
Coordinates of a point on Y-axis $\rightarrow (0, Y)$

- General equation of line, $y = mx + c$

  Equation of line passing through origin $y = mx$

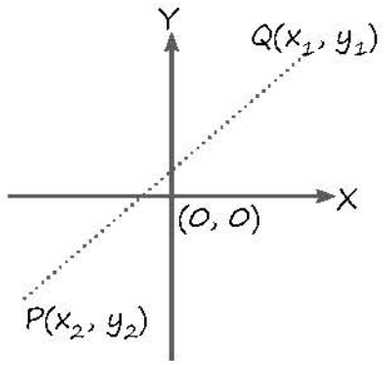
  Equation of x-axis $\rightarrow Y = 0$

  Equation of y-axis $\rightarrow X = 0$

Distance Formula

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance of a point (x, y) from origin = $\sqrt{x^2 + y^2}$



Section Formula

$$\text{Internal} = \left[\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right]$$

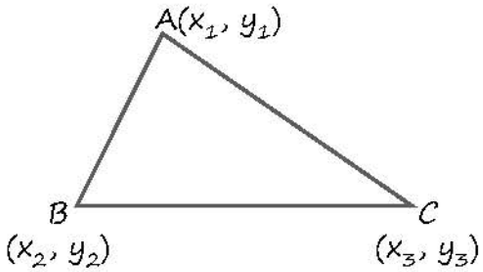
$$\text{External} = \left[\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}\right]$$

$$\text{Mid-point formula} = \left[\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right]$$

Area of Triangle

$$\text{Area of } \Delta = \frac{1}{2}\left|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right|$$

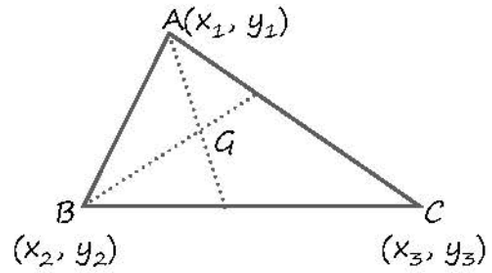
If points (x_1y_1), (x_2, y_2), (x_3, y_3) are collinear, then Area of $\Delta = 0$



Centroid of Triangle

$$G = \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}$$

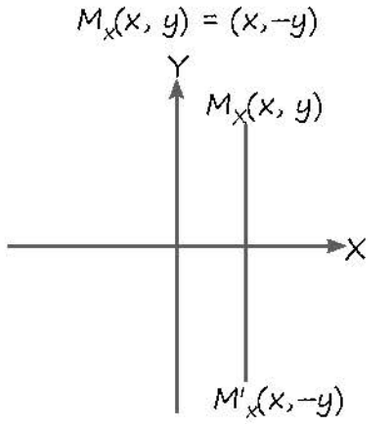
- Centroid divides the median into 2: 1.



Reflection

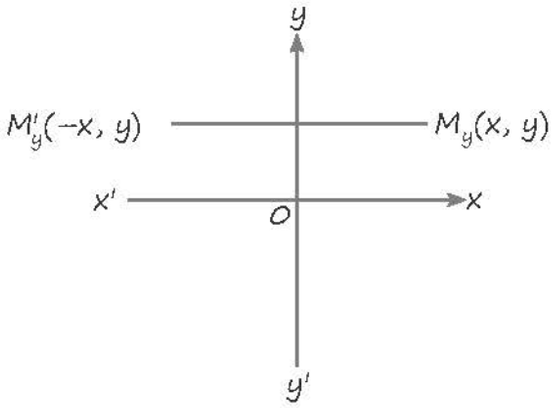
When an object is placed before a plane mirror, the image formed is at the same distance behind the mirror as the object is in front of it.

- Reflection in the line Y = 0, i.e., x-axis



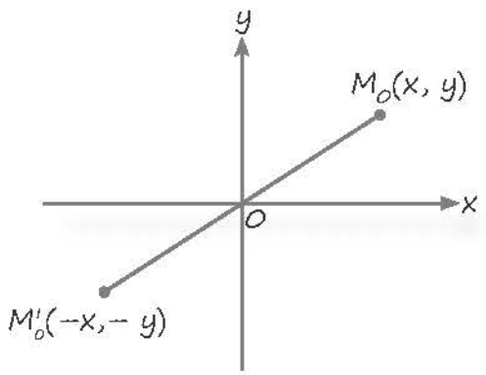
- Reflection w.r.t. line X = 0, i.e., Y-axis

$M_y(x, y) = (-x, y)$



- Reflection w.r.t. Origin

$M_o(x, y) = (-x, -y)$



9 CHAPTER

Similarity

Similarity

Figures which have exactly the same shape and the same size are said to be congruent.

- Figures which have exactly the same shape, but not necessarily the same size are said to be similar.
- All line segments are similar.
- All squares are similar.
- All equilateral triangles are similar.
- All circles are similar.

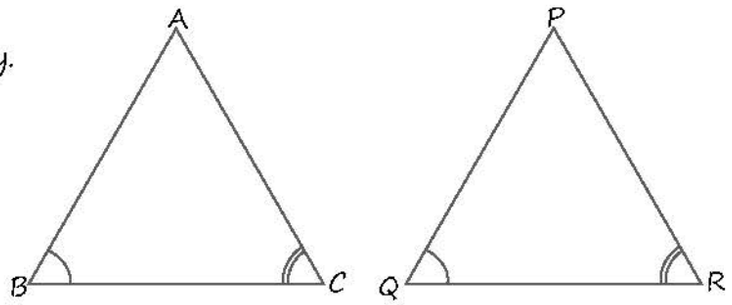
Rules For Similarity of Triangles

- AA (Angles-Angle) rule for similarity.

$\angle B = \angle Q$

$\angle C = \angle R$

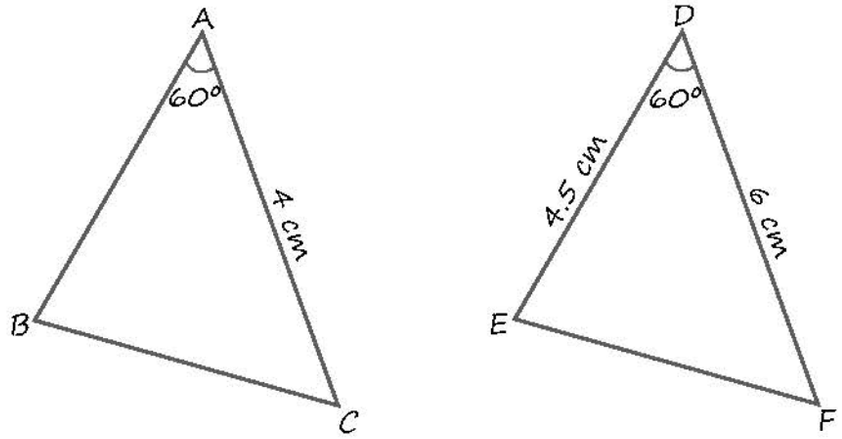
$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$$



$\Delta ABC \sim \Delta PQR$

If two angles of a triangle are equal to two angles of another triangle, then the two triangles are similar.

- SAS (Side-Angle-Side) rule of similarity $\frac{AB}{DE} = \frac{AC}{DF} = \frac{2}{3}$ & $\angle A = \angle D$

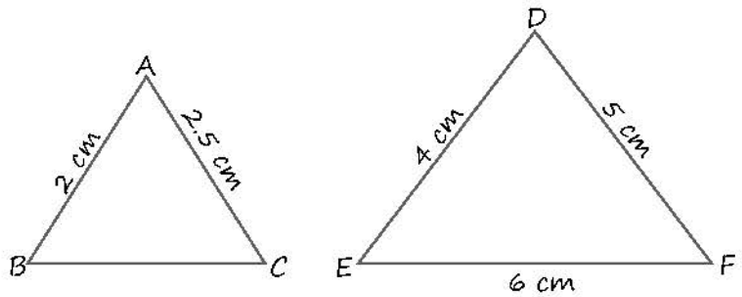


$\Delta ABC \sim \Delta DEF$

If one angle of triangle is equal to one angle of another angle & the sides including the angle are proportional, then two triangles are similar.

- SSS (side-side-side) rule of similarity.

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$



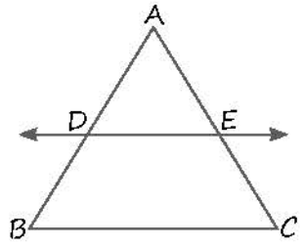
- If tow triangle have their three pairs of corresponding sides proportional, then the triangle are similar.

Basic Proportional Theorem (Thales Theorem)

If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divides in the same ratio.

If $DE \parallel BC$

$$\frac{AD}{DB} = \frac{AE}{EC}$$

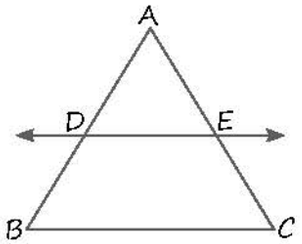


Converse of Basic Proportionality Theorem

If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.

If $\frac{AD}{DB} = \frac{AE}{EC}$

Therefore, $DE \parallel BC$



Relation between the areas of two similar triangles:

The areas of two similar triangle are proportional to the squares on their corresponding sides.

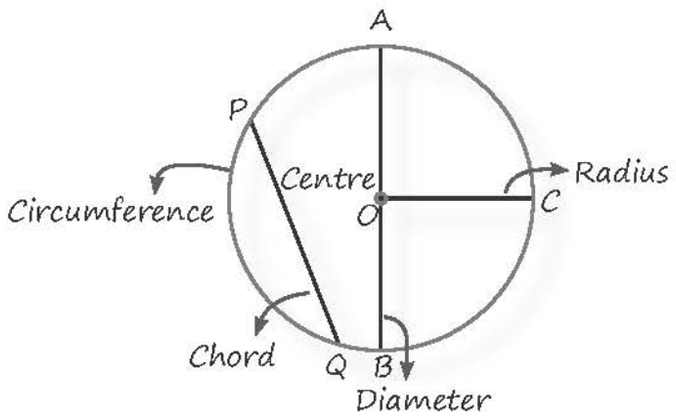
If $\Delta ABC \sim \Delta DEF$, then

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta DEF} = \frac{AB^2}{DE^2} = \frac{BC^2}{EF^2} = \frac{AC^2}{DF^2}$$

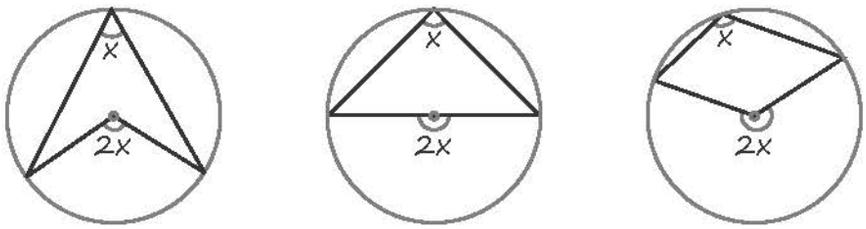
10 CHAPTER

Circles

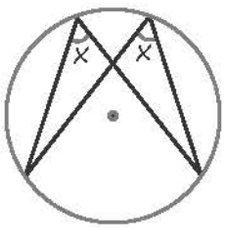
Circles



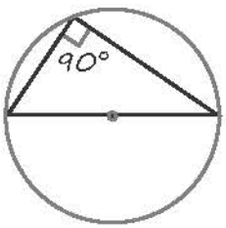
- Angle at centre and segment



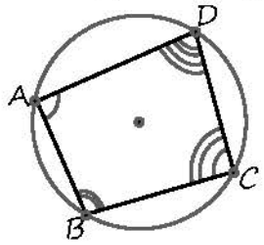
- Angle in same segment → Equal



- Angle in semicircle → 90°

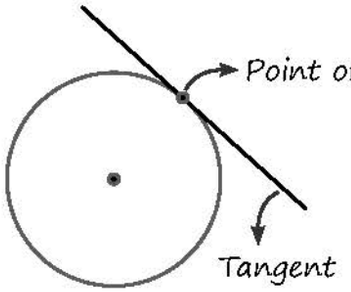


- Cyclic Quadrilateral → Sum of opposite angles of a cyclic quadrilateral is 80°

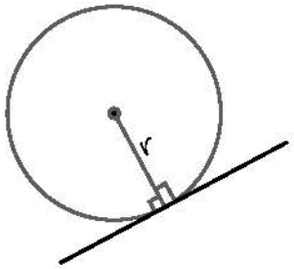


$\angle A + \angle C = 180°$
$\angle B + \angle D = 180°$

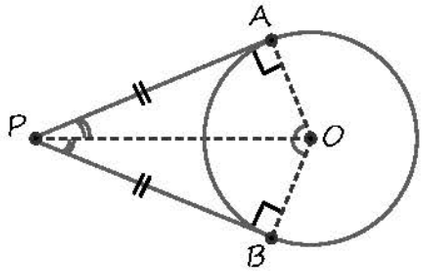
- Tangent → Line that touch circle at only one point.



- Radius is Perpendicular to POC of tangent.

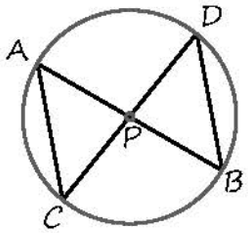


- Tangent from external point.



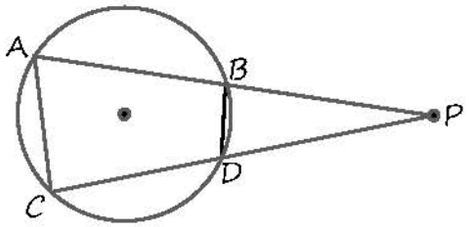
$\Rightarrow PA = PB$
$\Rightarrow \angle APO = \angle BPO$
$\Rightarrow \angle POA = \angle POB$

- Intersection of Chords
  - Internally



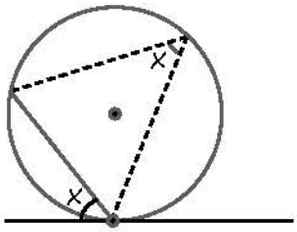
$\boxed{PA \times PB = PC \times PD}$

- Externally

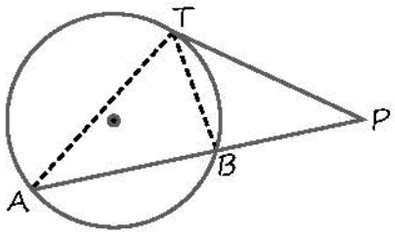


$PA \times PB = PC \times PD$

- Alternate segment theorem



- Tangent – Secant Theorem



$PA \times PB = PT^2$

11 CHAPTER

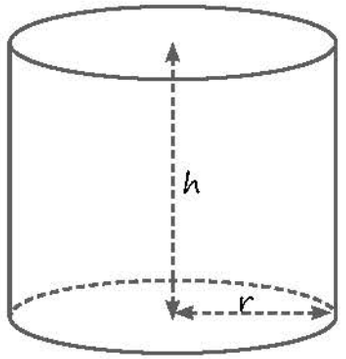
Mensuration (Surface Area and Volume)

Surface Area and Volume

- The sum of the areas of all the surfaces of a solid is called its SURFACE AREA or its TOTAL SURFACE AREA.
- The surface area of a solid excluding the areas of its top & bottom is called LATERAL or curved SURFACE AREA.
- The space occupied by a solid is called its VOLUME.
- If a solid is cut perpendicular to its length (breadth or height), the plane surface so obtained is called its CROSS-SECTION.

Cylinder

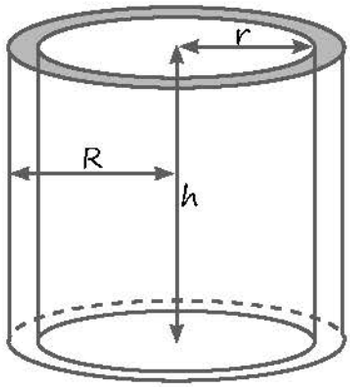
A solid which has uniform cross-section is called a cylinder. Let r be the radius & h be the height.



- Area of cross-section = πr^2
- Perimeter of cross-section = $2\pi r$
- Curved (Lateral) Surface Area = $2\pi r.h$
- Total surface area = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$
- Volume = $\pi r^2 h$

Hollow Cylinder

Let R be the external radius & r be the internal radius & h be its height.

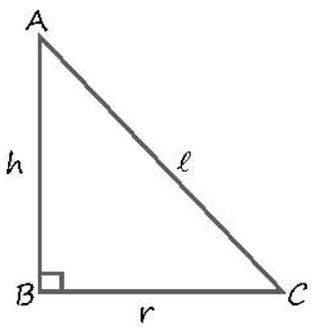


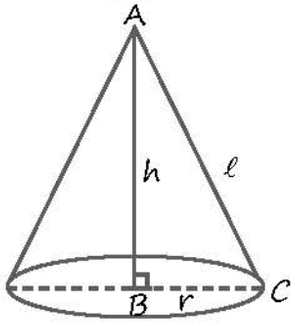
- Thickness of the wall = R – r
- Area of cross-section = $\pi(R^2 - r^2)$
- External curved surface area = $2\pi Rh$
- Internal curved surface area = $2\pi rh$
- Total surface area = $2\pi Rh + 2\pi rh + 2\pi(R^2 - r^2)$
- Volume of material = $\pi(R^2 - r^2).h$

Cone

The solid obtained on revolving a right angled triangle about one of its sides (other than hypotenuse). is called a right circular cone.

In ΔABC, $l^2 = h^2 + r^2$

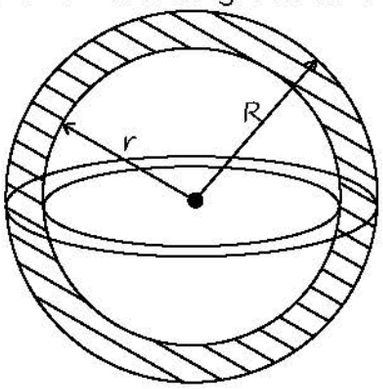




- Volume = $\frac{1}{3}\pi r^2 h$
- Curved or lateral surface area = πrl
- Total surface area = $\pi rl + \pi r^2 = \pi r(l + r)$

Sphere

A sphere is a solid obtained on revolving a circle about any diameter of it.

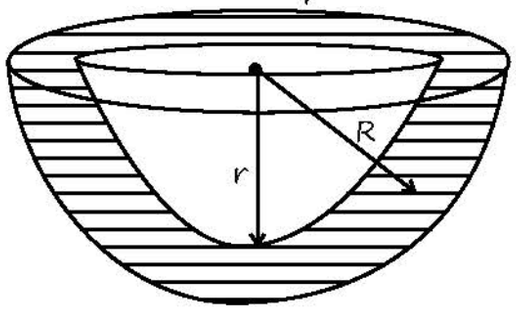


- Volume of sphere = $\frac{4}{3}\pi R^3$ = Volume of material in sphere
- Surface Area of Sphere $=4\pi r^2$

Spherical Shell

It is a solid enclosed between two spheres.
Thickness of shell = R – r

- Volume = $\frac{4}{3}\pi\left(R^3 - r^3\right)$ = Volume of material in spherical shell.



Hemisphere

- Volume of hemisphere = $\frac{2}{3}\pi R^3$
  Curved surface area = $2\pi r^2$
- Total surface area = $3\pi R^2$

Hemispherical shell

- Thickness of shell = R – r
- Area of bass = $\pi(R^2 - r^2)$
- External curved surface area = $2\pi R^2$
- Internal curved surface area = $2\pi r^2$
- Total surface area = $2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$
- Volume of material = $\frac{2}{3}\pi\left(R^3 - r^3\right)$

Note

- The total volume of the solids to be concerted is equal to the total the volume of the solid into which the given solid is to be concerted. although, the surface area may change.

12 CHAPTER

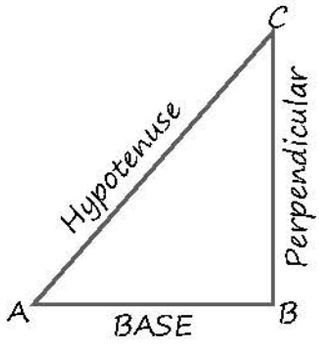
Trigonometry

Trigonometry

$\sin A = \frac{BC}{AC}$ $\quad$ $\operatorname{cosec} A = \frac{AC}{BC}$

$\cos A = \frac{AB}{AC}$ $\quad$ $\sec A = \frac{AC}{AB}$

$\tan A = \frac{BC}{AB}$ $\quad$ $\cot A = \frac{AB}{BC}$



$\sin A = \frac{1}{\operatorname{cosec} A} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$

$\cos A = \frac{1}{\sec A} = \frac{\text{Base}}{\text{Hypotenuse}}$

$\tan A = \frac{1}{\cot A} = \frac{\sin A}{\cos A} = \frac{\text{Perpendicular}}{\text{Base}}$

Trigonometric Identities

$\sin^2 A + \cos^2 A = 1, \quad 0° \le A \le 90°$

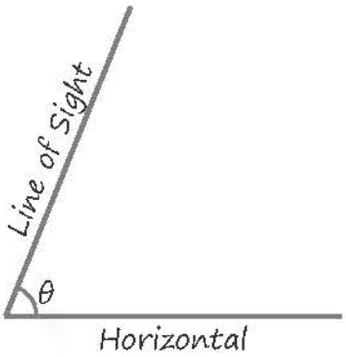
$\sec^2 A - \tan^2 A = 1, \quad 0° \le A \le 90°$

$\operatorname{cosec}^2 A - \cot^2 A = 1, \quad 0° \le A \le 90°$

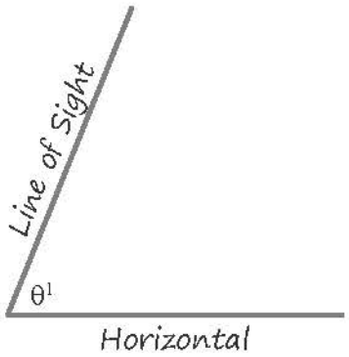
Trigonometric Ratio Table

	0°	45°	60°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not Defined
$\operatorname{cosec}\theta$	Not Defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not Defined
$\cot\theta$	Not Defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

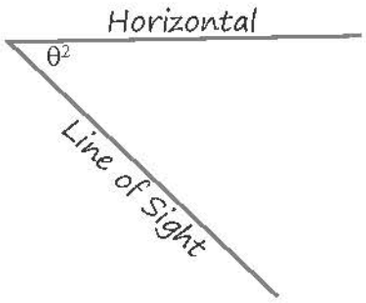
Heights & Distance



Angle of Elevation



Angle of Depression



13 CHAPTER

Statistics

Statistics

- Mean = $\frac{\text{Sum of observation}}{\text{No. of observation}}$
- Class mark $x_i = \frac{\text{Upper limit} + \text{lower limit}}{2}$

  - Direct Method

Classes	f_i	x_i	$f_i x_i$

$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$

  - Short-Cut Method

Classes	f_i	x_i	$4_i = \frac{x_i - A}{h}$	$f_i d_i$
		A		

$\bar{x} = \frac{\Sigma f_i U_i}{\Sigma fi} + h$ where A = Assumed Mean.

  - Step Deviation Method

Classes	f_i	x_i	$4_i = \frac{x_i - A}{h}$	$u_1 + di/n$	$f_i U_i$
		A			

$\bar{x} = A + \frac{\Sigma f_i U_i}{\Sigma fi} \times h$ where h = class size

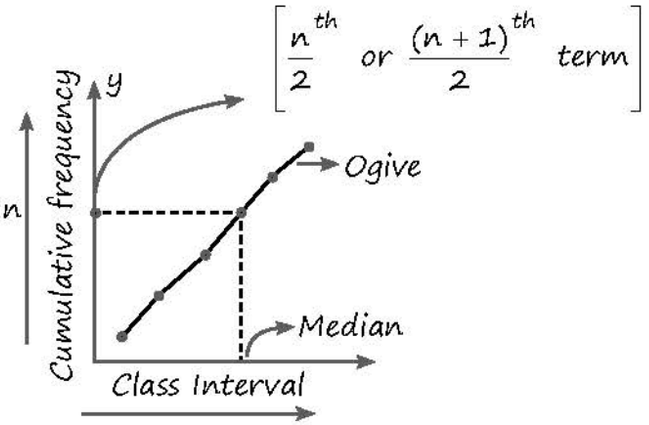
- Median - Middle Value of data arranged in ascending or descending order. If these are 'n' observations, then

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ observation, if } n \text{ is odd} \\ \dfrac{\left(\frac{n}{2}\right)^{th} \text{ observation} + \left(\frac{n}{2}+1\right)^{th} \text{ observation}}{2}, \text{ if } n \text{ is even} \end{cases}$$

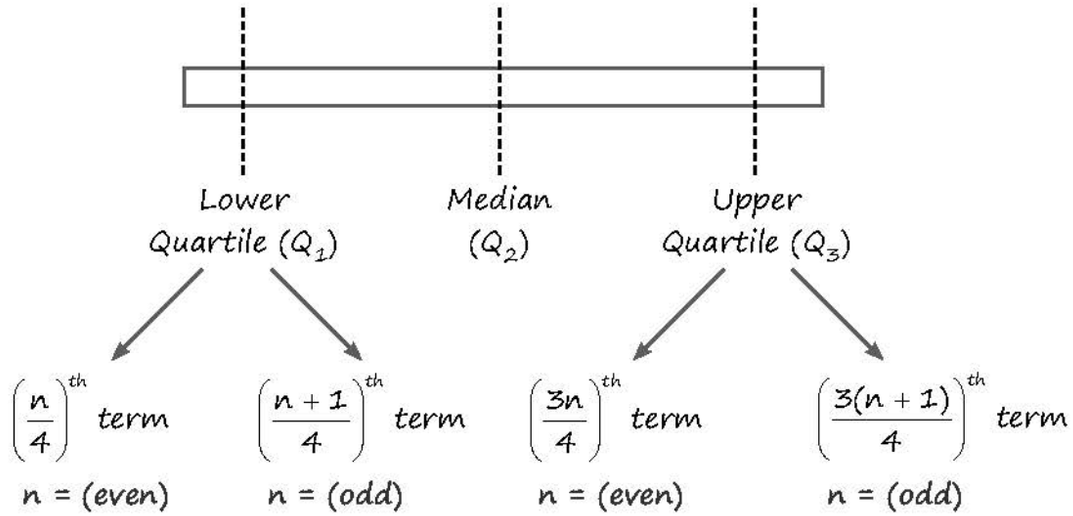
- **Tabulated Data** – Make a column for cumulative frequency
  n = total no. of observations
  Depending on n is even or odd find the median

- **Grouped Data**–

1. Make CF column
2. Draw a cumulative frequency curve (ogive)
3. It there be n terms in the ogive distribution then by the help of given find $\left(\frac{n}{2}\right)^{th}$ or $\left(\frac{n+1}{2}\right)^{th}$ term which will be the median of the given distribution.

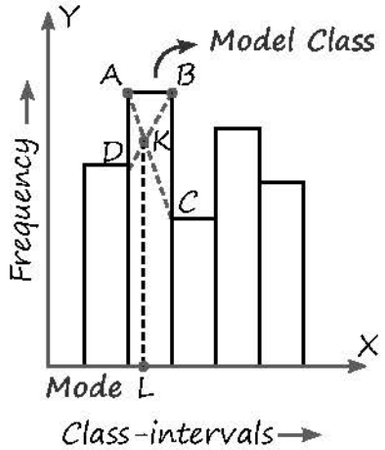


Quartiles



- Inter Quartile range = $Q_3 - Q_1$, which is always positive as $Q_3 > Q_1$.

- **Mode** – The value with maximum frequently in a given data set.
  - **Grouped Data** – Histogram



1. Draw a histogram of the given distribution.
2. Inside the highest rectangle, which is of maximum frequency (Modal class), Draw two lines AC and BD diagonals from the upper corners C and D of adjacent rectangles.
3. Let K be the point of intersection of diagonals AC and BD.
4. Draw KL perpendicular to horizontal axis.
5. The value of point L on the horizontal axis represents the value of mode.

BIOLOGY

1 CHAPTER

Cell-The Structural and Functional Unit of Life

The cell is the structural and functional unit of life. All living beings develop from pre-existing cells.
Robert Hooke (1665) discovered the cell. He observed cork cells of a tree bark.
Cell theory: All organisms start their life with a single cell. Each cell is capable of carrying out various metabolic process and independent existence.

Cell Number

Unicellular Organisms	Made of single cell. **Example:** Amoeba
Multicellular Organisms	Consists of many cells adapted to perform different functions. **Example:** Plants, Human Being

Cell Size

The size of cell ranges from $1/10^{th}$ to $1/100^{th}$ mm.

Largest cell in the world	Egg of an Ostrich.
Smallest cell	Mycoplasma gallisepticum
Largest human cell	Female Ovum
Smallest human cell	Red blood cell
Longest cell	Nerve cell

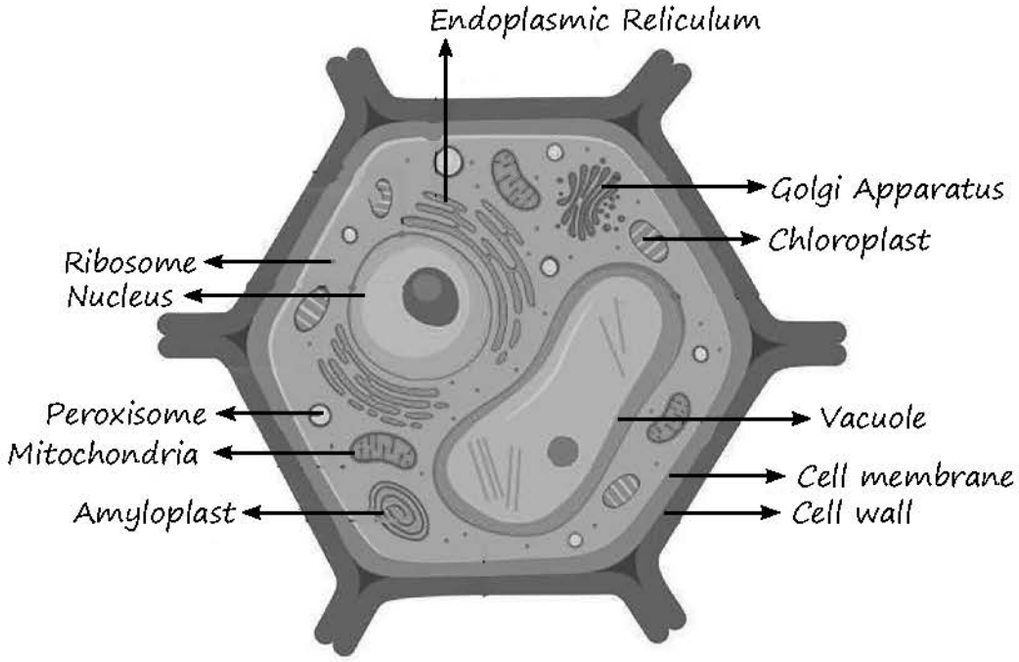
Cell Shape

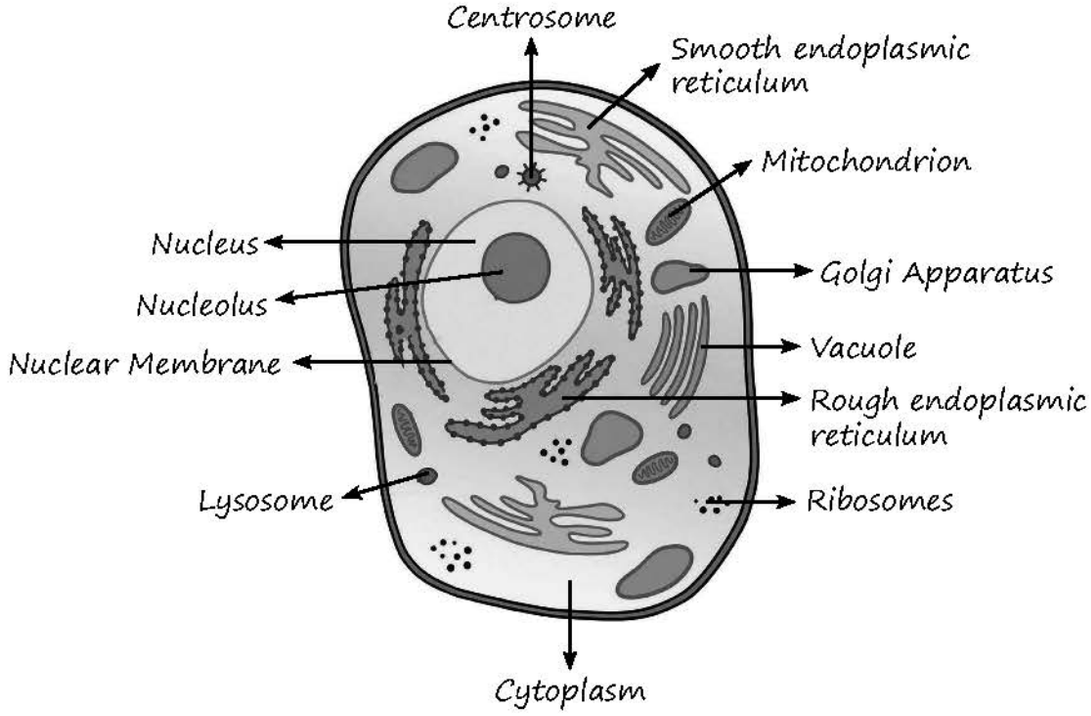
The shape of cells are often related to their functions.

- Human red blood cells are circular and biconcave for easy passage through blood capillaries and to transport oxygen.
- White blood cells are amoeboid bearing pseudopodia so that they can squeeze through blood capillaries and destroy pathogens.
- Nerve cells are slender and long to carry impulses.
- Guard cells are bean-shaped so that they can control the opening and closing of the stomata.

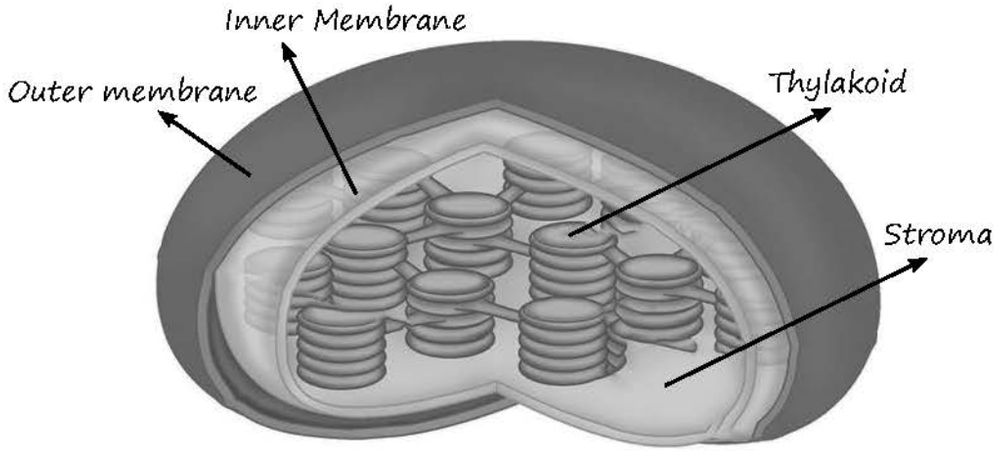
Microscope

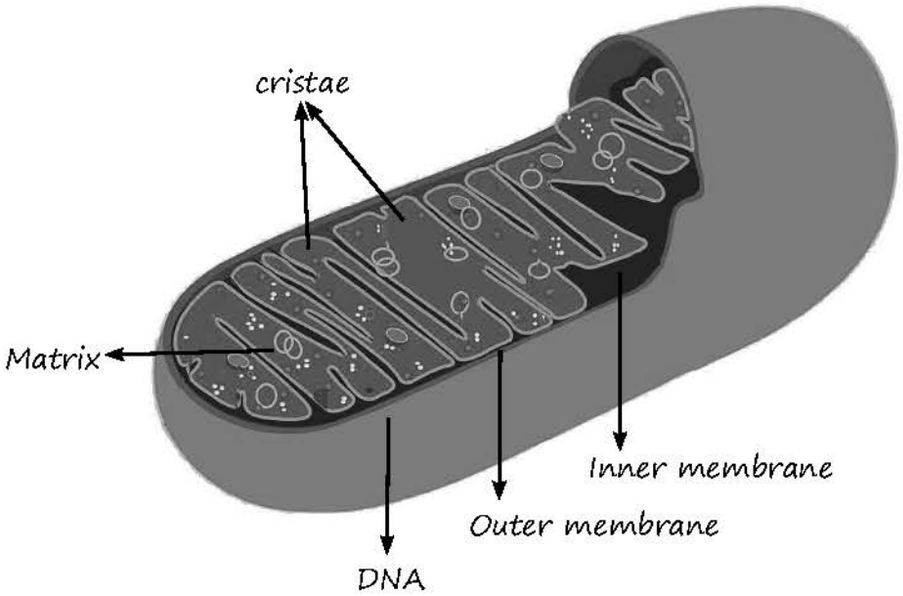
A microscope is an instrument used to magnify objects to be studied.





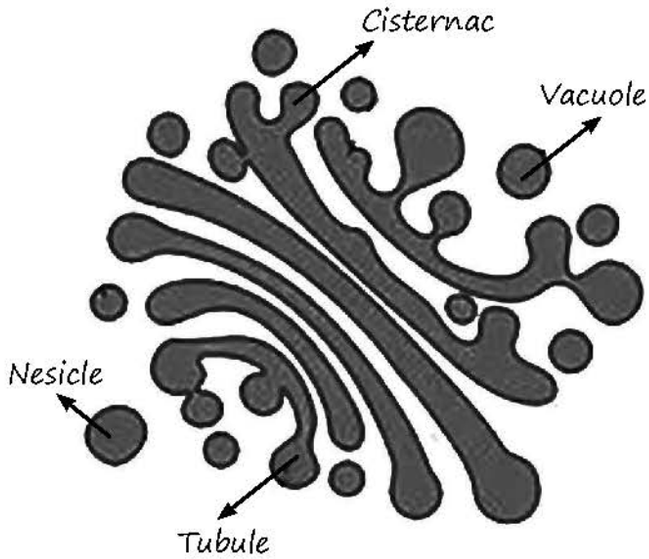
Part of cell	Characteristics	Functions
❑ Cell membrane	❑ Living membrane. ❑ Outermost in animals. ❑ Semi-permeable.	❑ Separates cell content from surrounding. ❑ Maintains cell shape in animal cells.
❑ Cell Wall	❑ Present only in plant cell ❑ Composed of cellulose.	❑ Gives the plant cell rigidity and shape. ❑ Provides Protection.

❑ Cytoplasm	❑ Semi-liquid substance. ❑ Cell Organelles lies in the cytoplasm.	❑ Site of enzyme dependent metabolic reactions.
❑ Endoplasmic reticulum	❑ Two types: Smooth endoplasmic reticulum and endoplasmic reticulum.	❑ Supportive framework of the cell.
❑ Ribosomes	❑ Small granular structures. ❑ Scattered in cytoplasm, nucleoplasm or on the endoplasmic reticulum.	❑ Site for protein synthesis.
❑ Lysosomes	❑ Membranous sacs	❑ Destroy foreign substances. ❑ Destroy old or damaged cells, so also known as suicidal sacs.
❑ Centrosome	❑ Found only in animal cells. ❑ A region surrounding the centrioles.	❑ Initiates and regulates cell division.
❑ Plastids	❑ Two kinds: Leucoplasts (colourless plastids) and chromoplasts (coloured plastids). ❑ Chloroplasts is a green-coloured pigment.	❑ Leucoplasts store starch. ❑ Chloroplasts are sites for photosynthesis.
❑	 Structure of Chloroplast	
❑ Mitochondria	❑ Lined by a double membrane. ❑ Have their own DNA known as mitochondrial DNA.	❑ Synthesis of respiratory enzymes. ❑ Site for aerobic respiration and energy storage, so also known as the powerhouse of the cell.



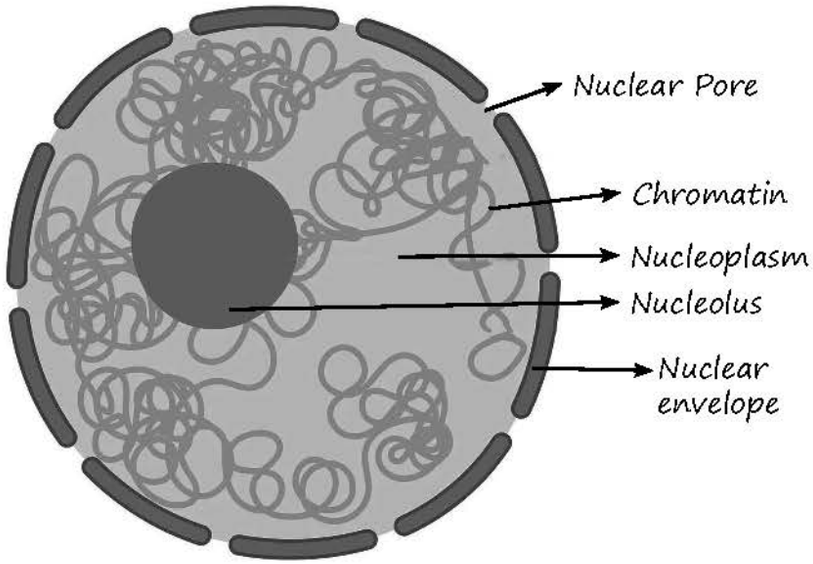
Structure of Mitochondria

❑ Golgi Apparatus	❑ Stacks of flattened membrane sacs.	❑ Synthesis and secretion of enzymes and hormone.



Structure of Golgi Apparatus

❑ Nucleus	❑ Largest cell organelle. ❑ Contains chromatin network.	❑ Regulates cell functions, so also known as control centre of the cell.



Structure of Nucleus

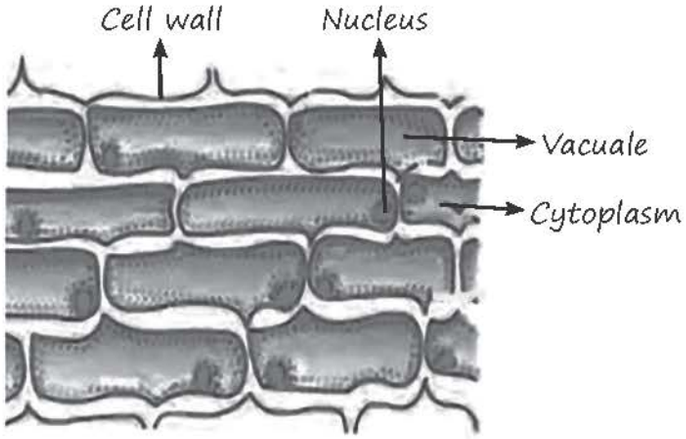
❑ Nucleolus	❑ Present inside the nucleus.	❑ Synthesis of ribosomes.
❑ Chromatin fibres	❑ Contains genetic material i.e. DNA.	❑ Carry hereditary formation.
❑ Vacuoles	❑ Sac-like structures which contain cell sap. ❑ Large and permanent in plant cell. ❑ Small and temporary in animal cells.	❑ Storage of water and substances dissolved in water.
❑ Granules	❑ Small particles, crystal or droplets such as fat-containing granules, starch containing in plant cells and glycogen containing granules in animal cells.	❑ Serve as food for cells.

Differences between Plant Cell and Animal Cell

	Plant Cell		Animal Cell
1.	Cell wall is present and is made of cellulose.	1.	Cell wall is absent.
2.	Centrosome is absent.	2.	Centrosome is present.
3.	Plastids are present.	3.	Plastids are absent.

Microscopic Examination of Onion Peel

1. The cell are firmly bound together.
2. The nucleus is placed towards one side which is usually the case in plant cells.



Nucleus is Essential for Normal Life

1. The nucleues is essential for life.
2. Amoeba divides by normal cell division.
3. If the nucleus is removed, then amoeba does not survive.
4. If the nucleus from another amoeba is transplanted in enucleated amoeba, the recipient amoeba survives and can divide, while the donor amoeba (enucleated) dies.

Differences between Eukaryotic Cells and Prokaryotic Cells

Eukaryotic Cells	Prokaryotic Cell
Membrane bound cell organelles are present.	Membrane bound cell organelles are absent.
The contents of nucleus are separated from the cytoplasm.	Nuclear material lies in the cytoplasm.
Example: Bacteria	Examples: Animal Cell, Plant Cell

2 CHAPTER

Structure of Chromosomes (Cell Cycle and Cell Division)

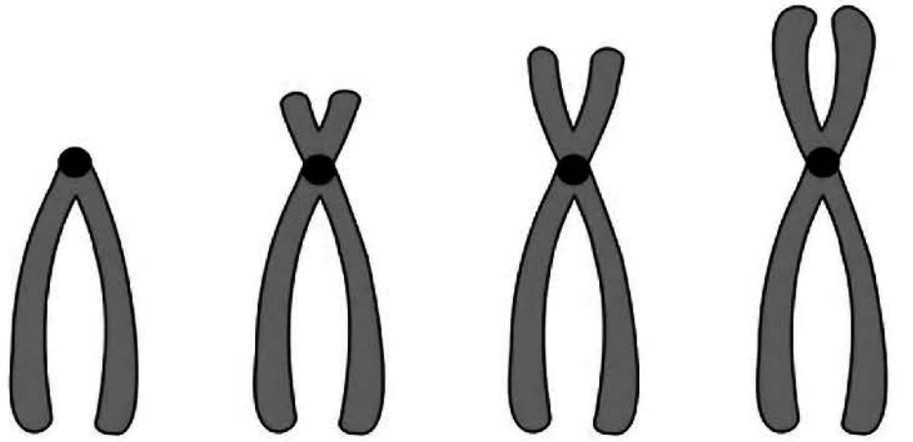
Structure of Chromosomes

Chromosomes plays a vital role in hereditary, variation and evolutionary development of other species.

They were discovered by the German scientist **Walther Fleming** in 1881.

Each chromosome has a centromere, which holds together the two sister chromatids.

Based on the position of the centromere, there are four morphological types of chromosomes.



Telocentric Acrocentric Submetacentric Metacentric

Molecular Structure of DNA

- Based on the X-Ray diffraction data produced by **Maurice Wilkins** and **Rosalind Franklin, James Watson** and **Francis Crick** proposed a very simple but famous double helix model of DNA in 1953 which was seconded by the findings of **Erwin Chargaff** for a double stranded DNA. The findings were:
  1. The purines and pyrimidines are always equal in amount i.e. A + G = T + C or A + G/T + C = 1.
  2. The amount of adenine will always be equal to thymine and the amount of guanine will always be equal to cytosine i.e. A + T = G + C.
  3. Adenine is joined to thymine with 2H -bonds while guanine is joined to cytosine by 3H-bonds.

Packaging of the DNA Helix

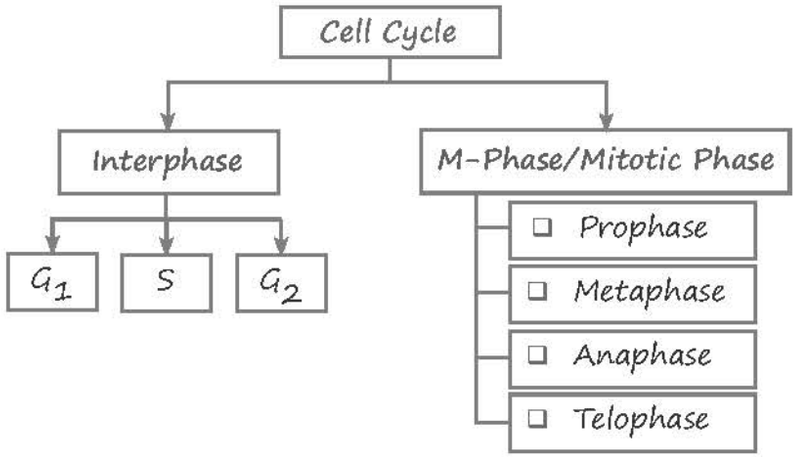
- In eukarytoes, the DNA is packed inside a membranous structure called the nucleus. Inside the nucleus, the DNA is wrapped around a unit of 8 molecules of positively charged protein

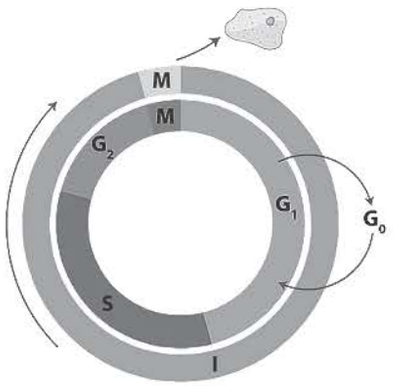
histone in the farm of histone octamer to form a strucuture called **Nucleosome** that contains around 200 bp of DNA helix. The nucleosome in turn form repeating units like "beads-on string' structures in the nucleus called chromatin.

- Chromatin is packaged to form chromatin fibres which are further coiled and condensed at the metaphase stage of cell division to form chromosomes.

Cell division and cell cycle

1. Cell division is a process by which a mature cell divides and forms daughter cells which resemble the parent cell in several characteristics.
2. Cell cycle – The sequence of events by which a cell duplicates its genome, synthesis the other constituents of the cell and eventually divides into two daughter cells is termed cell cycle.
3. Phases of cell cycle: The cell cycle consists of a long non-dividing interphase and a short dividing mitotic phase ar M-phase.





Cell Cycle

Interphase

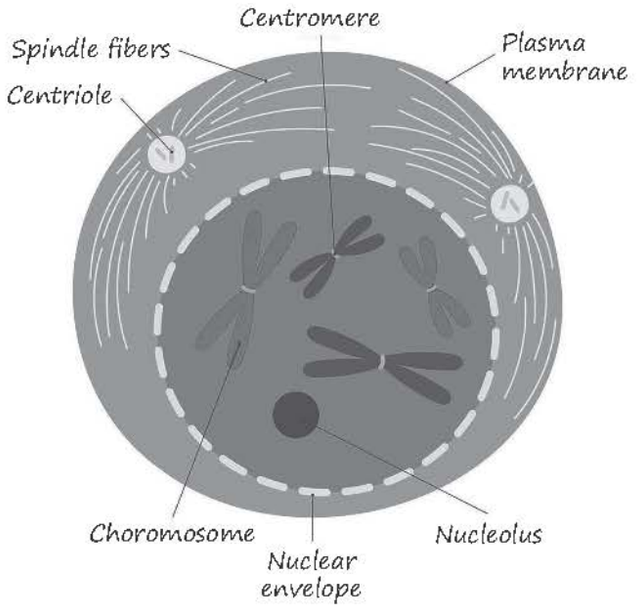
1. G1 Phase (Gap 1)
   - Interval between mitosis and initiation of DNA replication.
   - Metabolically active phase where the cell grows but does not replicate its DNA.
2. S Phase (Synthesis)
   - Period of DNA synthesis or replication.
   - DNA amount doubles progressing from 2C to 4C, but chromosomes number remains the same (2N).
3. G2 Phase (Gap 2)
   - Phase during which proteins are synthesised in preparation for mitosis.
   - Cell growth continues during G2 phase.

M-Phase (Mitosis Phase)

- Represents the phase of actual cell division.
- Begins with nuclear divisions (Karyokinesis), involving the separation of daughter chromosomes.
- Typically concludes with the division of the cytoplasm (cytokinesis).

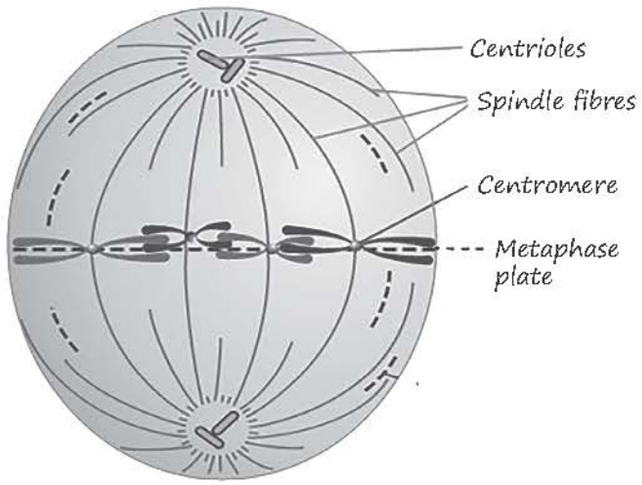
Phase of Mitosis/Karyorinesis

1. Prophase
   - The nuclear membrane begins to disintegrate.
   - The strands of chromosomes begin to condense into discrete observable structures.
   - The nucleolus disappears.
   - In the cytoplasm, the mitotic spindle begins to form, extending from one centrosome to the other.
   - Prophase is the longest phase of mitosis.
2. Metaphase
   - The chromosome line up in a single file located on the equator or metaphase plate.
   - Centrosomes are at opposite poles of the cell.
   - Spindle fibers run from the centrosomes to the kinetochores in the centromeres.



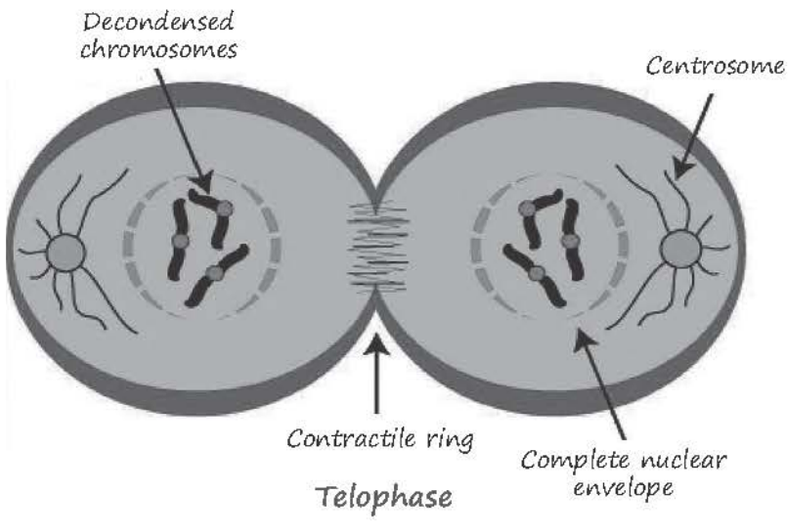
Prophase

3. Anaphase
   - Centromeres of each chromosome separate, and the spindle fibers pull apart the sister chromosomes.
   - This is the shortest phase of mitosis.
4. Telophase
   - Chromosomes cluster at opposite ends of the cell, and the nuclear membrane reforms.
   - The supercoiled chromosomes begin to unravel and return to their normal, pre-cell division condition as long, thread like strands.



Metaphase

Anaphase



Telophase

Significance of Mitosis

1. It is important for the growth of organisms.
2. Repair of damaged and wounded tissues occurs through mitosis.
3. Old or worn out cells are replaced during normal body functioning.

Meiosis I

- Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the s phase.
- Meiosis involves pairing of homologous chromosomes and recombination between non-sister chromatids of homologous chromosomes.

Two haploid cells are formed at the meiosis I each containing one set of chromosomes (a single copy of each chromosome).

Meiosis II

- Meiosis II is functionally the same as mitosis and consists of the same phases: prophase II, metaphase II, anaphase II, telephase II and cytokinesis..
- The chromosome number remains haploid, and daughter cells are genetically distinct to the parent cell. Each parent cell contains four chromosomes, the resulting process produces four haploid daughter cells, each with two chromosomes (half the number of the original parent cell)
- Four haploid cells are formed at the end of meiosis II.

Significance of Meiosis

1. Chromosome number is halved in gametes,so that on fertilisation,the normal number (2n) is restored.
2. Meiosis provide variation in the next generation.
3. Meiosis provide the basis for evolution.

Difference between Mitosis and Meiosis

	Mitosis		Meiosis
(i)	It occurs in somatic cells.	(i)	It occurs in reproductive cells only.
(ii)	Cell divide only once at a time.	(ii)	The cells divide twice in two successive divisions i.e. meiosis I and meiosis II.
(iii)	It is also called equational division because the daughter cells have identical genetic material as the parent cell.	(iii)	It is also called reductional division, because the daughter cells have half the number of chromosomes as the parent cell.
(iv)	Two diploid daughter cells are formed.	(iv)	Four haploid daughter cells are formed.

3 CHAPTER

Genetics - Some Basic Fundamentals

Genetics

It the study of transmission of characters from parents to offspring and the laws relating to such transmission.

Heredity

The phenomenon of passing of characters from parents to progency through successive generations is called heredity.

Variation

The variation in traits or characteristics among individuals within a species is termed variations. For e.g.,: In the majority of individuals, the earlobe is free-hanging, termed as a free earlobe, whereas in some individuals, it is closely attached to the side of the head, referred to as an attached earlobe.

Chromosomes

- The chromosomes is a thread like structure in the nucleus of a cell. It is formed of DNA and carries the genes. The chromosome number is constant for the individuals of a species. For example, human have 46 chromosomes. Some more example are given in the following table.

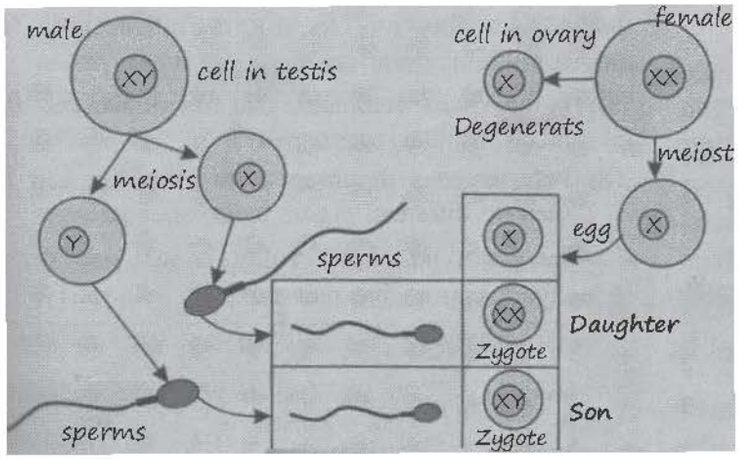
S. No.	Organism	Number of Chromosomes
(i)	Mouse	40
(ii)	Gorilla	48
(iii)	Monkey	54
(iv)	Dog	78
(v)	Onion	16
(vi)	Maize	20

- A pair of corresponding chromosomes of the same shape, and size, one from each parent, is known as homologous chromosomes.

Sex-Determination

- The process by which the sex of a person is determined is called sex determination.

- Sex chromosomes consists of two types: the X chromosome and the Y chromosome, denoted as XY.
- In females, the chromosomal composition include two X chromosomes, denoted as XX.
- If an X chromosome - carrying sperm fertilizes an egg, which also carries the X chromosome, the resulting combination of sex chromosomes will be XX, resulting in the birth of a female (girl).
- If a sperm carrying the Y chromosome fertilizes an egg, the resulting combination of sex chromosomes will be XY, leading to the birth of a male (boy).

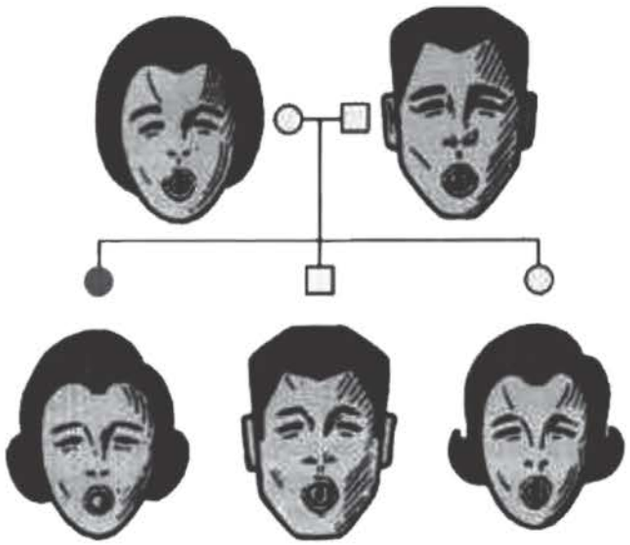


Sex-Determination

- Thus, the male (father) is responsible for determining the gender of the baby.

Genes

- Genes are specific DNA segment on a chromosome which determine hereditary characteristics.
- They play a role in passing traits from parents to their offspring.
- Genes work in pairs.
- Each genes possesses two alternative forms for a trait, which result in distinct effects, these alternate forms are known as alleles.
- The alternative forms of a character are known as traits.
- The gene which decides the appearance of an organism even in the presence of an alternative gene is known as dominant gene.
- The gene which decides the appearance of an organism only in the presence of another identical gene is called a recessive gene.
- The combination of genes present in an organism is called a genotype.
- Traits that are genetically determined and observable in an organism are referred to as its phenotype.
- In the trait of tongue rolling, individuals can exhibit one of two characters: they can either roll their tongue or not.


Tongue Rolling

An Example of Inheritance: Tongue Rolling

- The dominant gene for tongue rolling is denoted by 'R', while the recessive gene for non-rolling is represented by 'r'.
- Therefore, in case of the tongue rolling phenotype, there are two types of genotypes: homozygous

dominant, consisting of a pair of identical alleles (RR), and heterozygous dominant, consisting of a pair of different alleles (Rr).

- Hollow symbol denote the standard expressed traits, specifically tongue rolling, while solid symbols indicate the atypical expressed traits namely non-rolling of the tongue.
- In the provided chart, both the father and mother exhibit tongue rolling. Among their children, two demonstrate tongue rolling while one does not possesses this ability.
- The presence of the recessive allele 'rr' in the third child indicating non-rolling, implies that the child inherited one 'r' allele from each parent. Consequently, both parents must possess a heterozygous genotype tongue rolling (Rr).

Sex-linked Inheritance

Sex-linked inheritance is the appearance of a trait which is due to the presence of an allele exclusively either on X chromosome or on the Y chromosome.

- **X-linked inheritance:** Defect occur due to recessive genes, which are on the X-chromosome. For e.g., Colour blindness and Haemophilia
- **Haemophilia:** In individual with haemophilia, there is a risk of bleeding excessively or experiencing delayed clotting, potentially leading to fatal outcomes.
- **Colour blindness:** A person with colour blindness is unable to differentiate between red and green colours.

Criss-Cross Inheritance

- X-linked inheritance is also called criss-cross inheritance.
- This is because the son may inherit a trait from the normal carrier mother.

Y-Linked Inheritance

Traits resulting from dominant genes located on the Y chromosome are exclusive to males. Examples of such traits include hypertrichosis (excessive hair growth on ears) and baldness.

Mendel's Experiment on Inheritance

- Gregor Mendel meticulously investigated the inheritance patterns, focus on the transmission of traits from one generation to the next. He is widely recognized as the father of Genetics.
- Mendel conducted breeding experiment on Pisum Sativum (garden pea). He selected the pea plant for three reason:
  - There were many varieties available in alternative forms of a character.
  - Pure forms which breed true be produced the same type generation after generation.
- The following are the seven pairs of contrasting characters in the garden pea studies by Gregor Mendel.

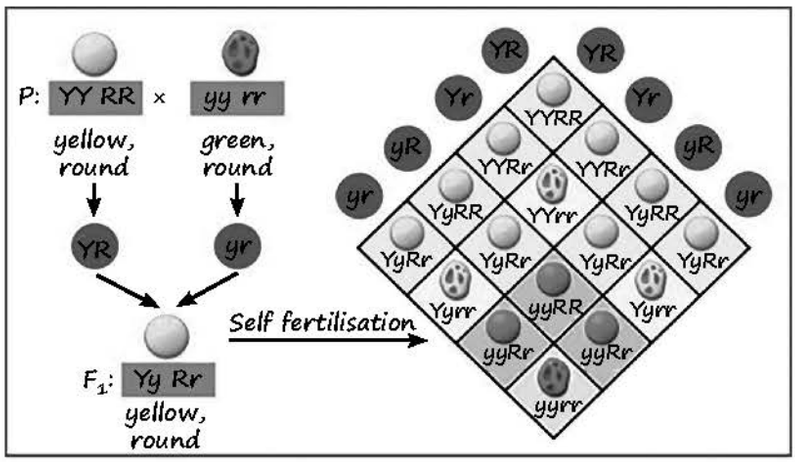
	Flower color	Flower position	Seed color	Seed shape	Pod shape	Pod color	Stem length
P	Purple × White	Axial × Terminal	Yellow × Gree	Round × Wrinkled	Inflated × Constricted	Green × Yellow	Tall × Dwarf
F_1	Purple	Axial	Yello	Round	Inflated	Green	Tail

Monohybrid Cross

- Mendel initially conducted cross-breeding experiments focusing on a single characters.
- Example: Pure tall (TT) pea plant was crossed with dwarf (tt) plant.
- F_1 generation: All plants obtained were heterozygous tall. Plants from the F_1 generation were self-pollinated to obtain the F_2 generation.
- F_2 generation: One homozygous dwarf plant were obtained.
- The ratio resulting from a cross involving two different traits of a single character is referred as the mono hybrid ratio. (3 : 1)

Dihybrid Cross

- Mendel carried out the cross breeding by taking two character together. Example: Plants with round yellow seed (RRYY) was crossed with the plants bearing wrinkled green seeds (rryy).
- He obtained only round-yellow seeds in the F1 generation. This indicated that round shape and yellow colour of seeds are dominant in nature.
- Meanwhile, the wrinkled shape and green colour of seeds are recessive traits. Then, F_1 progeny was self-pollinated. This resulted in four different combinations of seeds in the F_2 generation. They were wrinkled-yellow, round-yellow, wrinkled-green seeds and round-green in the phenotypic ratio of 9 : 3 : 3 : 1.



- The phenotypic ratio was 9 : 3 : 3 : 1.
- The ratio obtained by breeding two pairs of contrasting characters is called dihybrid ratio.

Mendel Law of Inheritance

1. Law of Dominance: Among a pair of contrasting characters coexist together, only one is capable of expressing itself, while the other remains suppressed.
2. Law of Segregation: Two members of a pair of factors separate during the formation of gametes.
3. Law of Independent Assortment

Mutations

Mutations refers to the abrupt alteration in one or more genes, or in the number of structure of chromosomes.

Sickle Cell anaemia: It is a blood disease caused due to gene mutation. The mutation cause changes in DNA and results in the production of sickle shaped RBCs.

Down's Syndrome: It is mental deficiency caused due to one extra chromosome (chromosome no. 21)

Radioactive Radiations

Radioactive radiations alter the gene structure. The atomic bombing that took place during World War II in 1945 in Japan resulted in various deformities in the bodies of animal and plants.

Modern Application of Genetics

Genetic engineering modifies an organism's genome by introducing new genes. This technique create genetically modified organisms (GMOs). One of its earliest applications was producing insulin.

4 CHAPTER

Absorption By Roots: The Process involved

Plant physiology is the branch of botany which deals with the study of metabolic activities as life processes of plants.

Absorption by Roots

(i) Water: For photosynthesis, transpiration, transportation, and mechanical stiffness.

(ii) Mineral Nutrients: Constituents of cells and cell organelles, synthesis of enzyme etc.

Characteristics of roots for absorbing water	
Surface Area	Large surface area provided by rootlets and root hairs. Eg. Balsam plant.
Cell Sap	Cell sap of root hair is of high concentration as compared to surrounding water for the occurrence of osmosis.
Thin walls of root hairs	The cell wall is thin and permeable while the cell membrane is very thin and semi-permeable. This is the main characteristic of root for water absorption.

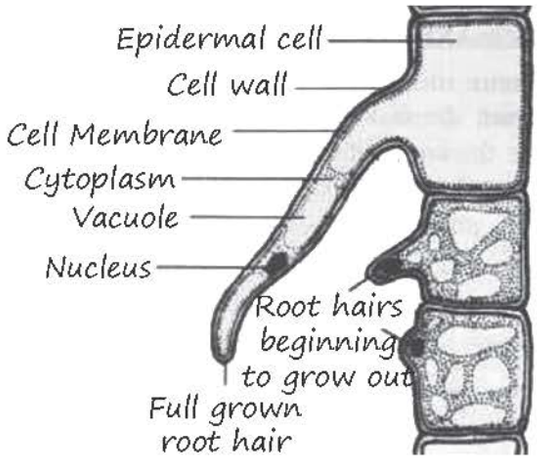
The processes in absorption and conduction of water and minerals

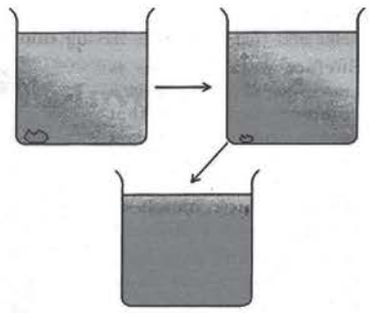
(A) Imbibition: It is a phenomenon in which living or dead cells of plants in their dry or semi-dry state absorb water by surface tension. It results in an increase in volume, liberation of heat, and development of pressure.

+ It is the passive absorption of water by substances such as cellulose and starch.

(B) Diffusion: The movement of molecules of a substance (gas, liquid, or solid) from the region of higher concentration to the region of lower concentration when are in direct contact (as shown is diagram).

+ Diffusion pressure (DP) is directly proportional to the number of diffusing particles.



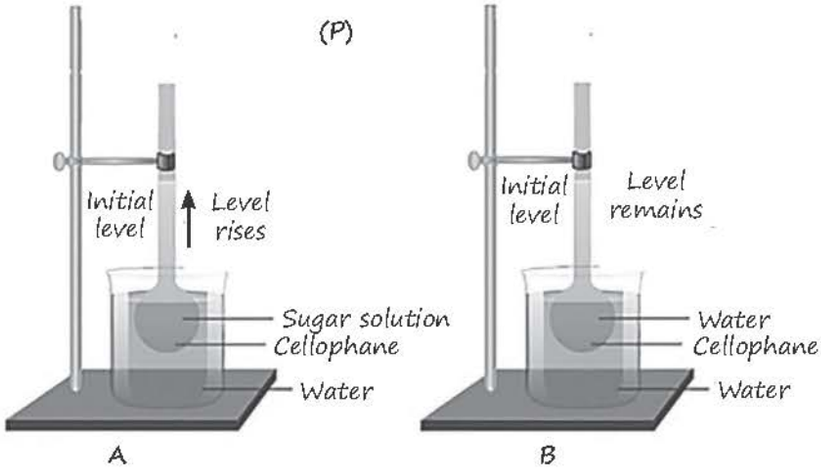


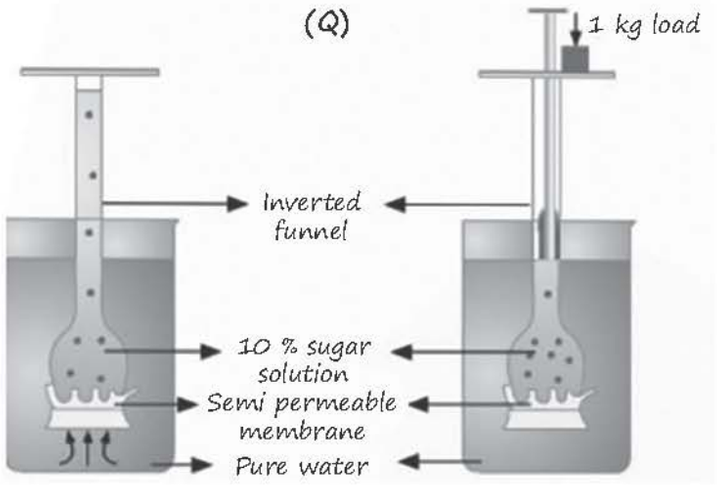
Diffusion of a soluble dye in water, forming a solution (schematic).
A –Diffusion started,
B –Diffusion progressing,
C– Diffusion completed, making a homogeneous solution; molecules of the solute are evenly distributed in the solvent.

- ✦ It is important for gaseous exchange of plants with the atmosphere through stomata and lenticels.

(C) Osmosis and osmotic pressure.

- ❑ Osmosis is the movement of water from the region of higher chemical potential (dilute solution) to the region of its lower chemical (concentrated solution) when diffusion of solute is prevented by a semipermeable membrane.
- ❑ Endosmosis: is the osmotic entry of water into a cell or system, when placed in pure water or hypotonic solution. It makes the cell swell up.
- ❑ Exosmosis: is the osmotic withdrawal of water from a cell or system, when placed in hypertonic solution. It makes the cell shrink.
- ❑ Osmotic Pressure: is the minimum pressure that must be exerted to prevent the passage of pure solvent into the solution when the two are separated by a semi-permeable membrane.



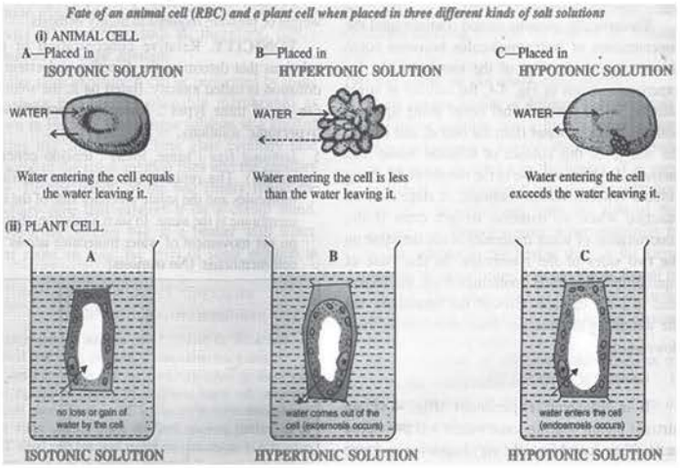


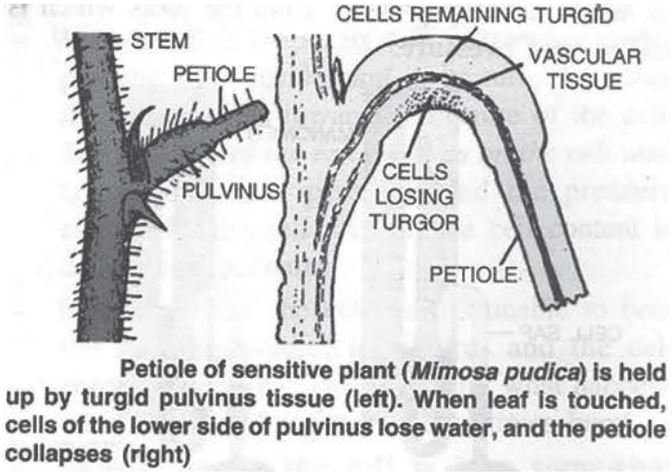
Experiment to show osmosis by thistle funnel containing sugar solution

Conclusion

(a) Some water has passed through cellophane paper to enter the funnel A, and sugar from thistle funnel A has not passed to the beaker.

(b) The cellophane paper acts as a semi-permeable membrane that allows only water to pass, not sugar, because of the concentration difference.

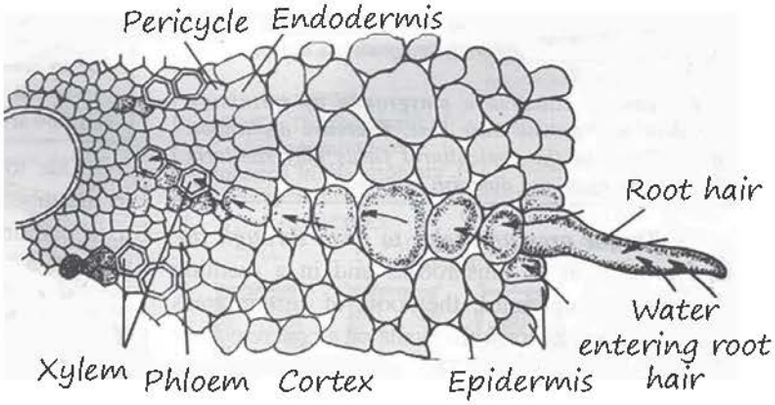




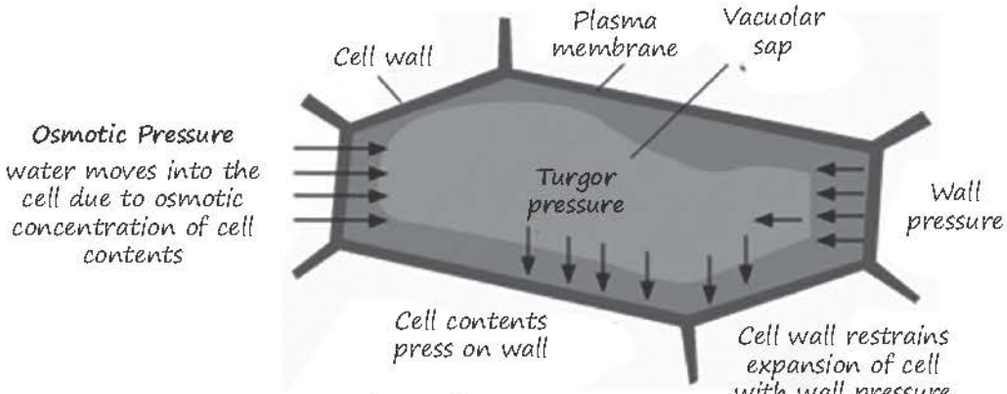
Petiole of sensitive plant (*Mimosa pudica*) is held up by turgid pulvinus tissue (left). When leaf is touched, cells of the lower side of pulvinus lose water, and the petiole collapses (right)

Experiment to show osmotic pressure by thistle funnel and airtight piston

- Conclusion: Osmotic pressure applied by piston did not allow water to pass through i.e., no osmosis take place.
- Tornicity: It is the relative concentration of the solutions that determines the direction and extent of diffusion.



Isotonic Solution	Hypotonic Solution	Hypertonic Solution
If the external solution has a similar solute or solvent concentration as the cell. Cell shape and size remain unchanged.	If the external solution has low solute (or high solvent) concentration than the cell. Cell slightly enlarges or even bursts.	If the external solution has a high solute (or low solvent) concentration than the cell. Cell shrinks in size and loses shape.



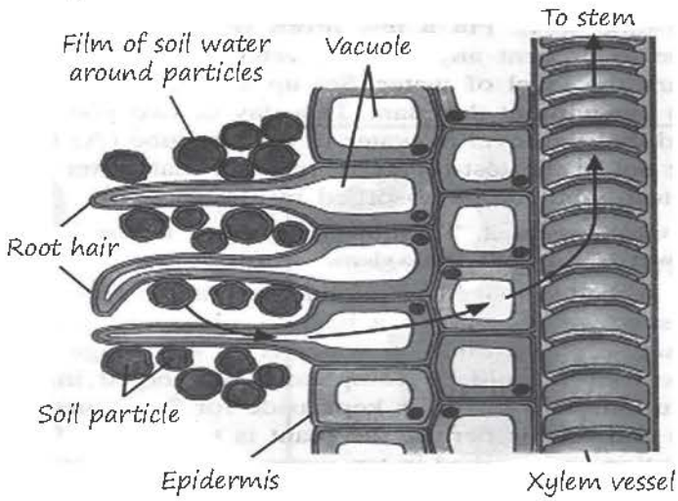
A plant cell showing osmotic pressure, turgor pressure and wall pressure represented together

Active Transport

- The passage of a substance (salts or ions) from its lower to higher concentration using energy (ATP) from the cell, through a living cell membrane (opposite of diffusion).
- Passive transport doesn't require energy. r.g., transpiration and diffusion.

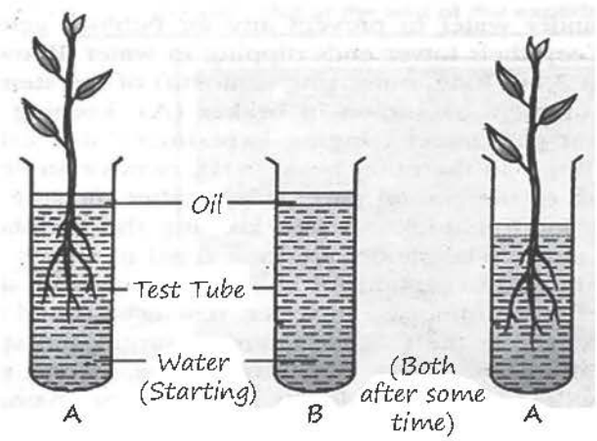
Turgidity and Flaccidity

- Turgidity is a condition of being fully distended due to endosmosis. Healthy plant cells are turgid.



Root pressure builds up due to osmosis and turgidity or foot cells, and up into the xylem vessel.

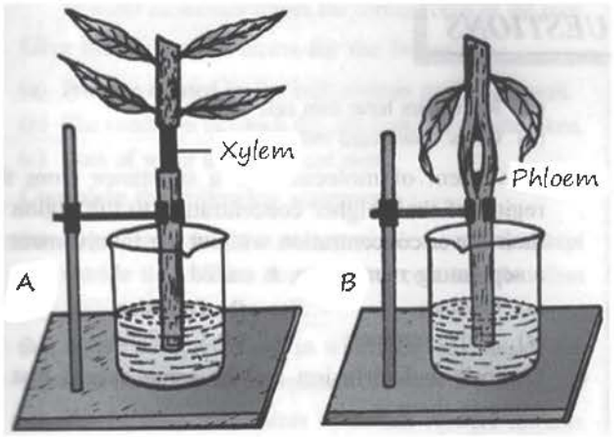
- The pressure of cell contents on the cell wall is known as turgor pressure, and the pressure of the cell wall on the cell contents is known as wall pressure.
- Flaccidity is a condition of absence of turgidity in which the cell loses water from its cytoplasm due to exosmosis. Such a cell is known as a flaccid cell, e.g., wilting property of leaves when the plant is exposed to the sun.
- Plasmolysis is the shrinkage of protoplast from the cell wall due to exosmosis caused by a hypertonic solution. Permanent plasmolysis causes death.



- Deplasmolysis is when a plasmolyzed cell is kept in water or a hypotonic solution. Water enters the cell due to endosmosis, and the cell becomes turgid again.

Uses of Turgidity

1. It provides rigidity to soft tissues like leaves.
2. Turgor pressure helps to push through the hard ground, e.g., mushrooms and seedlings.
3. Turgor in the opening and closing of stomata.
4. Turgor movements in Mimosa pudica plants.

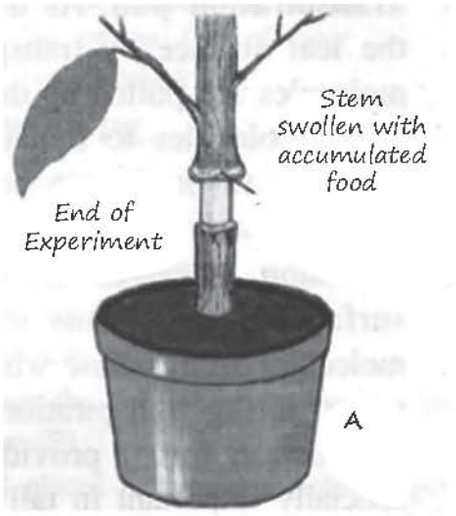


Root Pressure

- The pressure developed in the roots due to the inflow of water, brought about due to the alternate turgidity and flaccidity of the cells of the cortex and the root hair cell, which helps in pushing the plant's sap upwards.
- It is the maximum during spring in tropical regions and minimum in summer.

Experiments:

1. Cut off part shoot shows water droplets.
2. Water droplets along the margins of leaves due to excess root pressure (guttation). Eg tomato, grass, banana, ferns. etc.



Experiment to show absorption and conduction of water in plants

(A) To show the root absorbs water.

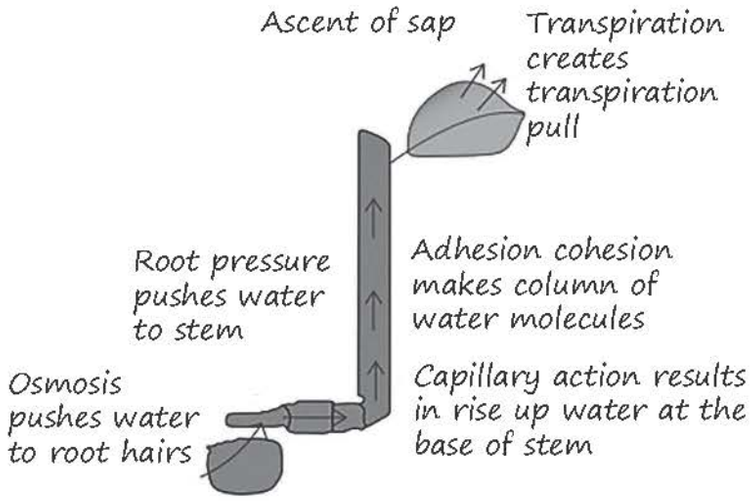
+ Conclusion: The level of water in test tube A falls but not in test tube B, proving that water lost in test tube A was absorbed by roots.

(B) To show that water is conducted upwards through the xylem

+ **Conclusion:** Water and salt travel upwards mainly in the xylem (stained part), and food substances travel up and down in the phloem.

(C) To show conduction of water through xylem

+ **Conclusion:** In balsam plant, after the experiment of girdling, the leaves in the first twig remain turgid and stand almost normally, but in the second twig, they get wilted and droop down.



(D) To show that food from the leaves is conducted downwards through the phloem in the stem

+ **Conclusion:** In girdling, the stem above the ring has grown in diameter, and the stem below the girdle has stopped growing and may even die. It proves that leaves continue to get a supply of water through the deeper located xylem.

Forces contributing to Ascent of Sap

- Upward transport of water to aerial parts along with the dissolved mineral salts from roots to the aerial parts against the downward pull of gravity is called the ascent of sap.
- Ascent of sap takes place through xylem.
- Mechanism: Vital force theory, Root pressure theory, Capillary force theory, and Cohesive force theory.

(A) **Root pressure:** Build up of sufficient force to push the sap in the xylem vessels up to a certain height and may be enough for herbaceous plants.

(B) **Capillarity:** (Narrow diameter) xylem vessels cause the water from a lower level to rise to fill up the vacuum created by the loss of water due to transpiration from the leaves by capillary force.

(C) **Transpiration pull:** More water molecules are pulled up due to the tendency of water molecules to remain joined (Cohesion) and produce a continuous column of water through the stem.

(D) **Adhesion:** Causes the water to stick to the surface of cells, thus drawing more water molecules from below when the leaf cells lose water during transpiration.

Eg: Pines (tall trees).

- Downward movement of sap mainly because of the force of gravity.

5 CHAPTER

Transpiration

The loss of water in the form of water vapour from the aerial parts such as the leaves and the green shoots of the plant is called transpiration.

Types of Transpiration

Depending upon the parts of the plant involved, there are three types of transpiration.

1. Stomatal Transpiration
   - This kind of transpiration occurs through stomata. Each stomata (singular) is surrounded by two guard cell. These guard cell regulate the opening and closing of stomata.
   - When stoma is opened, transpiration take place and when it is closed, transpiration stops.
2. Cuticular Transpiration
   - The loss of water vapour from the cuticle, is called cuticular transpiration. The lesser the thickness of the cuticle, the greater the transpiration. Cuticle transpiration accounts for about 10% of the total amount of water transpired from the leaves.
3. Lenticular Transpiration
   - In woody plants and in many fruits, transpiration also take place through lenticles, which remain open all the time because they do not have guard cells.
   - Lenticles are small openings in the bark of the woody stem, underground items, twigs and fruits. It also helps in diffusion of respiratory gases.

Mechanism of Stomatal Transpiration

- Water absorbed by the roots is conducted to the mesophyll cell of leaves through the xylem elements.
- These cells have large inter cellular spaces. Water evaporates from the surface of the mesophyll cells, accumulates in the intercellular space and then escapes the outer atmosphere through stomata.
- Most of the water gets transpired by stomata found on the leaves of plants. The stomata open for a certain period in the daytime for intake of carbon dioxide for photosynthesis and during the dark they remain partly closed. The guard cells at stomatal openings regulate the transpiration.

Measurement of Transpiration

1. Weighing method

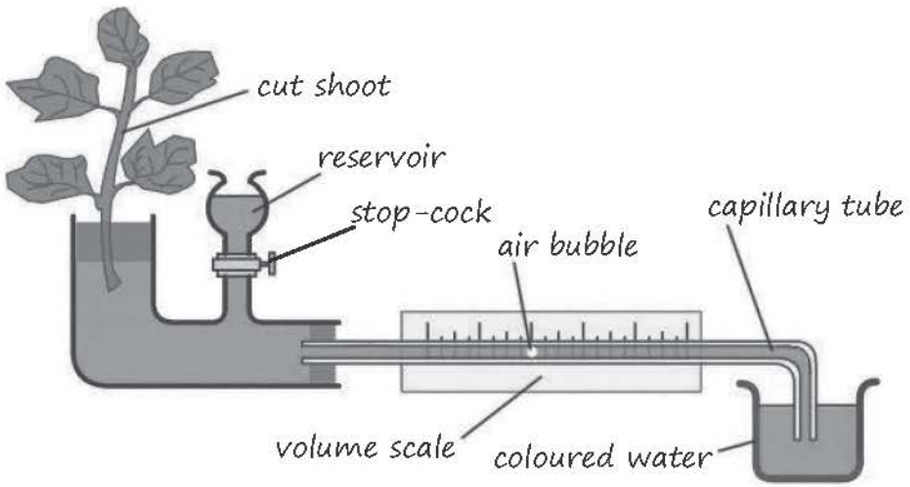
- The relation between transpiration and absorption can be worked out for the various hours of the day and under diverse external conditions by a weighing method.
- It should be noted that the experiment also shows separately the rate of absorption and the rate of transpiration which are almost equal.
- In this experiment, a leafy shoot is inserted in a test tube filled with water and some oil is poured on the surface of water to prevent loss of water by evaporation.

A B

- The test tube is weighed by keeping it in a beaker. The test tube is again weighed after a few hours. The difference in the weight shows the loss of water by the shoot.

2. Potometer method

- A glass device which is used to measure and compare the rate of transpiration in different conditions is known as potometer.
- There are different types of potometer: Ganong's potometer, simple potometer, farmer's potometer, Base's potometer etc.



Precautions to be taken during setting up of a Potometer

- The twig should be cut obliquely while dipped inside the water.
- The apparatus should be watertight.
- The air bubbles should be set on zero point by opening the stopper of the reservoir carefully.
- The capillary tube should be properly dipped in the water.
- Once the coloured water starts moving into the capillary tube, introduce the air bubbles.
- The potometer is best for regarding effects of different environmental factors on transpiration, such as air, light, temperature etc.

Limitations of potometer

- In a potometer, a cut shoot of a plant is used which gets shrivel up after few regarding.
- Also, the rate of transpiration may differ from that of the shoot attached to an intact plant due to the absence of the root system.

Cobalt chloride paper method

A B C

In this method, transpiration is indicated by change in colour of the cobalt chloride paper. The cobalt chloride paper is blue when dry and pink when moist. The method is suitable for comparing the rate of transpiration in case of dorisentral or dicot leaf.

Factors affecting the rate of transpiration

1. **Temperature:** The rate of transpiration increases as the temperature increases. This is because at higher temperature, there is more evaporation of water from mesophyll cells.
2. **Humidity:** The higher the humidity of the air, the less is the rate of transpiration.
3. **Velocity of wind:** On a windy day, water evaporates more quickly than on a fine day. This is because strong wind takes away the moisture laden air from the trees, then leaves transpire more water because air can held more moisture.
4. **Light:** High intensity and long duration of light increases the rate of transpiration. This is because stomata usually open in the light as a result of photosynthesis and close at night when there is no photosynthesis.
5. **Available soil water:** If there is less uptake of water by roots, the guard cells become flaccid, closing the stomata, and the rate of transpiration is decreased.
6. **Percentage of carbon dioxide:** An increase in the percentage of carbon dioxide i.e. more than 0.03% results in decrease of rate of transpiration because it combines with moisture.
7. **Atmospheric pressure:** The lesser the atmospheric pressure, the more is the rate of transpiration.

Adaptations to reduce excessive transpiration

The important structural modifications areas follows:

- The presence of thick cuticle on the upper side than on the lower side and multiple epidermis in certain leaves to check excessive loss of water e.g., banana, rubber plant, nerium and banyan leaves.
- The presence of sunken stomata e.g. prickly pear and nerium.
- The number of stomata gets reduced e.g. upper surface of dicot leaf.
- Rolling, folding and dropping of leaves e.g. Pteris (fern) and desert grass.
- Reduction in leaf area by their modification into spines or scales e.g. prickly poppy or cactus (xerophytes).

- The presence of hairs, scales, etc. on the surface of the leaf minimizes transpiration to a great extent e.g. petunia and hally hock.

Importance of Transpiration

The excess water is eliminated from the plant body by transpiration:

- Transpiration also helps in the distribution of minerals and water throughout the plant body. Transpiration secures concentration of the cell sap and thereby helps in the process of osmosis.
- As a result of transpiration, in summers, plant's body becomes cool since a considerable amount of latent heat is lost in converting water from liquid to a gaseous state.
- As a result of transpiration from the leaf surface, a suction force is generated by decreasing water content of leaf cells, which draws water from the cell below it in the form of a pull which helps in the ascent of water to the tip of the trees.
- As the water evaporates, hygroscopic salts are left on the surface of the leaf. These salts absorb moisture from the atmosphere and do not allow the leaf or the plant as a whole to dry up.

Guttation or Exudation

The excess water is eliminated from the plant body by transpiration:

- The excess of water is also eliminated by many herbaceous plants by a process, commonly called exudation or guttation.
- Thus, in balsam, rose, water lettuce, grapevine, many weeds, garden nasturtium, sunflower, chrysanthemum, canna, many grasses etc. It is seen that the drops of water accumulate at the apex or the margin of the leaves in the early morning.
- The water escapes through the hydathodes.
- Exudation normally takes place during warm and damp nights.
- In some plants, a considerable quantity of water exudes every night. Exudation does not take place at very low temperatures.
- Oozing out of the cell sap due to root pressure is called bleeding. e.g. Latex of rubber tree is collected by making a deep cut in the main stem.

6 CHAPTER

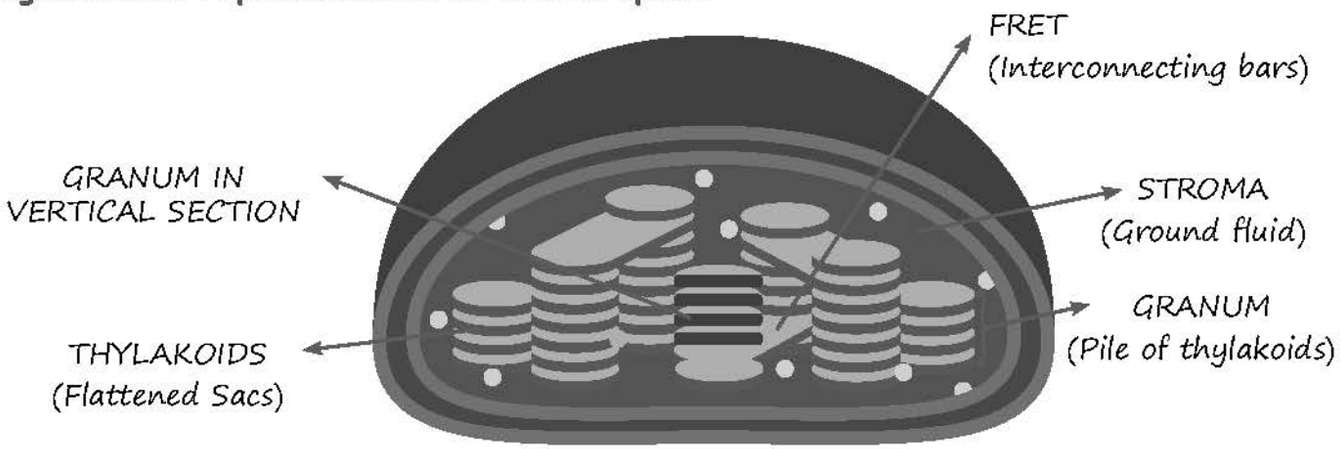
Photosynthesis

Photosynthesis: Provider of food for all

Photosynthesis is a physiological process by which plant cells containing chlorophyll produce food in the form of carbohydrate by using carbon dioxide, water and light energy. Oxygen is released as a byproduct.

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 \uparrow + 6H_2O$$

Diagrammatic representation of a chloroplast



Stomata

Stomata are the structures mainly responsible for the gaseous exchange in the process of photosynthesis, respiration and transpiration.

Stomata open in light when guard cell are turgid and close when they are flaccid.

These turgor changes lead to the opening and closing of stomata. There are two theories for the opening and closing of stomata:

1. Sugar concentration theory

According to this theory during daytime, as guard cells photosynthesise and produce sugar, the cells become turgid. This causes the stomata to open. If exosmosis happen then the guard cells become flaccid and stomata is closed.

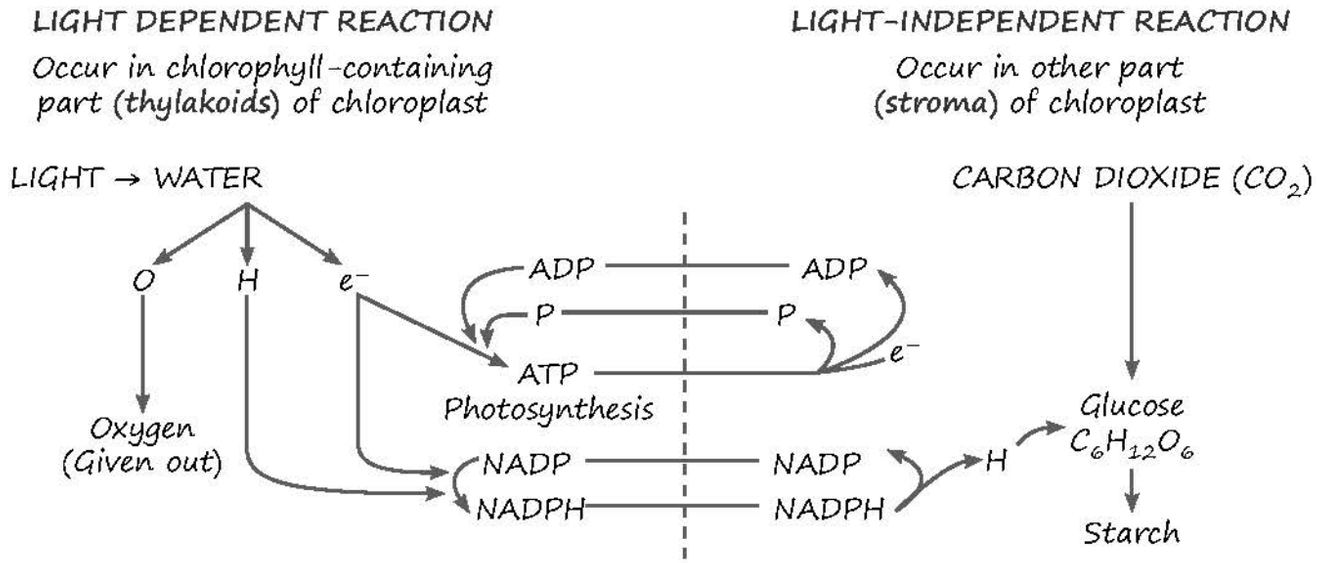
2. Potassium ion exchange theory

According to malate K^+ ions pump hypothesis, the turgor changes that open and close the stomata are due to the reversible absorption and loss of potassium ions (K^+).

(A) The stomata open when guard cells absorb K^+ ions from surrounding epidermal cells.
(B) The intake of K^+ ions by guard cells take place through H^+ and K^+ ion exchange.
(C) The H^+ ions appear in the guard sell due to dissociation of malic acid.
(D) The malic acid dissociates into malate anions and H^+ ions in the guard cells.
(E) During night, the K^+ ion concentration decreases in the guard cells. This lowers the osmotic pressure of the guard cells which results in the closing of stomata due to exosmosis

Mechanism of Photosynthesis

Photosynthesis consists of two stages: the light phase and the dark phase.



Light Reaction	Dark Reaction
It take place only in the presence of light.	It can take place in the presence or absence of sunlight.
It is a photochemical phase.	It is a biochemical phase.
It takes place in the grana of the chloroplast.	It takes place in the stroma of the chloroplast.
NADP utilizes H^+ ions to form NADPH	The hydrogen of NADPH combines with CO_2.
The end products are ATP and NADPH	Glucose is the end product. ATP and NADPH help in the formation of glucose.
The water molecules split into Hydrogen ions and oxygen.	Glucose is produced, CO_2 is utilized in the dark reaction.

Different Experiment on Photosynthesis

1. Test for Photosynthesis

Leaf is killed in boiling water (5–10 min), dried, decolourised in warm spirit, moistened and dipped in iodine solution. Blue black colour indicates starch.

2. CO_2 is necessary for Photosynthesis/Moll's Half Leaf Experiment

One half of the destarched leaf is inserted in an airtight wide mouthed bottle having a small quantity of KOH (for absorption of CO_2) and illuminated. Starch test after one hour indicates absence of CO_2 in the inserted half and presence in the outer half (where CO_2 is available).

3. Light is Necessary for Photosynthesis

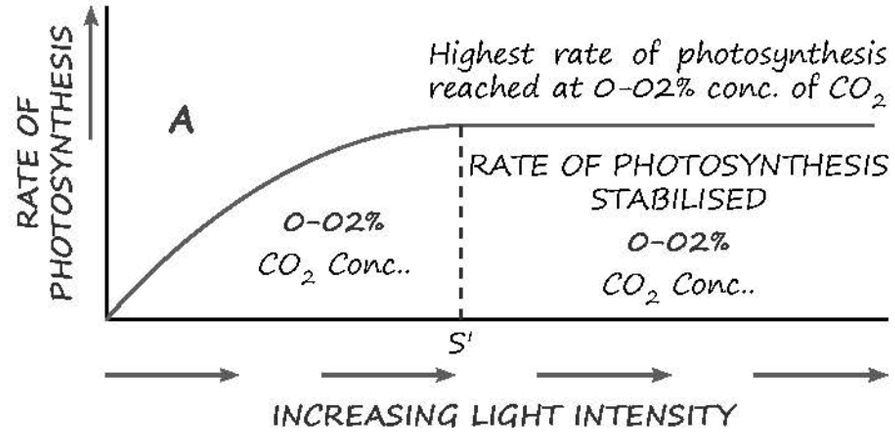
Intact leaf of a destarched plant is fitted in Ganong's light screen with a designed cut in its lid. The same is exposed to light for sometime and then tested for starch. Only the design through which light falls on the leaf becomes blue coloured.

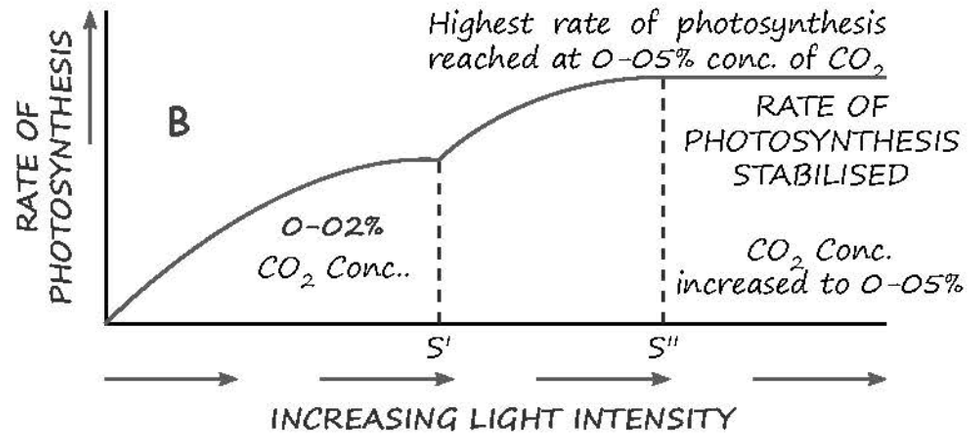
4. Chlorophyll is essential for photosynthesis

Illuminated variegated leaf of the coleus plant is tested for starch. Only those areas turn blue which had chlorophyll.

Factors Affecting Photosynthesis

1. External Factors





(A) **Light**: Photosynthesis is successfully accomplished in the visible light (380–760 nm wavelength) of the spectrum. The rate of photosynthesis is maximum in red light, average in blue light and minimum in green light. A moderate light intensity is favourable for high rate of photosynthesis.

(B) **Carbon dioxide**: When the concentration of CO_2 increases, up to a certain point (in this case, 0.05%), the rate of photosynthesis also increases. This is because more CO_2 is available for the plants to convert into glucose during the photosynthesis process. The increase in the rate of photosynthesis continues as CO_2 concentration rises until it reaches the threshold of 0.05%. Beyond this concentration, the benefits start to diminish.

(C) **Temperature**: Generally, the photosynthesis increases with an increases in temperature in the range of 10–35°C. Beyond the 85°C, the rate of photosynthesis decrease.

(D) **Water**: About 1% total water absorbed is used in photosynthesis. Water deficiency reduces the rate of photosynthesis.

2. Internal Factors

(A) **Chlorophyll**: It is essential for photosynthesis to occur. The rate of photosynthesis per unit of chlorophyll decreases with the age of the leaf.

(B) **Accumulation of end product of photosynthesis**: The rate of photosynthesis falls with the accumulation of food synthesised by photosynthesis.

(C) **Anatomy of leaf**: Photosynthesis rate is greatly influenced by the internal structure of the leaf, thickness of the cuticle and epidermis number, structure and distribution of stomata.

Adaptation of a leaf for photosynthesis

1. Large surface area to maximise light harvesting.
2. Thinning of leaves to reduce distance for CO_2 to diffuse through the leaf and to ensure that light penetrates into middle of the leaf.
3. Arrangement of chloroplast on the upper surface of leaves so as to receive maximum amount of light.
4. The presence of more stomata to allow rapid exchange of gases (CO_2 and O_2).
5. Loosely packed spongy mesophyll cells help to trap carbon: dioxide gas, allowing some photosynthesis even when the stomata are closed.

7 CHAPTER

Chemical Coordination in Plants

- Stimuli is a change in either the external or internal environment of an organism.
- A response occurs when actions or movements are prompted by stimuli.
- Plant sensitivity and coordination is controlled by some chemicals produced by them.
- Plant react to changes by producing special chemicals compound called "hormone" that act like messengers.
- Plant growth can be be either stimulated or inhibited by specific hormone.
- The primary sites for growth, where cell divisions occur in plants, are meristem, which respond to hormone.
- The hormone aids in expanding the cellulosic wall of meristematic cells to assist in division.
- Plants are sensitive to light and respond to the direction of light. Also sensitive to water, temperature, gravity etc.
- The shoot grows in the direction of light, while the roots grow opposite to it.
- The term phytohormone was created to differentiate hormones found in plants from those in animals.

Plants hormones

1. Auxin

- It is the first growth hormone which was discovered in plants.
- Auxins are powerful growth, stimulants and remain effective even at very low concentration.
- IAA(Indole-3-Acetic acid) is the main natural auxin in plants.
- It can be found in various plant parts such as the shoot open and lateral meristems.

2. Gibberellins

- The different plants produce various forms of gibberellins including GA_1, GA_2, GA_3 and others.
- Auxins are powerful growth, stimulants and remain effective even at very low concentration.
- The gibberellic acid (GA_3) is the most studied plant hormone.
- The stimulates internode growth through elongation of cells.
- It is present in a meristematic region like stem apex, root-open, buds, seeds etc.

3. Cytokinins

- It has a specific effect on cell division.
- It promotes plant growth through cell division, unlike auxin and gibberellins, which stimulate growth through cell elongation.
- It is widely distributed in plants.

- It is produced in the tips of roots, it is then transported via the xylem.
- It is found at germinating seeds, developing fruits embryo etc.

4. Ethylene

- It is the only hormone which is a gas at ordinary temperature.
- The location where it is made and where it works is the same.
- It is a hormone which causes the ripening of fruits.
- It is produced in higher plants and fungi.
- The meristematic tissue produce greater amount of ethylene.

5. Abscisic Acid (ABA)

- It is a growth-retarding hormone.
- It is found in angiosperms, gymnosperms, pteridophytes and some mosses.
- It is found in chloroplasts of leaves.
- The fruit and seeds contain the highest amount of ABA.
- It is also known as stress hormone.

Plant hormone	Site of synthesis	Functions
Auxin	Shoot apical meristem and young leaves.	Promote cell elongation. Suppress the growth of lateral buds. Delay fall of leaves. Induces parthenocarpy.
Gibberellins	Meristem of apical buds and roots.	Induce formation of parthenocarpy fruits. Helps in stem elongation. Break dormancy of seeds and buds. Delay senescence induce parthenocarpy.
Cytokinins	Roots and transported to other organs.	Stimulate cell divisions and cell enlargement. Prevent ageing of plants parts. Inhibit apical dominance.
Ethylene	Senescent leaves and flowers, germinating seeds and ripening fruits.	Induce fruit ripening. Promote senescence.
Abscisic acid	Green fruits and seeds at the beginning of wintering periods.	Induce dormancy of buds and seeds. Inhibit seed germination and development. Stimulates closing of stomata.

Tropic Movements in Plants

- The root is required to extend vertically into the soil to access the necessary support, water and minerals vital for the plant.
- Shoot must grown up towards the light source so that developing leaflets could carry out photosynthesis in others to produce food for the growing plant.
- The movement of these plant parts occurs in direct-reaction to external stimuli.
- The direction of response to the direction from which the stimulus comes. Such a response is known as tropism.
- The growth movements occurring in response to unidirectional external stimuli in a plant part are called tropic movements.

I. Phototropism

- The photo tropism is the ability of a plant to grow in the direction of light.
- Shoots exhibit positive photo tropism, while roots display negative photo tropism.
- Auxins play an important role in photo tropism.
- Auxins aid in the bending of shoots towards a light source by accumulating in the shaded region of the shoot, leading to rapid cell elongation.

II. Geotropism

- The term geotropism refers to the phenomenon of plants growing towards the earth's gravity. This process is also known as gravito tropism
- Root tips exhibit positive geotropism, whereas shoot tips display negative geotropism.

III. Hydrotropism

- Hydro tropism is the term used to describe the movement of plant part in response to water and moisture.
- When plant parts grow towards a source of moisture, it's referred to as positive hydrotropism.
- Roots are positively hydrotropic as water is available in the soil.
- When the plant part grows away from the source of moisture is called negatively hydro tropic (shoot).

IV. Thigmotropism

- Thigmotropism is the term used to describe the growth movements of plant in response to touch stimuli.
- Plants such as sweet peas, *cuscuta* and wines have tendrils which coil around other plants in response to one side touch.
- The stimulus is initially sensed by the tips of the tendrils and then transmitted to their basal parts. This process renders the entire tendril sensitive to the stimulus.

Chemotropism

- Growth of plant organs in response to chemicals.
- Eg.-Movement of pollen tube of angiosperm and gymnosperm towards sugar and peptones secreted by neck canal cells of female gametophyte.

8 CHAPTER

The Circulatory System

Body Fluids

Every organ in our body requires the involvement of the circulating fluids.

Blood

Plasma

Light-yellow coloured, alkaline liquid (pH: 7.3 to 7.45), water (90-92%), proteins (7-8%), inorganic salts (Sodium chloride, Sodium bicarbonate - 1%) other substances (glucose, amino acids, fibrinogen, hormones, urea etc.)

Serum Plasma without fibrinogen.

RBCs (Erythrocyte)

Red blood cells in human male are about 5 million and in females are about 4.5 million.

Minute (about 7 micron in diameter) biconcave disc-like structure (provide a large surface area for absorbing oxygen), flat in the centre and thick and rounded at the periphery.

Haemoglobin-Respiratory pigment in stroma of RBCs formed of an iron-containing part(haemin) and a protein part (globin). It combines with oxygen to form **oxyhaemoglobin** and with carbon dioxide forms **carbaminohaemoglobin**.

- **Carbon monoxide poisoning**: Haemoglobin has a very strong affinity for carbon monoxide (furnaces burning in closed area) forming carboxyhaemoglobin (HbCo) results in even death due to oxygen depletion in the body.
- In adults and children (after 5 years of age) RBC are produced in bone marrow (long bones like ribs, breast-bone, ilium of ilium of hip girdle). In embryo, produced in liver and spleen. On maturation; they become enucleated and average life is about 120 days.
- Destroyed in spleen, liver and bone marrow.
- **Polycythaemia**: Abnormally increased number of RBCs.
- **Erythropenia**: Abnormally decreased number of RBCs.
- White blood cells are about 4000-8000 per mm^3 of blood, amoeboid and can produce pseudopodia that can squeeze through the walls of the capillaries into the tissues (**diapedesis**). Produced in red bone marrow, lymph node, sometime liver and spleen.
- The average life is about two weeks.

Two major categories of WBCs	Five Cell Types (Abundance)	Distinguishing feature	Functions
(A) Granular: Cytoplasm contains granules. Nucleus usually constricted into lobes	1. Neutrophils	❑ Nucleus with 3–4 lobes. Granular cytoplasm. ❑ Stain with neutral dyes.	Engulf bacteria (phagocytosis)
	2. Eosinophils (2–3% count) increases in allergies	❑ Nucleus with 2 lobes. ❑ Cytoplasmic granules large and stain dark red with eosin (acid dye).	❑ Engulf bacteria ❑ Secrete antitoxins ❑ Associated allergy
	3. Basophils (0–4% count)	❑ Nucleus large indistinctly lobe. ❑ Granules stain with basic dyes (e.g., methylene blue)	Release chemical (histamine) for inflammation. which dilate blood vessels.
(B) Non-Granular: Cytoplasm without granules. A single large nucleus	4. Lymphocyte	❑ Smallest of WBCs. ❑ Single large nucleus	Produce antibodies.
	5. Monocytes	❑ Nucleus large kidney shape. ❑ At the site of infection transform into macrophages	Ingest germs

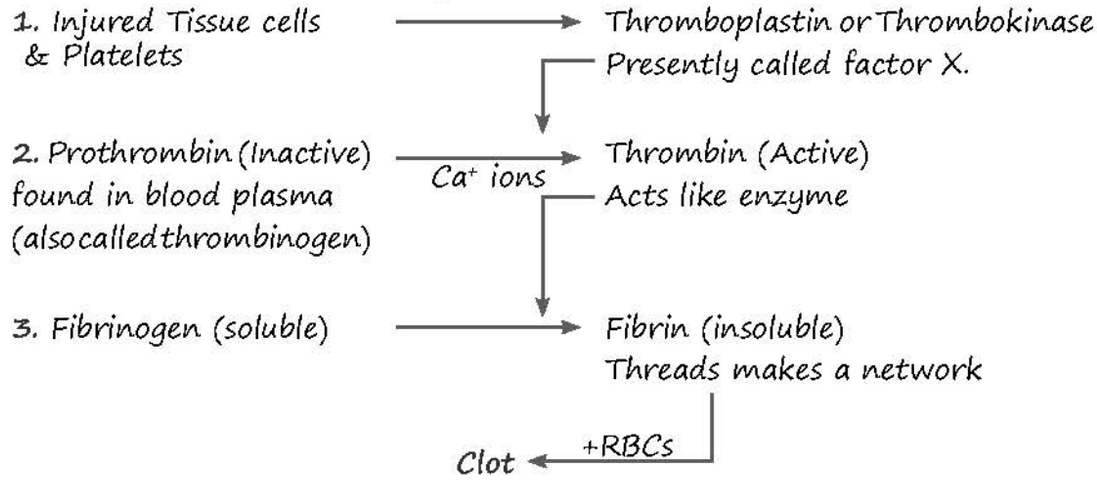
Leukemia: Cancer in which WBCs count increases at the cost of RBCs

Leucopenia: Abnormal decrease in the number of WBCs.

Platelets (Thrombocytes)

- ❑ Minute ovular round structures, non-nucleated, floating in the blood.
- ❑ In adult, these are about-200000 to 400000 per mm^{-3} of blood.
- ❑ Derived from megakaryocytes in the red bone marrow.
- ❑ Life span is 3–5 days and is destroyed in the spleen.
- ❑ Helps in clotting of blood (coagulation).

Main Reactions in Blood Clotting



Blood Transfusion and Blood Groups

In surgeries, blood is taken from a healthy person and injected to the patient's veins so blood should be compatible with the recipient's blood type.

Discovered by Karl Landsteiner. RBCs of human have specific proteins (antigens) on their surface and plasma has its complementary antibodies.

ABO System

There are four blood groups A, B, AB and O. O type blood (universal donor) can be given to persons of all types of blood i.e. O, A, B and AB. AB type blood (universal recipient) can receive blood from all types i.e., AB, A, B and O.

Rh system

Blood of most people contains Rh factor, Rh stands for Rhesus, discovered in our common ancestor 'Monkey.'

An Rh-ve women may become sensitive if she carries on Rh+ve child in her uterus (Husband is Rh+ve). The first Rh+ve child will be normal but if it sensitizes the mother, the second positive child if conceived soon can lead to foetal death and abortion.

Tissue Fluid

As the blood flows in the capillaries of the tissues, the plasma and the leucocytes "leak out" through their walls is known as intercellular or extracellular fluid.

This tissue fluid helps the cells to absorb oxygen and other substances in turn give out carbon dioxide and other wastes back to it.

Lymph and lymphatic system

Most tissue fluid entering into lymph vessels is known as lymph.

Lymph drains into lymph nodes and fresh lymph channels arises and lymph pour into the major anterior veins close to their entry into right auricle and again in circulation.

- Composition: Cellular leukocytes (mainly lymphocytes), non-cellular water (94%), solids (6% proteins, fats etc.)
- Functions: Supplies nutrition and oxygen to those parts where blood cannot reach and drain excess tissue fluid and metabolites. Lymphocytes and monocytes helps in the defense mechanism of the body.

Spleen

It is a large lymphatic organ, about -the size of a clenched fist, reddish brown in colour and situated in the abdomen behind the stomach and above the left kidney.

Functions

- It acts like a blood reservoir in an emergency (haemorrhage, carbon-monoxide poisoning etc.).
- It produces lymphocytes.
- It destroys worn-out-red blood cells (with the liver).
- In an embryo, it produces RBCs.

Blood Circulatory System

It consists of heart, arteries, veins and capillaries.

The blood vessels

Arteries, capillaries and veins.

Arteries	Veins
Blood vessels which carry blood away from the heart and into an organ.	Blood vessels which carry blood away from an organ and towards the heart. Except-Hepatic portal vein.
Progressively branched, decreasing in size.	Progressively unite increase in size.
The smallest artery breaks into arterioles.	The smallest vein arises from venules.
Have thick and more muscular walls.	Have thin and less muscular walls.
Walls are elastic.	Walls are non-elastic.
Have narrower lumen.	Have wider lumen.
Have no valves in their inner linings.	Have valves in their lining to prevent backward flow of blood.
Can constrict or dilate to control blood flow.	Cannot constrict.
Usually deeper placed.	Usually more superficial (nearer the skin).
Do not collapse when empty.	Collapse when empty.
Carry fully oxygenated blood (except pulmonary artery).	Carry partially deoxygenated CO_2 laden blood (except pulmonary vein).
Blood flows with jerks and under great pressure.	Blood flows continuously and under very little pressure.

Capillary

Very narrow tube (8 micrometers in diameter) The wall consists of a single layer of squamous epithelial cells and has no muscles.

Venules

Smallest united common branch of capillaries.

Functions: Allows outward diffusion of oxygen and inward diffusion of carbon- dioxide with power of vasodilation and vasoconstriction that affect the blood supply to the body parts. Allow inward and outward diffusion of substances like glucose, amino acids, urea etc. Allow leucocytes to squeeze out through capillary walls.

The Heart

- Right in the centre between two lungs and above diaphragm.
- The narrow end of the roughly triangular heart is pointed to the left side.

Covering and Chambers

- About the size of closed fist, protected by a double walled membranous covering called pericardium.

- Pericardial fluid reduces friction during heartbeat and protect from mechanical injuries.
- Upper two atria have thinner walls because major function is to receive blood from the body and pump it into the very next ventricles.
- Lower two ventricles have thick walls because they have to pump blood lo long distances like up to the toes or up to the brain against gravity.

Blood Vessels

- Anterior vena cava (superior vena cava) brings deoxygenated blood from the anterior or upper region of the body including head, chest and arms.
- Posterior vena cava (inferior vena cava) brings the deoxygenated blood from the posterior or lower region of the body including abdomen and legs.
- Four pulmonary veins (two from each lung) bring oxygenated blood lo the left atrium.

Blood Vessels Leaving the Heart

- Pulmonary artery arises from the right ventricle and carries deoxygenated blood to the lungs.
- Aorta arises from the left ventricle and carries oxygenaled blood to supply it to all parts of the body.
- Two coronary arteries (right and left) arise from the base of the aorta and supply the heart muscles.
- **Myocardial infarction/Heart attack** - Dead of the corresponding area of heart due to blockage of any coronary arteries or their branches.
- **Angina pectoris:** Chest pain due to insufficient supply of blood to the heart muscle.

Valves

- **Right atrio-ventricular/tricuspid valve** - Between the right auricle and the right-ventricle.
- Chordae lendinea arises from the papillary muscle of the ventricular wall and helps the flaps in handling position.
- **Left atrio ventricular/bicuspid/mitral valve** - Between the left auricle and the left ventricle.
- **Pulmonary semilunar values (three)** - At the opening of the right ventricle into pulmonary artery.
- **Aortic semilunar valves (three)** - At the point of origin of aorta from the left ventricle.

Circulation of Blood in the Heart

Contraction of two atria starts and ventricles are dilating and empty so blood passes from the atria into the ventricles. ⇒	Ventricles contract and atria relax, cuspid valve protect the return of blood. The ventricular blood enters the pulmonary artery and the aorta respectively. Semilunar valve face away from the ventricles so these become flat and get a clear passage in between. ⇒	Ventricle dilates the blood from pulmonary artery and the aorta tends to return, blood fills the pockets of valves and the closes these passage.

Heartbeat

Each full beat of the human heart lasts for about 0.85 seconds. The sequence of events in the heart beat are called cardiac cycles.

Heart Sound

LUBB sound when atrio-ventricular valves get closed sharply at the start of ventricular systole.

DUP sound when semilunar valves get closed at the beginning of ventricular diastole.

The pulse

- The alternate expansion and elastic recoil of the wall of artery during ventricular systole.
- Counting of the pulse is indirectly the counting of the heartbeat E.g., Pulse rate in radial artery (wrist).

Blood Pressure

- The pressure which the blood flowing through the arteries exerts on their walls.
- Systolic Pressure: Upper limit, the pressure when fresh flood is pushing through the artery in ventricular contraction of the heart.
- Diastolic Pressure: Lower limit, the pressure when the wave has passed over.
- Normal blood pressure: 100-140 mmHg (systolic) and 60-80 mmHg (diastolic)
- Hypertension: Rise in blood pressure above 140/90 mmHg.

Pacemaker

- Sino atrial node (SAN): Located in the walls of right auricle. Impulse is relayed to the ventricles by special conducting fibres.
- Atrio-ventricular node (AVN): Located near the interauricle septum near the tricuspid value.
- Bundle of HIS: It begins from AVN and extends to the interventricular septum. It consists of Purkinjie fibres together all these forms a system which creates an impulse and conducts it to every part of the heart. If any trouble hapens in the heart, the artificial pacemaker may fixed it.

Double Circulation-Pulmonary and Systemic

Blood flows twice in the heart before it completes one full round.

Pulmonary circulation: It starts from the right ventricle and splits into two branches, each entering a lung. In the lungs, blood gets oxygenated. The pulmonary veins then collect this oxygenated blood and carry it back to the left atrium of heart.

Systemic circulation: It begins with aorta which arises from the left ventricle. The aorta branches into multiple arteries that supply oxygen-rich blood to various body parts and tissues. Deoxygenated blood is then collected by veins and returned to the heart through the vena cave, emptying into right atrium.

Hepatic Portal System.

- The veins originating from the stomach and intestines first merge into hepatic portal vein, which carries nutrient-rich blood lo the liver. Unlike typical veins, the hepatic portal vein branches into a network of capillaries within the liver. These capillaries then reunite to form the hepatic vein, which drains into posterior vena cava, returning blood to the heart.
- The liver acts as a regulatory organ, controlling the quantity of nutrients entering the general bloodstream. For example, excess glucose is stored as glycogen for later use.

9 CHAPTER

The Excretory System

Excretion

Excretion is the removal of harmful and unwanted substances, especially nitrogenous wastes from the body.

Substances to get rid off

1. CO_2 and H_2O: CO_2 is eliminated through the lungs. H_2O becomes the part of the rest of the water in the body.
2. Nitrogenous metabolic wastes: Urea is excreted through the kidneys.
3. Excess salts: Salts are excreted through the kidneys.
4. Water: The excess quantity of water is removed through the kidneys.
5. Bile pigments: Some of the bile pigments are excreted in the urine.

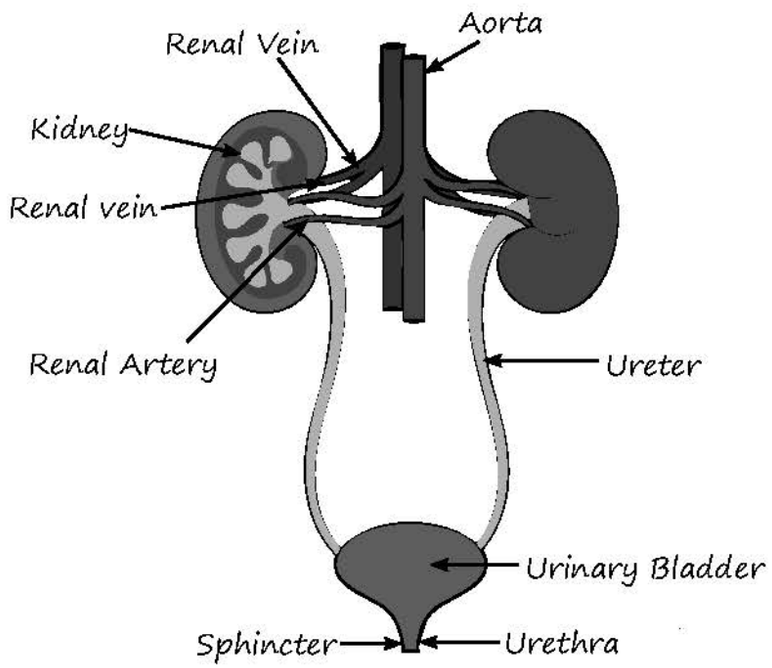
The Excretory Organs

1. Kidneys: Primary excretory organs.
2. Sweat glands: Excrete sweat which contains nitrogenous waste.
3. Lungs: CO_2 is expelled by the lungs.

The Urinary System

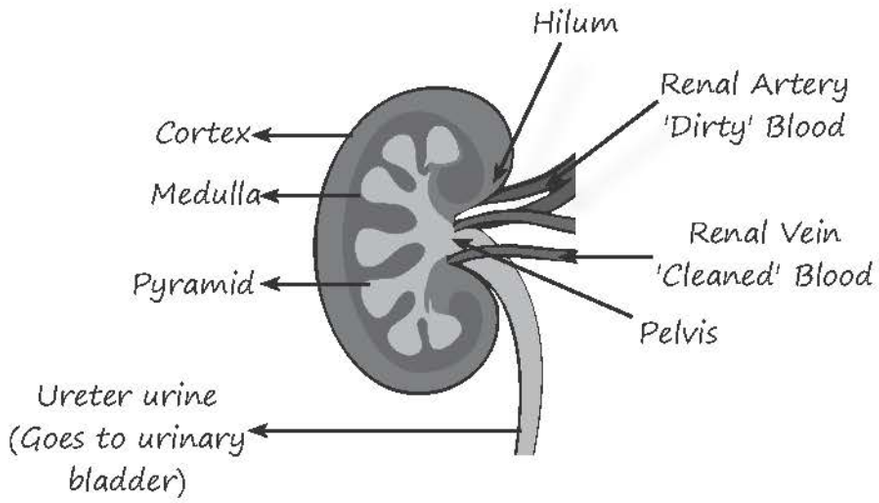
The urinary system consists of

1. 1 pair of kidneys
2. 1 pair of ureters
3. Urinary bladder
4. Urethra

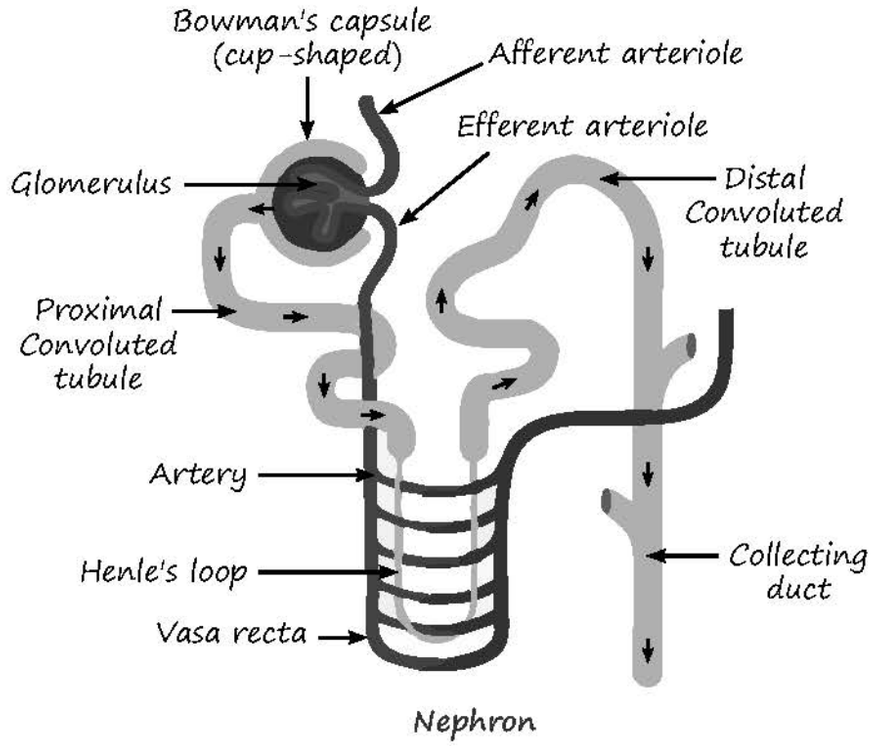


1 Pair of kidneys	❑ Dark red, bean shaped, 10 cm long 6 cm wide. ❑ The right kidney is slightly lower than the left kidney. ❑ The renal artery supplies oxygenated blood to the kidneys. ❑ The renal veins take away deoxygenated blood from the kidneys.
1 Pair of ureters	❑ Ureters are the tube-like structures which arise from notch i.e., hilum of each kidney. ❑ The ureters connect behind the urinary bladder. ❑ The ureters carry the urine produced to the urinary bladder.
Urinary bladder	❑ Muscular sac-like structure. ❑ Stores urine temporarily.
Urethra	❑ Short muscular tube which expels urine out of the body. ❑ It is long in males and very short in females. ❑ The opening is guarded by sphincters which open at the time of urination.

- ❑ The longitudinal section of the kidneys shows two regions-an outer dark cortex and an inner light medulla.
- ❑ The medulla is composed of conical pyramids.
- ❑ The apex of each pyramid i.e., papilla projects into pelvis.



Uriniferous Tubule

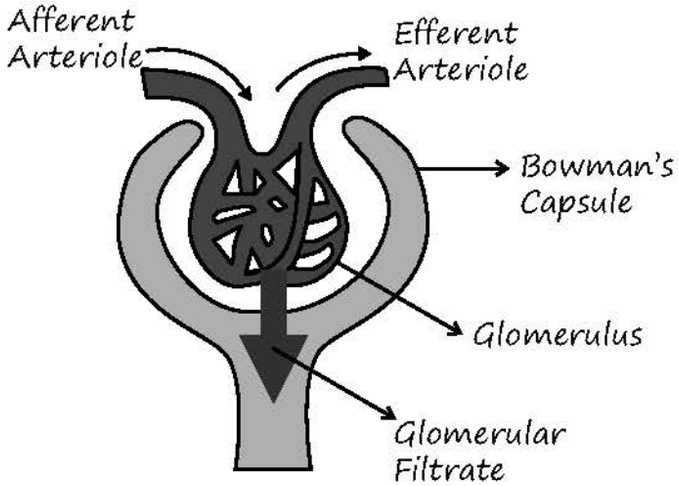


Nephron

The kidneys have an enormous number of uriniferous tubules. They are also known as nephrons, renal tubules or kidney tubules. Uriniferous tubules are the structural and functional units of the kidney.

Bowman's capsule

- It is a thin walled, cup like depression.
- A knot-like mass of blood capillaries called glomerulus is located in the concave depression of the Bowman's capsule.
- The Bowman's capsule and the glomerulus together are called malpighian corpuscles or renal corpuscles.



Proximal Convoluted Tubule (PCT)

- It is also known as the first convoluted tubule.
- Nearly all the essential nutrients and 70–80% of electrolytes and water are reabsorbed by this segment.
- PCT lies in the cortex region of the Kidney.

Loop of Henle

It is the U-shaped middle part of the tubule.

Distal Convoluted Tubule

- It is the end part of the kidney tubule.
- It opens into the collecting duct.
- Conditional reabsorption of Na^+ and water takes place on this segment.

Blood Supply to the Kidney

Dorsal Aorta

↓

Renal Artery

↓

Afferent arteriole

↓

Glomerulus

↓

Efferent arteriole

↓

Secondary capillary

↓

Vasa recta

↓

Renal vein

↓

Posterior vena cava

Formation of Urine

The process of urine formation occurs in three major steps:

(A) Ultra filtration

- Due to the hydrostatic pressure built in the glomerulus, the liquid part of the blood filters out from the glomerulus and passes into the Bowman's capsule.
- This filtration under extraordinary force is called ultra filtration.
- The filtrate is known as glomerular filtrate.

(B) Reabsorption

- The glomerular filtrate passes down the tubule, water and other substances required by the body are reabsorbed. This entire process is called selective reabsorption.
- K^+ ions and certain substances such as penicillin are passed into the urine through the DCT.

(C) Tubular Secretion

The cells of the walls of DCT are involved in bringing back into the renal tubule K^+ ions and other substances; hence, this process is known as tubular secretion.

Parts of the Renal Tubule	Activity
1. Glomerulus	Ultrafiltration
2. Bowman's Capsule	Receives glomerular filtrate
3. Proximal Convoluted tubule	Reabsorbs water, glucose, Na^+ and Cl^- ions.
4. Loop of Henle	Absorption of H_2O & Na^+ ions
5. Distal Convoluted tubule	Reabsorption of Cl^- ions of H_2O

Urine Excretion

- The filtrate left after reabsorption and tubular secretion is called urine.
- The urine passes from the collecting duct to the pelvis of the kidneys. From there, it is sent to the urinary bladder through the ureters.
- By relaxing the sphincters present at the opening of the urethra, the urine is expelled from the body. This process is known as micturition or urination.

Physical Properties of urine

- Colour: Yellow, due to the pigment urochrome.
- Volume: 1–1.5 L (However, the volume can vary depending on the water intake of the person).
- pH: 5 to 8, slightly acidic.
- Odour: Faint smell ammonia like due to bacterial activity.
- Specific gravity: 3 to 1.035
- Urine is made of 95% water and 5% of solid wastes.

Abnormal Constituents of Urine

1. Haematuria: Due to infection in the urinary tract, kidney stone or tumour, blood passes out with the urine.
2. Glycosuria: Excess glucose passes out with urine due to diabetes mellitus.
3. Due to anaemia, hepatitis or liner cirrhosis bile is passed out in the urine.

Regulation of Urine Output

- The water in the urine is controlled by Antidiuretic hormone (ADH).
- Reduction in the secretion of ADH results in loss of water through the urine known as diuresis.
- Substances which increase the production of urine are called as diuretics.

Osmoregulation

- While moving urea from the blood, the kidneys also regulates the composition of the blood i.e., the water and salt concentration in the blood. This is. called as osmoregulation.
- Drinking enough water helps the Kidneys to function properly.
- In summer, we lose a considerable amount of water through perspiration which makes the urine thicker and concentrated. Hence the Kidneys have to reabsorb more water from the urine.

Artificial Kidney

- If one kidney is damaged or removed the other kidney is capable of fulfilling the excretory needs of the patient.
- However, the failure of both the kidneys would lead to death.
- Such a patient undergoes dialysis. The dialysis machine is an artificial kidney in Which the patient's blood is led from the radial artery through the machine Where excess salts and urea are removed.
- The purified blood returns to a vein in the same arm.

10 CHAPTER

The Nervous System

Introduction

(i) Living organisms have the ability to respond and react to their surrounding environment.

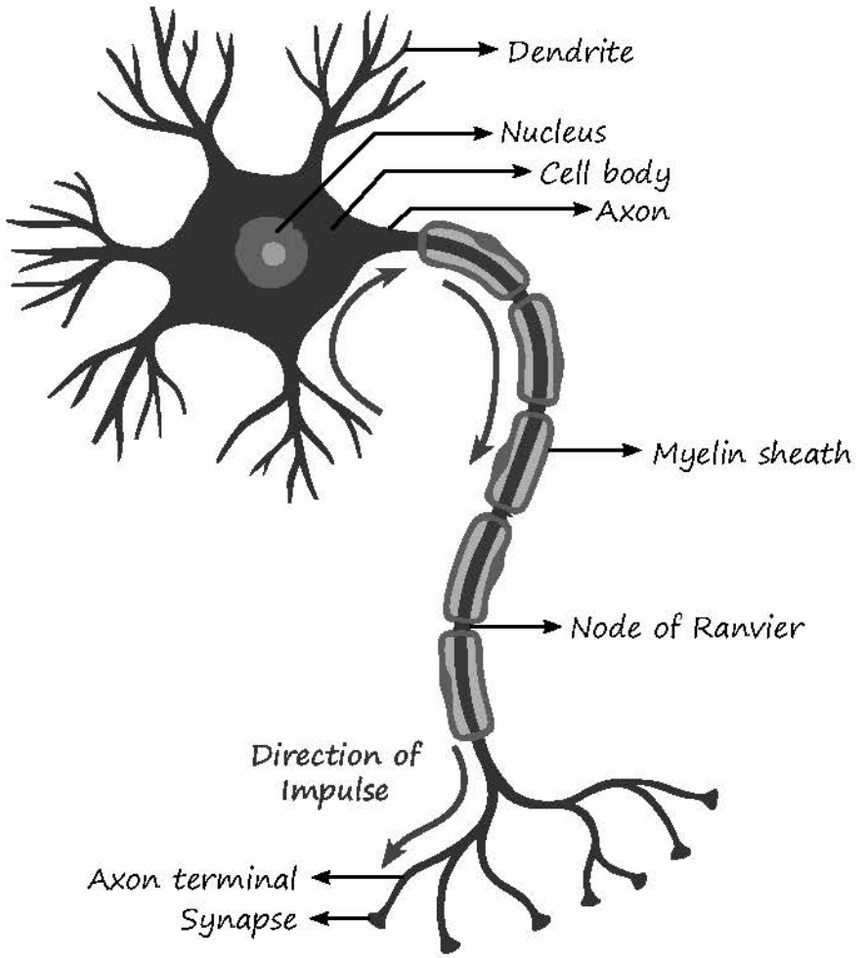
(ii) Plant responses are regulated by chemical substances called phyto-hormones.

(iii) The organ system in the animal body that helps in coordination and integration of body activities is the nervous system.

(iv) Nervous system of animals are formed by nervous tissue, which is composed of neuron.

Neuron

1. Living organisms have the ability to respond and react to their surrounding environment.
2. Neuron is the longest cell in the body.
3. A neuron has three components cell body, dendrites and axon.
4. A cell body is like a typical cell containing nucleus and granular cytoplasm. Dendrites are short and branched cytoplasmic projections.
5. Axon is a single, elongated fibre. Both axons and dendrite arise from the cell body.
6. There are two types of axons, namely myelinated (Covered by myelin sheath) and non-myelinated axon. The axolemma and neurolemma are continuous sheath. There is no centrosome in the cyton because the nerve cells have the ability to divide.



Some Basic Terms

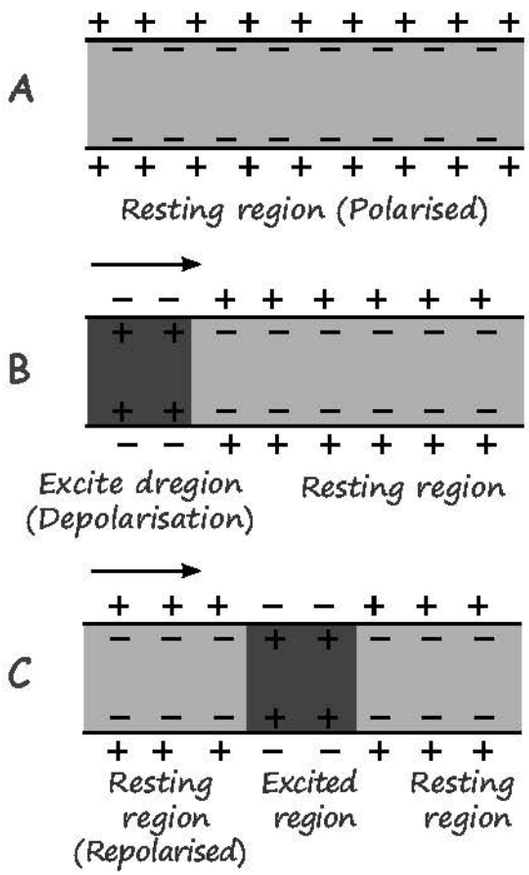
1. **Stimulus:** An agent or sudden changes of the external or internal environment which results in a change in an organism or any of its body parts.
2. **Response:** The change in organisms resulting from a stimulus.
3. **Impulse:** A wave of irritability i.e. an electrical disturbance, which sweeps over the nerve cell.
4. **Receptors:** The nerve cells which set up waves of impulses towards the central nervous system on receiving the stimulus.
5. **Effectors:** Muscles or glands which contract or secrete substances on receiving an impulse from the brain or the spinal cord.

Synapse

1. A synapse is the point of contact between the terminal branches of the axon of a neuron and the dendrites of another neuron.
2. As the nerve impulse reaches the axon terminal of one neuron, the neuro transmitter acetylcholine is released by the bulbs present in the axon.
3. Acetylcholine is then brokendown by an enzyme to ensure that the synapse is ready for the transmission of the next nerve impulse.

Transmission of Nerve Impulse

- In the resting condition, the outer side of the nerve fibre carries a positive charge, i.e., more Na^+ ions outside the axon membrane. This is called the polarised state or polarisation of the nerve fibre.
- On the stimulation, the axon membrane at the site of stimulation becomes more permeable to Na^+ ions. Thus, Na^+ ions move inwards and result in loss of polarisation which is known as depolarised state or depolarisation of the nerve fibre. Such a region of the nerve fibre is known as the excited region.
- The point of depolarisation becomes the stimulus for the next region of axon membrane which in turn become depolarised.
- The previous region of the membrane becomes repolarised due to the active transport of Na^+ ions to the outside of the membrane.

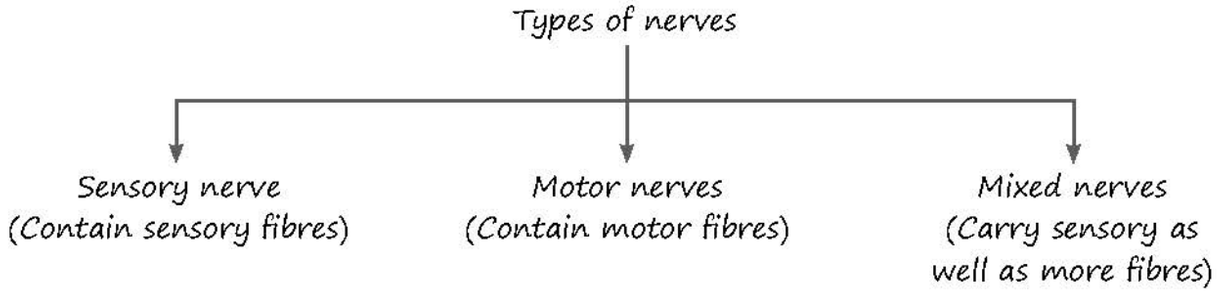


Types of Neurons

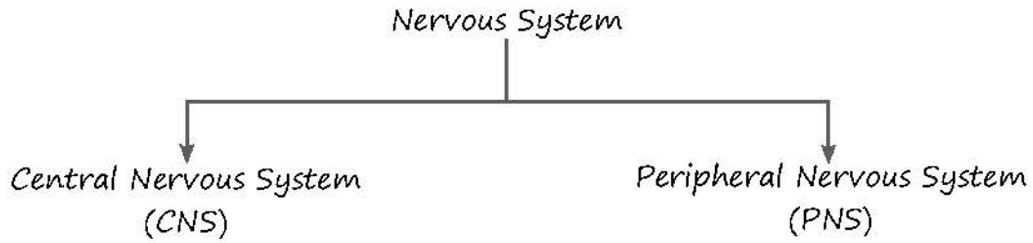
1. **Sensory neuron:** Convey the impulse from the receptor (sense organs) to the main nervous system (the brain or spinal cord).
2. **Motor neuron:** Carry impulse from the main nervous system to an effector i.e., muscle or gland.
3. **Associated Neurons:** They interconnect sensory and motor neurons.

Types of Nerve

A nerve is a bundle of nerve fibre (axons) of separate neurons enclosed in a tubular sheath. Ganglia are an aggregation of the nerves (cell bodies) from which the nerve fibres may rise or enter.



Division of the Nervous System



The Central Nervous System

The central nervous system includes the brain and spinal cord.

The Brain

1. The human brain is well protected inside the cranium or the skull. In adults, it weighs about 1.35 kg.
2. It is protected by three meninges - **dura mater, arachoid** and **pia mater.**
3. The space between the covering membranes, central spaces of the brain and the central canal of the spinal cord cosists of **cerebrospinal fluid** which protects the brain from shocks.

Three Primary Regions of the Brain

1. Forebrain:
   - The cerebrum is the centre of intelligence, memory, consciousness, will power and voluntary actions.
   - The thalamus relays pain and pressure impulses to the cerebrum.
   - The hypothalamus controls the body temperature and the activity of the pituitary gland.
2. Mid brain:
   - This small tube like part is responsible for reflexes involving the eyes and ears.
3. Hind brain:
   - The cerebellum coordinates muscular activity and balance of the body.
   - The pons carry impulses from one hemisphere to the other hemisphere and coordinate muscular movements on both sides of the body.
   - The medulla oblangata controls the activities of internal organs, heartbeat, breathing etc.

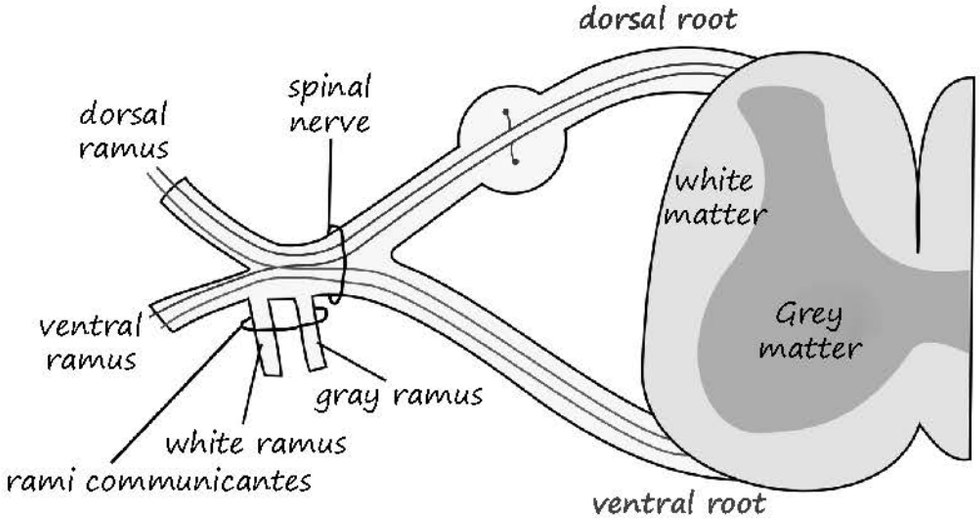
Cerebrum	❑ The largest portion of the brain. ❑ It is divided into two cerebral hemispheres connected to each other by the corpus callosum. ❑ The cortex contains cell bodies of the neuron and is greyish in colour, hence it is called grey matter. ❑ The grey matter has many folds (i.e. gyri) and grooves (i.e. sulci)
Cerebellum	❑ The medulla consists of the axons of the nerve fibres and is called white matter. ❑ Located at the base of the cerebrum in a median section, its white matter appear like a branching trees.
Medulla oblongata	❑ Located at the base of the skull. ❑ It is roughly triangular. ❑ It is continued behind as the spinal cord. ❑ Injury to the medulla oblongata results in death.

The Spinal Cord

1. It lies within the neural canal of the vertebrae.
2. The grey matter is one side and the white matter is on the outer side of the spinal cord.
3. Similar to the brain, it is covered with three meninges dura mater, arachnoid and pia mater.
4. Functions:
   (i) Responsible for the reflexes below the neck.
   (ii) Conducts sensory impulses from the skin and muscles to the brain.
   (iii) Conducts motor responses from the brain to muscles of the trunk and limbs.

Peripheral nervous system

The peripheral nervous system consists of nerves which carry impulses to and from the central nervous system.



Sympathetic Nervous System	Parasympathetic Nervous System
❑ Nerves arise from the spinal cord between the neck and waist regions.	❑ Located anteriorly in the head and neck while posteriorly in the sacral region.

❑ Sympathetic and parasympathetic systems have functions which are opposite to each other.

Opposite Effects of the Two Systems

Organ	Sympathetic system	Para sympathetic system
Heart	Accelerates heart beat	Retards heat beat
Intestines	Peristalsis decreased	Peristalsis increased
Lacrimal (Tear) Gland	Stimulates secretion	Inhibits secretion
Pupil of eyes	Dilation	constriction

Reflexes: The reflex action is an automatic, quick and involuntary action in the body brought about by a stimulus.

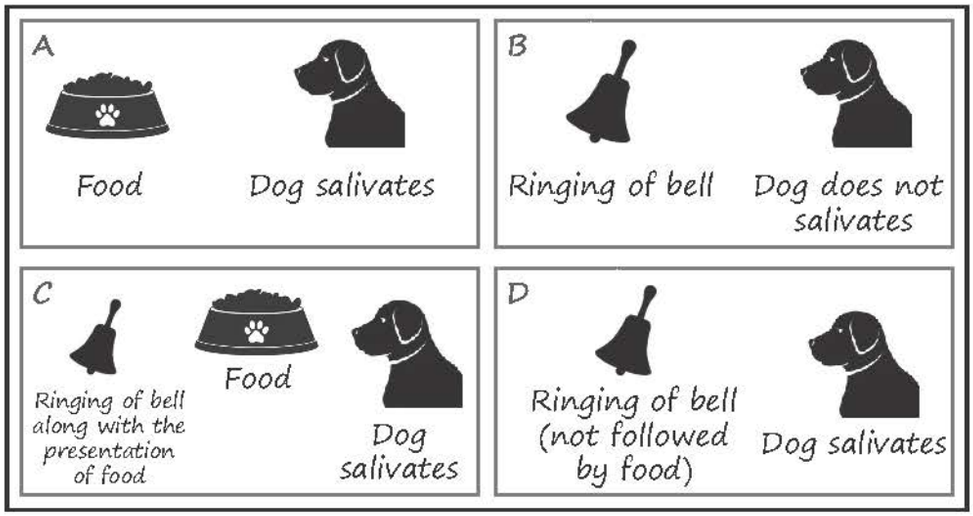
Differences between reflexes: Involuntary actions and voluntary actions.

Reflexes (Involuntary actions)	Voluntary actions
❑ Initiated by some stimulus such as touch, pain, pressure, heat, light etc.	❑ Initiated by a willing thought.
❑ Command originate in the spinal cord, autonomic nervous system and a few in the brain as well.	❑ Commands originate in the brain.

Types of Reflexes

Natural (Inborn) Reflex	Conditional (Acquired) Reflex
❑ Previous experience or learning is not required.	❑ Develops during lifetime due to experiences.
❑ Similar in all humans	❑ Differs from individual to individual as it depends on experience and learning.
❑ Salivation peristalsis, swallowing.	❑ Salivation just by the smell of food.

Pavlov's Experiment



Nervous pathways in reflexes - A reflex action must be quick to give response. Therefore, the pathway for receiving and sending information must be short. A reflex arc can be represented as follow: Stimulus → receptor in the sense organs → afferent (sensory) nerve fibre → CNS (spinal cord/brain) → efferent (motor) nerve fibre →muscle/gland → Response.

11 CHAPTER

Sense Organs

- The sense organs enable us to be aware of the condition of the environment.
- A receptor is any specialised tissue or a cell sensitive to a specific stimulus.

Mechanoreceptor

Receptors of touch, i.e. pressure on the skin due to mechanical change.

Chemoreceptors

Receptors of taste of the tongue and smell of the nose due to chemical influence.

Photoreceptor

Receptor of light present in rods and cones of the retina of eyes.

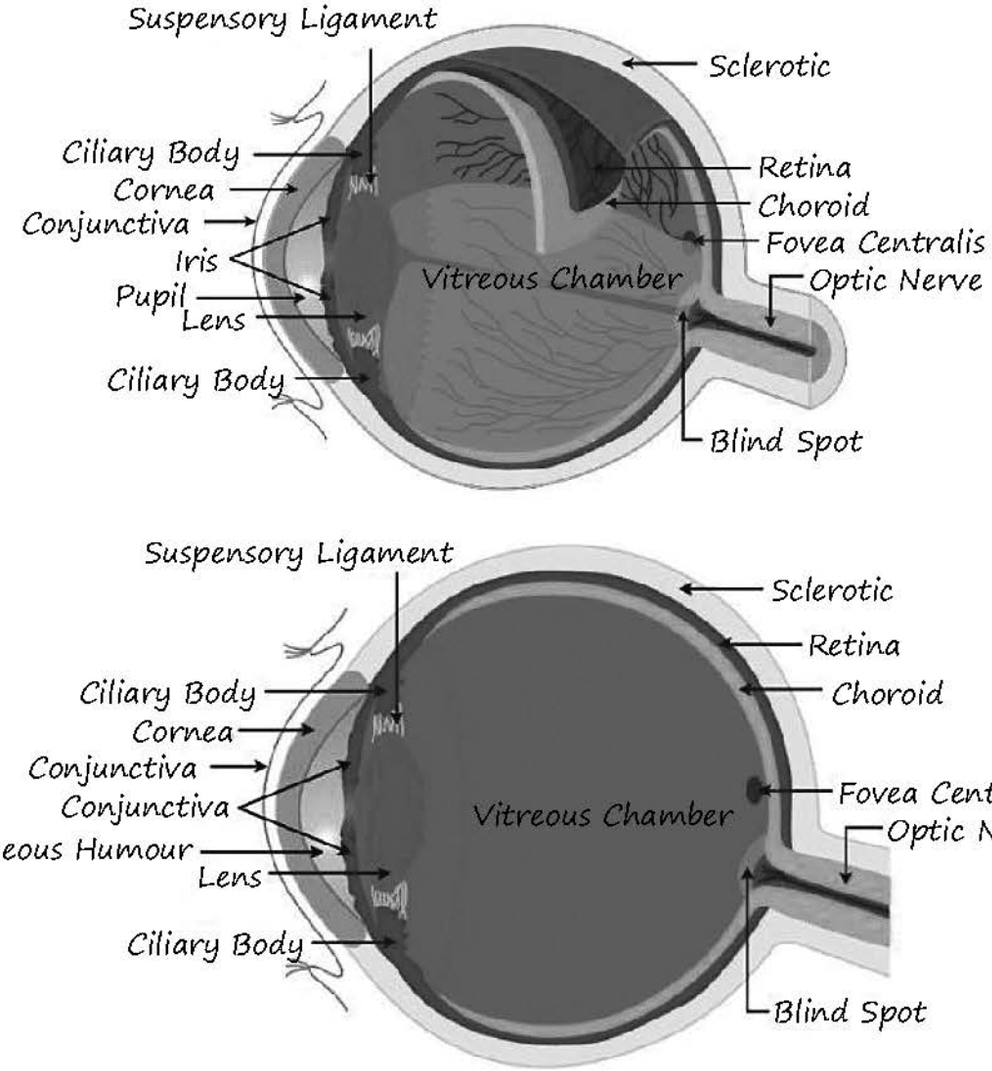
Thermoreceptors

Heat and cold receptors in the skin due to change in temperature.

The Eyes

- The two eyes are located in the deep sockets called orbits.
- The upper and lower moveable eyelids protect the front surface of the eyes.
- The eyebrow prevent the raindrops or the tricklings perspiration from getting into the eyes.
- Tear glands are also known as lacrimal glands.
- There are 6–12 tear glands.
- The functions of the tear gland are: Lubricate the surface of the eye, and wash away the dust particles.
- Tear glands posses antiseptic properties because of the enzyme lysozyme which effectively eliminates germs.
- A thin membrane which covers the, entire front part of the eyes is called conjunctiva.
- Due to viral infections of the conjunctiva, we suffer from eyes disease called conjunctivitis.

Structure of the Eyeball



- The wall of the eyeball is composed of the following three concentric layers:

1. **Sclerotic Layer (Outer Layer)**
   - The white visible portion of the eyeball is nothing but the sclera.
   - The sclera covers the coloured part of the eye, i.e., the cornea.
2. **Choroid layer (Middle layer)**
   - It is richly supplied with blood vessels to provide proper nourishment.
   - The choroid expands in the front of the eye to form a ciliary body.
   - The smooth muscles in the ciliary body alter the shape of the lens.
   - Iris is also a part of the choroid.
   - The iris partially covers the lens. It leaves a circular opening in the centre called a pupil.
   - The muscles of the iris regulate the size of the pupil. Thus, the pupil regulates the amount of light entering the eyes.

3. Retina (Inner layer)
   - It possesses two types of sense cells → rod and cones.
   - The rod cells are sensitive to dim light and do not respond to colour.
   - The cone cells are sensitive to bright light and are responsible for colour vision.

Comparison Between Rods and Cones

Rods	Cones
These cells are more in number.	These cells are less in number.
It is located at the periphery of the retina.	It is located in the centre of the retina.
They contain the pigment rhodopsin.	They contain the pigment iodopsin.

Yellow Spot and Blind Spot

Yellow spot	Blind spot
It contains a maximum number of sensory cells, particularly cones.	It does not contain any sensory cells.
This is the region of colour vision and the brightest vision.	This is the region of no vision.

Lens

- It is transparent, biconvex and crystalline.
- It is held by a suspensory ligament which attaches the lens to the ciliary body.

Aqueous and Vitreous Chambers

- The lens divides the inner cavity of the eyeball into two chambers.

Aqueous Chamber	Vitreous Chamber
The front chamber between the lens and the cornea.	Large chamber behind the lens.
It is filled with a clear, watery liquid called aqueous humour.	It is filled with transparent jelly like fluid called vitreous humour.
It refracts light.	It protects the retina and its nerve endings.

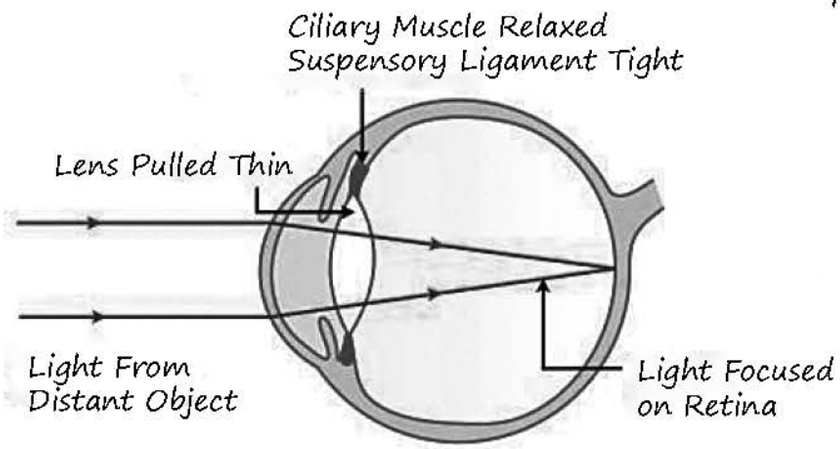
Four Major Steps in Seeing an Object

- Light rays reflected from the object enter the eyes through transparent structures.

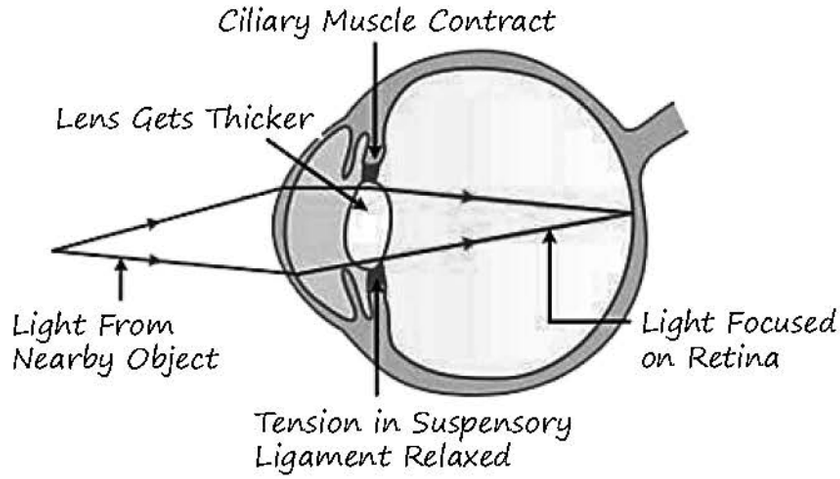
- First, the curvature of the cornea converges the rays to some extent, and then the lens converges them further.
- The image on the retina is inverted and real.
- The light energy produces chemical changes in rods and cones which send the nerve impulse. This nerve impulse is sent to the cerebrum through the optic nerve. The cerebrum gives the sensation of sight.
- The brain interprets the inverted image on the retina, and the object is seen upright.

Accommodation Vision

- The process of focusing the eyes at different distances is called accommodation. This is brought about by change in the curvature of the lens.
- For distant vision, the lens is more flattened. The lens remains stretched suspensory ligaments.



- For near vision, the lens becomes convex and rounded.



- When the ciliary muscles contract, they pull the ciliary body forward. This action releases the tension of suspensory ligaments, causing the lens to become convex and rounded.

Light and Dark Adaption

Dark adaption

When we move from a bright place to dark one, we might have trouble seeing objects for a bit. This is called dark adaption.

Light adaption

When we move from a dark place to a bright one, we might experience a moment of dazzling effect for a short period of time. This is known as light adaptation.

Common Defects of the Eyes

1. **Myopia (Short-sightedness):** The near objects are seen clearly, but distant objects appear blurred. The lens is too curved. Myopia is corrected by suitable concave lens.
2. **Hyperopia (Hypermetropia/long-sightedness):** The near objects are seen clearly, but distant objects appear blurred. The lens is too curved. Myopia is corrected by suitable concave lens.
3. **Astigmatism:** Some parts of the object are seen in focus, while other appears blurred.
4. **Presbyopia:** Observed in older people. The near objects cannot be seen clearly.
5. **Cataract:** The lens turn opaque and the vision is reduced.
6. **Night blindess:** Difficulty in seeing in dim light due to non-formation of rhodopsin in rod cells.
7. **Colourblindness:** Colour blind people cannot distinguish between certain colours such as red and green.
8. **Squint:** The eyes coverge leading to cross eye.
9. **Corneal opacities:** The cornea of patients get scarred and turns opaque and non-functional which can cause minor irritation, vision problems and ever blindness.

Stereoscopic Vision

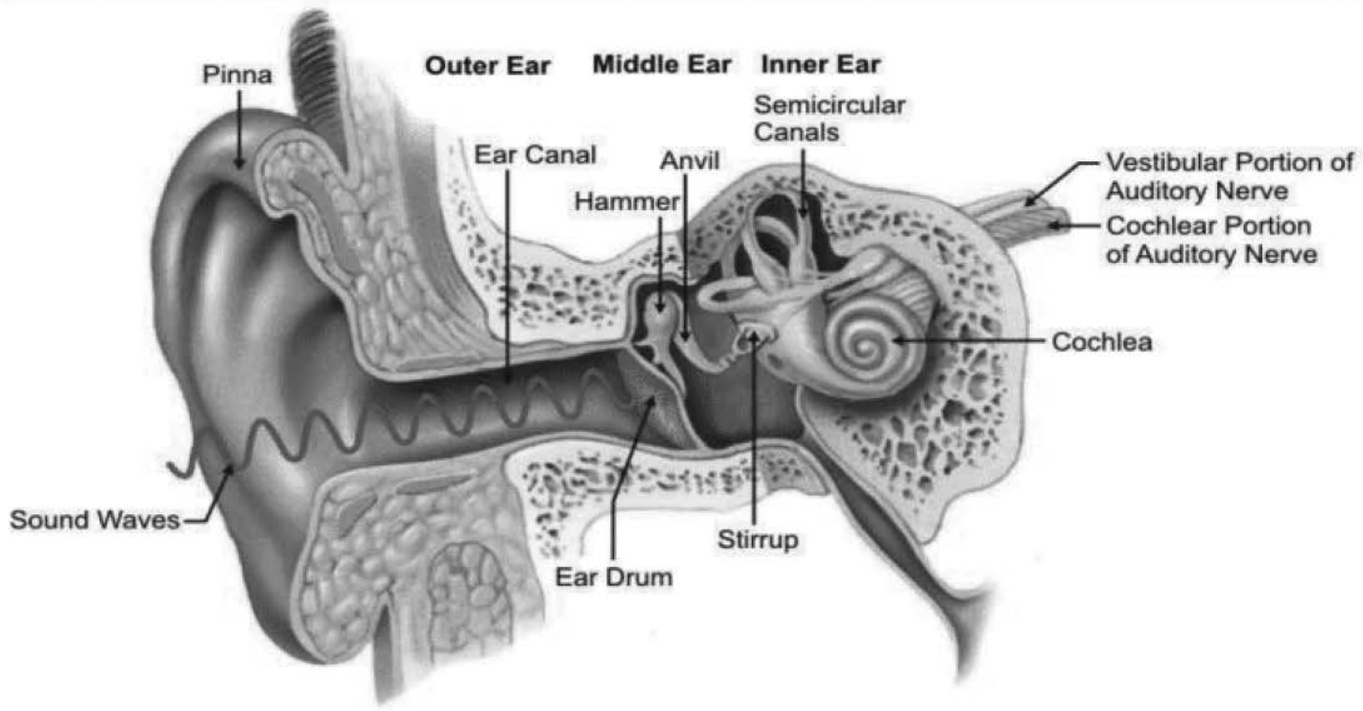
Humans, monkeys and apes are able to perceive depth or the relative distance of objects because they simultaneously focus on an object with both eyes. This overlapping of images from both eyes creates a three-dimensional effect.

After Images

When one looks at a brightly coloured objects and then looks at a dark surface, an image of the object in the same colour will persist. This is known as persistence image or after-image.

The Ear

The human ear has the three following main divisions:



Outer Ear

It consists of pinna/auricle and auditory canal.

Middle Ear

- *It contains three ear ossicles-malleus (hammer), incus (anvil) and stapes (stirrup) and the eustachian tube.*
- *The eustachian tube connects the cavity of the middle ear with the throat.*

Inner Ear

- *It is also known as membranous labyrinth.*
- *It contains cochlea and the semicircular canals.*
- *The cavity of cochlea is divided into three parallel canals. The middle canal consists of the organ of corti which is responsible for hearing.*
- *The cavity of cochlea is divided into three parallel canals. The middle canal consists of the organ of corti which is responsible for hearing.*
- *The ends of the semicircular canals widen to form an ampulla.*
- *The ampulla contains sensory cells.*
- *The short stem joining the bases of semi-circular canals to the cochlea is caled the vestibule.*
- *The vestibule contains two sacs-utriculus and sacculus.*

Functions of the Ear

Two functions of the ears are hearing and body balance.

Hearing

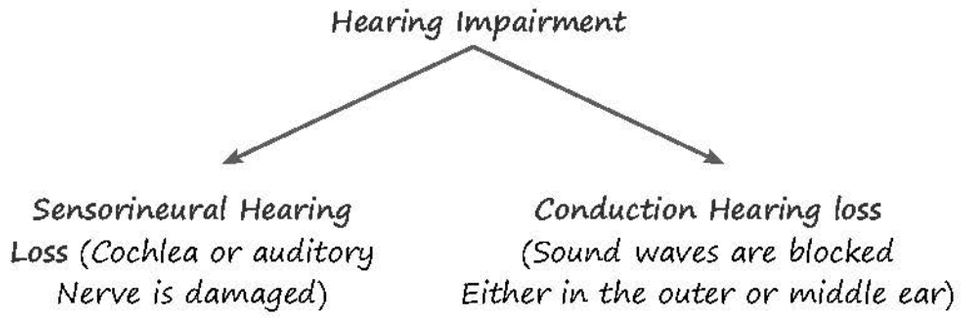
The pinna gathers sound waves and guide them into the external auditory canal. These waves then reach the eardrum, causing it to vibration.

1. Equalises the air pressure on either side of the ear drum.
2. Sets the three ear ossicles into vibration.
   - Transmits vibration to the membrane of the oval window.
   - Sets the fluid in the cochlear canal into vibration.
   - Sensory cells of cochlea transmit impulses to the auditory nerve.

Body Balance

- The sensory cells located in the semi circular canals are concerned with dynamic equilibrium, which occurs when the body is in motion.
- The sensory cells in vetriculus and sacculus are concerned with static equilibrium, i.e., when the body is stationary.

Hearing Impairment



12 CHAPTER

The Endocrine System

Endocrine System

Endocrine gland activate each others and work as a system of organs.

Hormone

A secretion from some glandular part of the body, which is poured into blood and which acts on the target organs or cell of the same individual, bringing about coordination between distant parts of the body.

Characteristics of Hormones

- Directly secreted into the blood, not in lymph.
- Regulate the physiological processes by chemical means like affect the enzyme systems.
- Act on the target organs or cells and similar influence in other species.
- Produced in small quantities and or biologically active.
- Chemically some are peptides (insulin), amines (adrenaline) and steroids (testosterone).
- Their excess or deficiency leads to serious consequence and not stored in the body.

Difference between Hormonal Control and Nervous Control of the Body

Hormonal Control	Nervous Control
1. Usually slow.	1. Immediate/Rapid
2. Transmitted chemically through blood.	2. Transmitted electro-chemically through the nerve fibres and chemically across synapses.
3. Affects different organs (widespread in the body).	3. Affects only the particular muscles or the gland (local).
4. Effect in short term or long lasting.	4. Effect only short-lived.
5. Can affect growth.	5. Cannot affect growth.
6. Can bring about specific chemical changes and regulates metabolism.	6. Does not influence chemical changes and cannot regulates metabolism.
7. Cannot be modified by learning from previous experience.	7. Can be modified to some extent by learning from previous experience.

Difference between Exocrine Gland and Endocrine Gland

Exocrine Gland	Endocrine Gland
❑ Glands with ducts as their secretory products are released into the ducts.	❑ Duct less glands as their secretory products are released into the blood stream.
❑ Excretory products are enzyme and sweat etc.	❑ Excretory products are hormones.
❑ Eg.: Sweat gland, mammary gland and lacrimal gland.	❑ Adrenal gland, thyroid gland etc.

Types of Endocrine Glands in the Human Body

- ❑ Adrenal Gland
- ❑ Thyroid
- ❑ Parathyroid gland
- ❑ Pancreas
- ❑ Pituitary gland
- ❑ Thymus gland

Endocrine Gland	Hormones Secreted	Actions	Hypersecretion	Hyposecretion
Adrenal Gland	Adrenal Medulla (Emergency hormone and fight flight and fright (3F) hormone)	❑ Stimulate heart beat blood glucose and uterine contraction during labour. ❑ Increasing the clotting capacity of blood. ❑ Water and electrolyte (Na^+ and K^+) balance. ❑ Synthesis of glycogen in liver, protein breakdown increased WBC etc. Regulate fat and protein metabolism.		
	❑ Adrenal cortex. mineralocorticoid (Aldosterone) ❑ Glucocarticoids (cortisal, cartic osterone) ❑ Sex corticoids (Adrenal androgens)		❑ Neuromuscular disturbance cushing syndrome. ❑ Adrenal virilism	❑ Nervous disorder addison's disease. ❑ Gynecomasita

Pancreas (Heterocrine gland)	❑ Islets of hanger-hans cells. ❑ Beta cells - Insulin ❑ Alpha cells - Glucagon ❑ Delta cell - Somatostalin	❑ Lowers of blood sugar level by promoting glucose utilization and deposition of extra glucose in the liver and muscles. ❑ Rises blood sugar level by stimulating liver to convert glycogen into sugar. ❑ Inhibits the secretion of insulin glycogen.	❑ Insulin shock (un consciousness) Nerve cell starvation.	❑ Diabetes Millitus (frequent urination with sugar in urine thirsty)
Thyroid	❑ Thyroxine	❑ Regulating the normal basal metabolism of the body, growth and differentiation. ❑ Metamorphism in insects life cycle.	❑ Ex ophthalmic Goitre/Grove's disease	❑ Simple goitre Gretinis ❑ Myxodema
	❑ Calcitonin	❑ Promotes movement of calcium ions from blood to bones.		
Pituitary ❑ Anterior Pituitary:	❑ Growth hormone (Somatotrophin; GH)	❑ Body growth (skeleton)	❑ Gigantism (childhood) Acromeglay (Adult)	❑ Dwarfism (Childhood)
	❑ Thyroid Stimulating Hormone (TSH)	❑ Stimulates thyroxine secretion		
	❑ Adrenocorticotropic Hormone (ACTH)	❑ Stimulates adrenal cortex	❑ Pigmentation of Addison's disease	
	❑ Gonadotrophic Hormones: ❑ Follicle-stimulating Hormone (FSH)	❑ Regulates activities of testes and ovaries		

	❑ Luteinizing Hormone (LH)	❑ In females: Growth of Graafian follicle and estrogen secretion. In males: Spermatogenesis		
	❑ Prolactin	❑ In females: Ovulation, maintenance of corpus luteum, and secretion of progesterone		
❑ Posterior Pituitary:	❑ Vasopressin / Antidiuretic Hormone (ADH)	Regulates absorption of water from kidney tubules, contraction of blood vessels	Elevation of blood pressure	Diabetes insipidus
	❑ Oxytocin	Stimulates smooth muscle contraction, especially of the uterus during childbirth		
	❑ Intermediate lobe: Melanocyte Stimulating Hormone (MSH)	Pigmentation of skin		

Control of Hormonal Secretion/Feedback Mechanism

- **Negative feedback mechanism:**

  Most hormones are controlled by negative feedback, in which the hormone feeds back to decrease its own production. It brings things back to normal state.

  Example: Regulation of blood sugar level, etc.

- **Positive feedback mechanism:**

  It occurs when the original effect of the stimulus is enhanced by the output. The conditions become extreme in this mechanism.

  Example: Uterine contraction during childbirth

13 CHAPTER

The Reproductive System

Reproduction is the formation of new individuals by sexual or asexual means which can repeat the process on their own.

Patterns of Reproduction

Asexual Reproduction	Sexual Reproduction
Involves reproduction through any part of the body. Male and female gametes are absent.	Involves the fusion of male and female gametes.
It is a simple forces of cell division. E.g., Binary fission in amoeba, budding in Hydra.	The fusion of male and female gamete form a zygote from which develops a new individual. This process is known as fertilisation. E.g., Reproduction in multicellular organisms.

Secondary Sexual Characters

Males and females show outwardly differentiating features called secondary sexual characters.

Males	Females
Distribution of hairs on the body and face (Beard and moustache).	Breasts in females.
Muscularity and strong built	Femininity and larger hips.
Deep voice	High pitch voice.

Reproduction in Humans

Male Reproductive System

The male reproductive system consist of the following organs:

1. Testes	**Location of Testes** ❑ Pair of oval glands located in a thin-walled sac called scrotum located outside the body cavity. ❑ The scrotal sac is suspended outside the body. Its temperature is 2-3°C lower than the body temperature. ❑ In abnormal conditions, when the testes do not descend into the scrotum, it results in sterility i.e., incapability of producing sperms.
	Structure of Testes ❑ Each testes is encased in a capsule. The capsule is internally divided into 15-20 lobules. ❑ Each lobule contains seminiferous tubules.
	Seminiferous Tubules Sperms are produced in the seminiferous tubules by the process spermatogenesis.
	Interstital cells (Leydig cells) They produce the male hormone i.e., testosterone.
	Epididymis ❑ The sperm from the seminiferous tubules pass into 12-14 efferent ducts. ❑ These ducts further join to form the epididymis. ❑ The epididymis stores the sperms when they mature.
	Vas deferens (Sperm duct) ❑ It is about 45 cm long. ❑ The two vas deferens loop over the ureters and join the urethra. ❑ Hernia: Due to pressure of the abdomen, the intestines bulges into scrotum through the inguinal canal.
	Ejaculatory duct The vas deferens and the seminal vesicles unite to form the ejaculatory duct.
2. Accessory Gland	**Seminal Vesicles** ❑ A pair of lobulated glands ❑ Produce and alkaline fluid. ❑ It constitute about 60% of the total volume of semen.
	Prostate Gland ❑ It is a bilobed structure. ❑ It pours an alkaline secretion into the semen. This secretion constitute about 13-33% of the semen.
	Bulbourethral Gland (Cowper's Gland) ❑ Two small ovoid glands. ❑ Its secretion serves as a lubricants.
3. Penis	❑ It is the passage for both urine and semen. ❑ It is highly vascular organs with erectile tissues and vascular spaces. ❑ Under the influence of sexual stimulation blood flows in large amount into the vascular spaces of the penis. Which makes it erect. Such condition is called erection.

Semen: It is the mixture of sperms and secretions from the prostate and cowper's gland. It is a milky fluid.

Female Reproductive System

Ovaries	Two ovaries are present in the pelvic cavity on each side of the uterus. Ovaries produce ova by the process of oogenesis.
Oviducts	❑ The two oviducts or fallopian tubes are 12 cm long and attached to the lateral ends of the ovaries. ❑ The open distal end of the ducts is funnel shaped and called oviduct funnel, in fundibulum or ostium. ❑ The infundibulum has finger like projections called finbriae which help to push the release ovum into the oviduct.
Uterus	❑ It is a hollow pear shaped muscular organ. ❑ The internal wall of the uterus is lined by tissue layers. The innermost layer is called endothelium which is lined by ciliated epithelium. ❑ It protects and nourishes the developing embryo.
Vagina	It is muscular tube about 10-15 cm long. It extends from cervix to the outside. It receives the male penis during copulation. It also serves as the passage during childbirth.
Vulva	It is the external female genitalia. It contains independent openings of urethra and vagina.

Ovulation

- Ovulation is the release of the mature ovum by the rupture of the Graafian follicle. The Graafian follicle bursts and release the ovum into the infundibulum of the fallopian tubes.
- The corpus luteum is an endocrine tissues. It secretes two hormones.
- **Oestrogen:** This harmone is secreted by the follicle before ovulation.
- **Progestrone:** It prepares the uterus for the implantation of the embryo.

Menstrual Cycle

The reproductive period of the human female starts from about 13 years and continues till about 45-50 years. This period is marked by a characteristic even repeated almost every month. It is called menstrual cycle.

Menstrual phase

It lasts for 3-5 days. During this phase, the blood is discharged.

Follicular phase

As the follicle grow, it finally matures into the graafian follicle. Oestrogen stimulates the thickening of the uterine endometrium.

Ovulatory phase

On about the 13^{th} or 14^{th} day, the follicle ruptures and the released ovum travels down the fallopian tube.

Luteal phase

It lasts for 15-28 days. The uterus lining thicken further. Empty follicle turns into corpus luteum. The corpus luteum secretes oestrogen and progesterone. If fertilisation does not occur, then the ovum disintegrates and the endometrial lining starts shedding on 28^{th} day.

- If fertilization occurs, then the embryo gets implanted in the uterine wall and there is no menstrual flow.
- The placenta does not allow the entry of germs from the mother to the foetus, but viruses (e.g., HIV) compass through the placenta if the mother is already infected.
- The placenta secretes the hormones oestrogen and progesterone.

Functions of Placenta

1. **From mother to foetus:** Oxygen, Amino acids, Vitamins, minerals, fats and lipids.
2. **From foetus to mother:** Carbon-dioxide, urea and waste products.

Amnion

- It is a sac which develops around the embryo even before the formation allantois.
- The amniotic fluid falls the space between the amnion and embryo.
- The amniotic fluid protects the embryo from physical damage by jerks and mechanical shocks.

Parturition

- The full term of the development of an embryo in the uterus is called gestation.
- In humans, it lasts for about 280 days.
- At the time of birth, the baby is pushed out by contractions of the muscles of the uterus.
- After sometime, the umbilical cord shrinks and can be tied and cut.
- After about 15 minutes, the placenta breaks from the uterus.

14 CHAPTER

Population

Population The Increasing Number And Rising Problems

- Organisms belong to a particular species inhabiting a definite area, which can reproduce among themselves and produce fertile offspring is called population.
- Humans are the most intelligent animal among the species inhabiting the world. He has learnt the way in which he can protect himself from natural calamities, diseases, etc. This leads to undue increase in human population.

World Population (Some important dates and figures)

- 11 July – Population crossed the 5.0 billion mark.
- 1995 – Population was nearly 5.6 billion.
- 12th October 1999 – Population crossed 6 billion.
- 11 July – World Population Day.

Rapid Rise in Population

Major reasons for the rise in population over the past years are as follows:

- Better Healthcare
- Better Medical Aid
- Availability of Food
- Improved Nutrition
- Large-scale Immunization
- More number of children
- Reaching reproductive age

Important Terms Related To Population Are As Follows:

1. Demography:
   - Statistical study of human population with reference to size, density, distribution, and other vital statistics.
2. Population Density:
   - The number of individuals per square kilometre at any given time.
   - Birth Rate: The number of live births per 1000 people population per year.
   - Death Rate: The number of deaths per 1000 people population (Mortality) per year.

- **Growth Rate of Population:** The difference between the birth rate and the death rate.
- When the birth rate exceeds the death rate, the population grows.
- When the birth rate is lower than the death rate, the population declines.
- **Vital Index:** The percentage ratio of natality and mortality in a unit population in a unit time is called the vital index.
- Formula for calculating vital index is:

$$\text{Index} = \left(\frac{\text{Natality}}{\text{Mortality}}\right) \times 100$$

- **Zero population growth:** When the individuals added to the population

Family Welfare

1. Family welfare is represented by an inverted red triangle.
2. Offices and hospitals with this symbol displayed are known as family welfare centres. These centres provide help or advice about family planning.

Three Aspects of Family Welfare Are:

1. Family planning in terms of having a small family.
2. Total welfare of the small family, including the diet and nutrition of the child and the pregnant mother.
3. Subsequent care of children, such as immunisation and oral rehydration therapy, to ensure the survival of young ones.

Methods of Contraception		
1.	Hormonal Method:	Various hormone preparations come in the form of tablets or pills, commonly called contraceptive pills.
2.	Barrier Method:	❑ **Condom:** Used by men, prevents sperm from being deposited in the vagina. ❑ **Diaphragm:** It is fitted deep in the vagina to prevent the entry of sperm into the uterus. ❑ **Spermicidals:** These chemicals are placed in the vagina near the cervix to kill the sperms.

15 CHAPTER

Pollution

- **Pollution:** The addition of any such constituent to air, water, or land which deteriorates the natural quality of the environment.
- Substances that are responsible for pollution are called pollutants.
- **Biodegradable pollutants:** These pollutants are degraded by decomposers. Eg. food, kitchen waste, etc.
- **Non-biodegradable pollutants:** These pollutants are not degraded or decomposed. Eg. DDT, Styrofoam, plastics, etc.

Types of Pollution

Based on the components of the environment being affected:

- Air pollution
- Water pollution
- Soil pollution
- Noise pollution
- Radiation
- **Brick Kilns:** Fire-heated enclosures used for making bricks that are used in construction purposes. The material produced by these enclosures are large quantities of ash and broken unused pieces of brick, which causes air pollution.
- **Water Pollution:** Any change in the water quality which makes it unsuitable for use by humans and other living organisms.

Five major sources are:

- **Household detergents:** Detergents and soaps used for household purposes i.e. for washing clothes and utensils and for bathing release dirty water.
- **Sewage:** Discharge of untreated and partially treated sewage into rivers, canals, tanks, etc. causing spreading of waterborne diseases like dysentery, typhoid, cholera, jaundice, etc.
- **Industrial waste:** Discharge of untreated waste (waste from fish processing industries, breweries, dyeing paper, pharmaceutical industries, etc.) of industries into nearby rivers or lakes.
- **Oil spills:** Spills from oil wells and refineries and washing of tankers during loading and unloading. These spills are always in danger of catching fire and spreading over the surface of water and killing planktons etc.
- There is reduced oxygenation of underlying waters. As a result, many aquatic life forms such as fishes, etc., are killed.

Thermal Pollution:

- Thermal power plants, oil refineries, and nuclear power plants use water for cooling their machinery.
- Harmful Effects:
- Interferes in communication and concentration of thought.
- Lowers efficiency of work, disturbs sleep, can damage eardrum and lead to deafness.
- Bird life gets disturbed by aircraft landing or taking off from airport.

Controls:

- Prohibiting blowing of horns and restriction on loudspeakers (during night especially).
- Planting trees by the road, doors closed in the house, and do not burn fire-crackers.

Radiations:

- It is the form of energy which consists of high-energy particles such as photons, X-Rays, etc.
- Sources of radiation are mainly X-Rays and radioactive rays released from nuclear power plants.
- X-Rays contain a huge amount of energy particles which may damage the body cells.

Examples:

- Explosion in Japan (11 March 2011, Fukushima Daiichi Nuclear plant) and Chernobyl (1986, Ukraine) used Iodine-131, which was responsible for hemorrhage and even thyroid cancer.
- Carelessly discarded radioactive material like Cobalt-60 caused severe problems in Mayapuri, Delhi (2010) like burns and even deaths.

Effects of Various Types of Pollution

1. Air Pollution: Damage of kidney, liver, brain, muscles, reproductive system, etc.

- smog leads to asthma and poor visibility exhaust SO_2 cause bronchitis carbon monoxide binds with haemoglobin and causes.
- Deforestation, reducing CO_2 absorption through photosynthesis.
- In the last 30 years, Earth's temperature has risen by 0.5°C causing
  - Helling lie caps rising sea levels.
  - Decline in agriculture fishery resources.

Ozone Layer Depletion

- Ozone is a triatomic molecule C of oxygen found in the upper atmosphere. In the stratosphere, it helps in absorbing harmful UV radiations (UV-B) coming from the sun.
- The absorption of UV radiation is directly proportional to the thickness of ozone shields.
- Global average ozone layer thickness is declining because of the escape of CFCs, CH_4, and N_2O.
- Ozone depletion is particularly marked over the Antarctic region.

- This led to the formation of a large area of thinned ozone layer, called the ozone hole!
- Chlorofluorocarbons break down into chlorine atoms, which in turn break down ozone into oxygen (O_2) and O.
- Increase in UV radiations causes skin cancer, including melanoma.
- UV radiation reduces the immune system.
- UV radiation spoils photosynthesis and obstructs the food chain, indirectly affecting humans.

Abatement of Pollution

- Pollution cannot be totally stopped, but several steps can be taken to curtail it:
  - Diesel buses should be replaced by CNG buses.
  - Use of unleaded petrol in cars and two-wheelers.

16 CHAPTER

Human Evolution

Evolution

- Evolution can be defined as the formation of more complex organisms from pre-existing simpler organisms over a certain period.
- It is a slow but progressive, natural, sequential development or transformation of animals and plants from ancestors of different forms and functions.
- Variations and heredity are the two basic factors of evolution.
- The selection of variants by environmental factors forms the basis of evolutionary processes.

Theories of Evolution

- Two modern theories have been put forward to explain the mode of evolution.

Theory of Lamarckism

- As per the theory of inheritance of acquired traits, changes in the structure or function of an organ acquired during an individual's lifetime in response to environmental changes are passed on to offspring and accumulate gradually over time.
- The theory of Lamarckism was revived with new facts, which led to the theory of Neo-Lamarckism, which stated that the acquired characters which become incorporated in the germplasm are heritable and accumulate generation after generation, resulting in the origin of new species.

Vestigial Organs

- Vestigial organs, also known as non-functional organs, are reduced or rudimentary organs that don't serve a purpose in the body.
- They help us learn about the history of evolution and the continuity of life.
- Unlike monkeys, humans do not have tails, but they do possess a rudimentary tail bone.
- Wisdom teeth typically emerge between the ages of 17 and 20 and are rarely utilized for chewing food.
- The vermiform appendix is a function-less organ in humans but is helpful for ruminants for digesting cellulose.
- The pinna, a vestigial organ, acts as a passage from the auditory canal to the eardrum. However, it is under-developed in humans.

Darwin's Theory of Natural Selection

- All organisms have the capacity to reproduce at a very high rate.
- The overproduction of organisms results in competition among.

- The overproduction of organisms results in competition among individuals within the population for resources such as food and space.
- Only organisms that are well-adapted to the changing environment are able to survive.
- Organisms which are unfit are eliminated and ultimately die.
- the struggle for existence, the organisms which develop new or favorable characteristics will survive in the long run. This idea is called survival of the fittest.
- Organisms that survive will pass on favorable traits to their offspring.
- These characters get accumulated and give rise to a new species.

Human Evolution

- The evolution of modern humans is considered one of the most significant developments in evolutionary history. Human evolution has been examined through methods including excavation, carbon dating, fossil analysis, and DNA sequencing to trace evolutionary relationships.
- A vast diversity has always been observed with respect to the human body and its features.
- The research indicates that early Homo sapiens originated in Africa approximately one hundred thousand years ago. Some of our ancestors migrated out of Africa, while others remained. Therefore, regardless of our current geographic location, all human species are ultimately descendants of African origins.
- The earliest human fossils include the Australopithecus genus, followed by Homo habilis, H. erectus, H. heidelbergensis, and eventually modern Homo sapiens.
- About 25-30 million years ago, ape-human or hominoid stock started descending from the trees and gradually became ground dwellers who evolved into apes and men.
- Subsequent separation of ape and human ancestors is regarded to begin about 5 million years ago, which led to the evolution of Pongidae (apes) and Hominidae (men).

Ramapithecus:

- Approximately 10-15 million years ago, the earliest fossil resembling humans was discovered in Africa and Asia.
- They were capable of walking erect on their feet, had small canines, and had a short face and small brain.

Australopithecus afarensis:

- They appeared in South Africa and had common traits to both human and ape features.
- He was about 1.5 meters tall and was mainly a terrestrial creature with bipedal locomotion.
- His cranial capacity was 500 cubic centimeters.

Australopithecus africanus:

- They appeared 2.5 million years ago in Africa.
- He has a low forehead, protruding face, lack of chin, and low brain capacity (350-450 cubic centimetres).

Homo habilis:

- He lived in Africa 2 million years ago and was about 1.5 to 1.8 meters tall.
- His cranial cavity or brain volume ranged from 650 to 800 cubic centimetres, larger than that of Australopithecus.

Homo erectus:

- Around 1.7 million years ago, they evolved from Homo habilis or Australopithecus. Fossils of these individuals have been discovered in Java, Peking, Heidelberg, and Europe.
- He stood approximately 5.5 feet tall and had a bowl-shaped pelvis. His foot had a pronounced arch to support body weight, and the ability to grasp with the foot was completely lost.
- The size of the cranial cavity ranged from 750 to 1100 cc.

Homo neanderthalensis:

- These early men arose 1,00,000 years ago and flourished in Europe and Asia but became extinct about 25,000 years ago.
- They possessed a flat cranium, a sloping forehead, protruding jaws, strong mandibles, and lacked a chin.
- They demonstrated intelligence, were skilled hunters, utilized animal skin for clothing, and practiced the burial of the deceased.

Homo sapiens fossils (Cro-Magnon man: Early Modern Man):

- These fossils, found approximately 5000 years ago, were discovered from caves in northwest Italy and rock shelter caves in France.
- The Cro-Magnon man was 1.8 meters tall with a sturdy body and less hair. The cranial capacity was about 1650 cc.

Homo sapiens sapiens (Modern Man):

- The living modern men have evolved from the Cro-Magnon man about 25,000 years ago.
- They had undergone minor morphological changes such as thinning of skull bones and slight reduction in cranial capacity (1300-1600 cc).

MEET OUR PW SUPERSTARS

AIR
12
MOHIT PRAVIN AGRAWAL
IIT BOMBAY
JEE ADVANCED 2025

AIR
3
MAJID MUJAHID HUSAIN
IIT BOMBAY
JEE ADVANCED 2025

AIR
48
RISHABH VIVEK IYER
IIT BOMBAY
JEE ADVANCED 2025

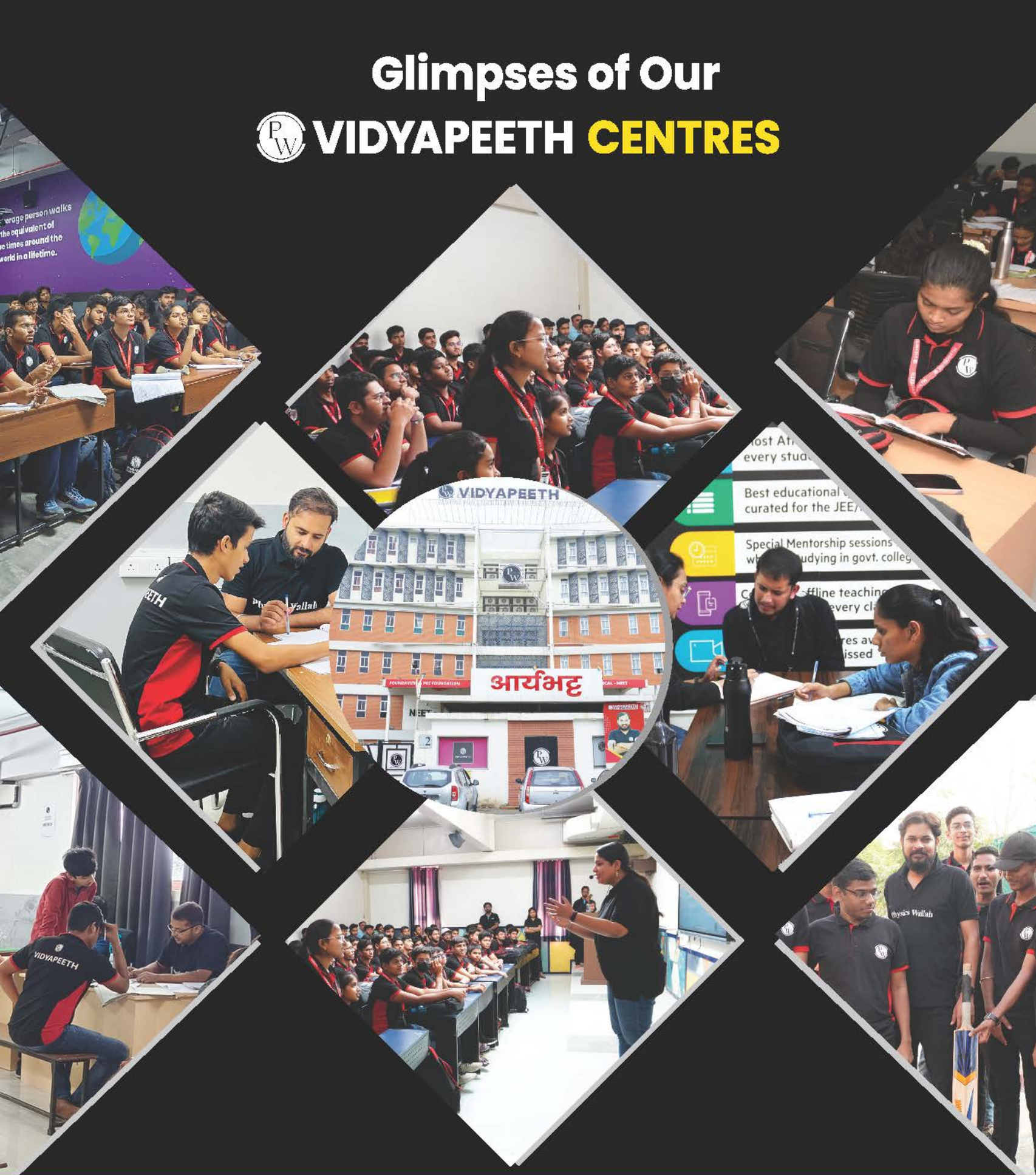
AIR
9
SHIVEN VIKAS TOSHNIWAL
IIT BOMBAY
MARKS IMPROVEMENT PROGRAM
JEE MAIN 2025

AIR
87
RAGHAV GOYAL
AIIMS JODHPUR
NEET (UG) 2025

AIR
100
MUSKAN
MAMC
MARKS IMPROVEMENT PROGRAM
NEET (UG) 2025

AIR
12
SAURAV
IIT KHARAGPUR
JEE MAIN 2025

Glimpses of Our
VIDYAPEETH CENTRES



ADMISSIONS OPEN

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